

Bottom-Up/Shift-Reduce Parsing

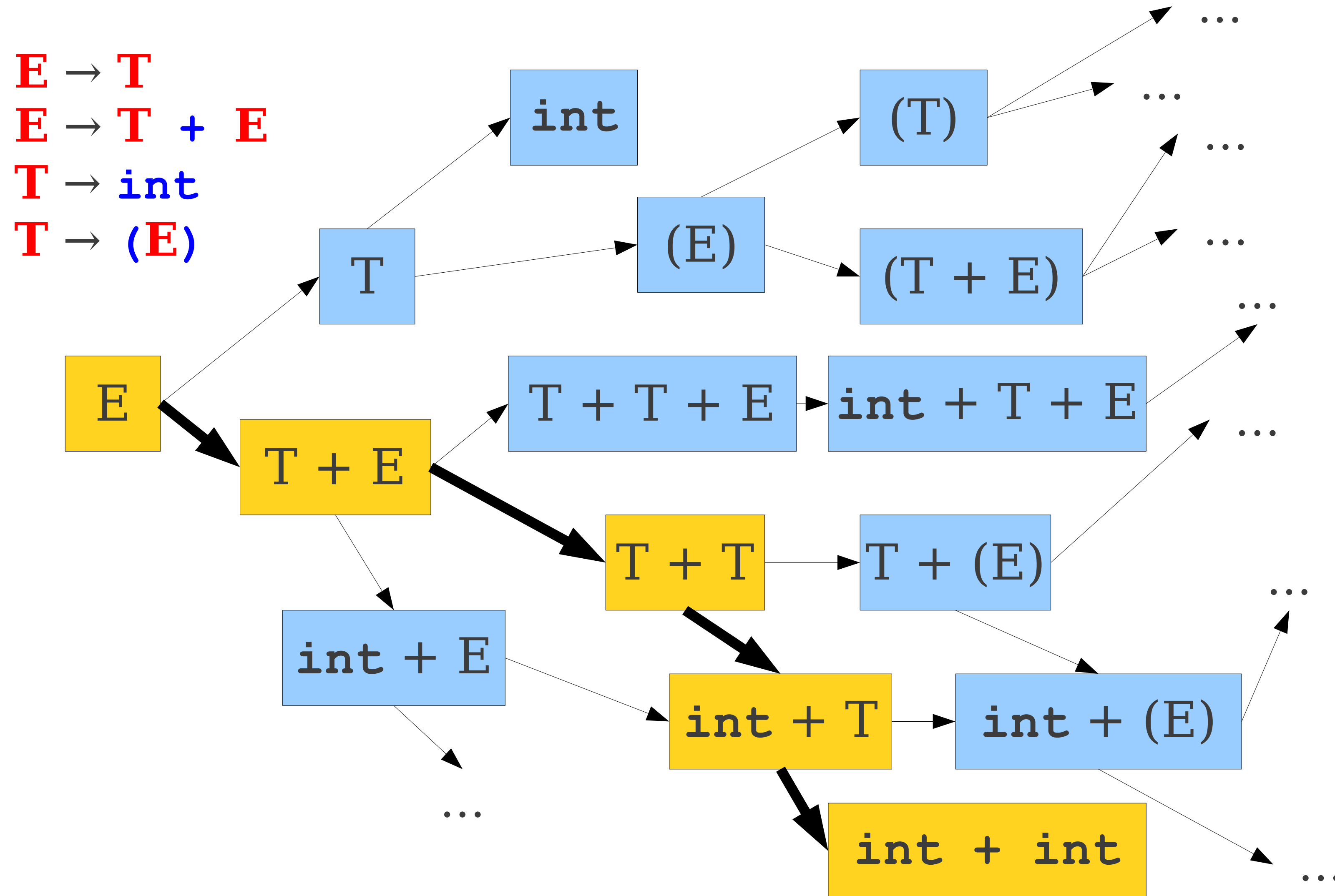
Parsing with CFGs

- Last Time: Top-down/Predictive/LL(1)
 - Fast, simple enough for hand-written parsers
 - Too weak for many practical languages
- This week: Bottom-up/Shift-Reduce/LR(0)/SLR/LR(1)
 - More Expressive
 - Too complicated for hand-written, instead use parser generators

Top-Down vs Bottom-Up

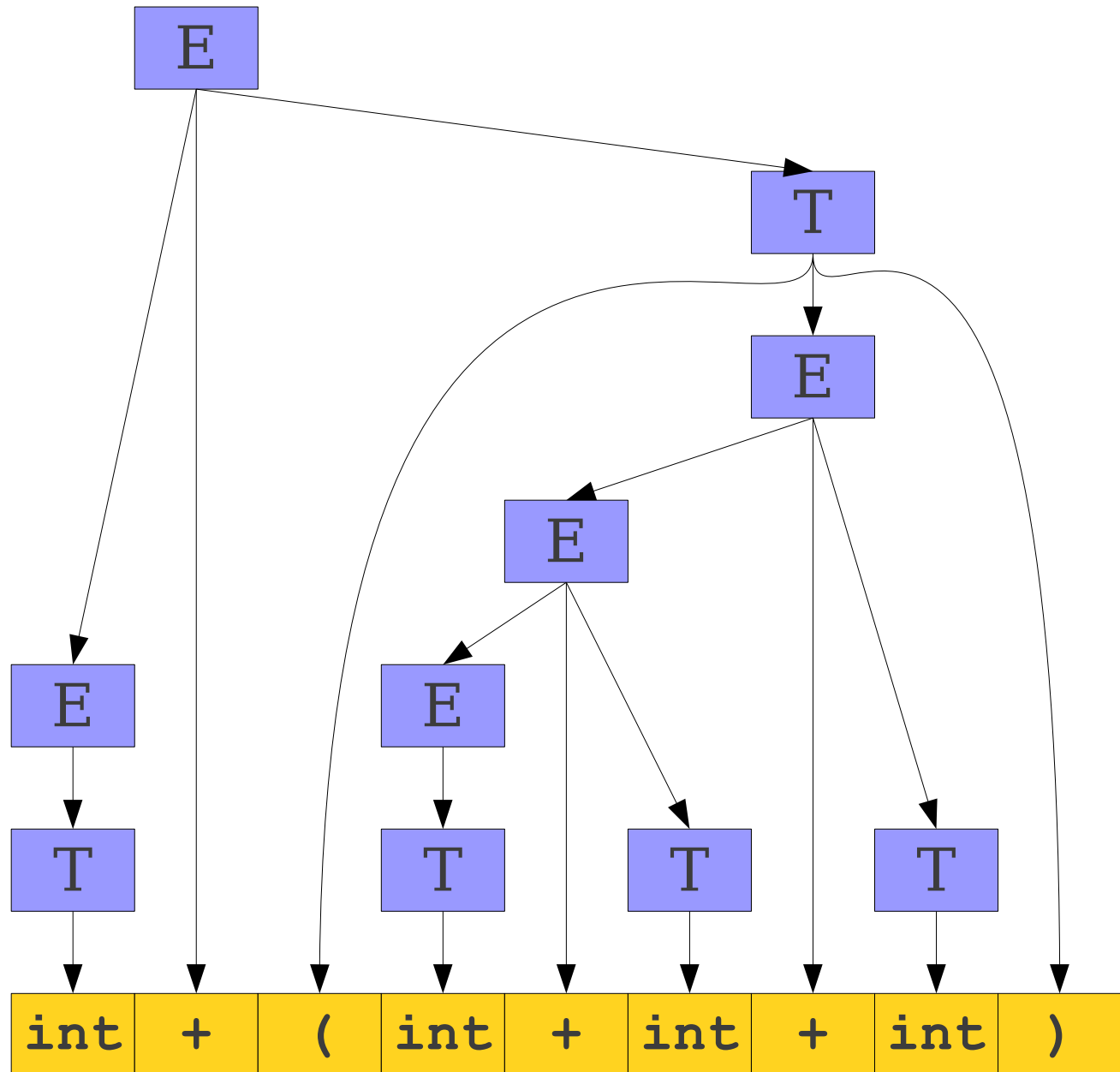
- Top-Down
 - Start from the start symbol S
 - Based on the input, rewrite to a new state $aSb \rightarrow agb$
 - Succeed when all input is read, the state is the input sequence
- Bottom-Up
 - Start from the input sequence w
 - Based on the input, rewrite
 - Succeed when all input is read, the state is the start symbol

Top-Down



One View of a Bottom-Up Parse

$E \rightarrow T$
 $E \rightarrow E + T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



A Second View of a Bottom-Up Parse

$E \rightarrow T$	$\text{int} + (\text{int} + \text{int} + \text{int})$
$E \rightarrow E + T$	$\Rightarrow T + (\text{int} + \text{int} + \text{int})$
$T \rightarrow \text{int}$	$\Rightarrow E + (\text{int} + \text{int} + \text{int})$
$T \rightarrow (E)$	$\Rightarrow E + (T + \text{int} + \text{int})$
	$\Rightarrow E + (E + \text{int} + \text{int})$
	$\Rightarrow E + (E + T + \text{int})$
	$\Rightarrow E + (E + \text{int})$
	$\Rightarrow E + (E + T)$
	$\Rightarrow E + (E)$
	$\Rightarrow E + T$
	$\Rightarrow E$

A left-to-right, bottom-up parse is a rightmost derivation traced in reverse.

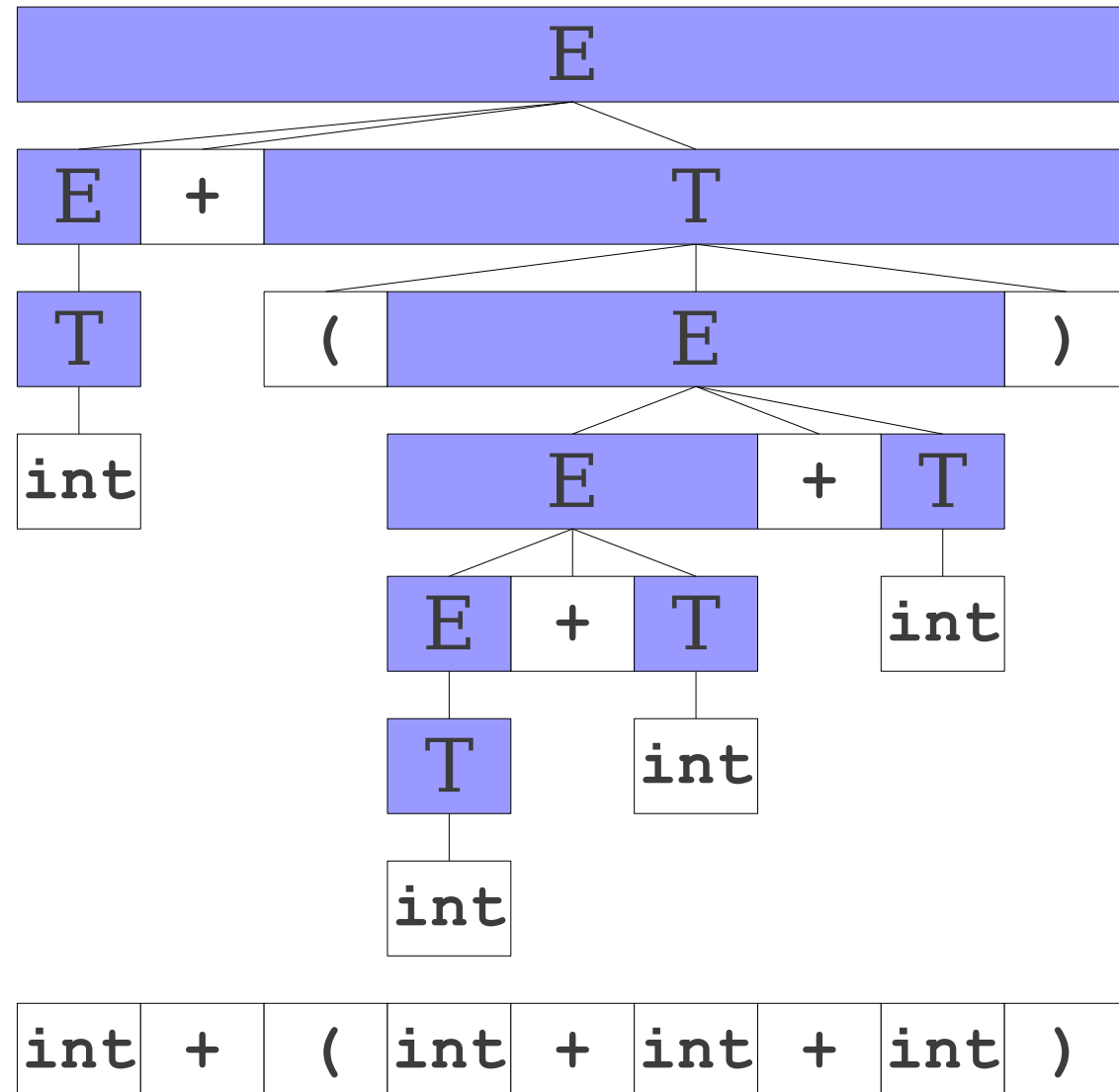
A Third View of a Bottom-Up Parse

`int + (int + int + int)`
⇒ `T + (int + int + int)`
⇒ `E + (int + int + int)`
⇒ `E + (T + int + int)`
⇒ `E + (E + int + int)`
⇒ `E + (E + T + int)`
⇒ `E + (E + int)`
⇒ `E + (E + T)`
⇒ `E + (E)`
⇒ `E + T`
⇒ `E`

Each step in this bottom-up parse is called a **reduction**.
We **reduce** a substring of the sentential form back to a nonterminal.

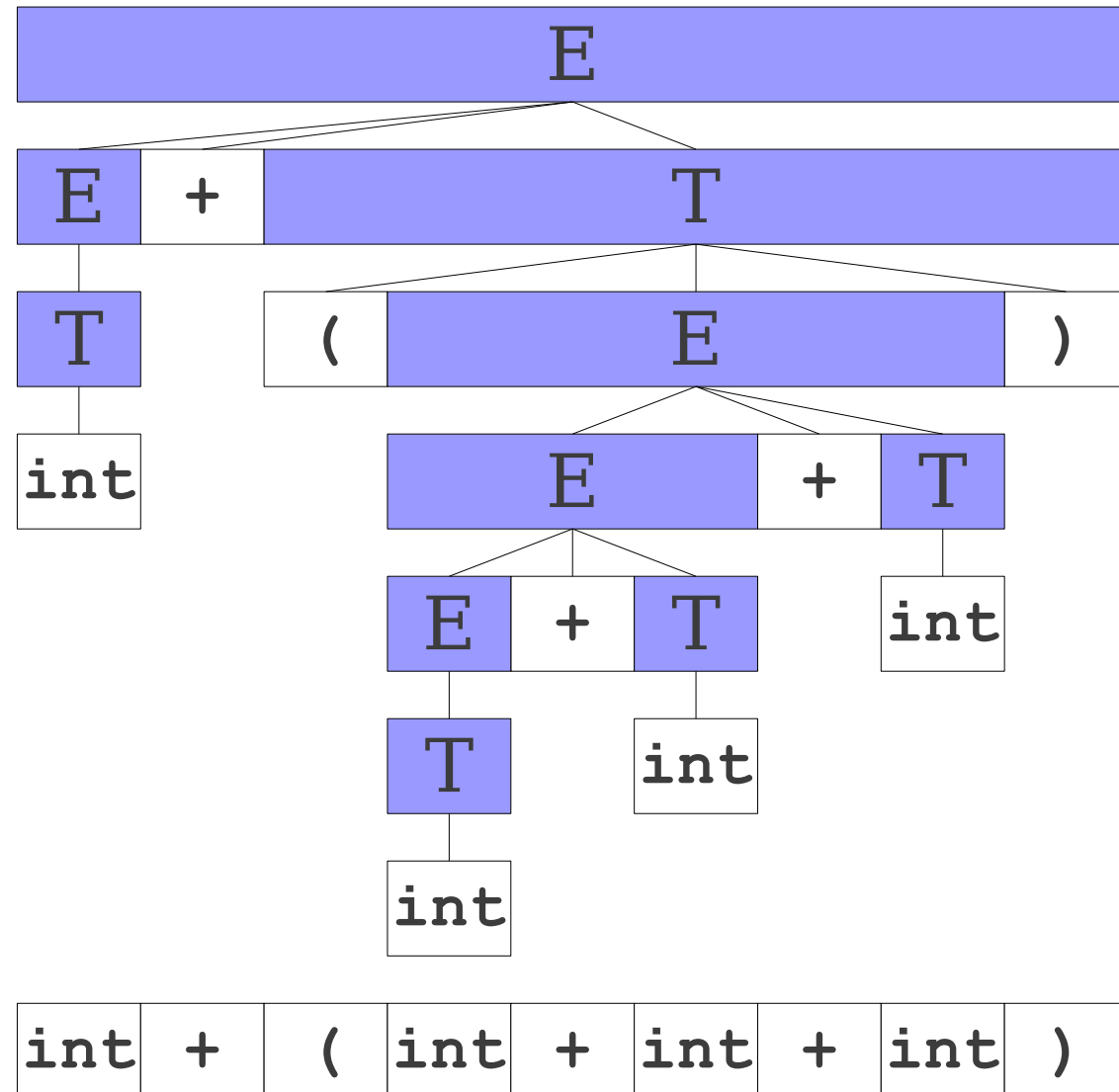
A Third View of a Bottom-Up Parse

`int + (int + int + int)`
⇒ **T** + (int + int + int)
⇒ **E** + (int + int + int)
⇒ **E** + (**T** + int + int)
⇒ **E** + (**E** + int + int)
⇒ **E** + (**E** + **T** + int)
⇒ **E** + (**E** + int)
⇒ **E** + (**E** + **T**)
⇒ **E** + (**E**)
⇒ **E** + **T**
⇒ **E**



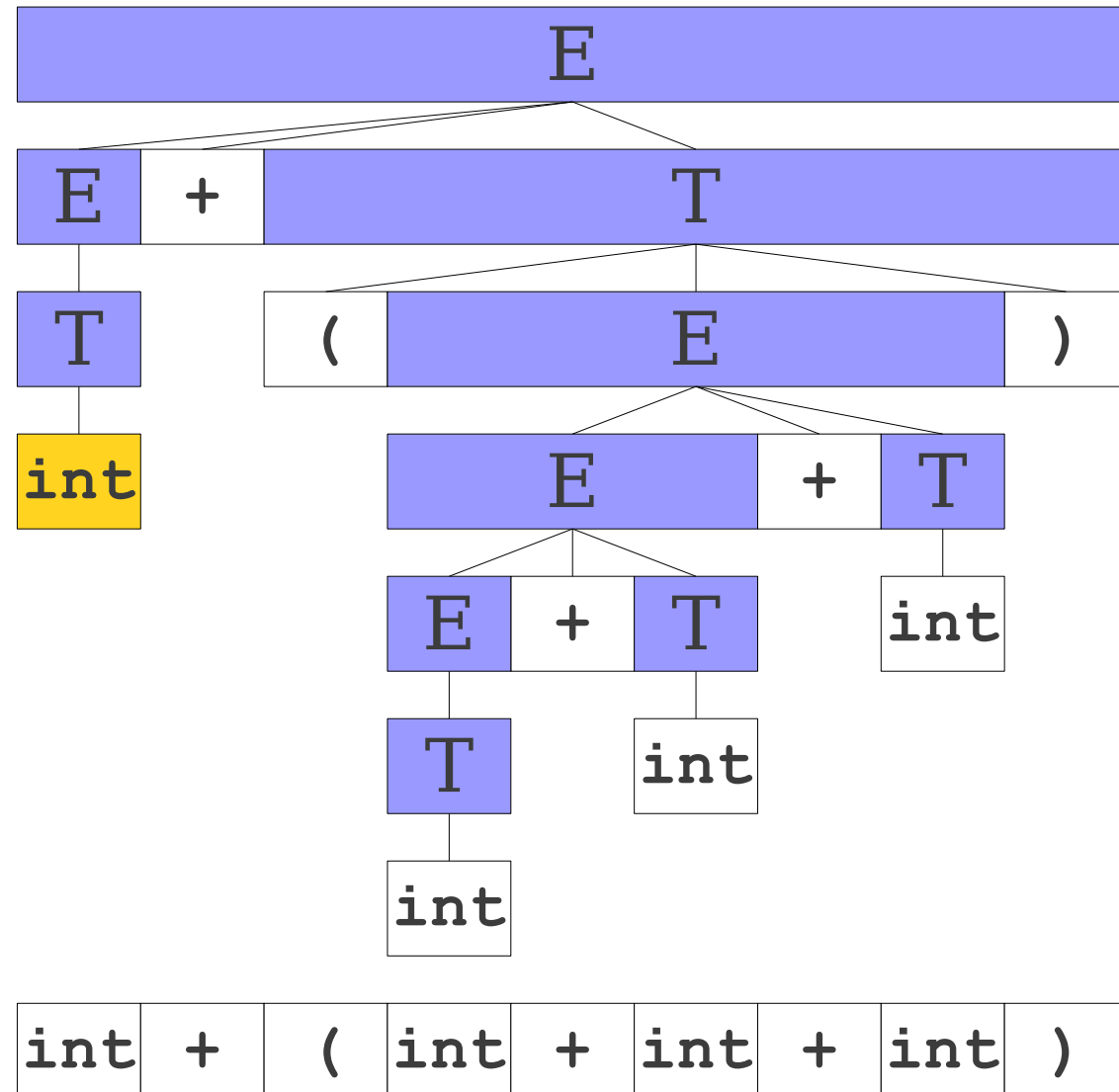
A Third View of a Bottom-Up Parse

int + (int + int + int)
⇒ **T** + (int + int + int)
⇒ **E** + (int + int + int)
⇒ **E** + (**T** + int + int)
⇒ **E** + (**E** + int + int)
⇒ **E** + (**E** + **T** + int)
⇒ **E** + (**E** + int)
⇒ **E** + (**E** + **T**)
⇒ **E** + (**E**)
⇒ **E** + **T**
⇒ **E**



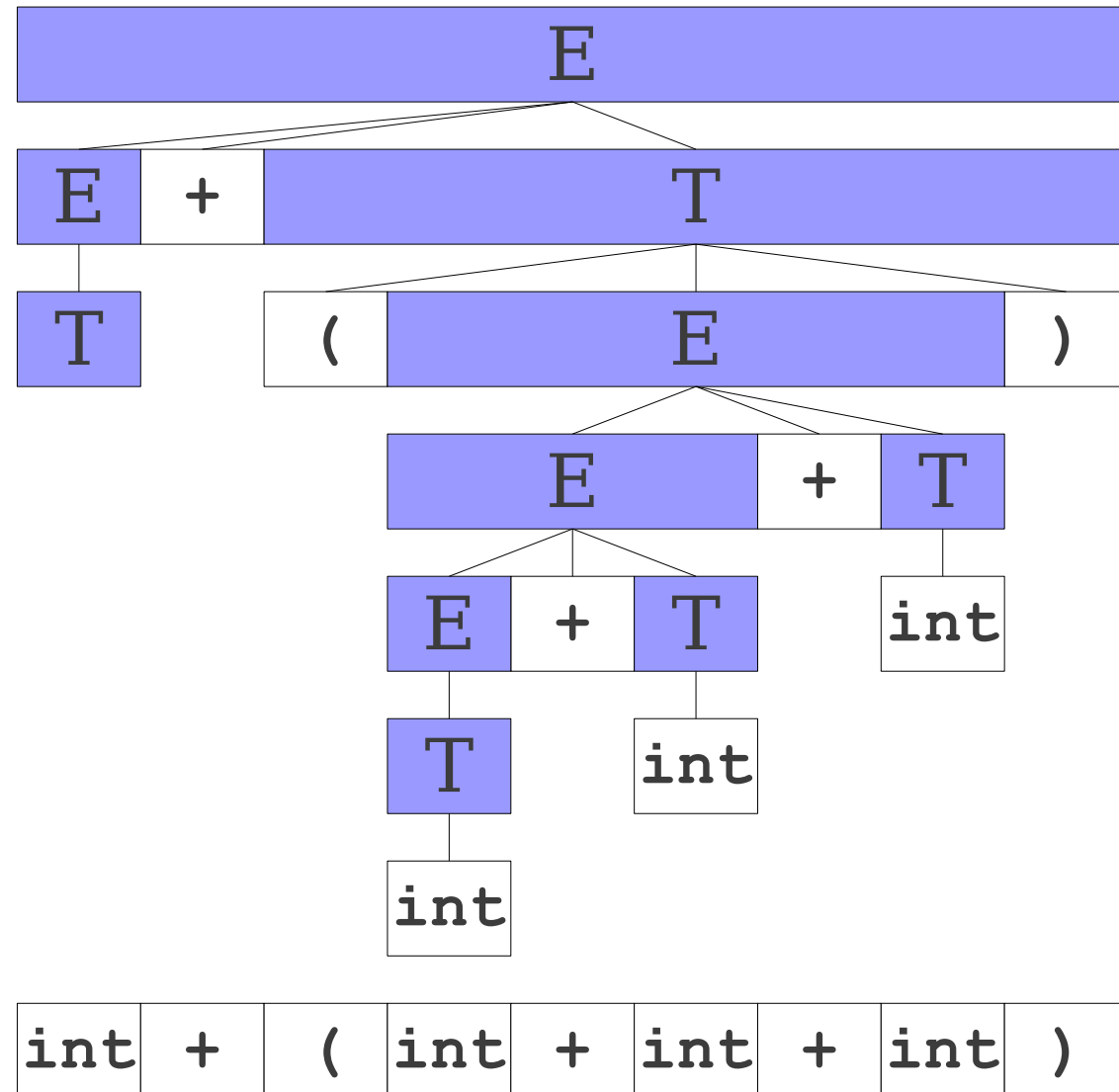
A Third View of a Bottom-Up Parse

int + (int + int + int)
⇒ **T** + (int + int + int)
⇒ **E** + (int + int + int)
⇒ **E** + (**T** + int + int)
⇒ **E** + (**E** + int + int)
⇒ **E** + (**E** + **T** + int)
⇒ **E** + (**E** + int)
⇒ **E** + (**E** + **T**)
⇒ **E** + (**E**)
⇒ **E** + **T**
⇒ **E**



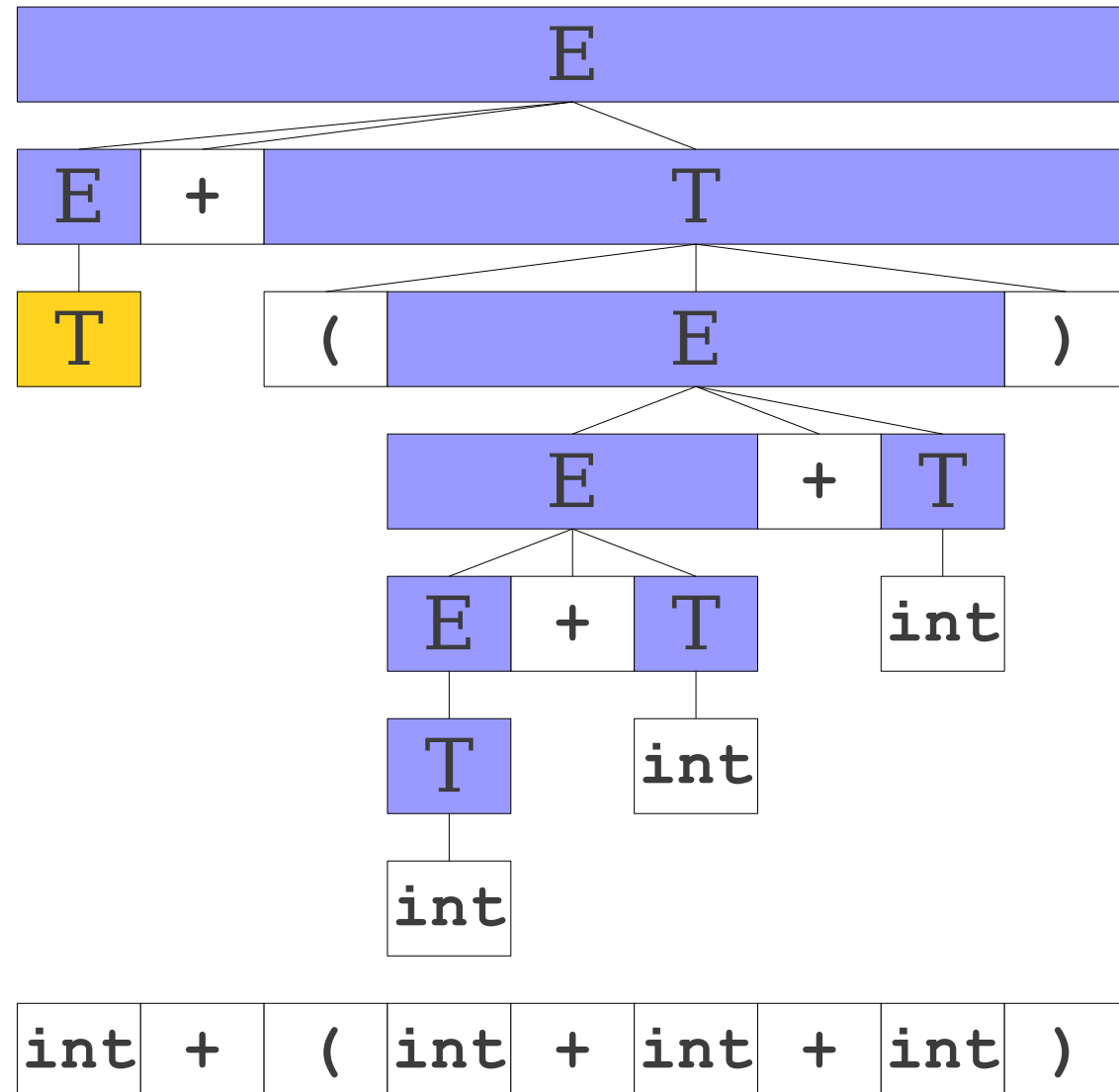
A Third View of a Bottom-Up Parse

$\Rightarrow T + (int + int + int)$
 $\Rightarrow E + (int + int + int)$
 $\Rightarrow E + (T + int + int)$
 $\Rightarrow E + (E + int + int)$
 $\Rightarrow E + (E + T + int)$
 $\Rightarrow E + (E + int)$
 $\Rightarrow E + (E + T)$
 $\Rightarrow E + (E)$
 $\Rightarrow E + T$
 $\Rightarrow E$



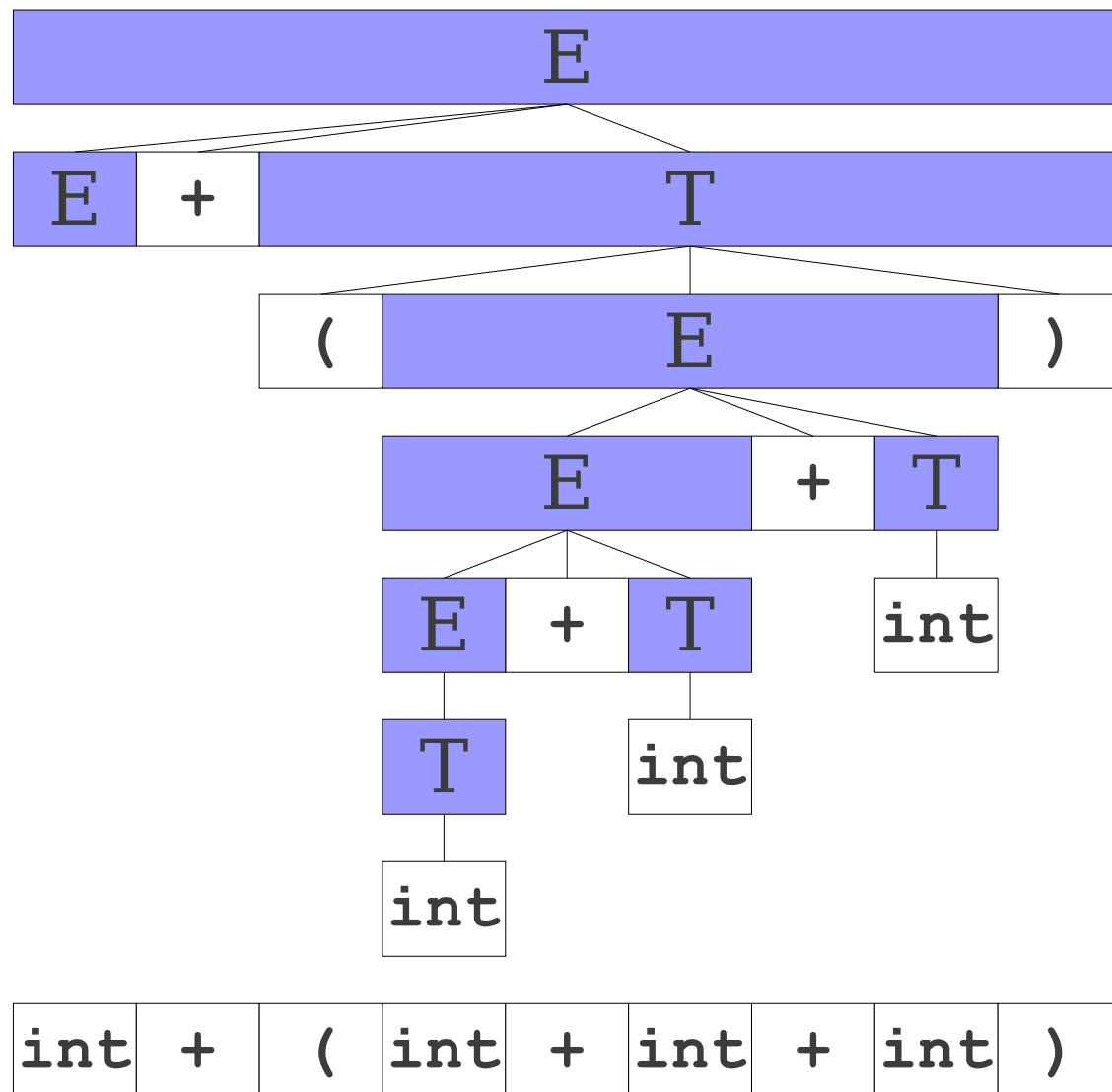
A Third View of a Bottom-Up Parse

$\Rightarrow$ **T** + (int + int + int)
 $\Rightarrow$ **E** + (int + int + int)
 $\Rightarrow$ **E** + (**T** + int + int)
 $\Rightarrow$ **E** + (**E** + int + int)
 $\Rightarrow$ **E** + (**E** + **T** + int)
 $\Rightarrow$ **E** + (**E** + int)
 $\Rightarrow$ **E** + (**E** + **T**)
 $\Rightarrow$ **E** + (**E**)
 $\Rightarrow$ **E** + **T**
 $\Rightarrow$ **E**



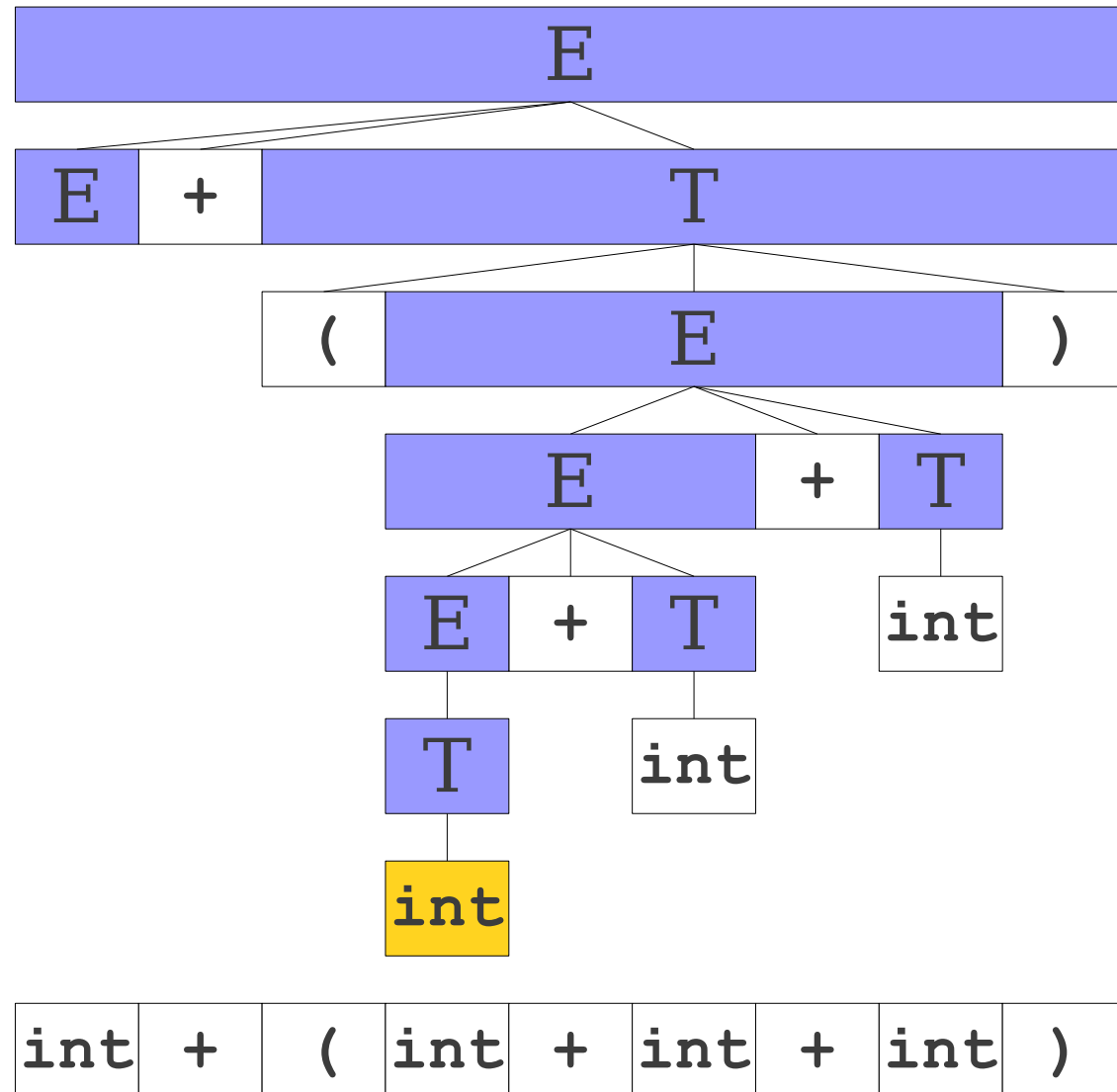
A Third View of a Bottom-Up Parse

$\Rightarrow E + (int + int + int)$
 $\Rightarrow E + (T + int + int)$
 $\Rightarrow E + (E + int + int)$
 $\Rightarrow E + (E + T + int)$
 $\Rightarrow E + (E + int)$
 $\Rightarrow E + (E + T)$
 $\Rightarrow E + (E)$
 $\Rightarrow E + T$
 $\Rightarrow E$



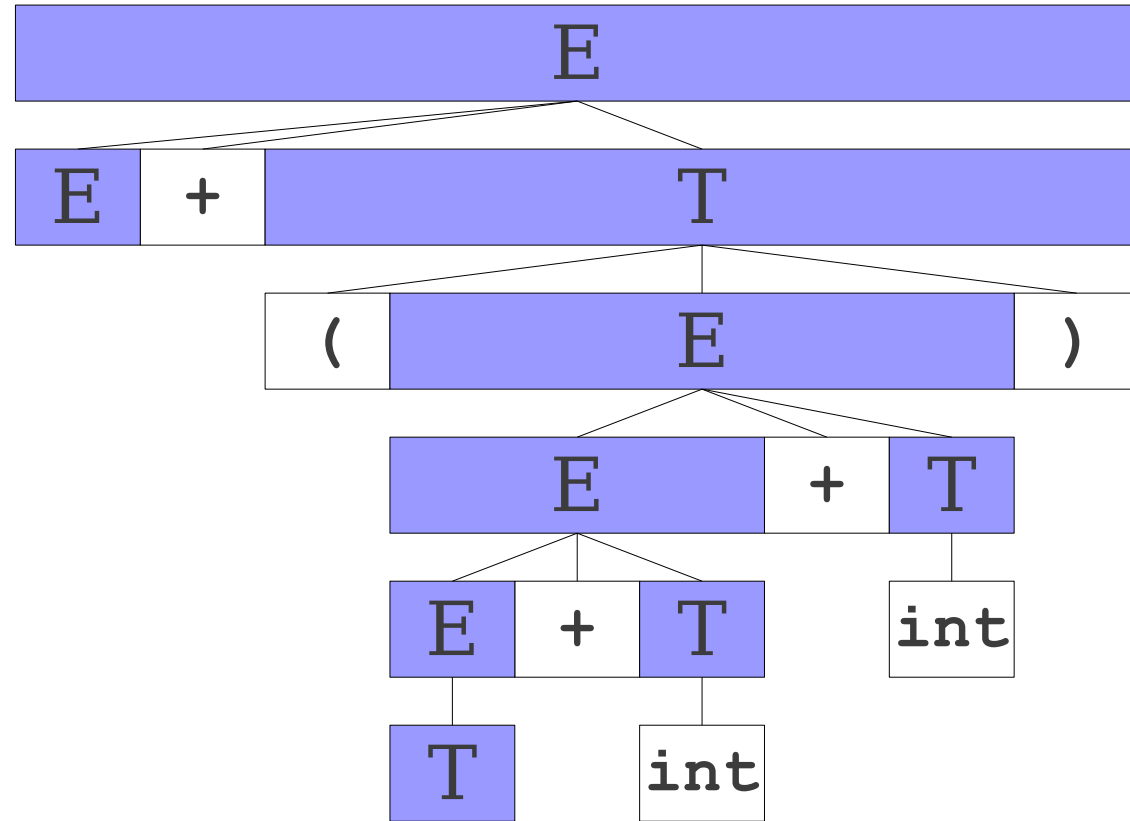
A Third View of a Bottom-Up Parse

$\Rightarrow E + (\text{int} + \text{int} + \text{int})$
 $\Rightarrow E + (T + \text{int} + \text{int})$
 $\Rightarrow E + (E + \text{int} + \text{int})$
 $\Rightarrow E + (E + T + \text{int})$
 $\Rightarrow E + (E + \text{int})$
 $\Rightarrow E + (E + T)$
 $\Rightarrow E + (E)$
 $\Rightarrow E + T$
 $\Rightarrow E$



A Third View of a Bottom-Up Parse

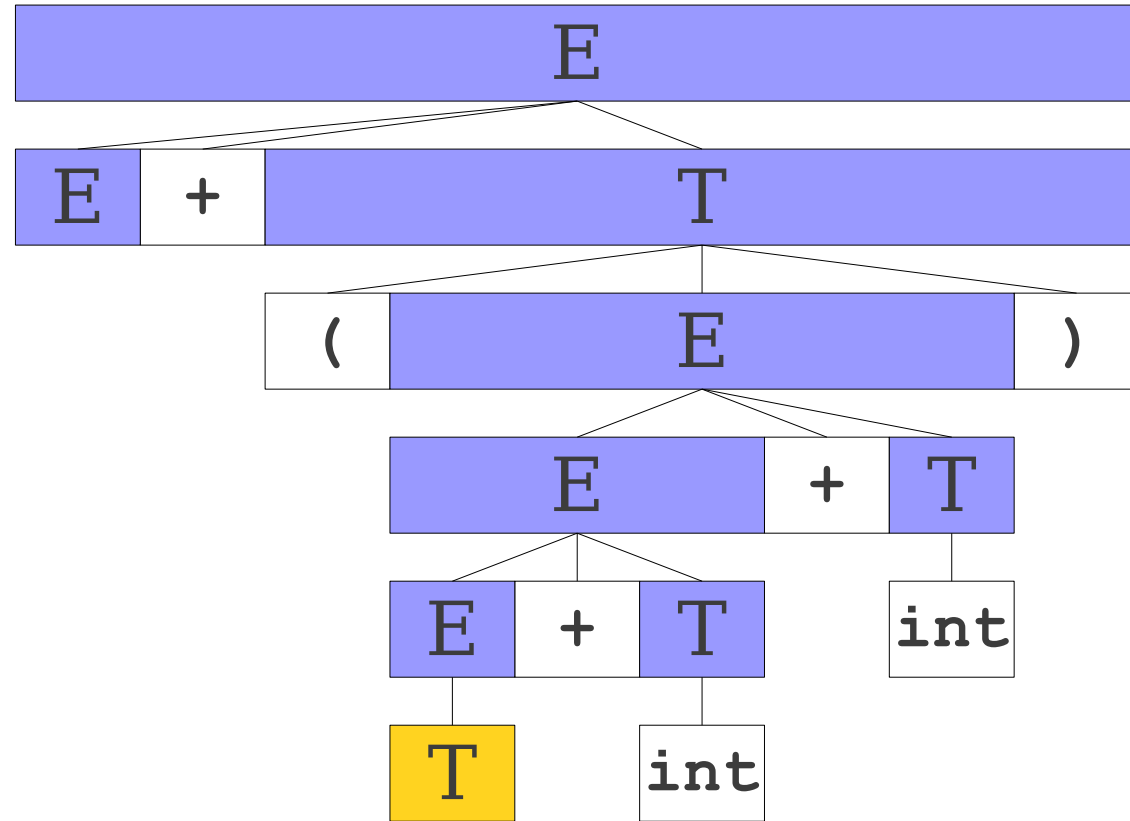
$\Rightarrow E + (T + \text{int} + \text{int})$
 $\Rightarrow E + (E + \text{int} + \text{int})$
 $\Rightarrow E + (E + T + \text{int})$
 $\Rightarrow E + (E + \text{int})$
 $\Rightarrow E + (E + T)$
 $\Rightarrow E + (E)$
 $\Rightarrow E + T$
 $\Rightarrow E$



int	+	(	int	+	int	+	int	)
-----	---	---	-----	---	-----	---	-----	---

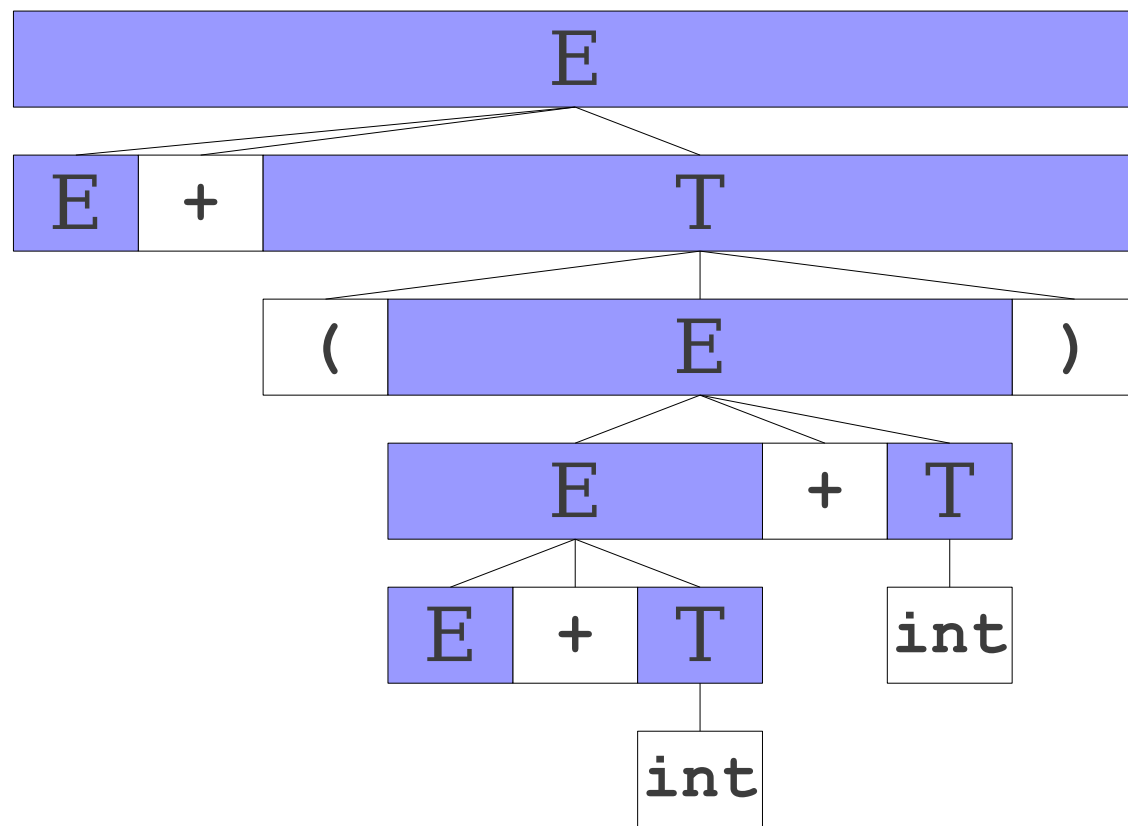
A Third View of a Bottom-Up Parse

$\Rightarrow E + (T + \text{int} + \text{int})$
 $\Rightarrow E + (E + \text{int} + \text{int})$
 $\Rightarrow E + (E + T + \text{int})$
 $\Rightarrow E + (E + \text{int})$
 $\Rightarrow E + (E + T)$
 $\Rightarrow E + (E)$
 $\Rightarrow E + T$
 $\Rightarrow E$



int	+	(	int	+	int	+	int	)
-----	---	---	-----	---	-----	---	-----	---

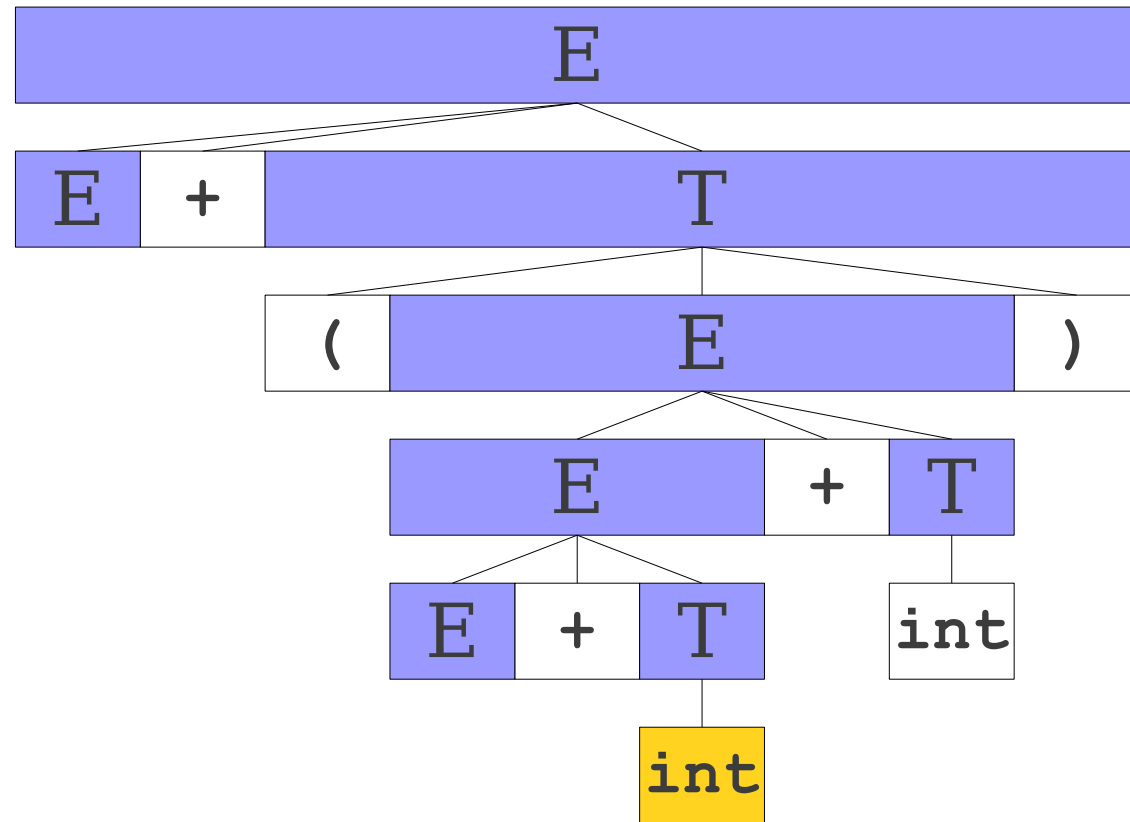
A Third View of a Bottom-Up Parse


$$\begin{aligned} &\Rightarrow E + (E + \text{int} + \text{int}) \\ &\Rightarrow E + (E + T + \text{int}) \\ &\Rightarrow E + (E + \text{int}) \\ &\Rightarrow E + (E + T) \\ &\Rightarrow E + (E) \\ &\Rightarrow E + T \\ &\Rightarrow E \end{aligned}$$

int	+	(	int	+	int	+	int	)
-----	---	---	-----	---	-----	---	-----	---

A Third View of a Bottom-Up Parse

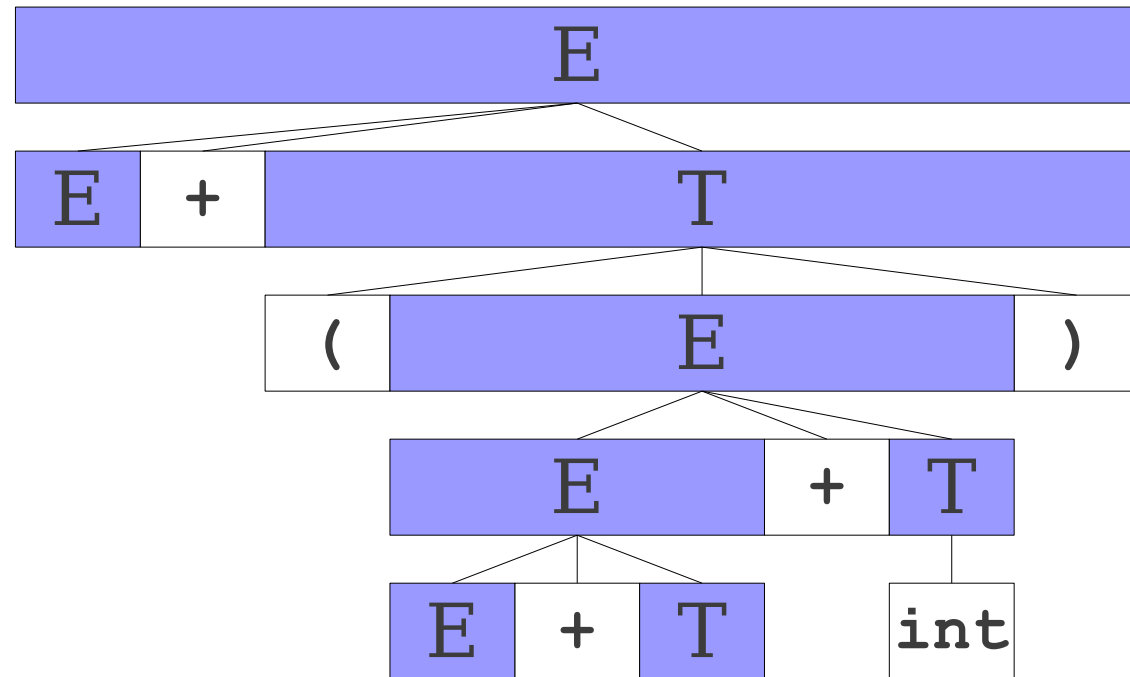
$\Rightarrow E + (E + \text{int} + \text{int})$
 $\Rightarrow E + (E + T + \text{int})$
 $\Rightarrow E + (E + \text{int})$
 $\Rightarrow E + (E + T)$
 $\Rightarrow E + (E)$
 $\Rightarrow E + T$
 $\Rightarrow E$



$\text{int} + (\text{int} + \text{int} + \text{int})$

A Third View of a Bottom-Up Parse

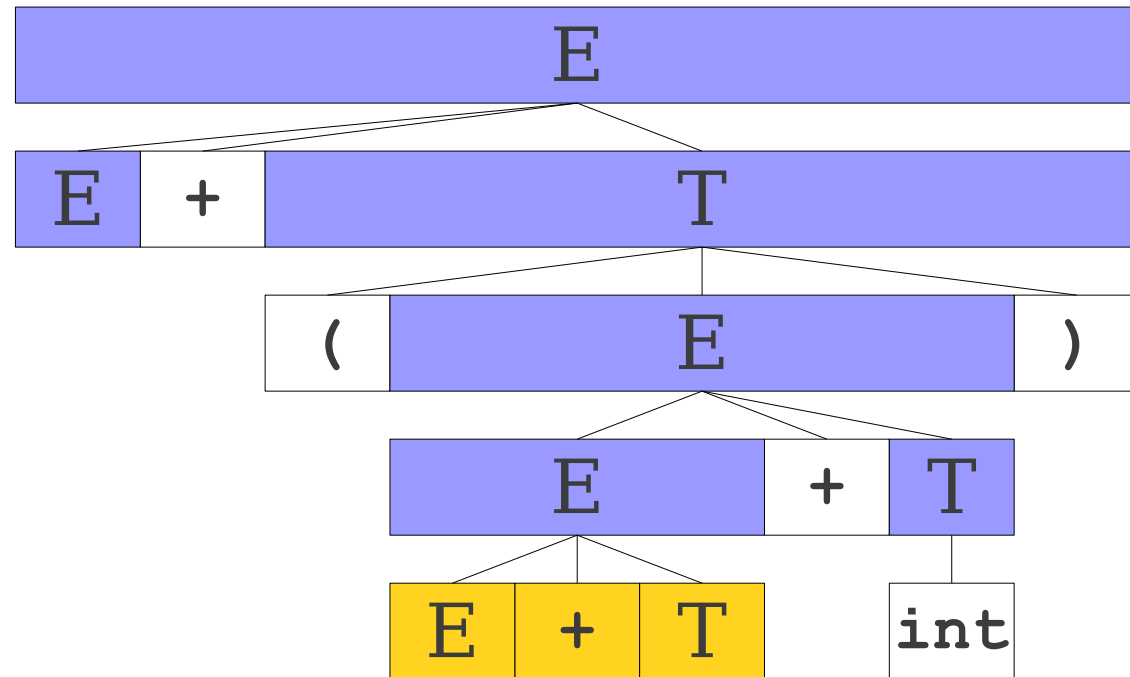
$\Rightarrow E + (E + T + \text{int})$
 $\Rightarrow E + (E + \text{int})$
 $\Rightarrow E + (E + T)$
 $\Rightarrow E + (E)$
 $\Rightarrow E + T$
 $\Rightarrow E$



int	+	(	int	+	int	+	int	)
-----	---	---	-----	---	-----	---	-----	---

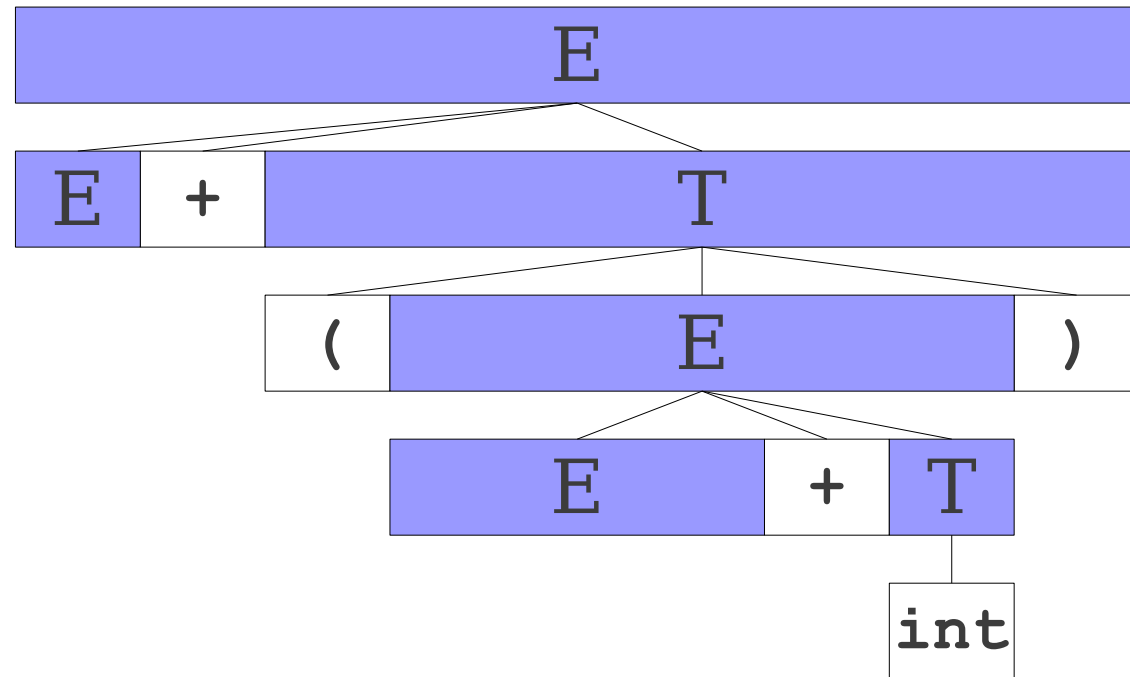
A Third View of a Bottom-Up Parse

$\Rightarrow E + (E + T + \text{int})$
 $\Rightarrow E + (E + \text{int})$
 $\Rightarrow E + (E + T)$
 $\Rightarrow E + (E)$
 $\Rightarrow E + T$
 $\Rightarrow E$



int	+	(	int	+	int	+	int	)
-----	---	---	-----	---	-----	---	-----	---

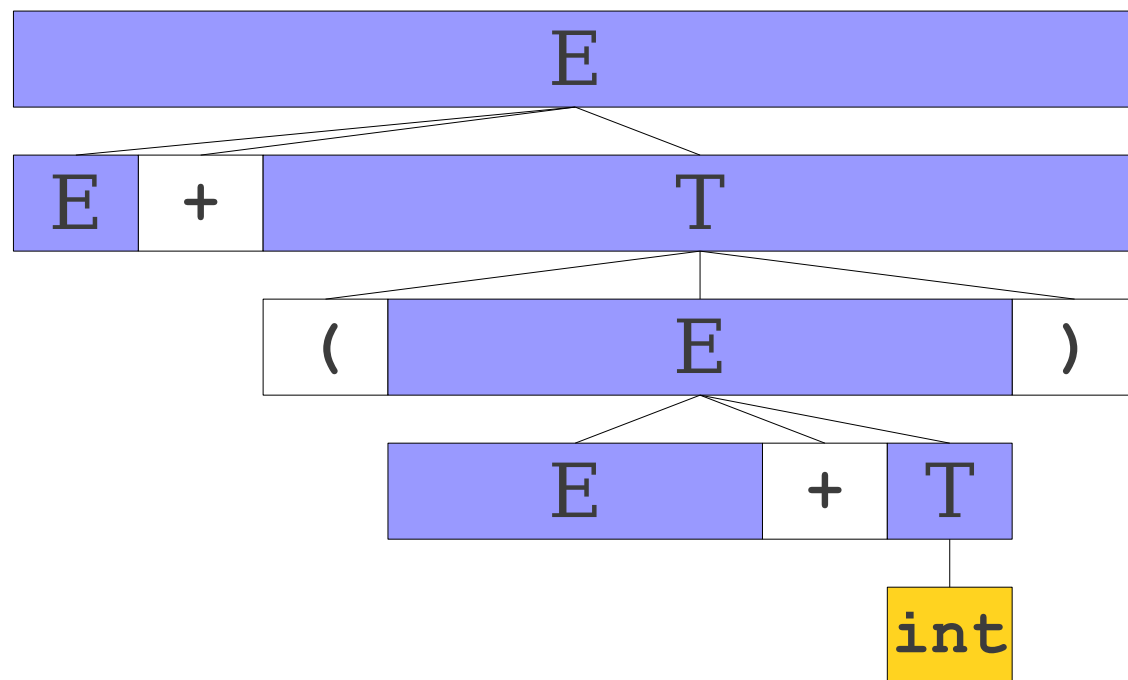
A Third View of a Bottom-Up Parse



$\Rightarrow \mathbf{E} + (\mathbf{E} + \mathbf{int})$
 $\Rightarrow \mathbf{E} + (\mathbf{E} + \mathbf{T})$
 $\Rightarrow \mathbf{E} + (\mathbf{E})$
 $\Rightarrow \mathbf{E} + \mathbf{T}$
 $\Rightarrow \mathbf{E}$

int	+	(	int	+	int	+	int	)
-----	---	---	-----	---	-----	---	-----	---

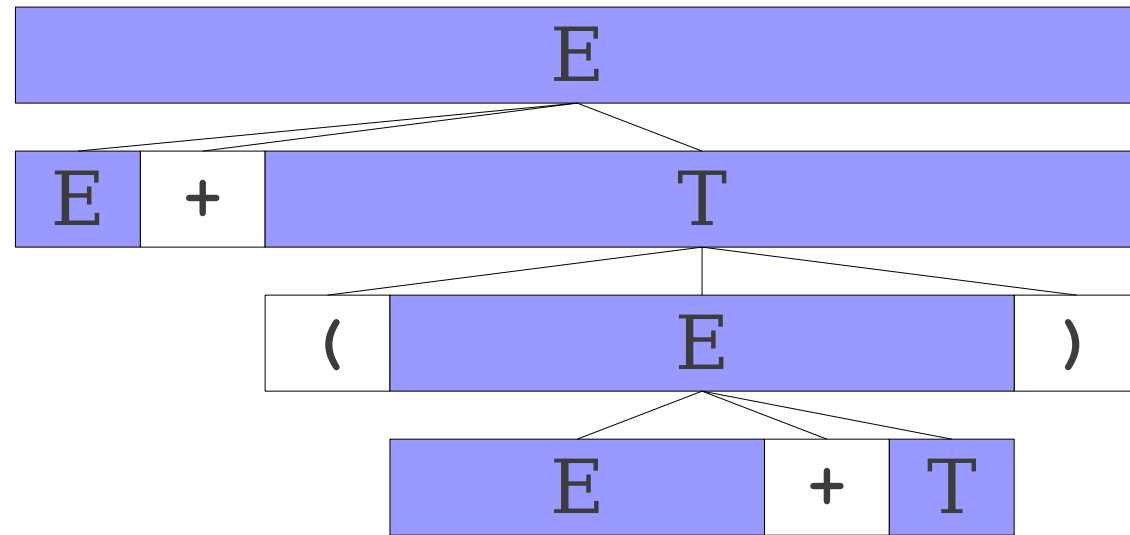
A Third View of a Bottom-Up Parse



$\Rightarrow E + (E + \text{int})$
 $\Rightarrow E + (E + T)$
 $\Rightarrow E + (E)$
 $\Rightarrow E + T$
 $\Rightarrow E$

int	+	(	int	+	int	+	int	)
-----	---	---	-----	---	-----	---	-----	---

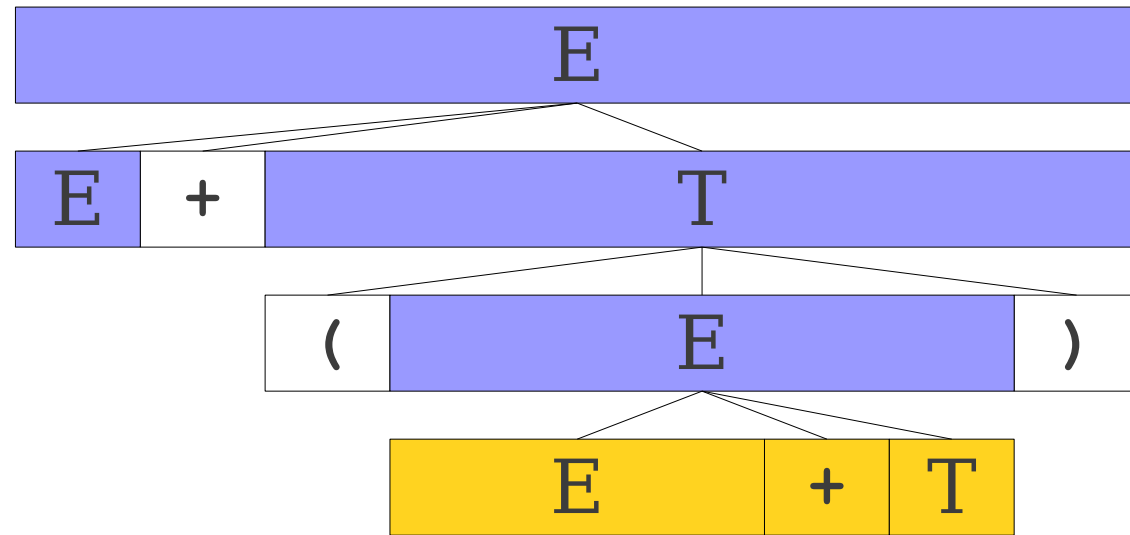
A Third View of a Bottom-Up Parse



$\Rightarrow \mathbf{E} + (\mathbf{E} + \mathbf{T})$
 $\Rightarrow \mathbf{E} + (\mathbf{E})$
 $\Rightarrow \mathbf{E} + \mathbf{T}$
 $\Rightarrow \mathbf{E}$

int	+	(	int	+	int	+	int	)
-----	---	---	-----	---	-----	---	-----	---

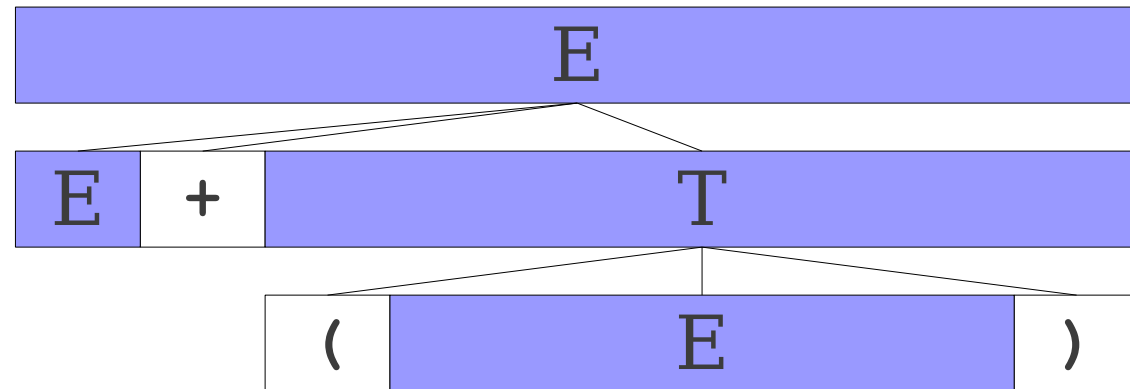
A Third View of a Bottom-Up Parse



$\Rightarrow \mathbf{E} + (\mathbf{E} + \mathbf{T})$
 $\Rightarrow \mathbf{E} + (\mathbf{E})$
 $\Rightarrow \mathbf{E} + \mathbf{T}$
 $\Rightarrow \mathbf{E}$

int	+	(	int	+	int	+	int	)
-----	---	---	-----	---	-----	---	-----	---

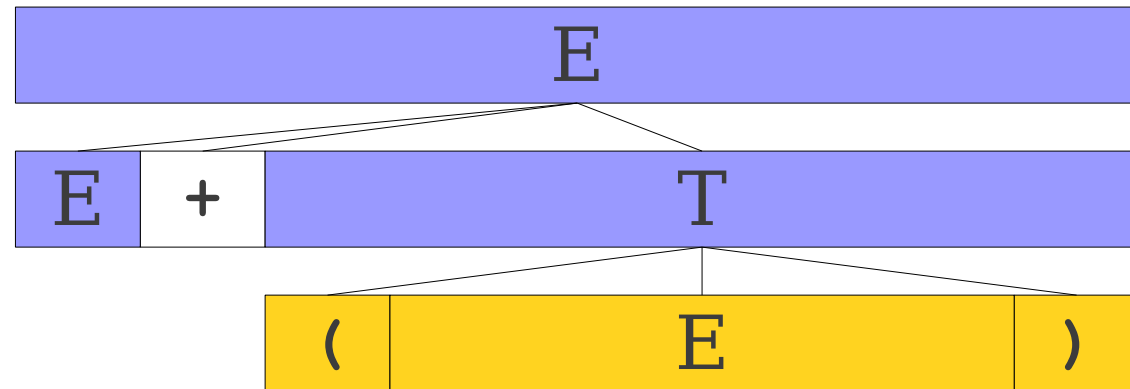
A Third View of a Bottom-Up Parse



$\Rightarrow \mathbf{E} + (\mathbf{E})$
 $\Rightarrow \mathbf{E} + \mathbf{T}$
 $\Rightarrow \mathbf{E}$

int	+	(	int	+	int	+	int	)
-----	---	---	-----	---	-----	---	-----	---

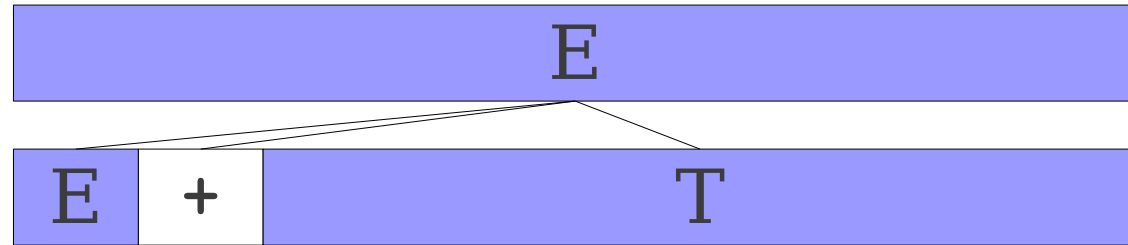
A Third View of a Bottom-Up Parse



$\Rightarrow \mathbf{E} + \mathbf{(E)}$
 $\Rightarrow \mathbf{E} + \mathbf{T}$
 $\Rightarrow \mathbf{E}$

int	+	(	int	+	int	+	int	)
-----	---	---	-----	---	-----	---	-----	---

A Third View of a Bottom-Up Parse

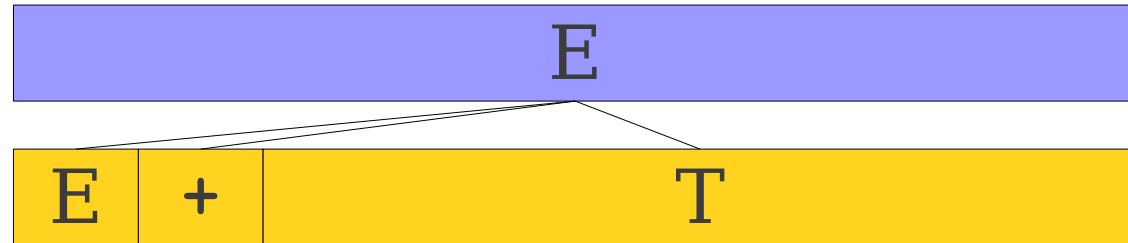


$\Rightarrow \mathbf{E} + \mathbf{T}$

$\Rightarrow \mathbf{E}$

int	+	(	int	+	int	+	int	)
-----	---	---	-----	---	-----	---	-----	---

A Third View of a Bottom-Up Parse



$\Rightarrow \mathbf{E} + \mathbf{T}$

$\Rightarrow \mathbf{E}$

int	+	(	int	+	int	+	int	)
-----	---	---	-----	---	-----	---	-----	---

A Third View of a Bottom-Up Parse

E

⇒ E

int	+	(	int	+	int	+	int	)
-----	---	---	-----	---	-----	---	-----	---

Handles

- Informally, a “**handle**” is a substring that matches the body of a production, and whose reduction represents one step along the reverse of a rightmost derivation.
- A left-to-right, bottom-up parse works by iteratively searching for a handle, then reducing the handle.

Shift/Reduce Parsing

- The bottom-up parsers we will consider are called **shift/reduce** parsers.
 - Contrast with the LL(1) **predict/match** parser.
- Idea: Split the input into two parts:
 - Left substring is our work area; all handles must be here.
 - Right substring is input we have not yet processed; consists purely of terminals.
- At each point, decide whether to:
 - Move a terminal across the split (**shift**)
 - Reduce a handle (**reduce**)

A Sample Shift/Reduce Parse

E → **F**

E → **E** + **F**

F → **F** * **T**

F → **T**

T → **int**

T → (**E**)



int	+	int	*	int	+	int
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A Sample Shift/Reduce Parse

E → **F**

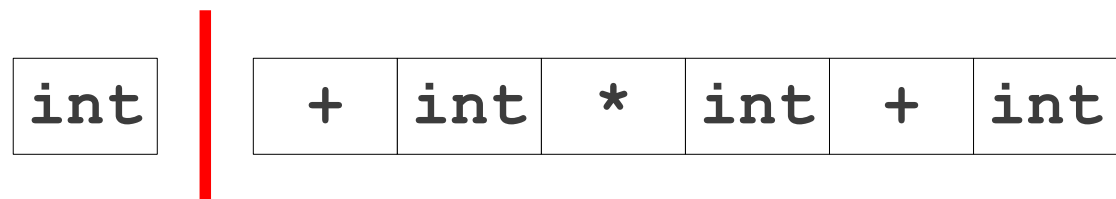
E → **E** + **F**

F → **F** * **T**

F → **T**

T → **int**

T → (**E**)



A Sample Shift/Reduce Parse

E → **F**

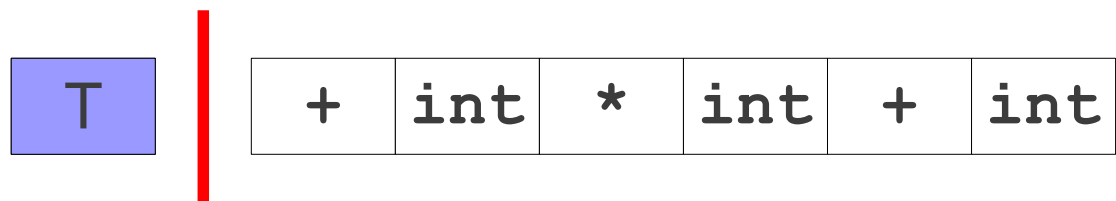
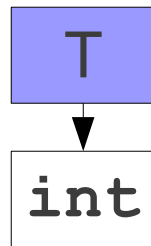
E → **E** + **F**

F → **F** * **T**

F → **T**

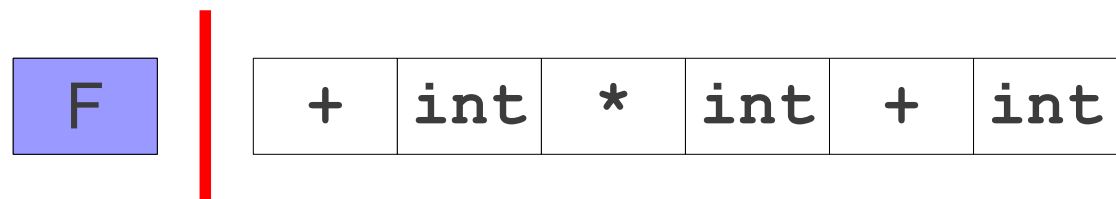
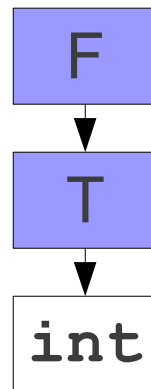
T → **int**

T → (**E**)



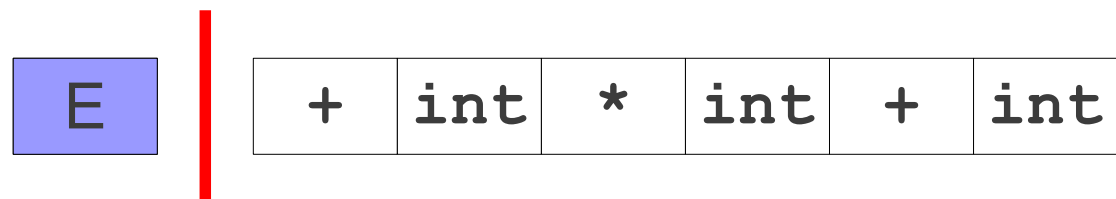
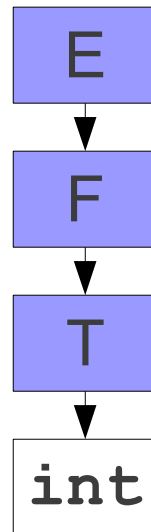
A Sample Shift/Reduce Parse

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



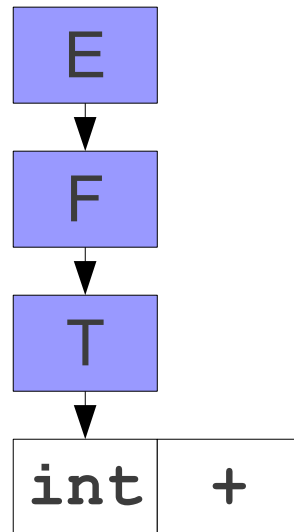
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 $T \rightarrow (E)$



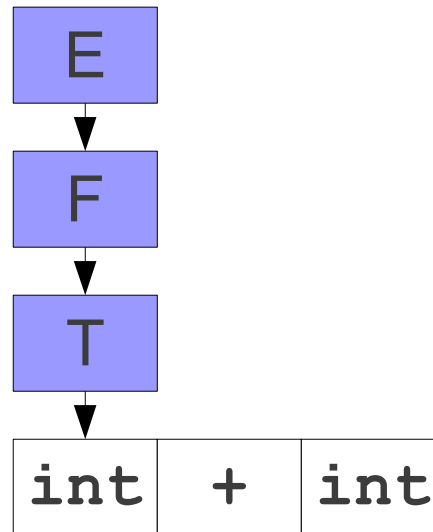
A Sample Shift/Reduce Parse

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A Sample Shift/Reduce Parse

$E \rightarrow F$
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 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

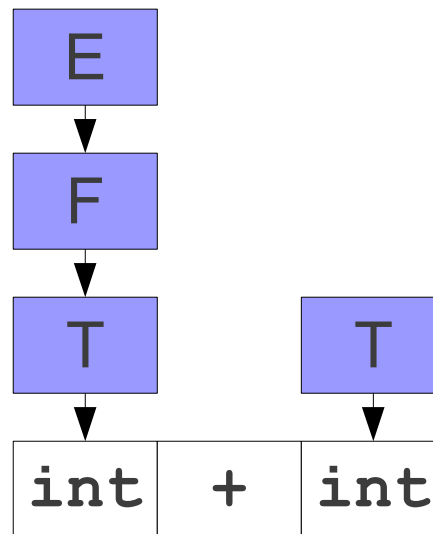


E	+	int
----------	---	-----

*	int	+	int
---	-----	---	-----

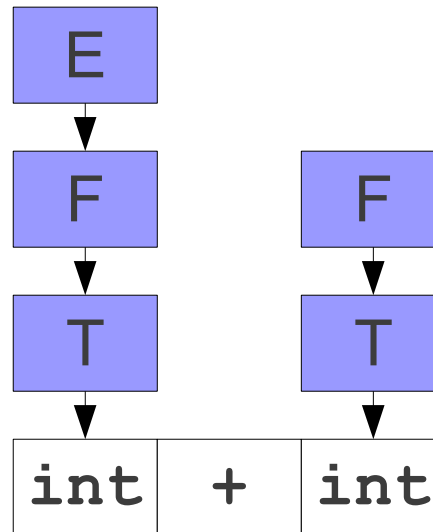
A Sample Shift/Reduce Parse

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 $F \rightarrow T$
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 $T \rightarrow (E)$



A Sample Shift/Reduce Parse

$E \rightarrow F$
 $E \rightarrow E + F$
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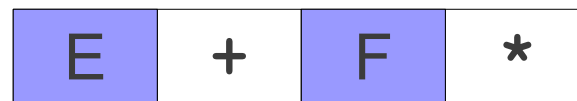
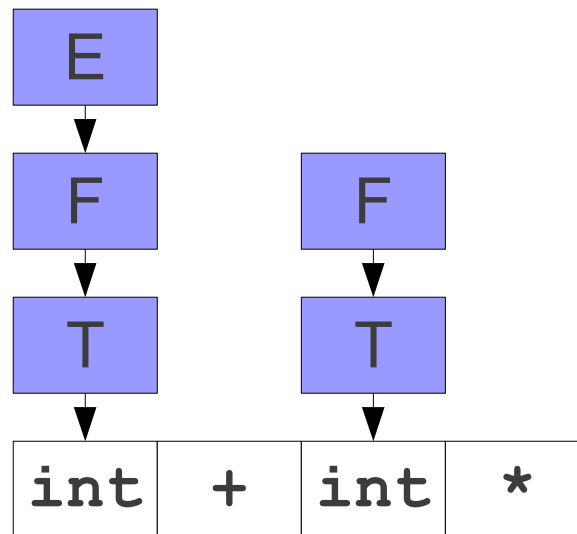


E	+	F
----------	---	----------

*	int	+	int
---	-----	---	-----

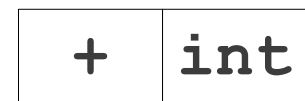
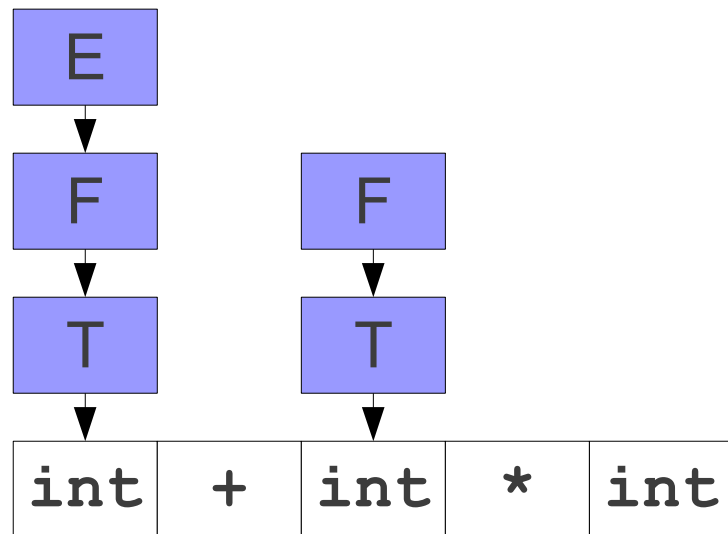
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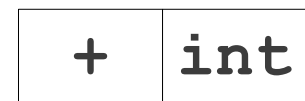
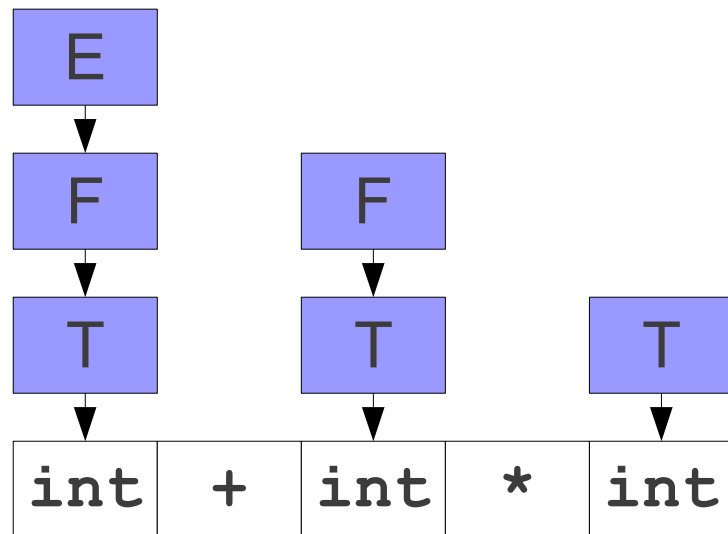
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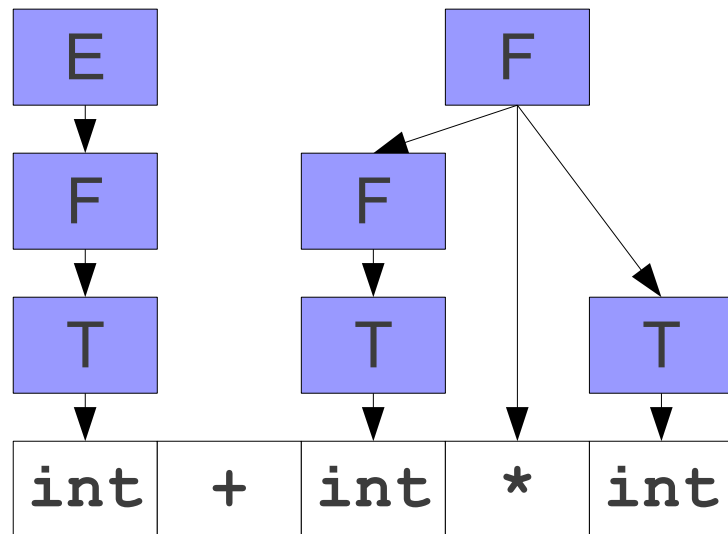
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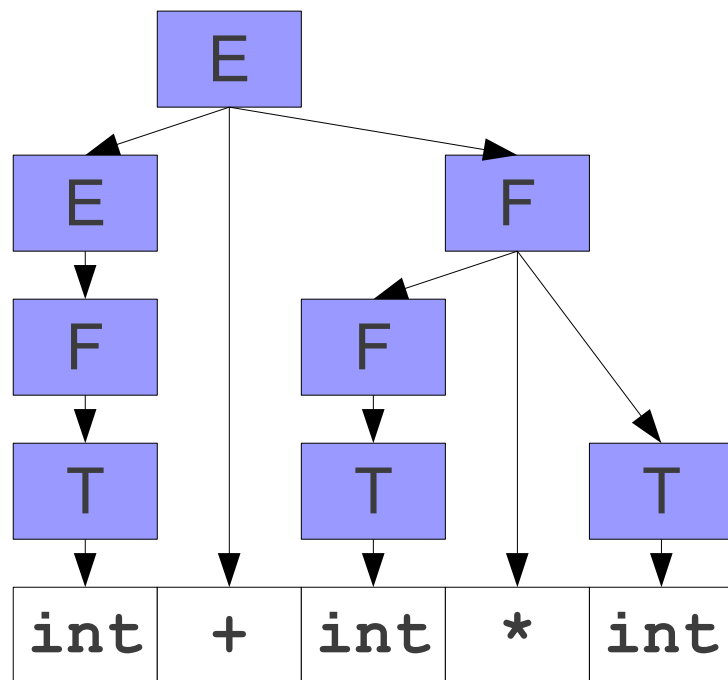
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 $T \rightarrow (E)$

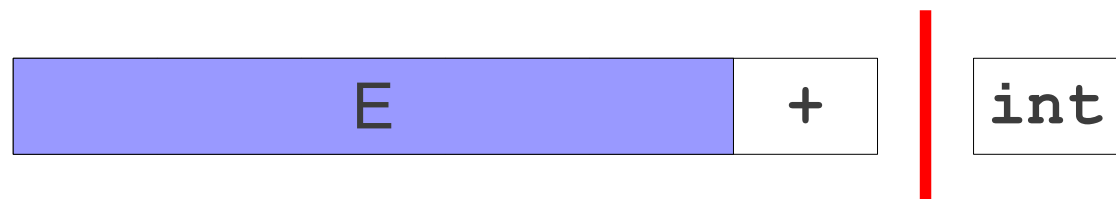
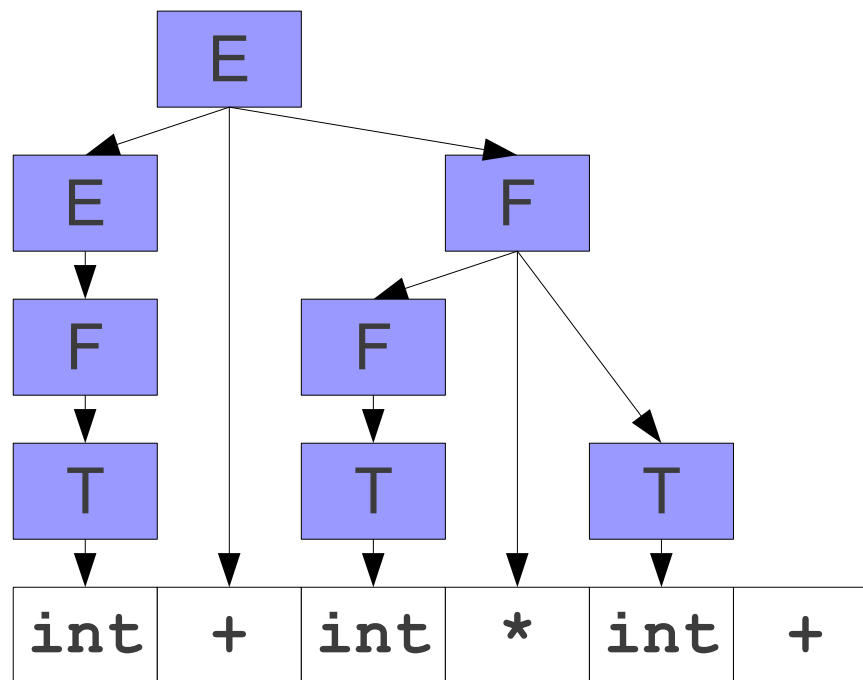


E

+ **int**

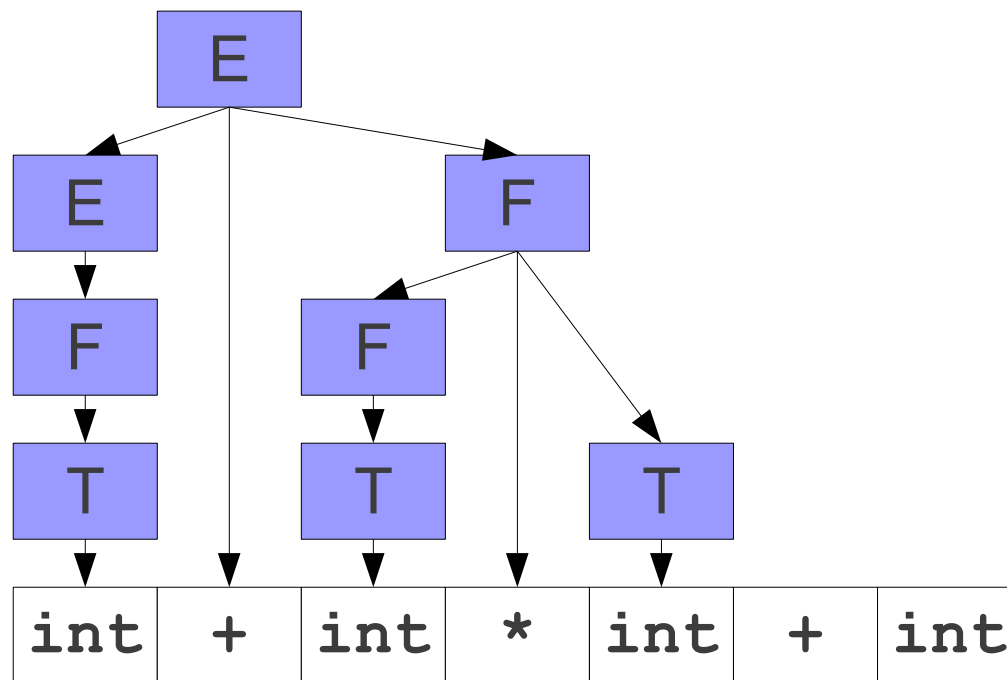
A Sample Shift/Reduce Parse

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



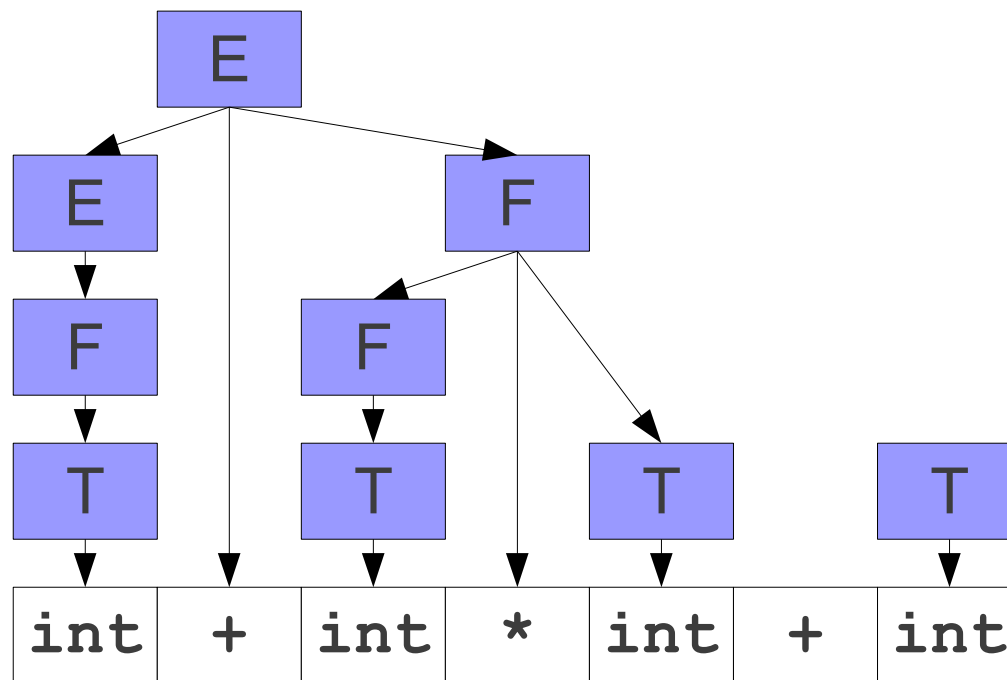
A Sample Shift/Reduce Parse

$E \rightarrow F$
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 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



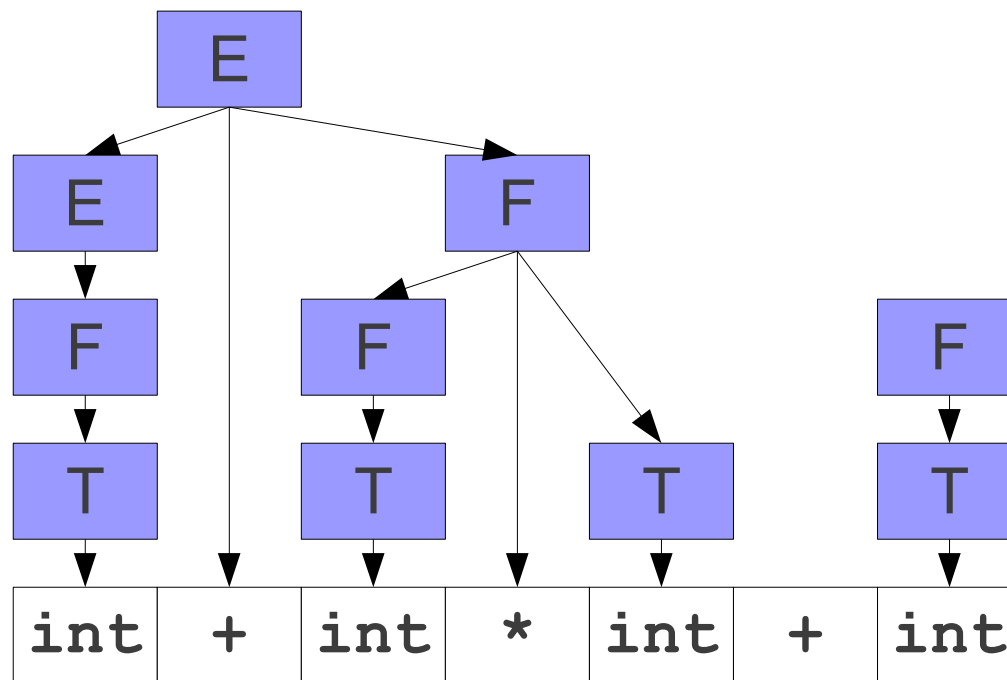
A Sample Shift/Reduce Parse

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



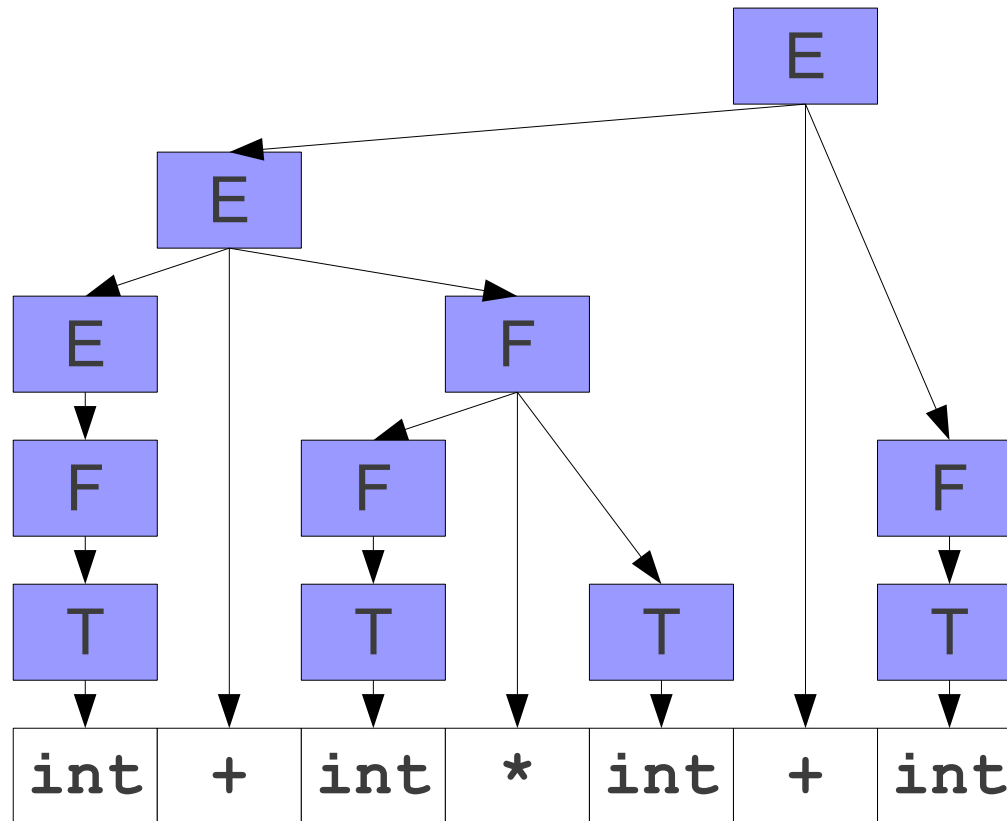
A Sample Shift/Reduce Parse

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



A Sample Shift/Reduce Parse

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



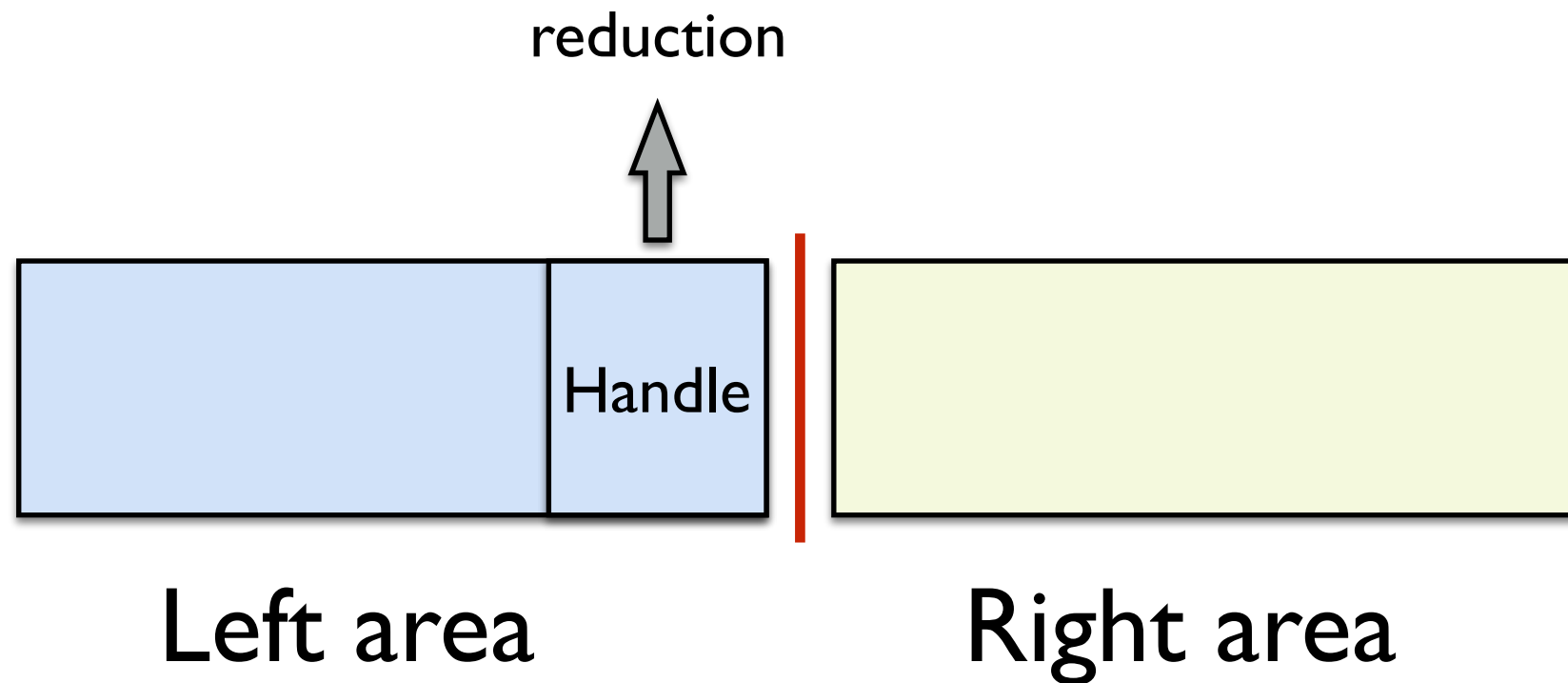
E



Simplifying our Terminology

- All activity in a shift/reduce parser is at the far right end of the left area.
- Idea: Represent the left area as a stack.
- Shift: Push the next terminal onto the stack.
- Reduce: Pop some number of symbols from the stack, then push the appropriate nonterminal.

- What's the pattern of the “left area” in a shift/reduce parser?
- How to recognize it?



Finding Handles

- Where do we look for handles?
 - **At the top of the stack.**
- How do we search for handles?
 - What algorithm do we use to try to discover a handle?
- How do we recognize handles?
 - Once we've found a possible handle, how do we confirm that it's correct?

Question Two:

How do we search for handles?

Searching for Handles

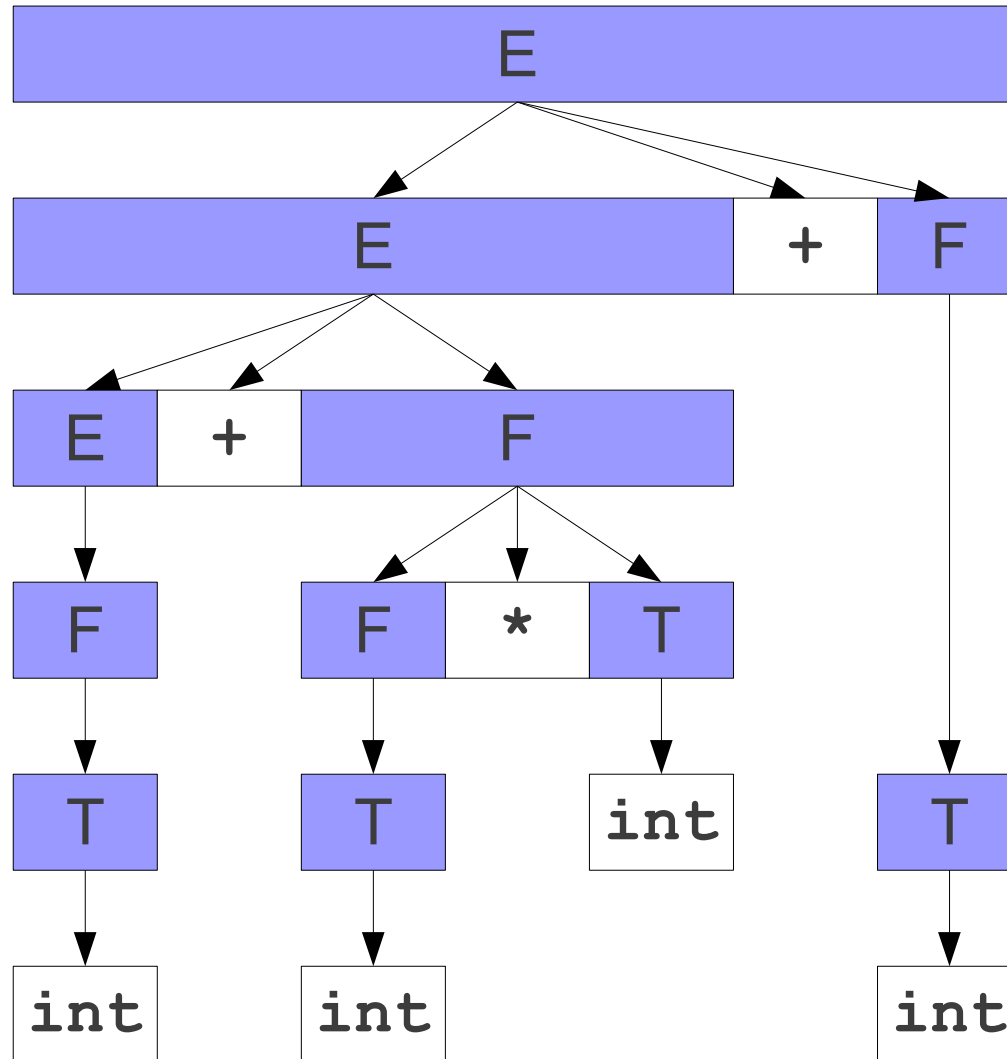
- When using a shift/reduce parser, we must decide whether to shift or reduce at each point.
- We only want to reduce when we know we have a handle.
- **Question:** How can we tell that we might be looking at a handle?

Exploring the Left Side

- The handle will always appear at the end of string in the left side of the parser.
- Can *any* string appear on the left side of the parser, or are there restrictions on what sorts of strings can appear there?
- If we can find a pattern to the strings that can appear on the left side, we might be able to exploit it to detect handles.

Another Look at Handles

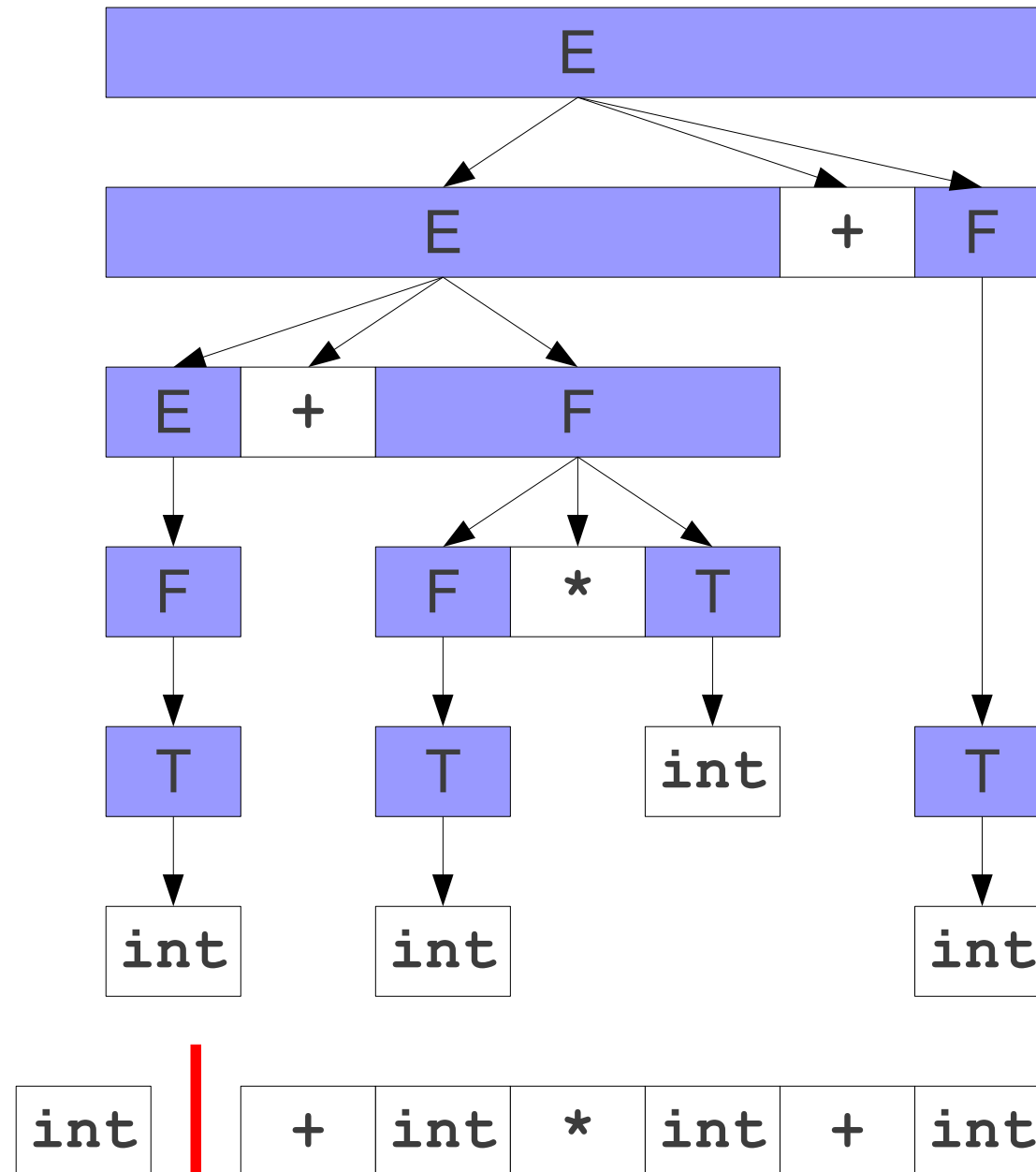
$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



int + int * int + int

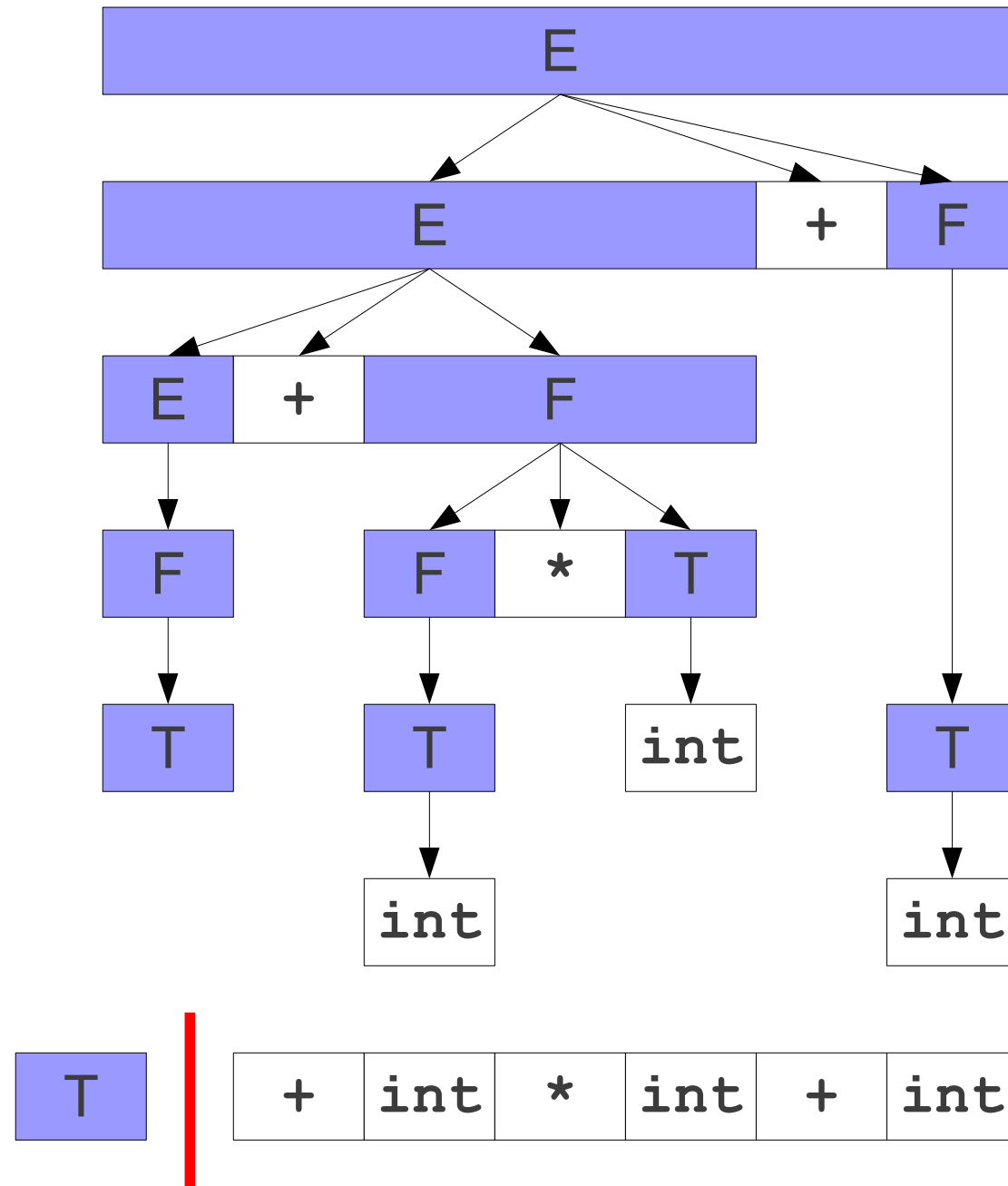
Another Look at Handles

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

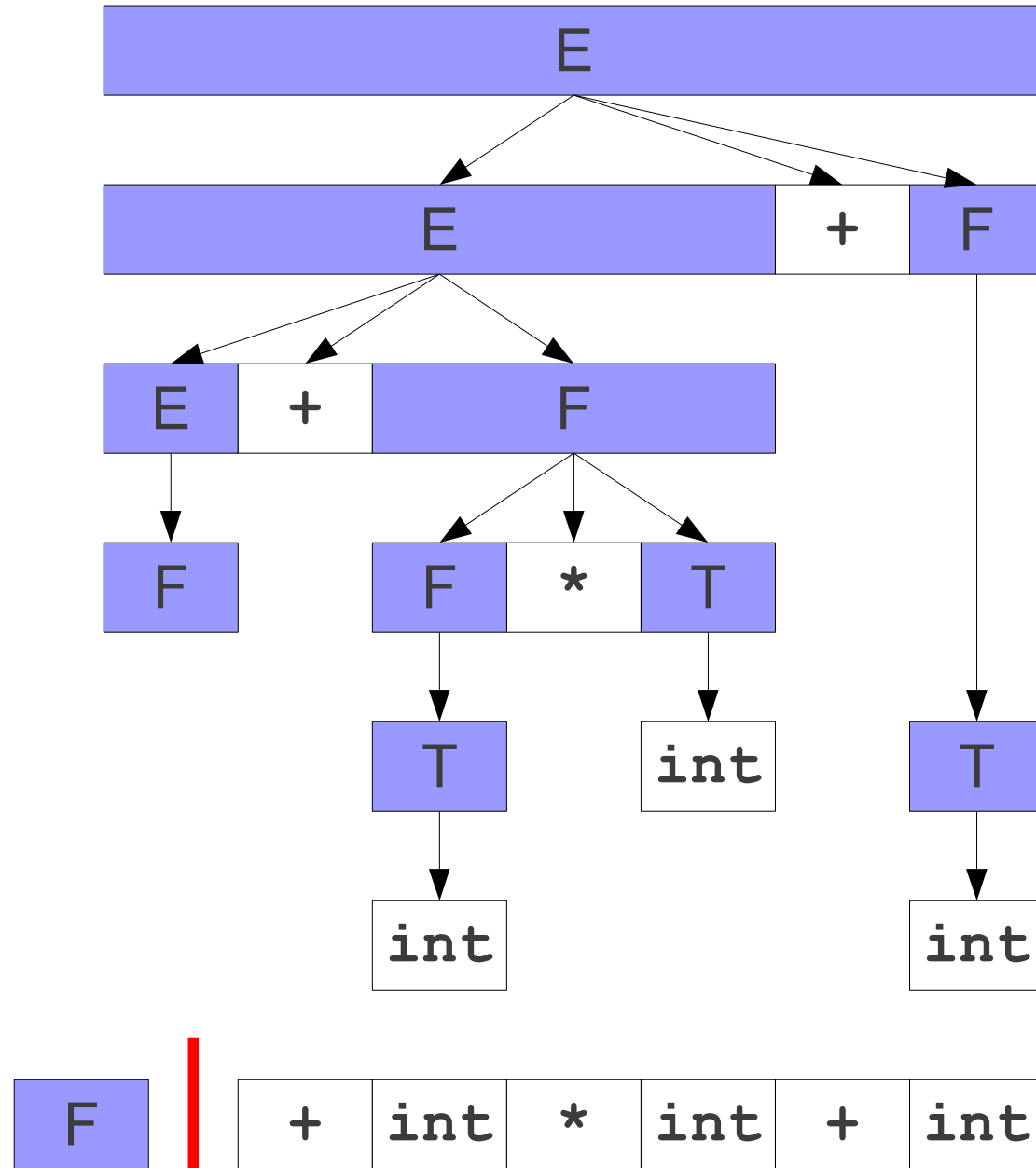
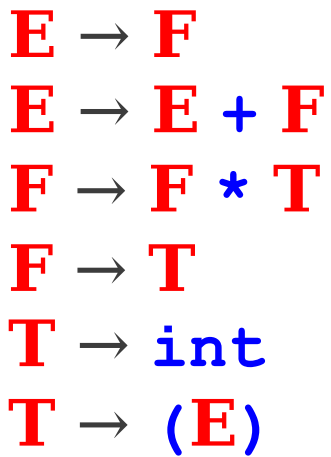


Another Look at Handles

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

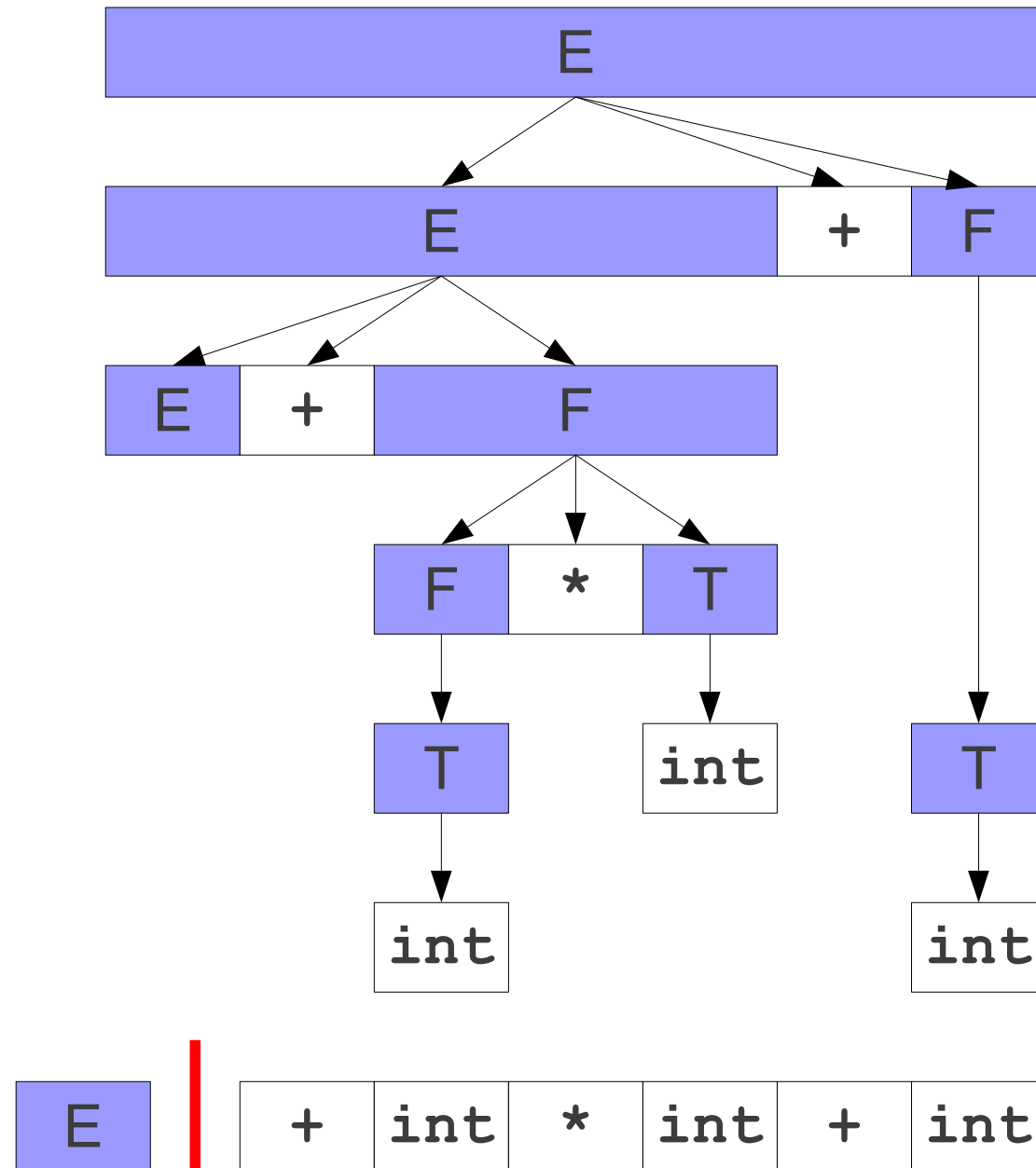


Another Look at Handles



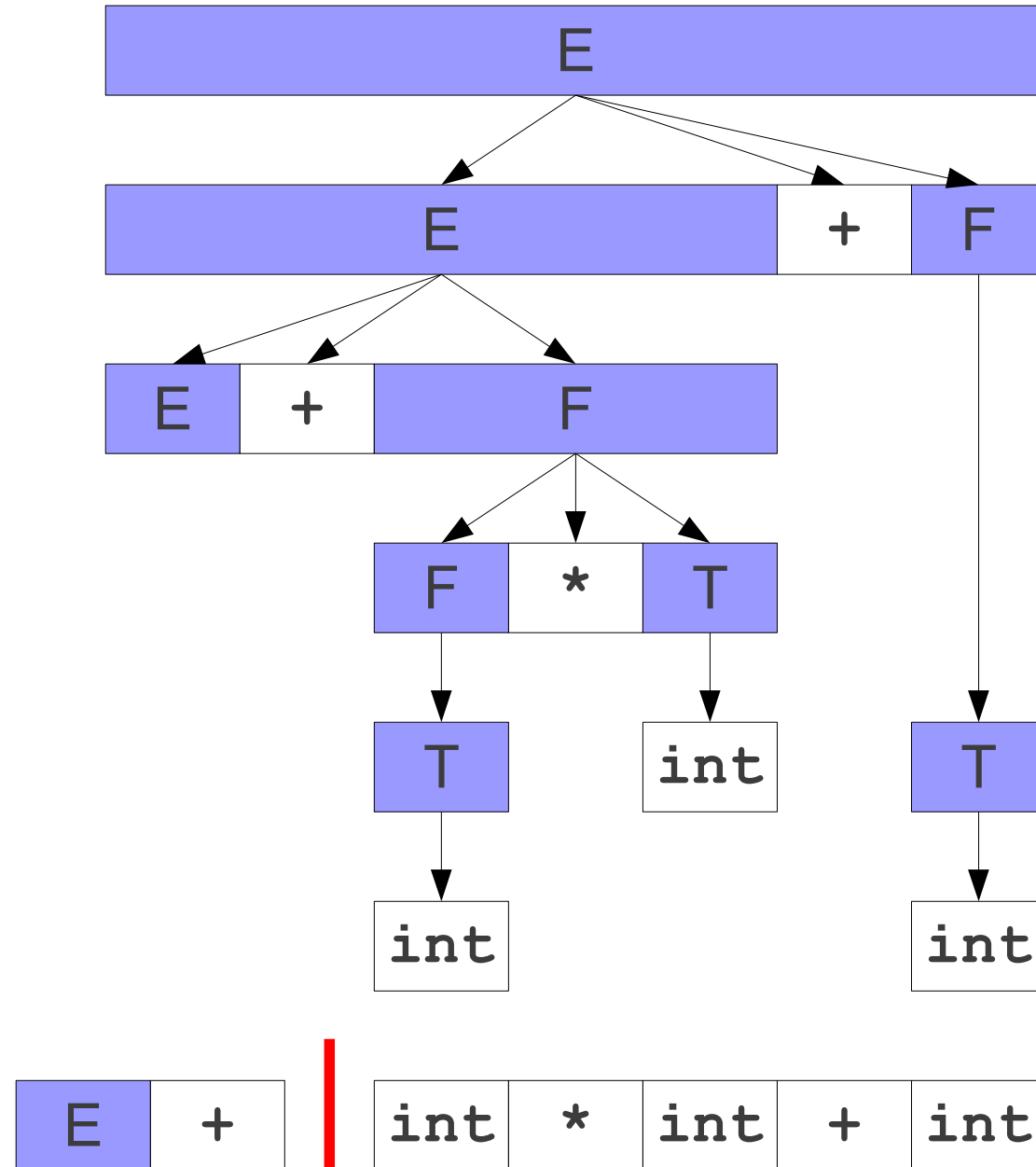
Another Look at Handles

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



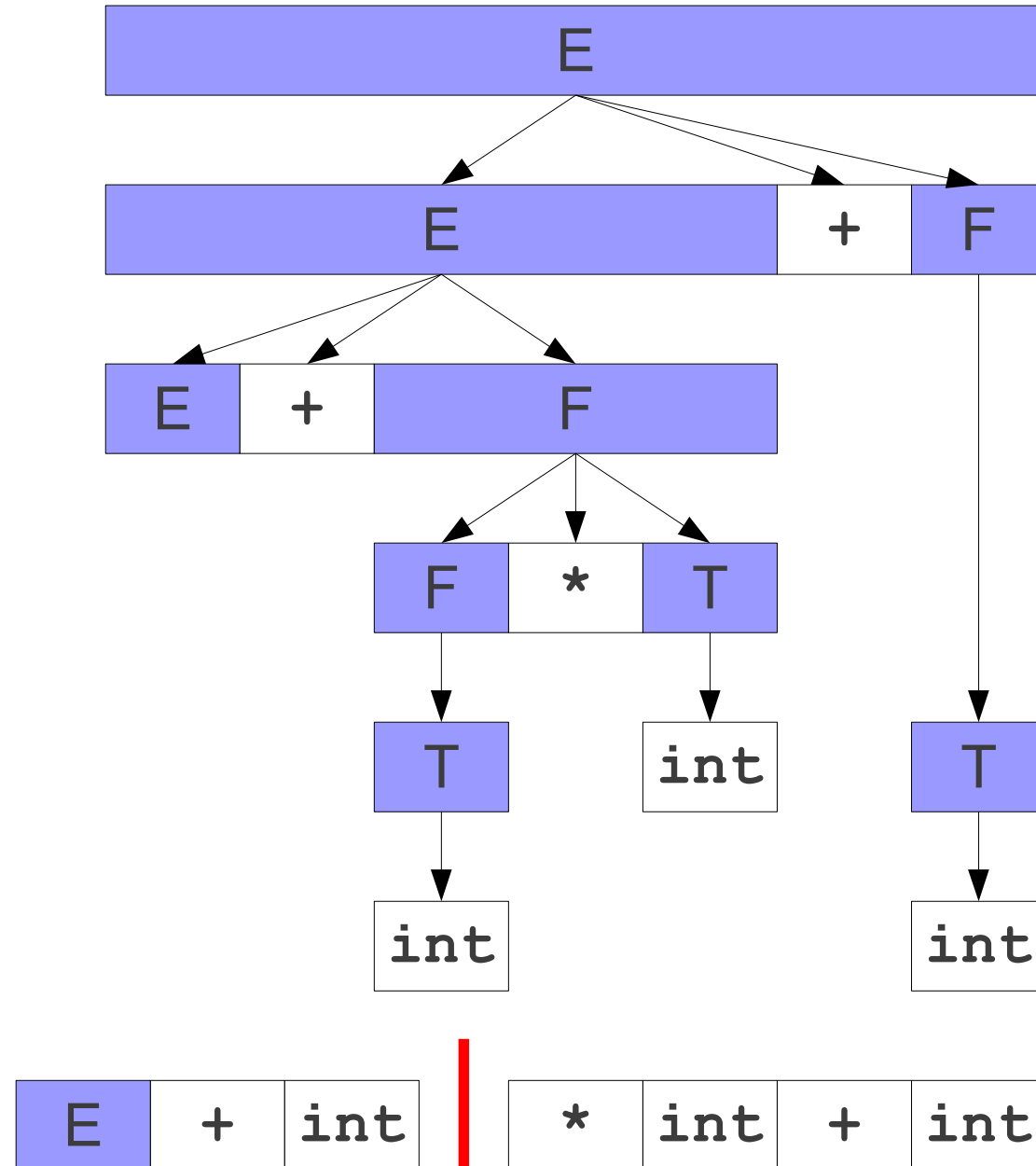
Another Look at Handles

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



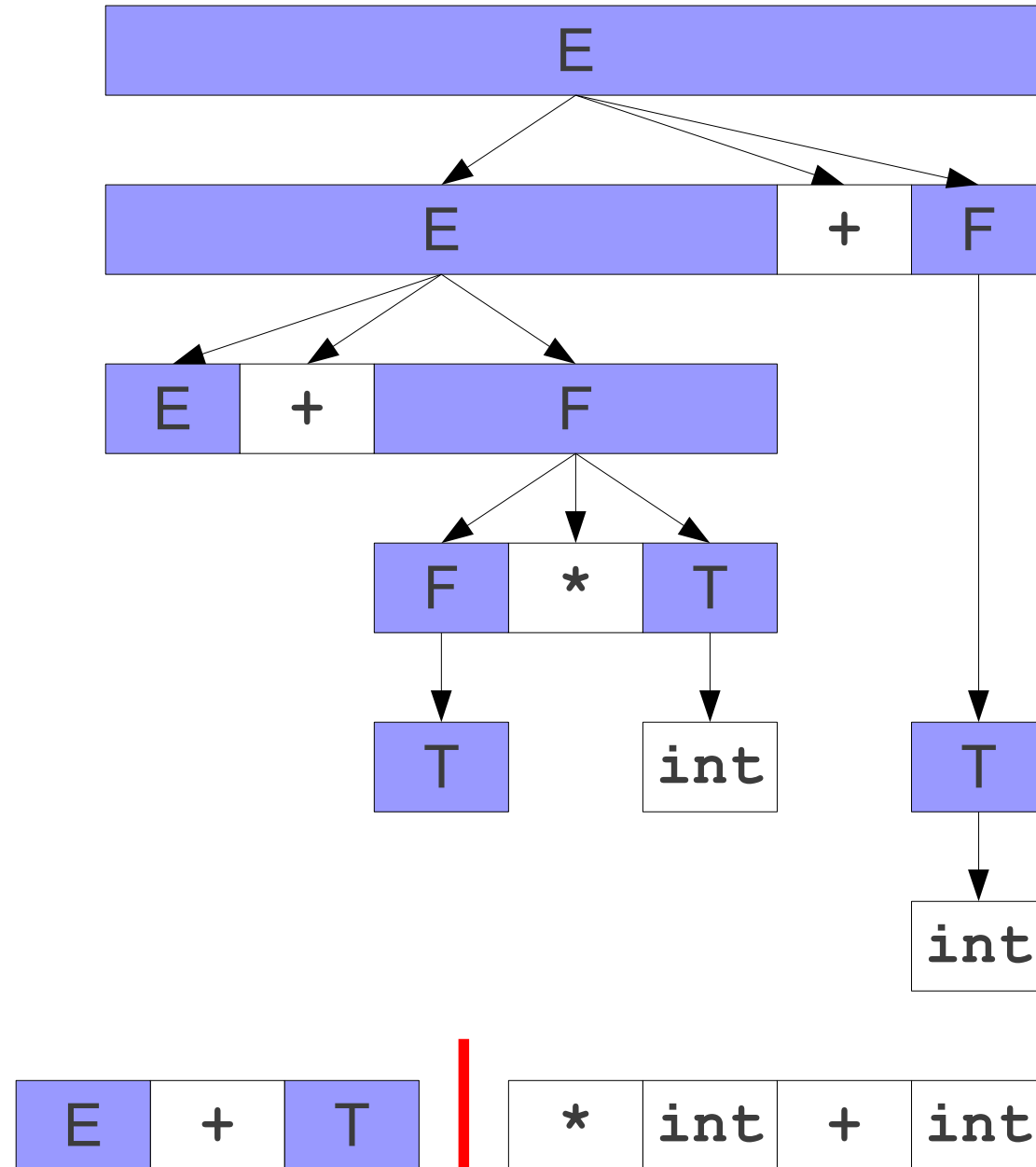
Another Look at Handles

$E \rightarrow F$
 $E \rightarrow E + F$
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 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



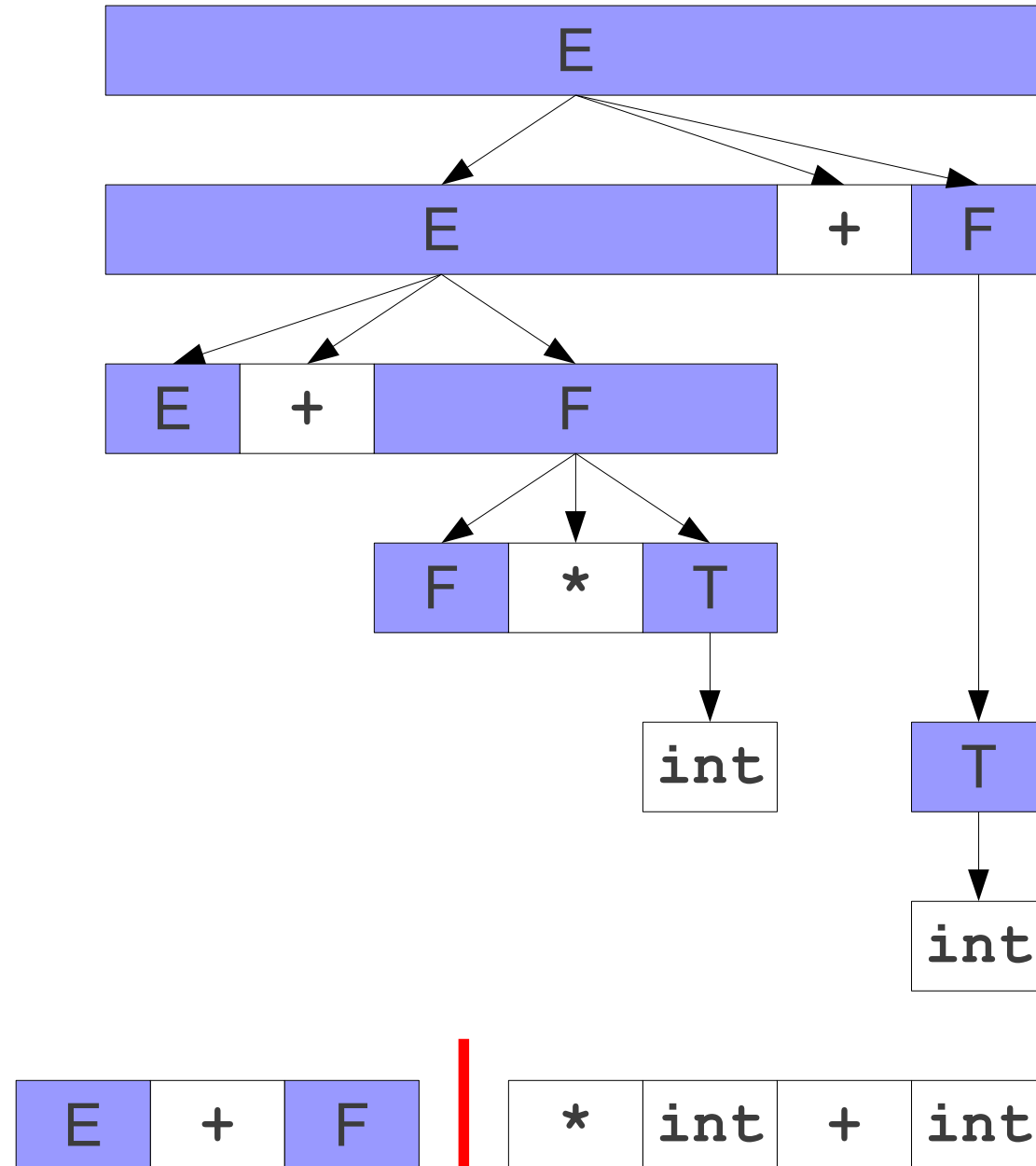
Another Look at Handles

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



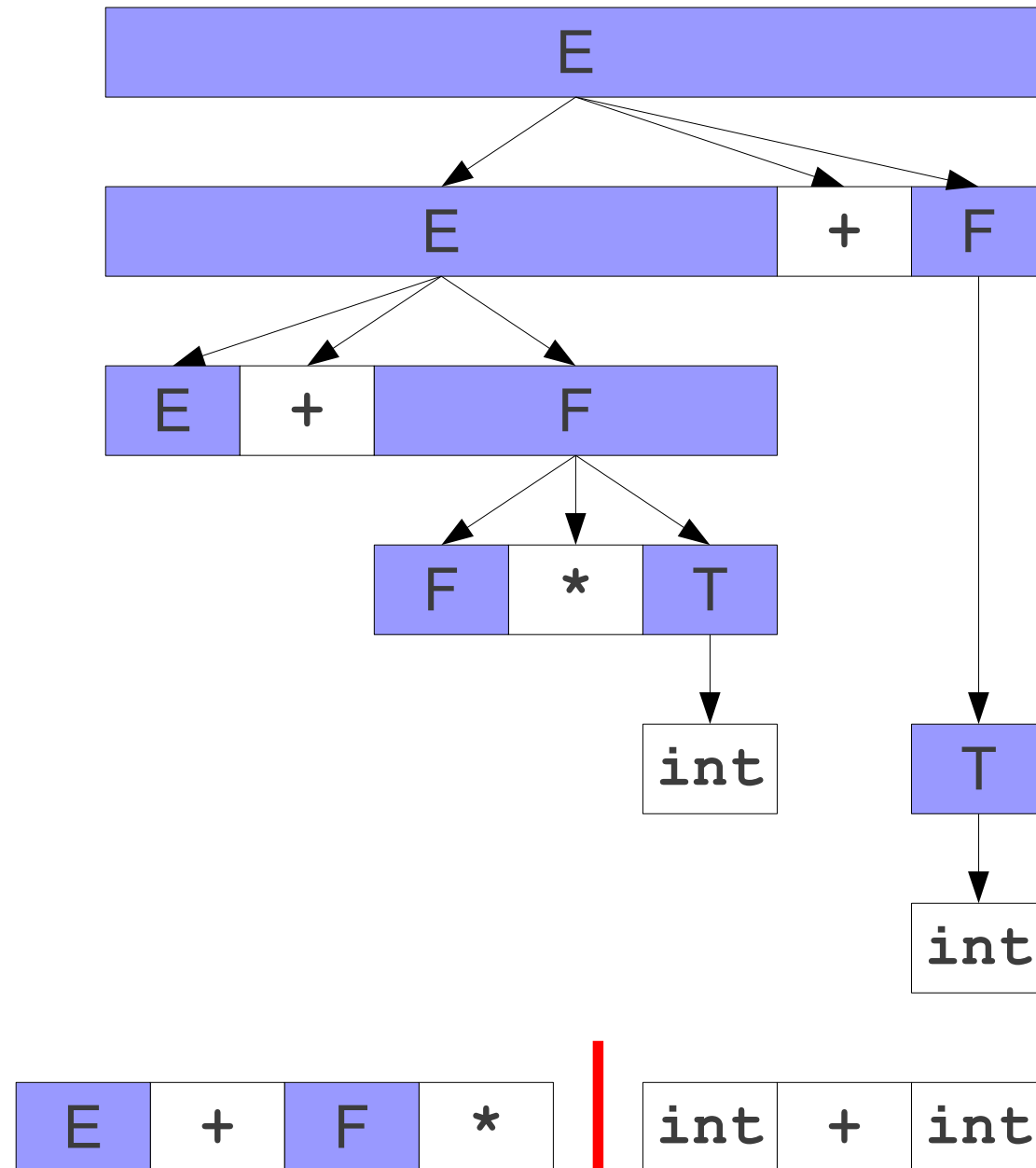
Another Look at Handles

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



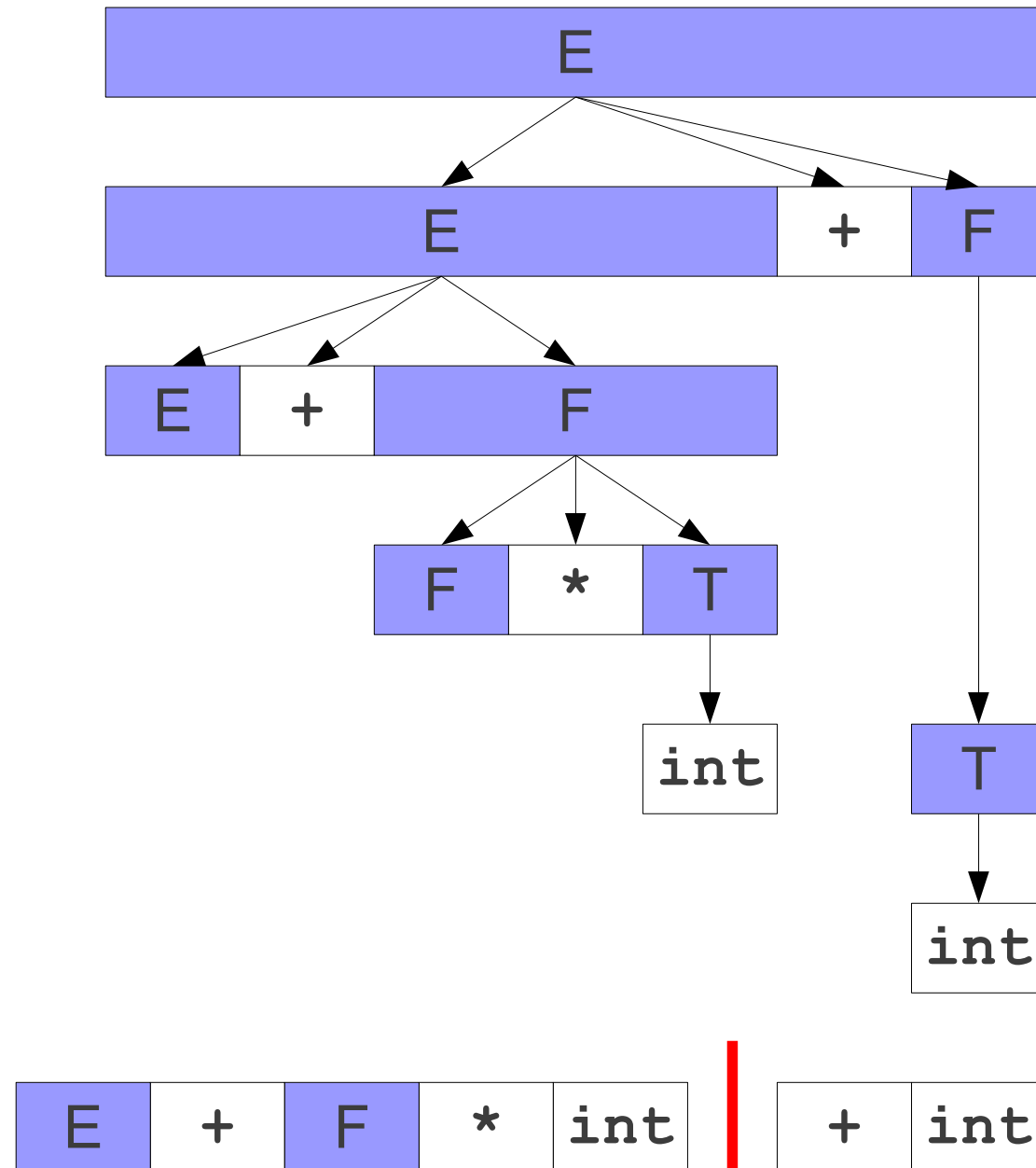
Another Look at Handles

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



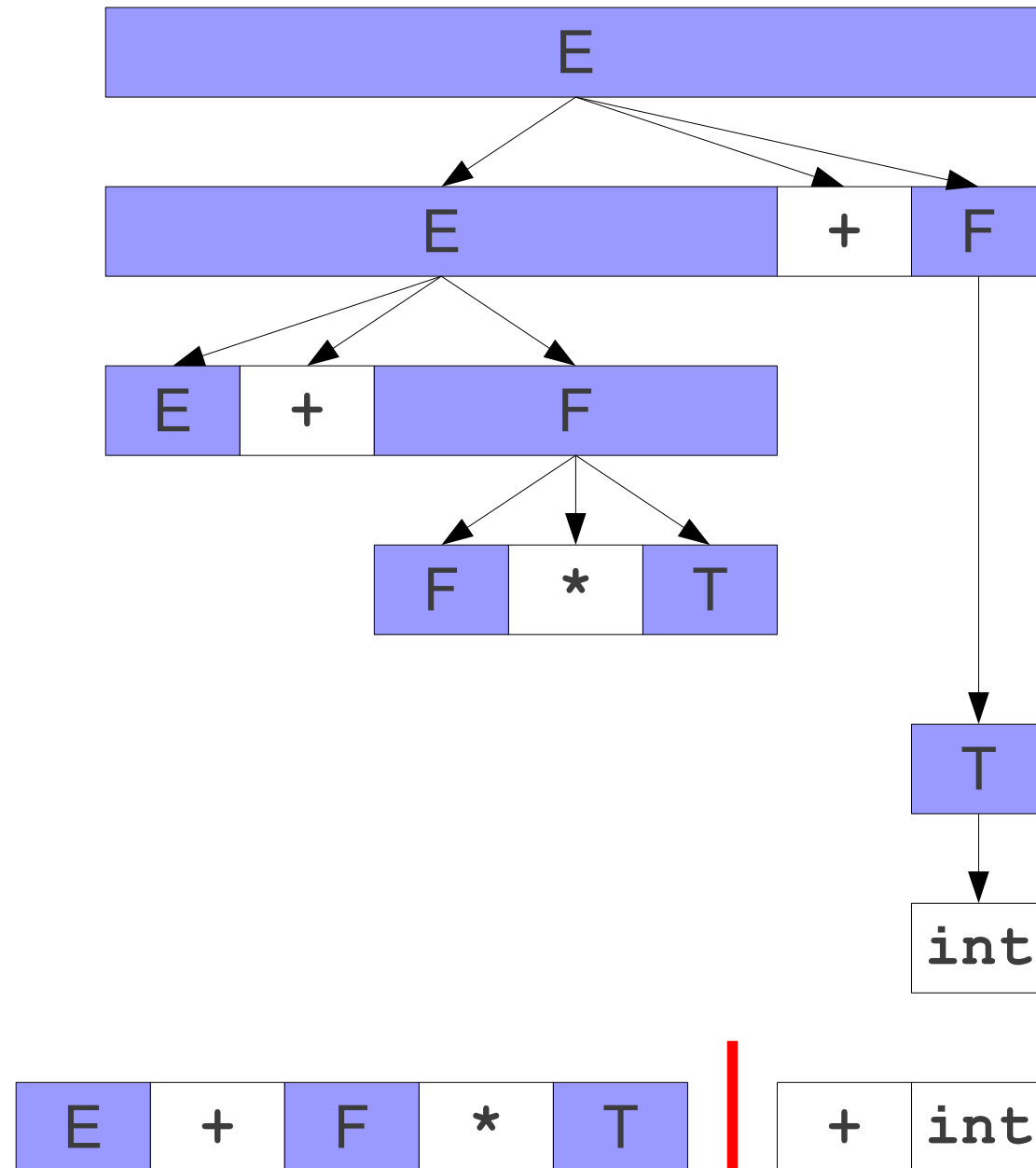
Another Look at Handles

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



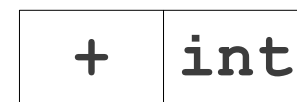
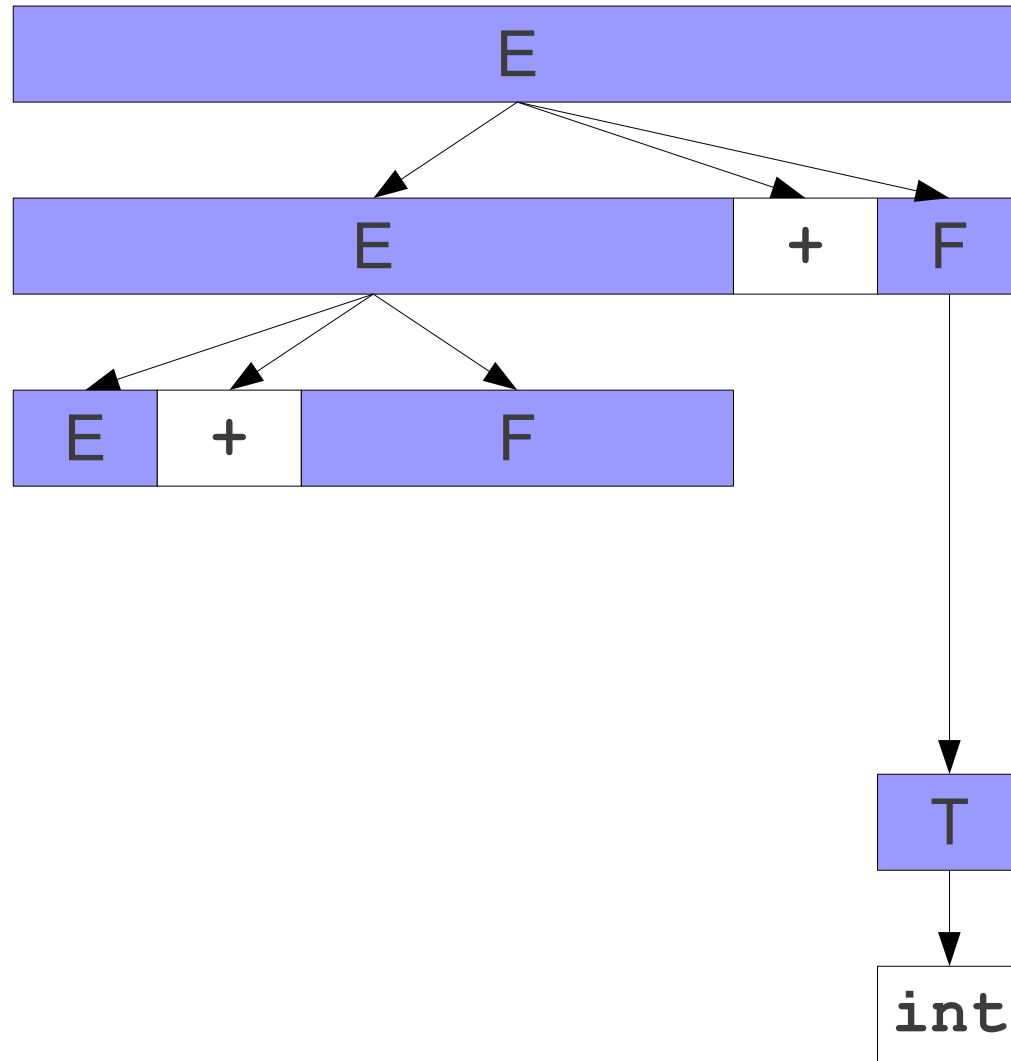
Another Look at Handles

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



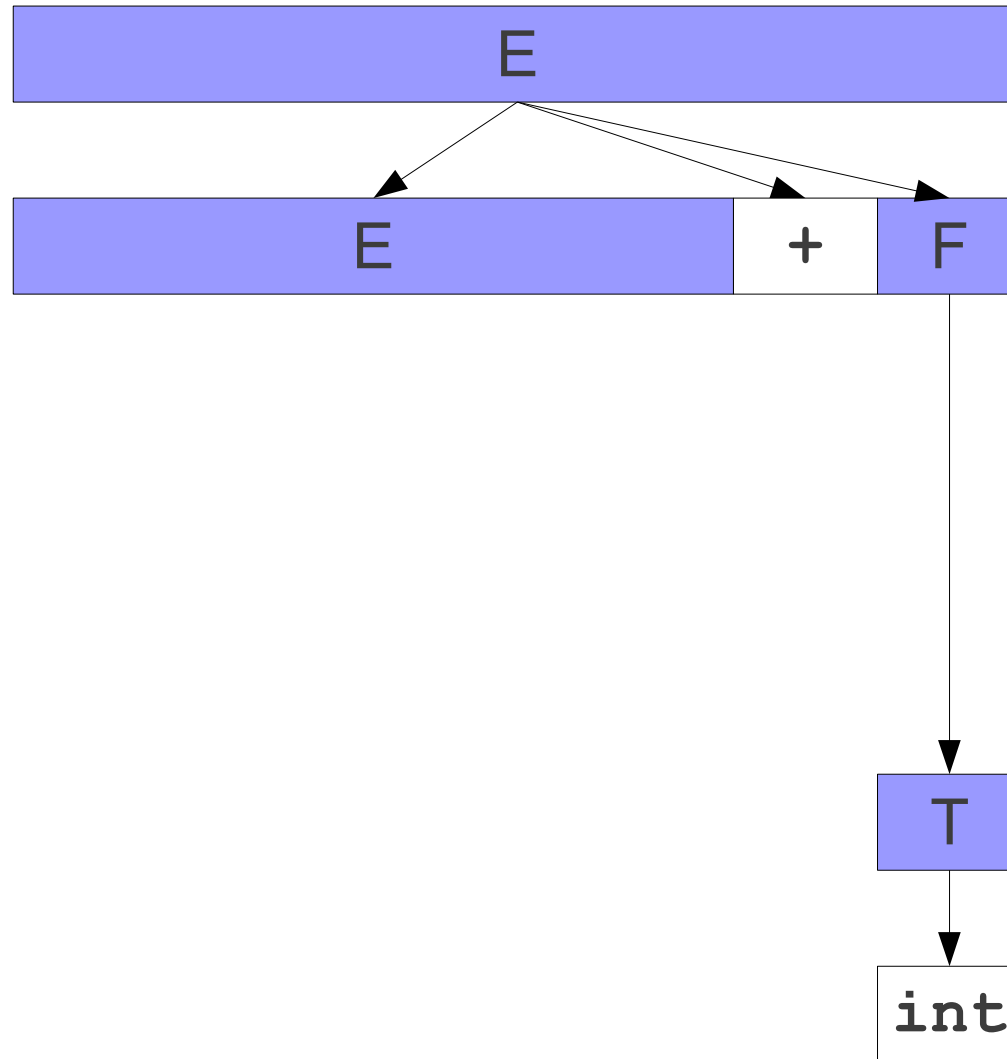
Another Look at Handles

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



Another Look at Handles

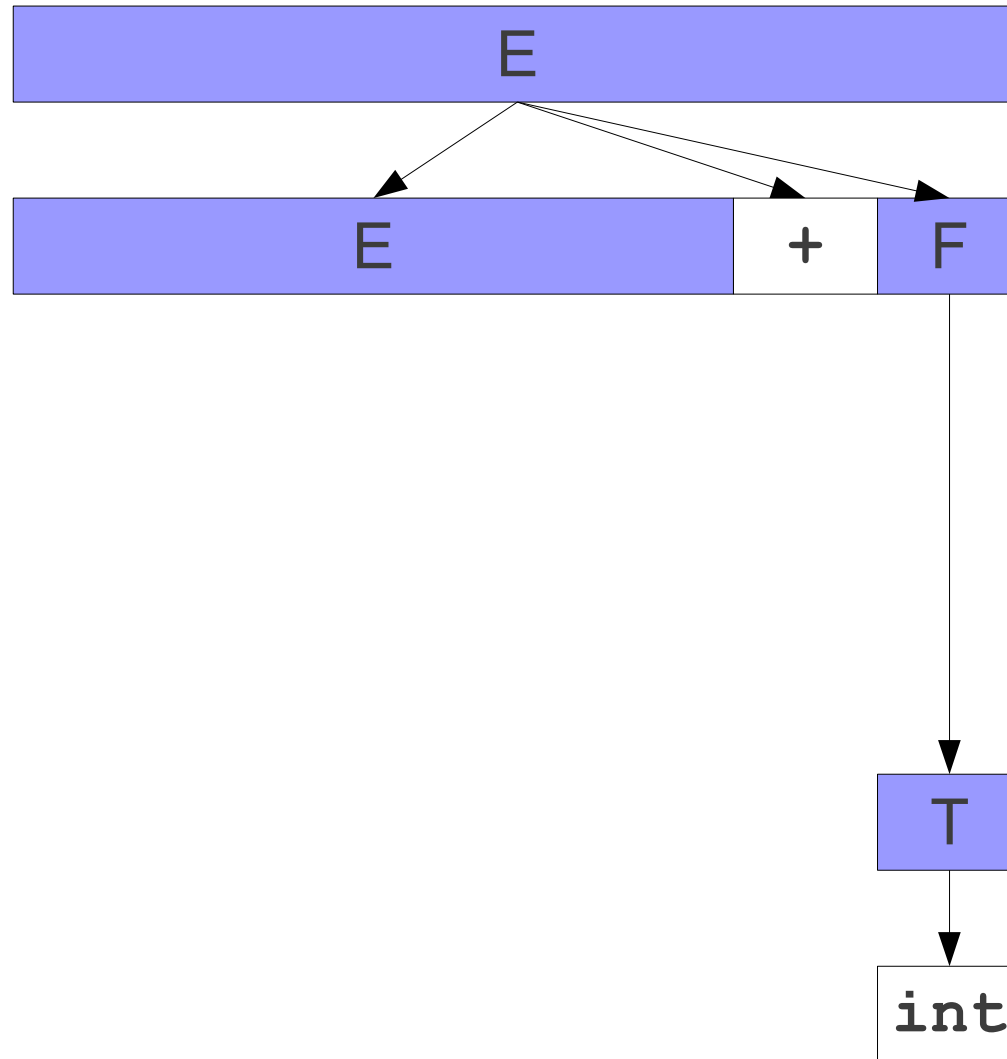
$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



E

+ int

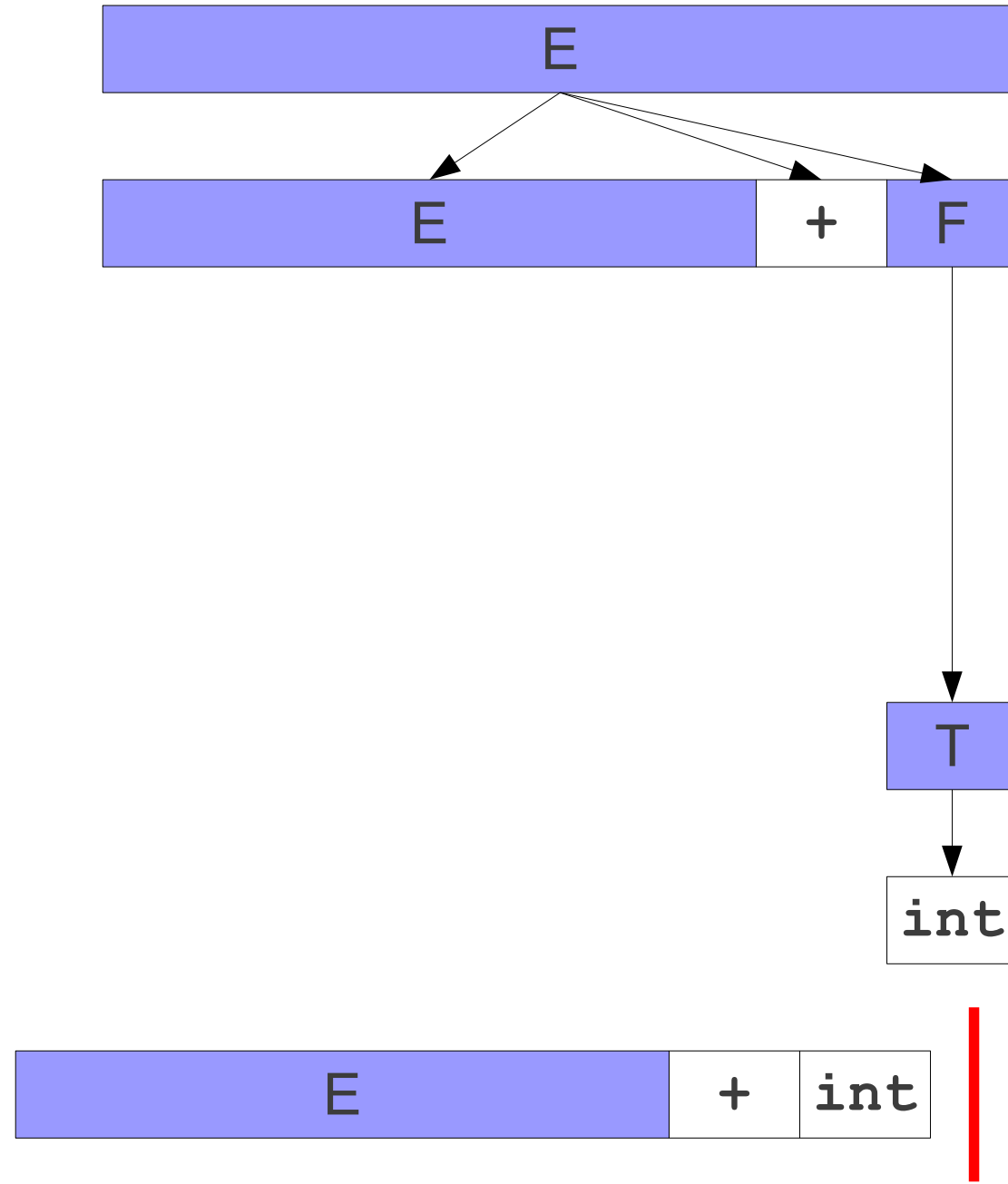
Another Look at Handles



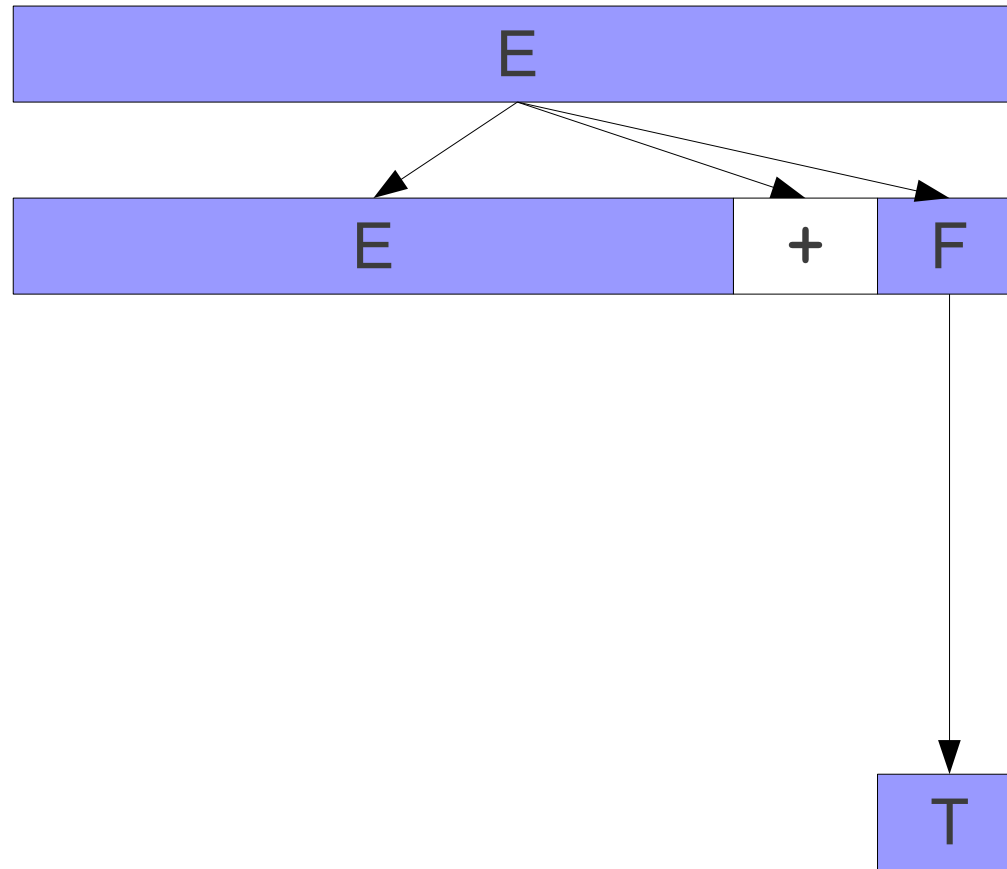
E → **F**
E → **E** + **F**
F → **F** * **T**
F → **T**
T → `int`
T → (**E**)

Another Look at Handles

$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



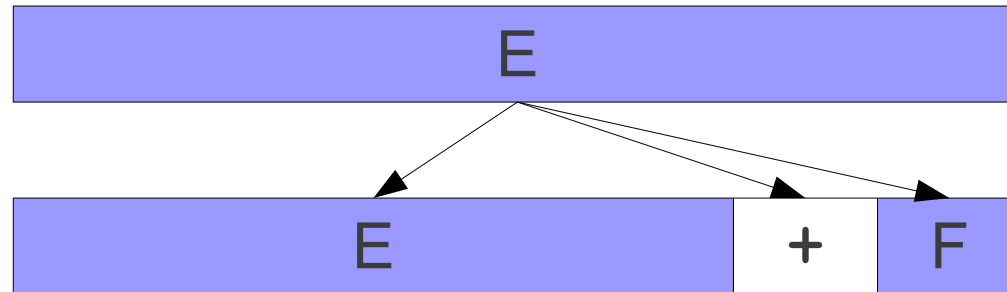
Another Look at Handles



$E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



Another Look at Handles



E → **F**
E → **E** + **F**
F → **F** * **T**
F → **T**
T → *int*
T → (**E**)



Another Look at Handles



E

E $\rightarrow$ **F**

E $\rightarrow$ **E** + **F**

F $\rightarrow$ **F** * **T**

F $\rightarrow$ **T**

T $\rightarrow$ *int*

T $\rightarrow$ (**E**)



E



Tracking Our Position

E → **F**

E → **E** + **F**

F → **F** * **T**

F → **T**

T → **int**

T → (**E**)

| int + int * int + int

Tracking Our Position

S → **E**

E → **F**

E → **E** + **F**

F → **F** * **T**

F → **T**

T → **int**

T → (**E**)

| int + int * int + int

Tracking Our Position

$S \rightarrow \cdot E$

$S \rightarrow E$

$E \rightarrow F$

$E \rightarrow E + F$

$F \rightarrow F * T$

$F \rightarrow T$

$T \rightarrow \text{int}$

$T \rightarrow (E)$

| int + int * int + int

Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$

| int + int * int + int

Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot E + F$

int	+	int	*	int	+	int
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Tracking Our Position

S → **E**
E → **F**
E → **E** + **F**
F → **F** * **T**
F → **T**
T → **int**
T → (**E**)

$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot F$

int	+	int	*	int	+	int
-----	---	-----	---	-----	---	-----

Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot F$
$F \rightarrow \cdot T$

int	+	int	*	int	+	int
-----	---	-----	---	-----	---	-----

Tracking Our Position

S → **E**
E → **F**
E → **E** + **F**
F → **F** * **T**
F → **T**
T → **int**
T → (**E**)

$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot F$
$F \rightarrow \cdot T$
$T \rightarrow \cdot \text{int}$

int	+	int	*	int	+	int
-----	---	-----	---	-----	---	-----

Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

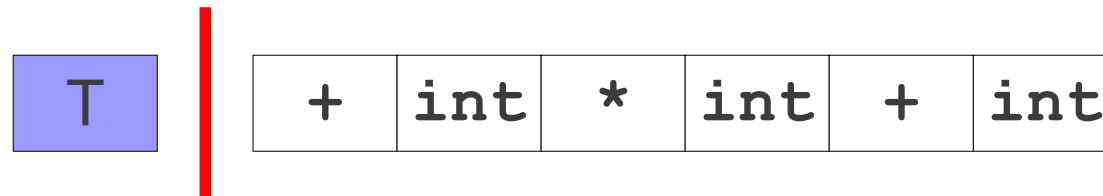
$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot F$
$F \rightarrow \cdot T$
$T \rightarrow \text{int} \cdot$

int		+	int	*	int	+	int
-----	--	---	-----	---	-----	---	-----

Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

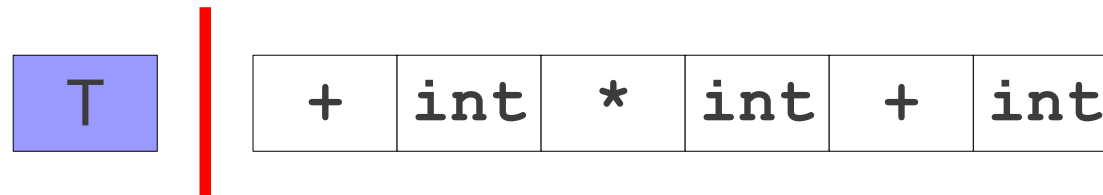
$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot F$
$F \rightarrow \cdot T$



Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
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 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

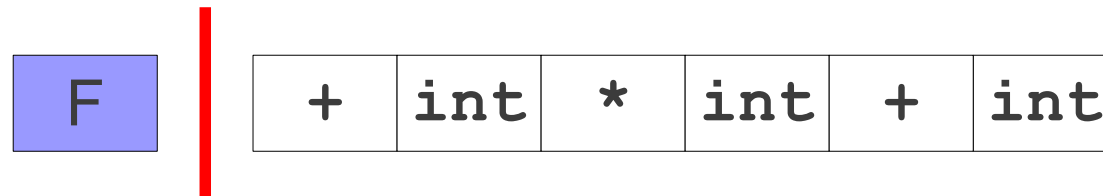
$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot F$
$F \rightarrow T \cdot$



Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

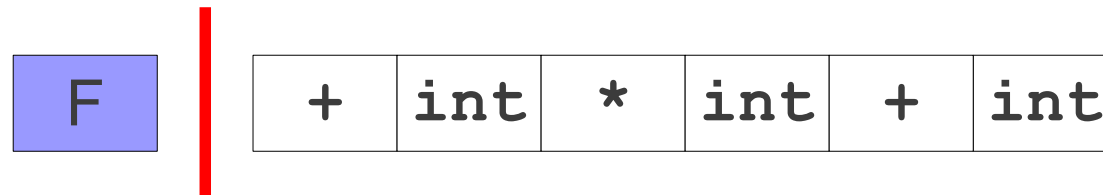
$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot F$



Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

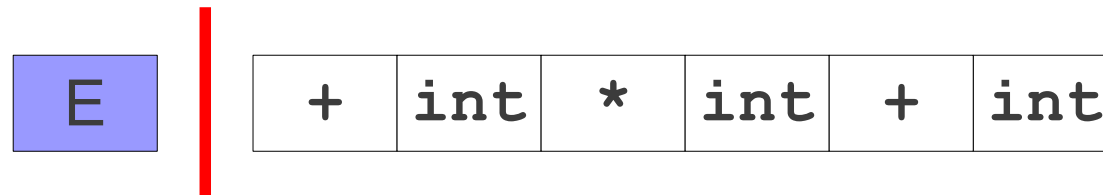
$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot E + F$
$E \rightarrow F \cdot$



Tracking Our Position

S → **E**
E → **F**
E → **E** + **F**
F → **F** * **T**
F → **T**
T → **int**
T → (**E**)

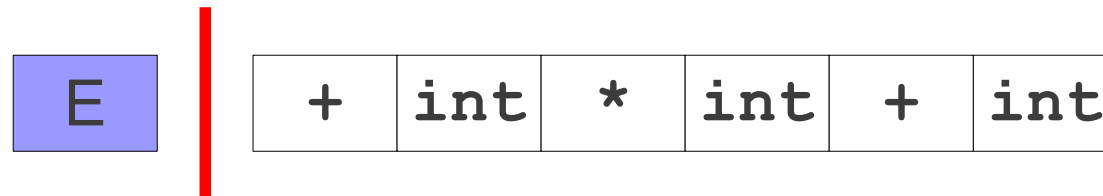
$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow \cdot E + F$



Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

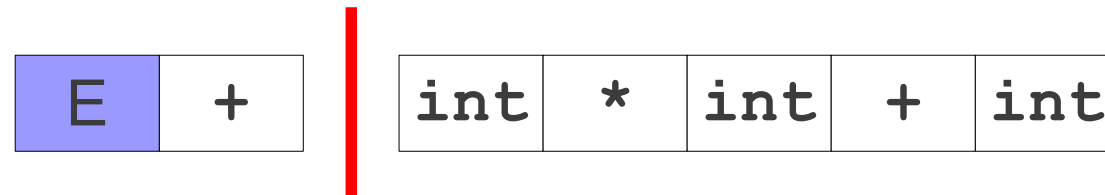
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$E \rightarrow \cdot E + F$
$E \rightarrow E \cdot + F$



Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

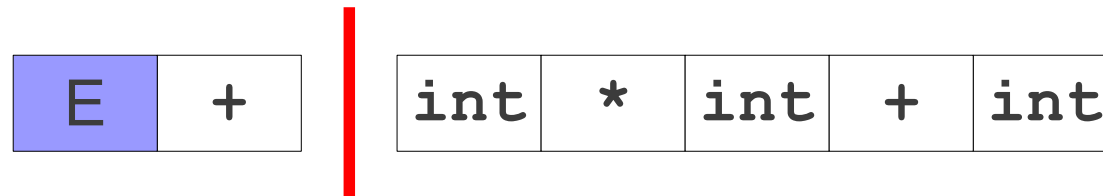
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$E \rightarrow \cdot E + F$
$E \rightarrow E + \cdot F$



Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

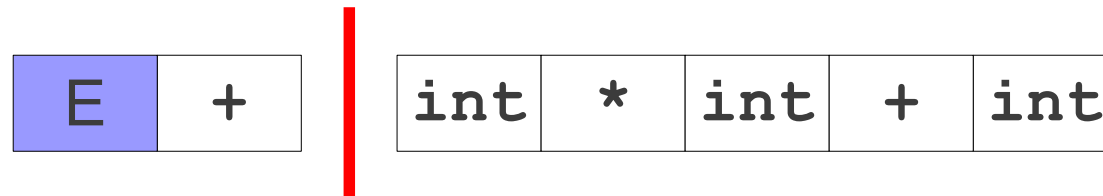
$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow E + \cdot F$
$F \rightarrow \cdot F * T$



Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow E + \cdot F$
$F \rightarrow \cdot F * T$
$F \rightarrow \cdot T$



Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow E + \cdot F$
$F \rightarrow \cdot F * T$
$F \rightarrow \cdot T$
$T \rightarrow \cdot \text{int}$

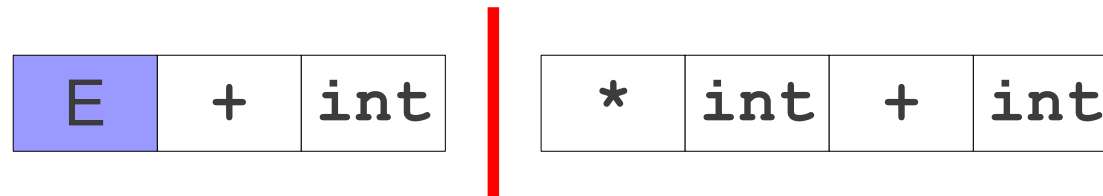
E	+
---	---

int	*	int	+	int
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Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

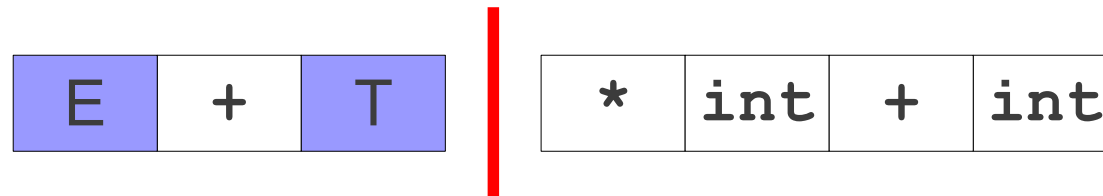
$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow E + \cdot F$
$F \rightarrow \cdot F * T$
$F \rightarrow \cdot T$
$T \rightarrow \text{int} \cdot$



Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

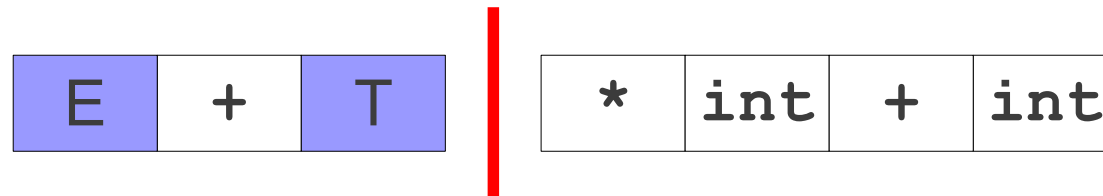
$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow E + \cdot F$
$F \rightarrow \cdot F * T$
$F \rightarrow \cdot T$



Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

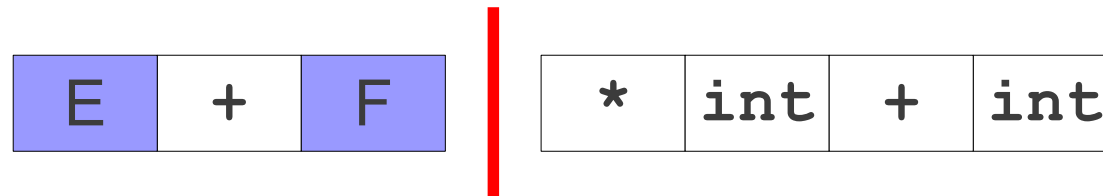
$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow E + \cdot F$
$F \rightarrow \cdot F * T$
$F \rightarrow T \cdot$



Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
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 $T \rightarrow \text{int}$
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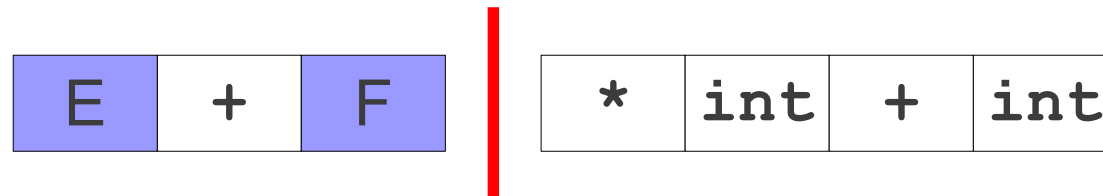
$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow E + \cdot F$
$F \rightarrow \cdot F * T$



Tracking Our Position

$S \rightarrow E$
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 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

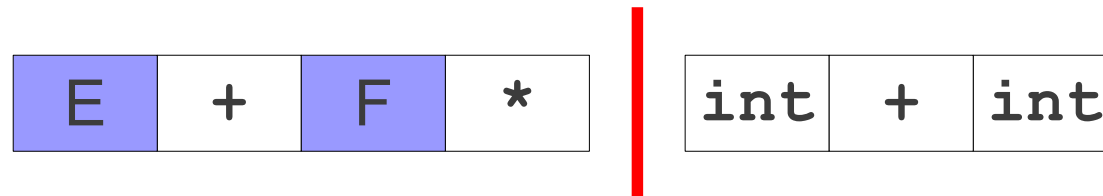
$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
$E \rightarrow E + \cdot F$
$F \rightarrow F \cdot * T$



Tracking Our Position

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

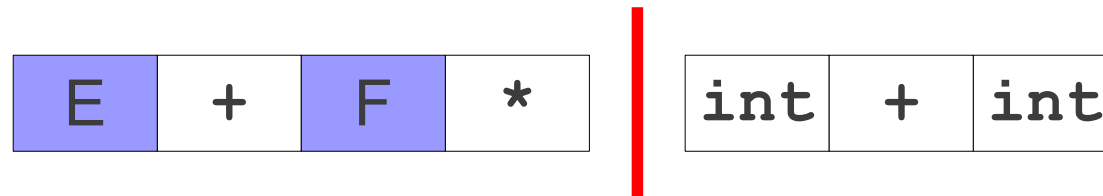
$S \rightarrow \cdot E$
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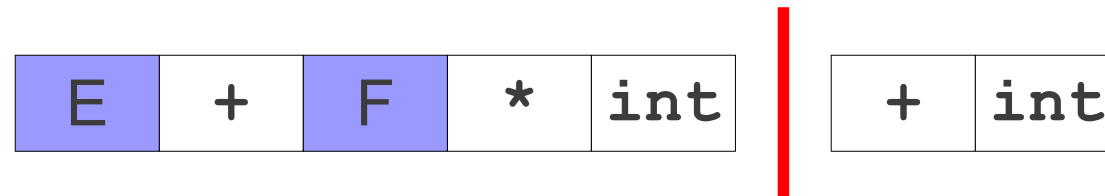
$S \rightarrow \cdot E$
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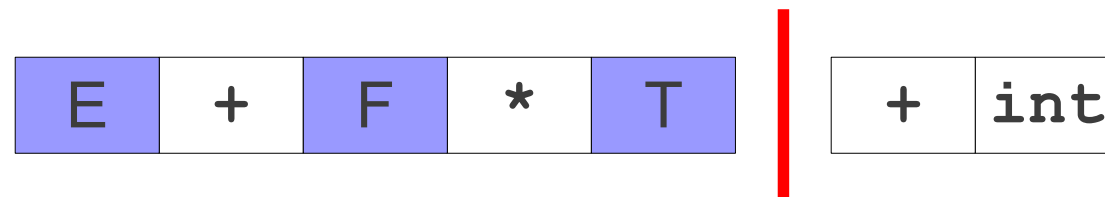
$S \rightarrow \cdot E$
$E \rightarrow \cdot E + F$
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 $F \rightarrow T$
 $T \rightarrow \text{int}$
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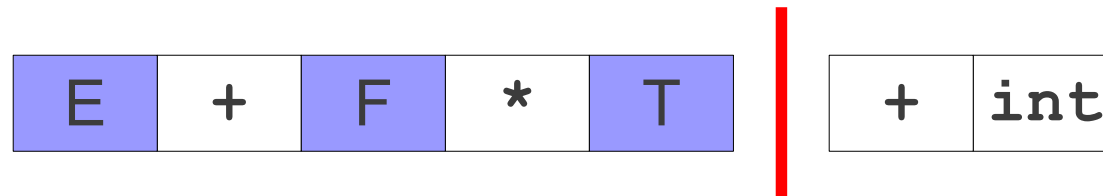
$S \rightarrow \cdot E$
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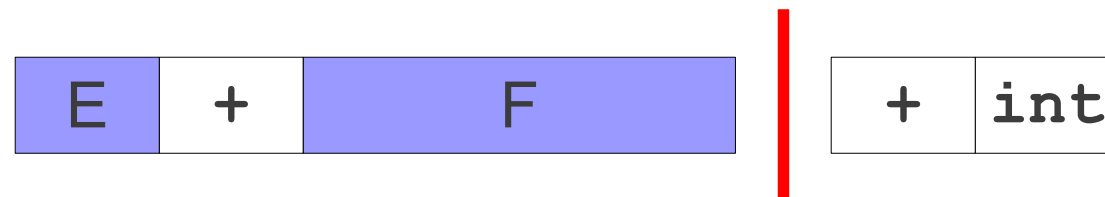
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$F \rightarrow F * T \cdot$



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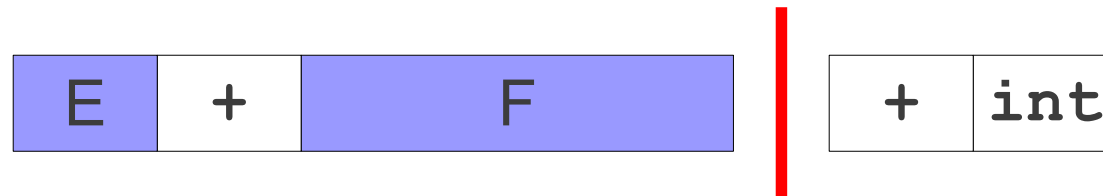
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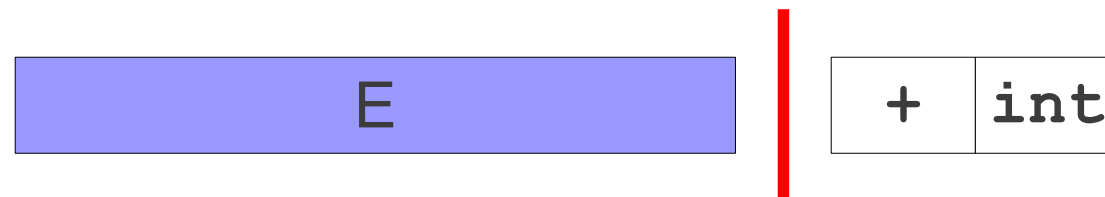
$S \rightarrow \cdot E$
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$E \rightarrow E + F \cdot$



Tracking Our Position

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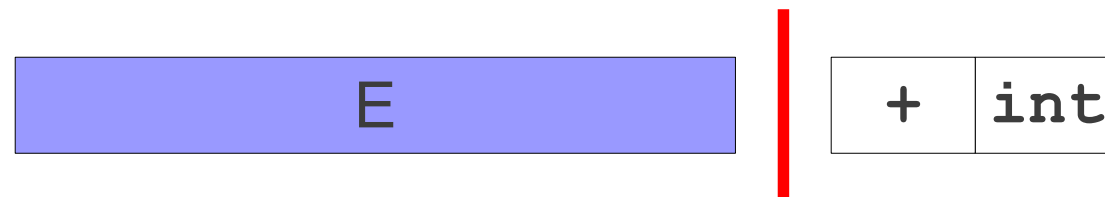
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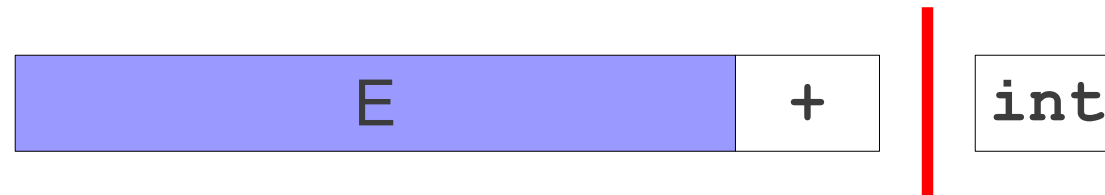
$S \rightarrow \cdot E$
$E \rightarrow E \cdot + F$



Tracking Our Position

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 $E \rightarrow E + F$
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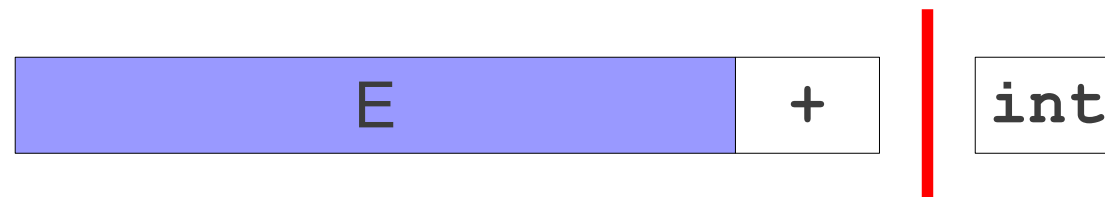
$S \rightarrow \cdot E$
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Tracking Our Position

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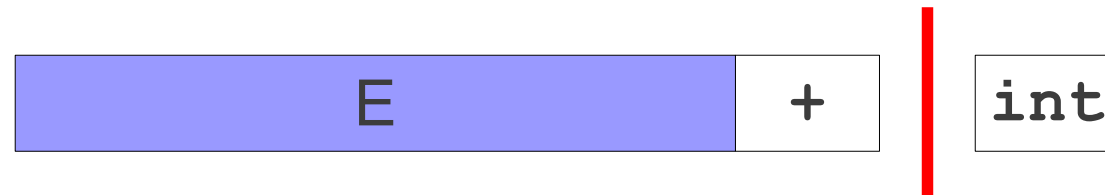
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Tracking Our Position

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$E \rightarrow E + \cdot F$
$F \rightarrow \cdot T$
$T \rightarrow \text{int} \cdot$

E	+	int
---	---	-----



Tracking Our Position

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$S \rightarrow \cdot E$
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$F \rightarrow T \cdot$

E	+	T
---	---	---



Tracking Our Position

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 $F \rightarrow T$
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$S \rightarrow \cdot E$
$E \rightarrow E + \cdot F$

E	+	F
---	---	---



Tracking Our Position

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 $F \rightarrow T$
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 $T \rightarrow (E)$

$S \rightarrow \cdot E$
$E \rightarrow E + F \cdot$

E	+	F
---	---	---



Tracking Our Position

$S \rightarrow \cdot E$

$S \rightarrow E$

$E \rightarrow F$

$E \rightarrow E + F$

$F \rightarrow F * T$

$F \rightarrow T$

$T \rightarrow \text{int}$

$T \rightarrow (E)$

E

Tracking Our Position

$S \rightarrow E \cdot$

$S \rightarrow E$

$E \rightarrow F$

$E \rightarrow E + F$

$F \rightarrow F * T$

$F \rightarrow T$

$T \rightarrow \text{int}$

$T \rightarrow (E)$

E

Generating Left-Hand Sides

- At any instant in time, the contents of the left side of the parser can be described using the following process:
 - Trace out, from the start symbol, the series of productions that have not yet been completed and where we are in each production.
 - For each production, in order, output all of the symbols up to the point where we change from one production to the next.

Recognizing Left-Hand Sides

- Given that we have a procedure for *generating* left-hand sides, can we build a procedure for *recognizing* those left-hand sides?
- Idea: At each point, track
 - Which production we are in, and
 - Where we are in that production.
- At each point, we can do one of two things:
 - Match the next symbol of the candidate left-hand side with the next symbol in the current production, or
 - If the next symbol of the candidate left-hand side is a nonterminal, nondeterministically guess which production to try next.

Recognizing Left-Hand Sides

S → **E**

E → **F**

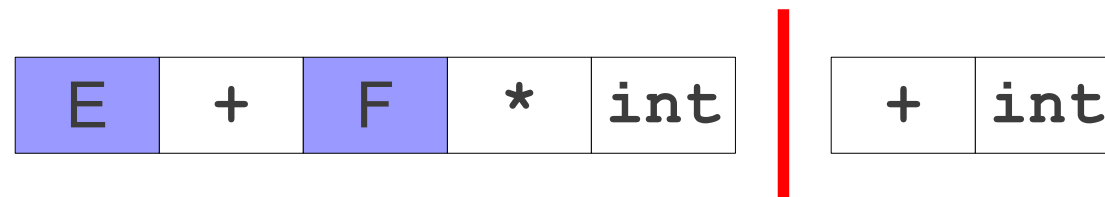
E → **E** + **F**

F → **F** * **T**

F → **T**

T → **int**

T → (**E**)



Recognizing Left-Hand Sides

$S \rightarrow \cdot E$

$S \rightarrow E$

$E \rightarrow F$

$E \rightarrow E + F$

$F \rightarrow F * T$

$F \rightarrow T$

$T \rightarrow \text{int}$

$T \rightarrow (E)$

E	+	F	*	int
---	---	---	---	-----

+	int
---	-----

Recognizing Left-Hand Sides

$S \rightarrow \cdot E$

$S \rightarrow E$

$E \rightarrow F$

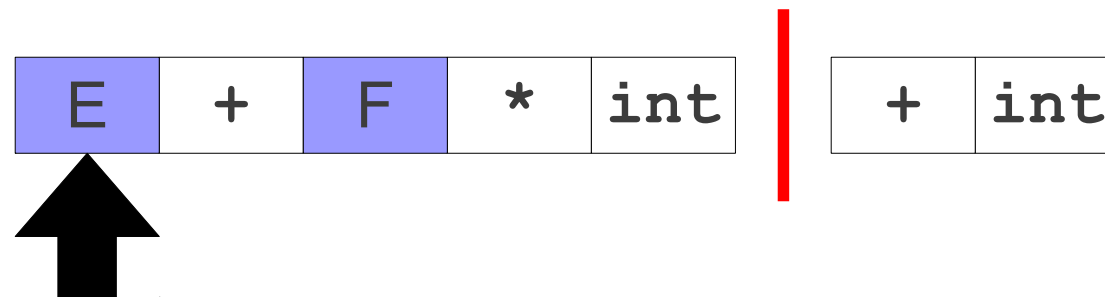
$E \rightarrow E + F$

$F \rightarrow F * T$

$F \rightarrow T$

$T \rightarrow \text{int}$

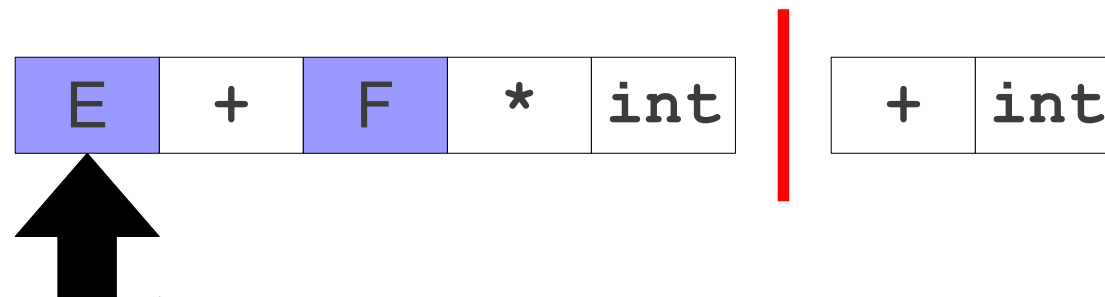
$T \rightarrow (E)$



Recognizing Left-Hand Sides

$S \rightarrow E$
 $E \rightarrow F$
 $E \rightarrow E + F$
 $F \rightarrow F * T$
 $F \rightarrow T$
 $T \rightarrow \text{int}$
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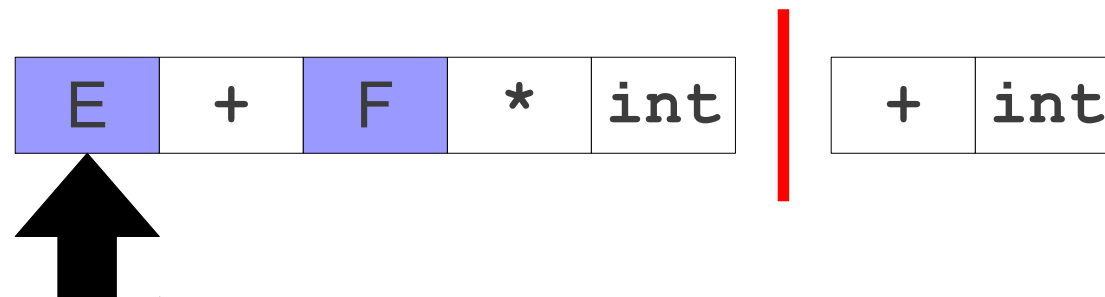
$S \rightarrow \cdot E$
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Recognizing Left-Hand Sides

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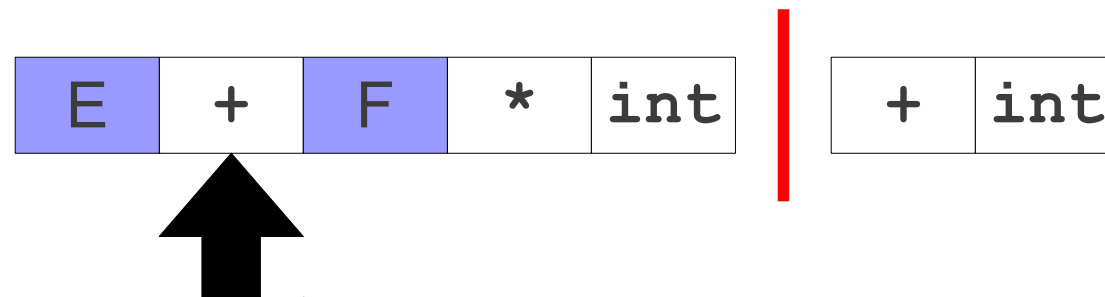
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Recognizing Left-Hand Sides

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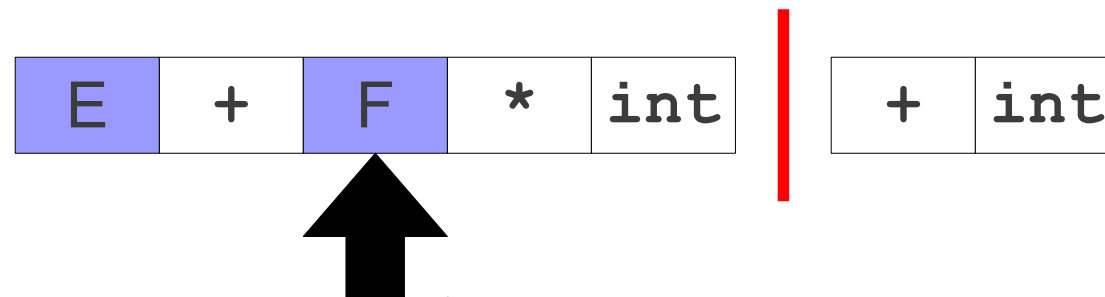
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$E \rightarrow E \cdot + F$



Recognizing Left-Hand Sides

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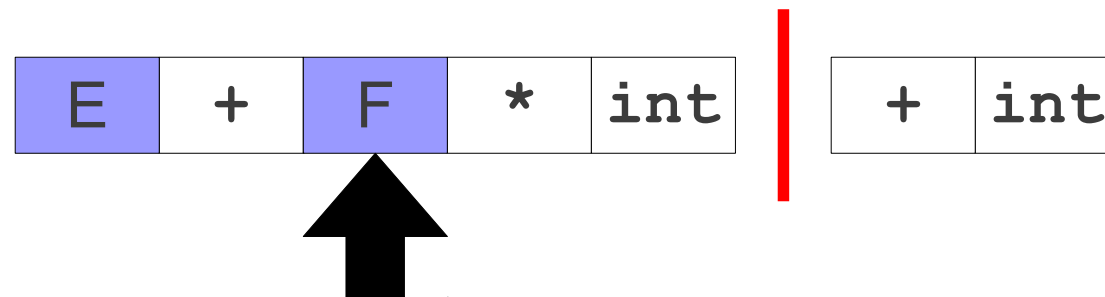
$S \rightarrow \cdot E$
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Recognizing Left-Hand Sides

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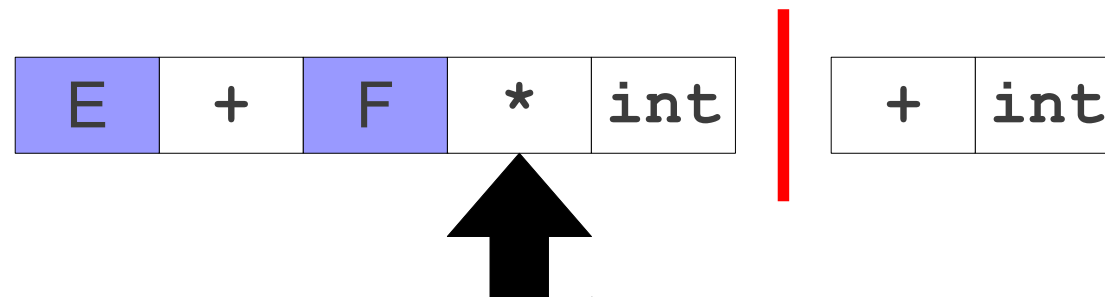
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$F \rightarrow \cdot F * T$



Recognizing Left-Hand Sides

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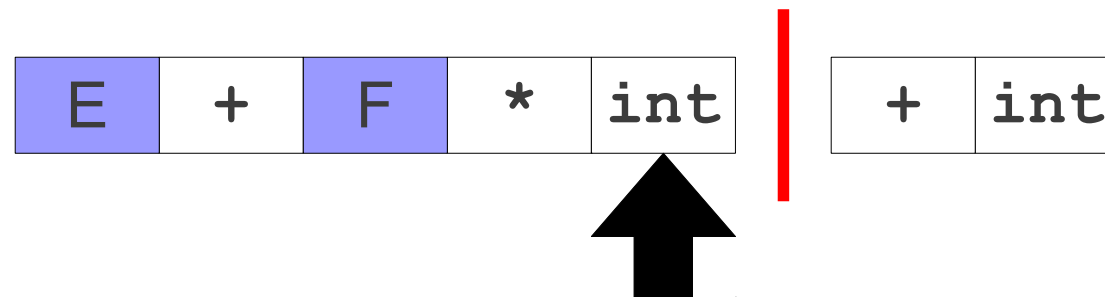
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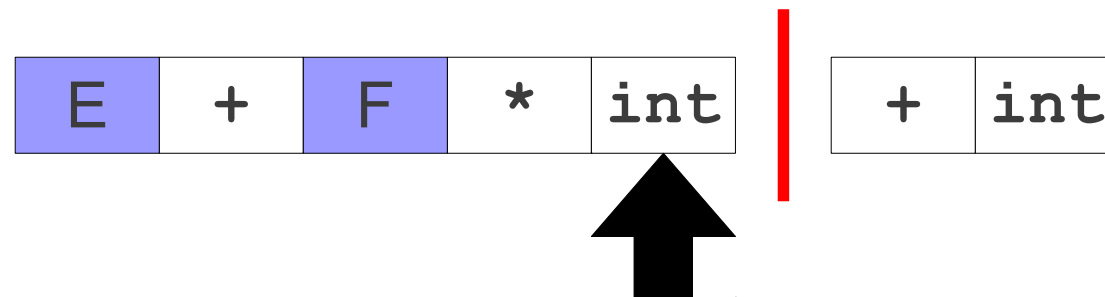
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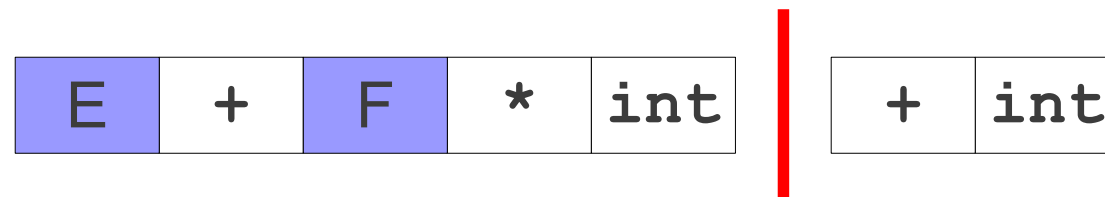
$S \rightarrow \cdot E$
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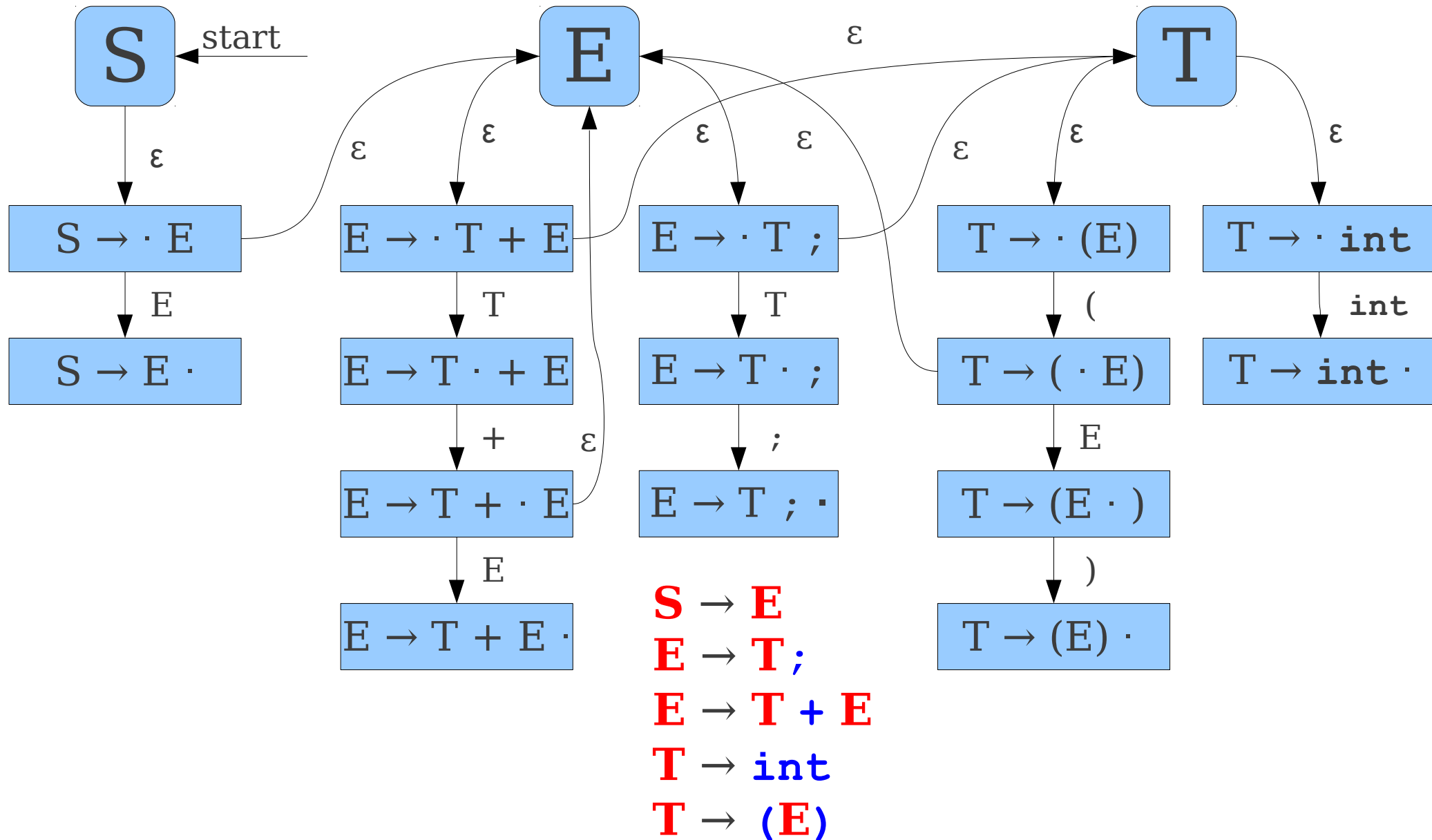
$S \rightarrow \cdot E$
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$F \rightarrow F * \cdot T$
$T \rightarrow \text{int} \cdot$



An Important Result

- There are only finitely many productions, and within those productions only finitely many positions.
- At any point in time, we only need to track where we are in one production.
- There are only finitely many options we can take at any one point.
- **We can use a finite automaton as our recognizer.**

An Automaton for Left Areas



Constructing the Automaton

- Create a state for each nonterminal.
- For each production $A \rightarrow \gamma$:
 - Construct states $A \rightarrow \alpha \cdot \omega$ for each possible way of splitting γ into two substrings α and ω .
 - Add transitions on x between $A \rightarrow \alpha \cdot x\omega$ and $A \rightarrow \alpha x \cdot \omega$.
- For each state $A \rightarrow \alpha \cdot B\omega$ for nonterminal B , add an ε -transition from $A \rightarrow \alpha \cdot B\omega$ to B .

Why This Matters

- Our initial goal was to find handles.
- When running this automaton, if we ever end up in a state with a rule of the form

$$\mathbf{A} \rightarrow \omega \cdot$$

- Then we might be looking at a handle.
- This automaton can be used to discover possible handle locations!

Adding Determinism

- Typically, this handle-finding automaton is implemented deterministically.
- We could construct a deterministic parsing automaton by constructing the nondeterministic automaton and applying the subset construction, but there is a more direct approach.

A Deterministic Automaton

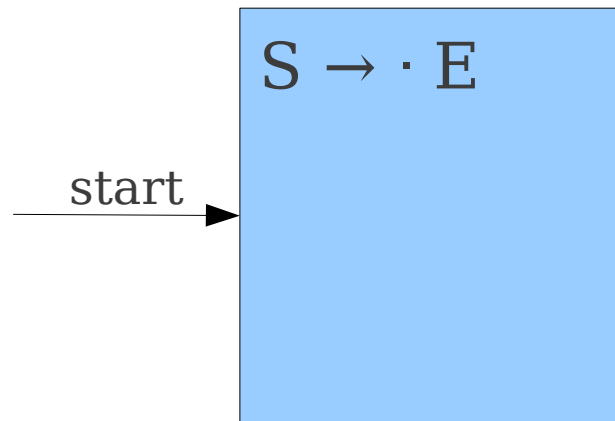
S → **E**

E → **T** ;

E → **T** + **E**

T → **int**

T → (**E**)



A Deterministic Automaton

S → **E**

E → **T**;

E → **T** + **E**

T → **int**

T → (**E**)

start →

S → · E
E → · T;
E → · T + E

A Deterministic Automaton

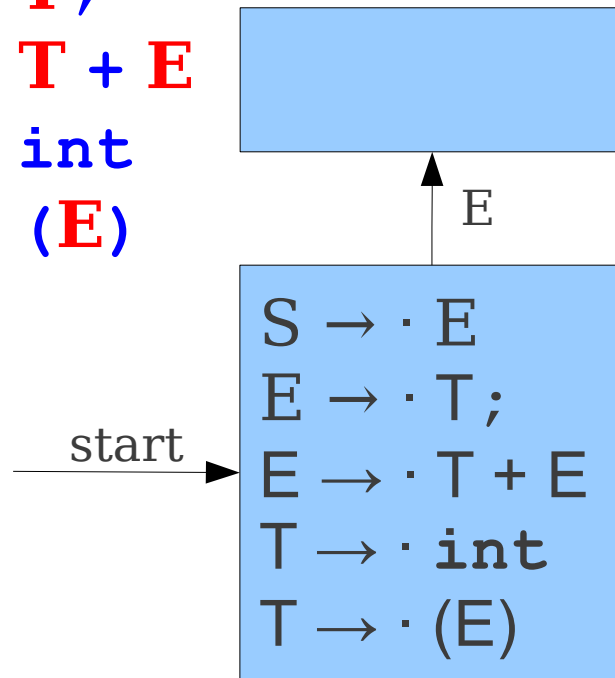
$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

start →

$S \rightarrow \cdot E$
 $E \rightarrow \cdot T;$
 $E \rightarrow \cdot T + E$
 $T \rightarrow \cdot \text{int}$
 $T \rightarrow \cdot (E)$

A Deterministic Automaton

S → **E**
E → **T**;
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A Deterministic Automaton

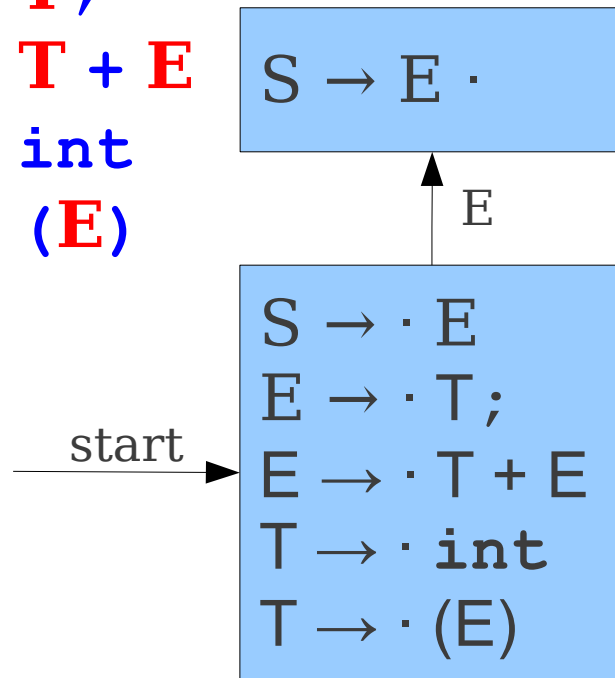
S → **E**

E → **T**;

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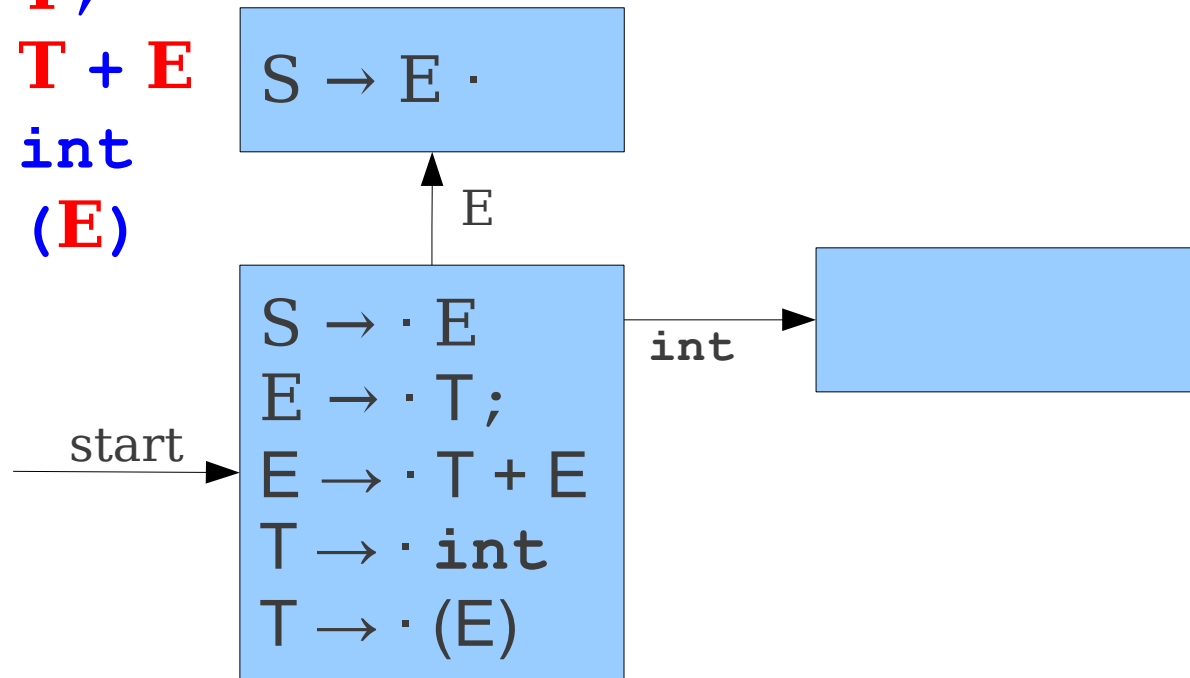
T → **int**

T → (**E**)



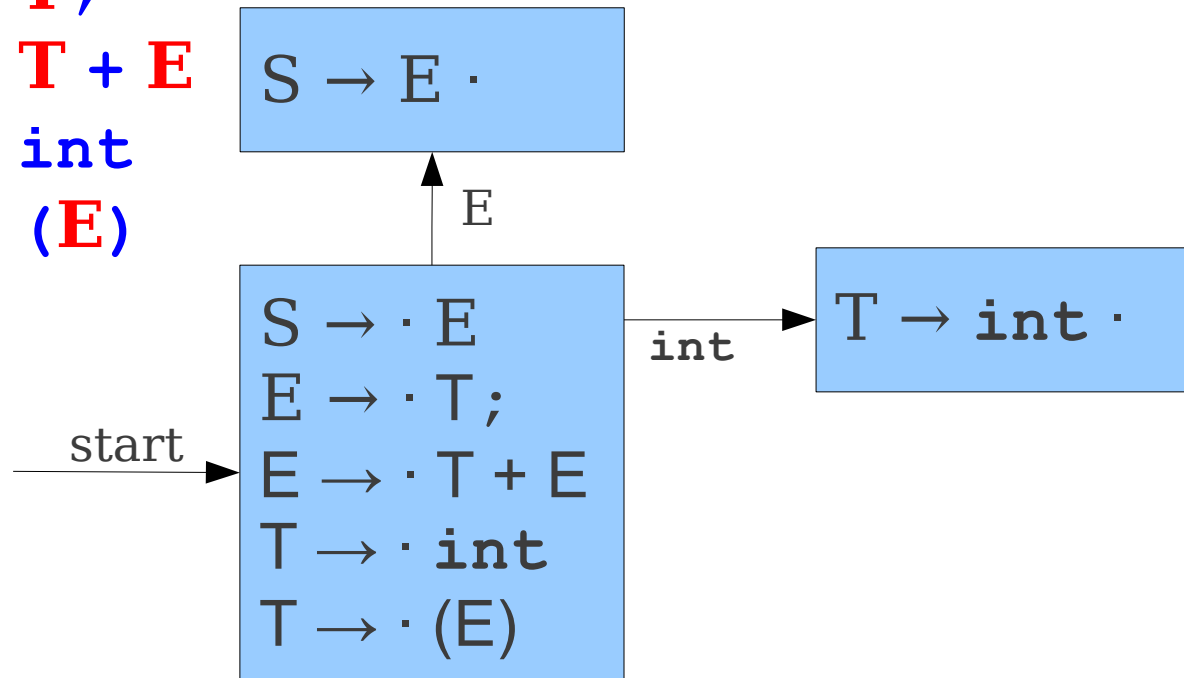
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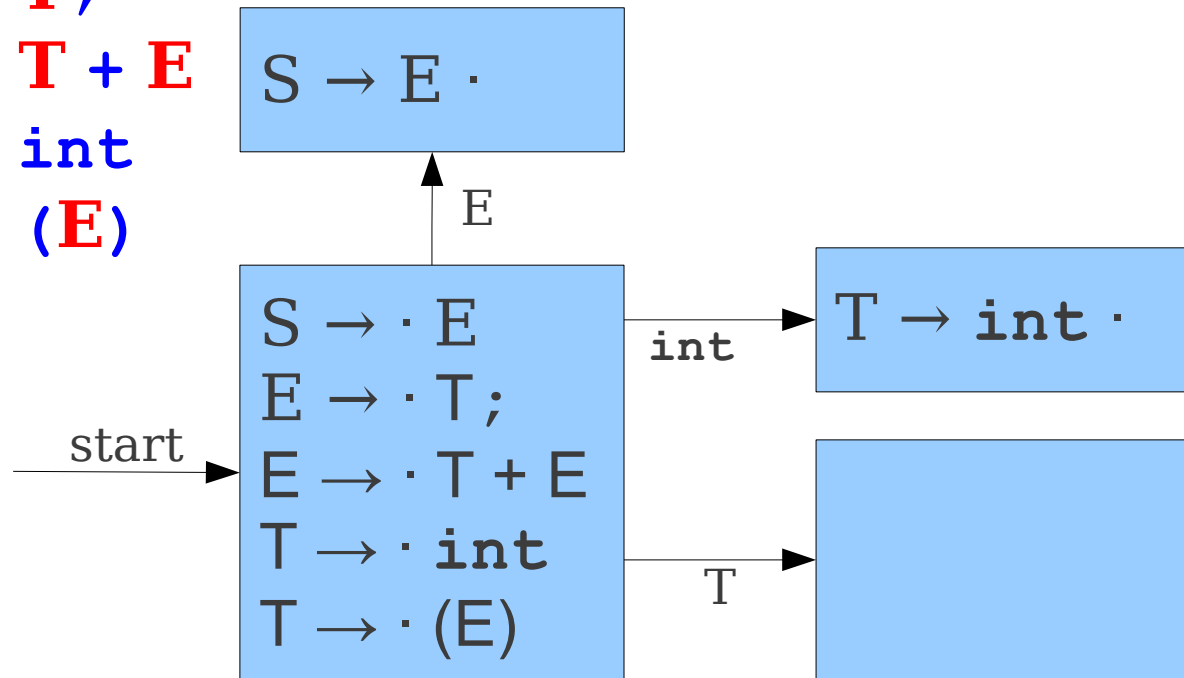
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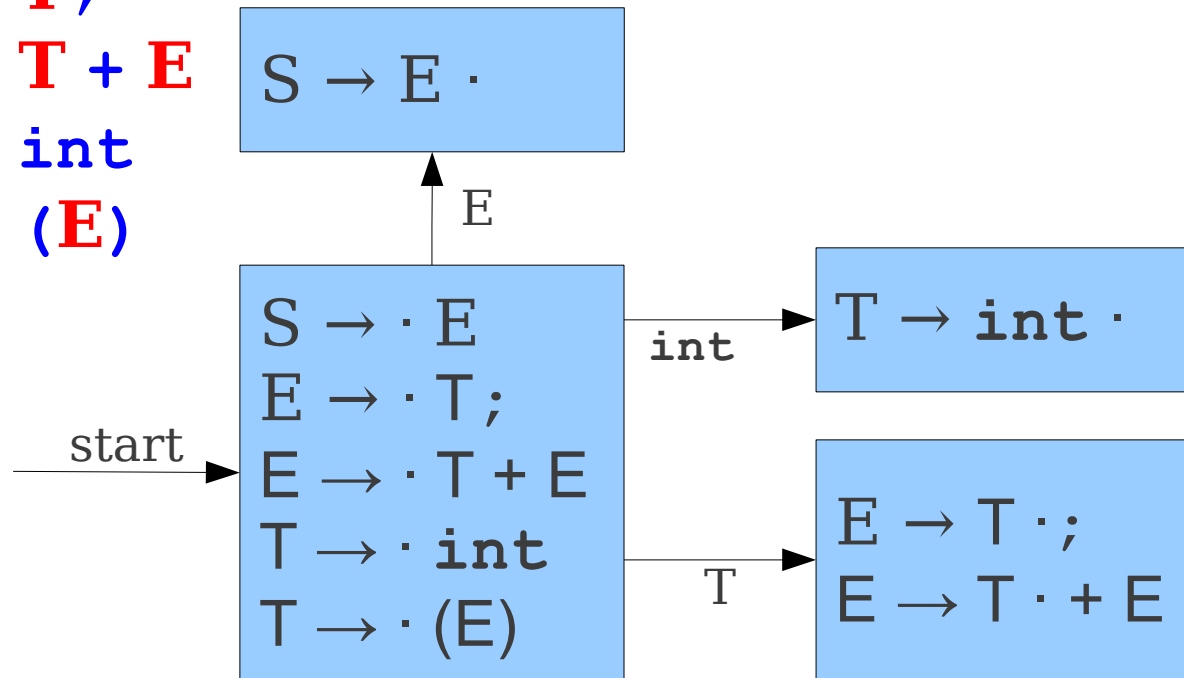
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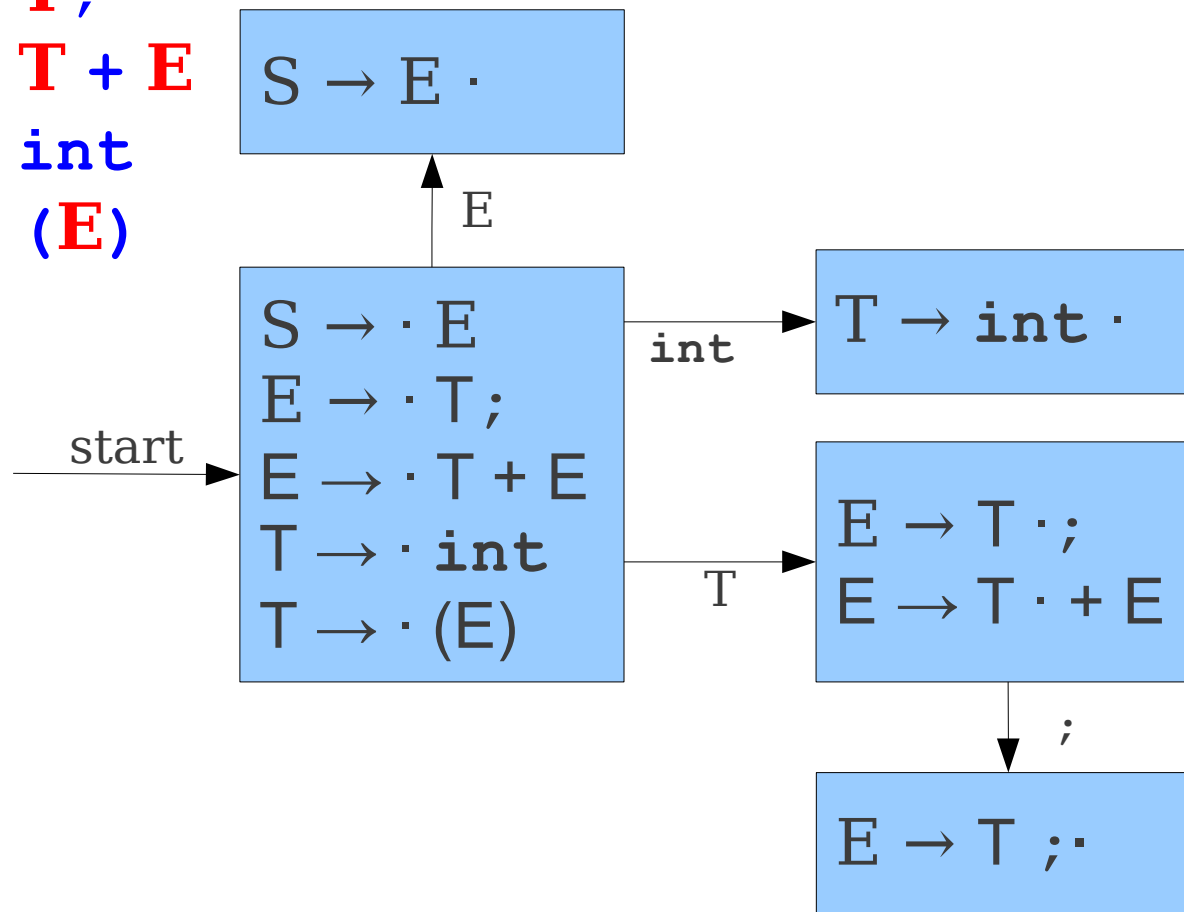
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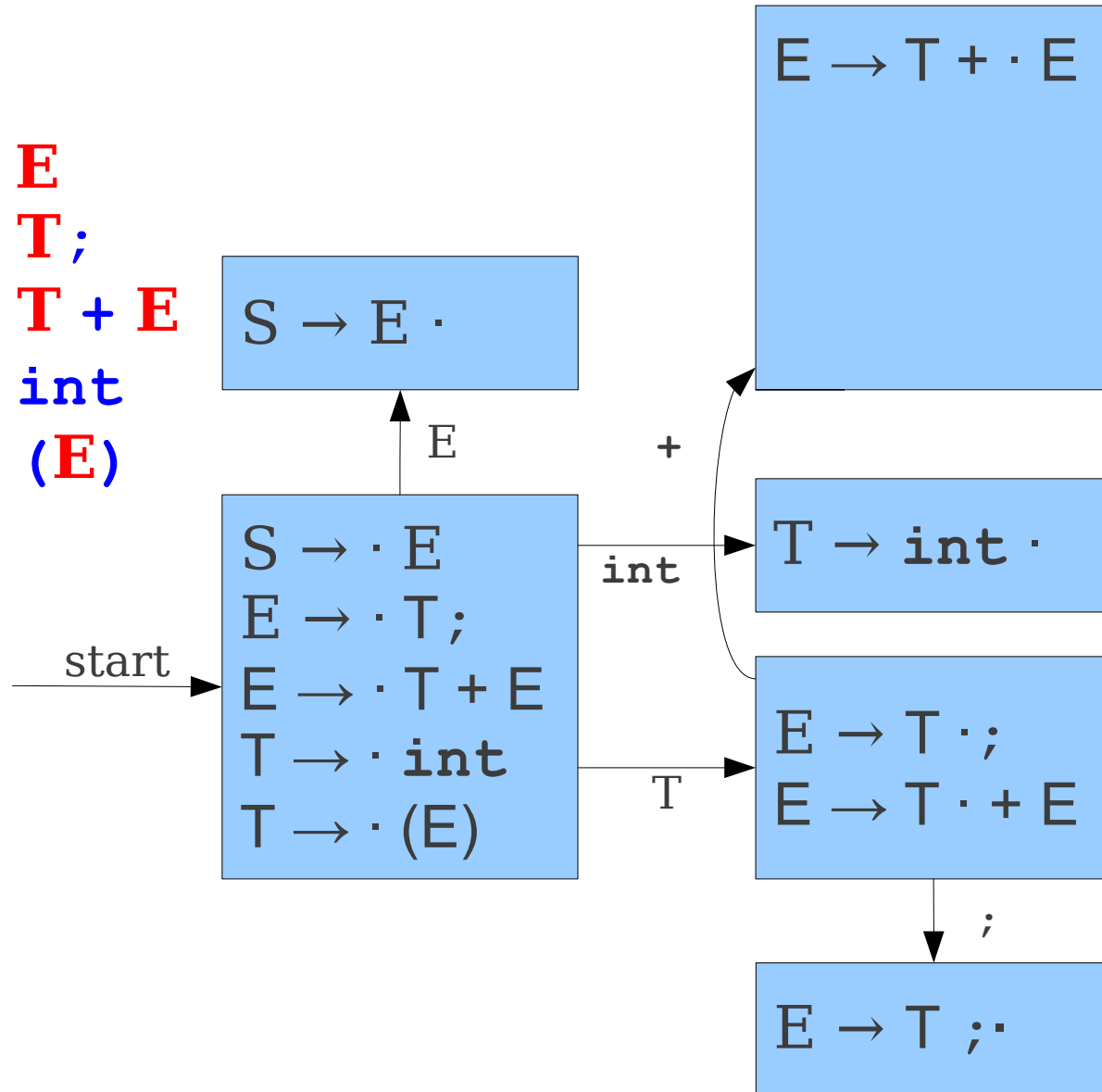
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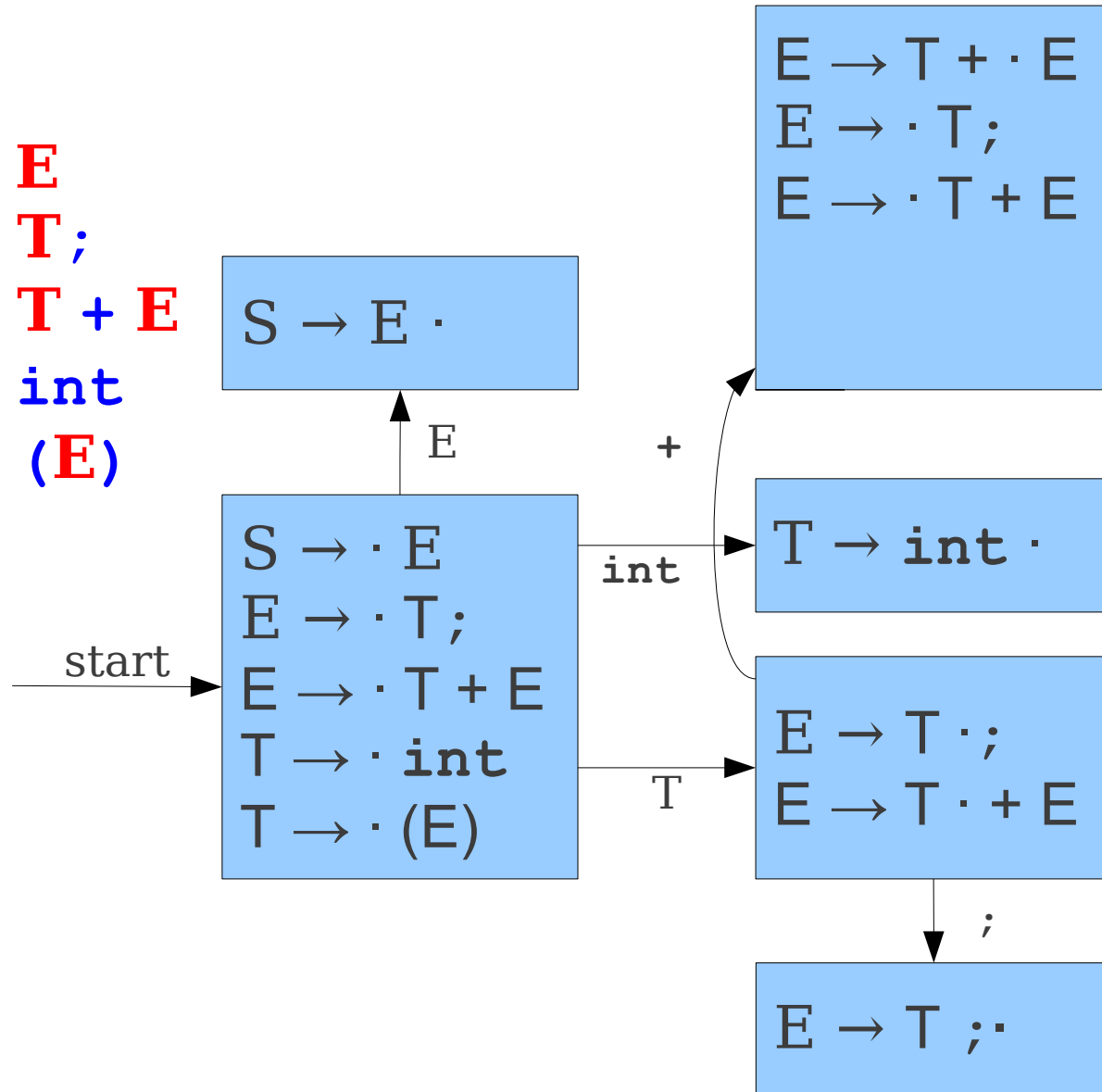
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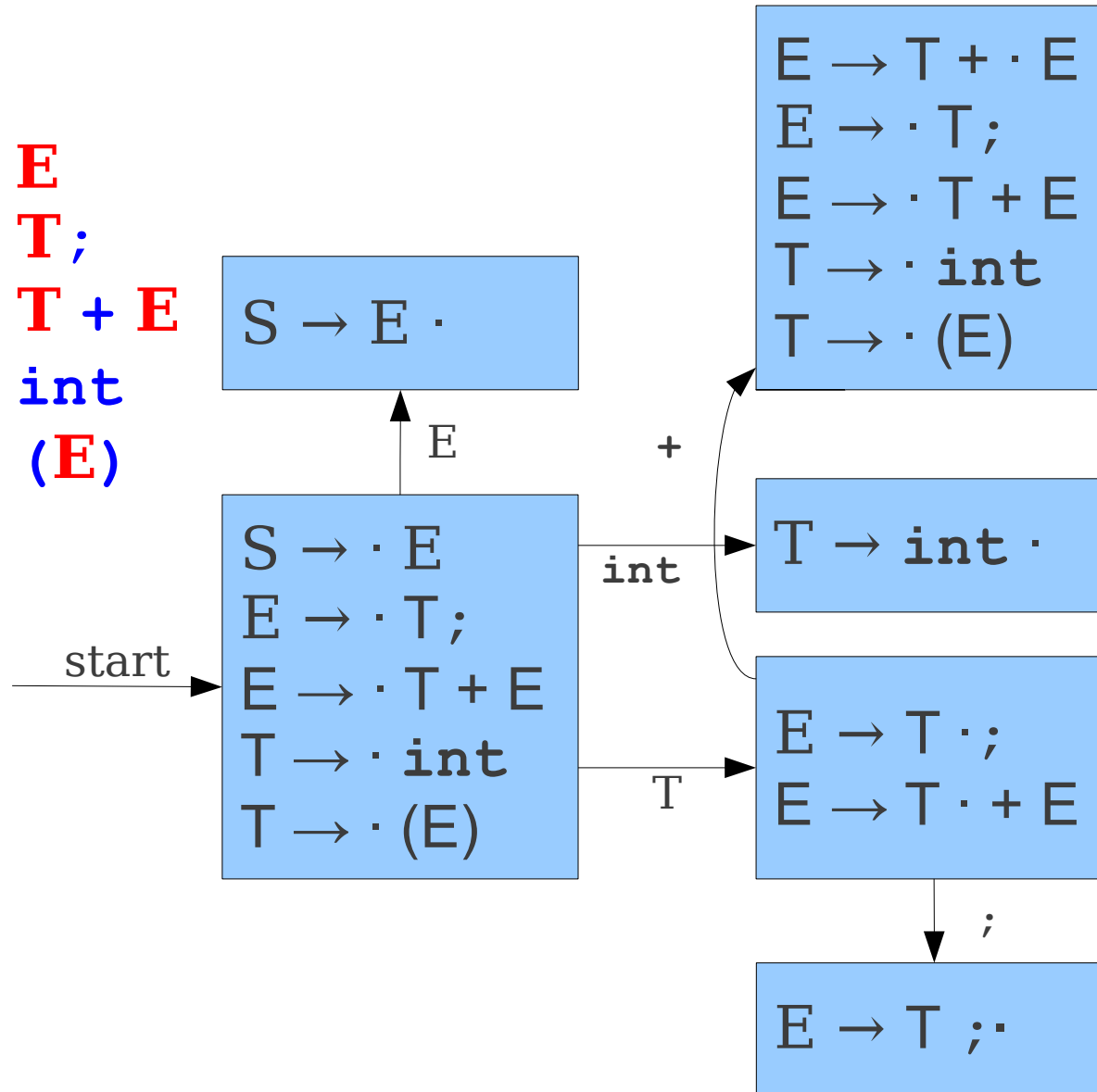
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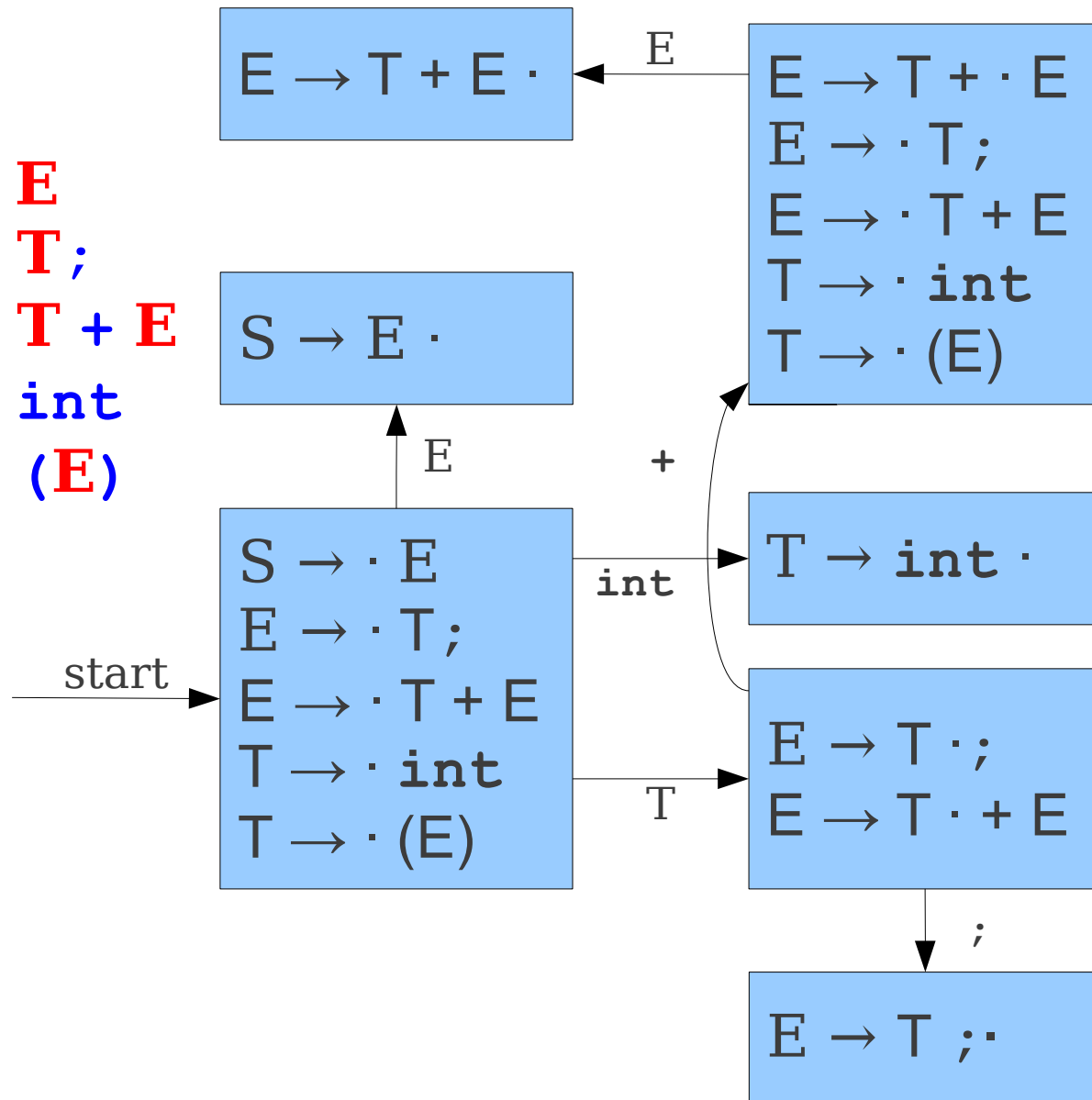
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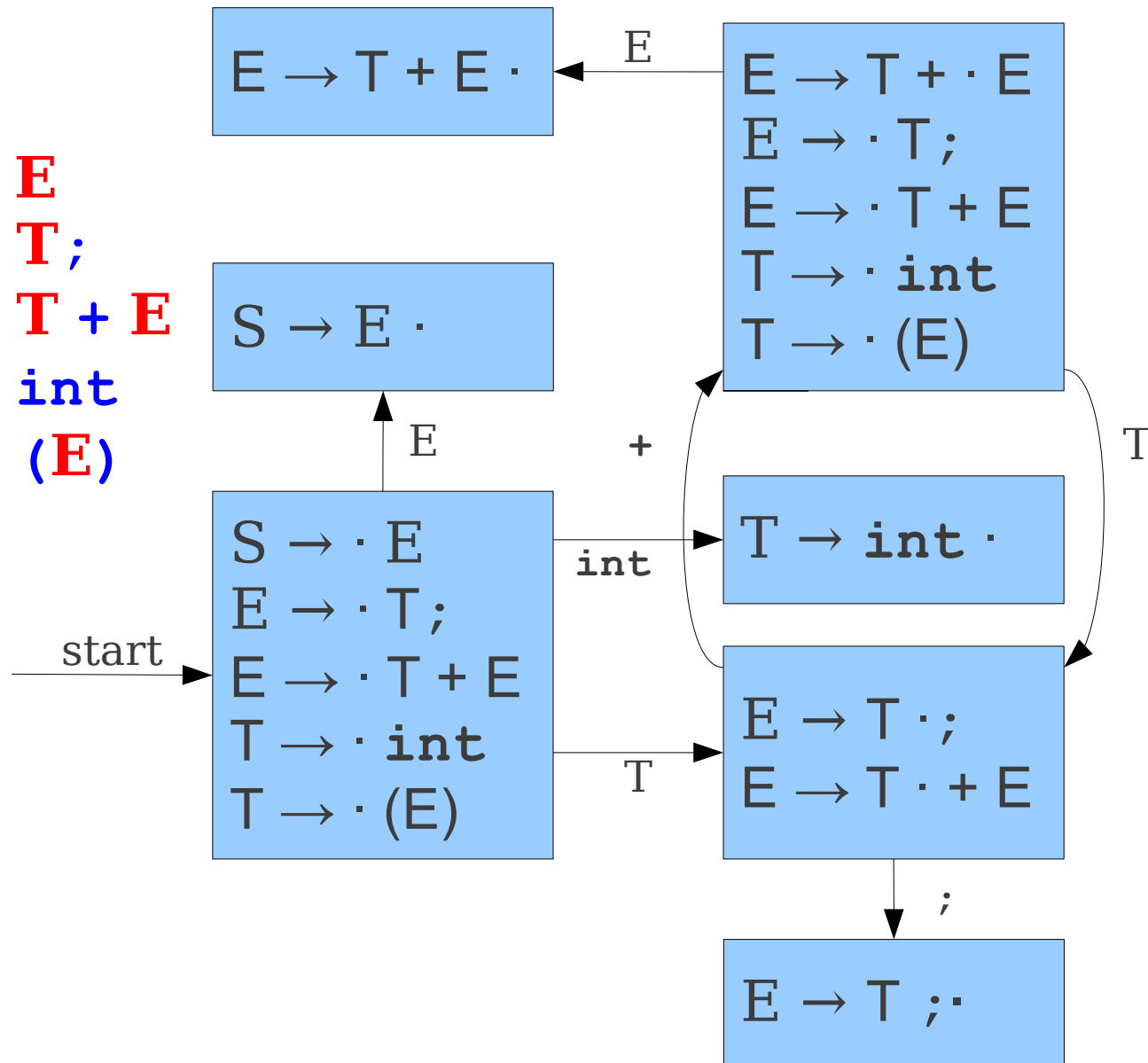
A Deterministic Automaton

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E → **T**;
E → **T** + **E**
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T → (**E**)



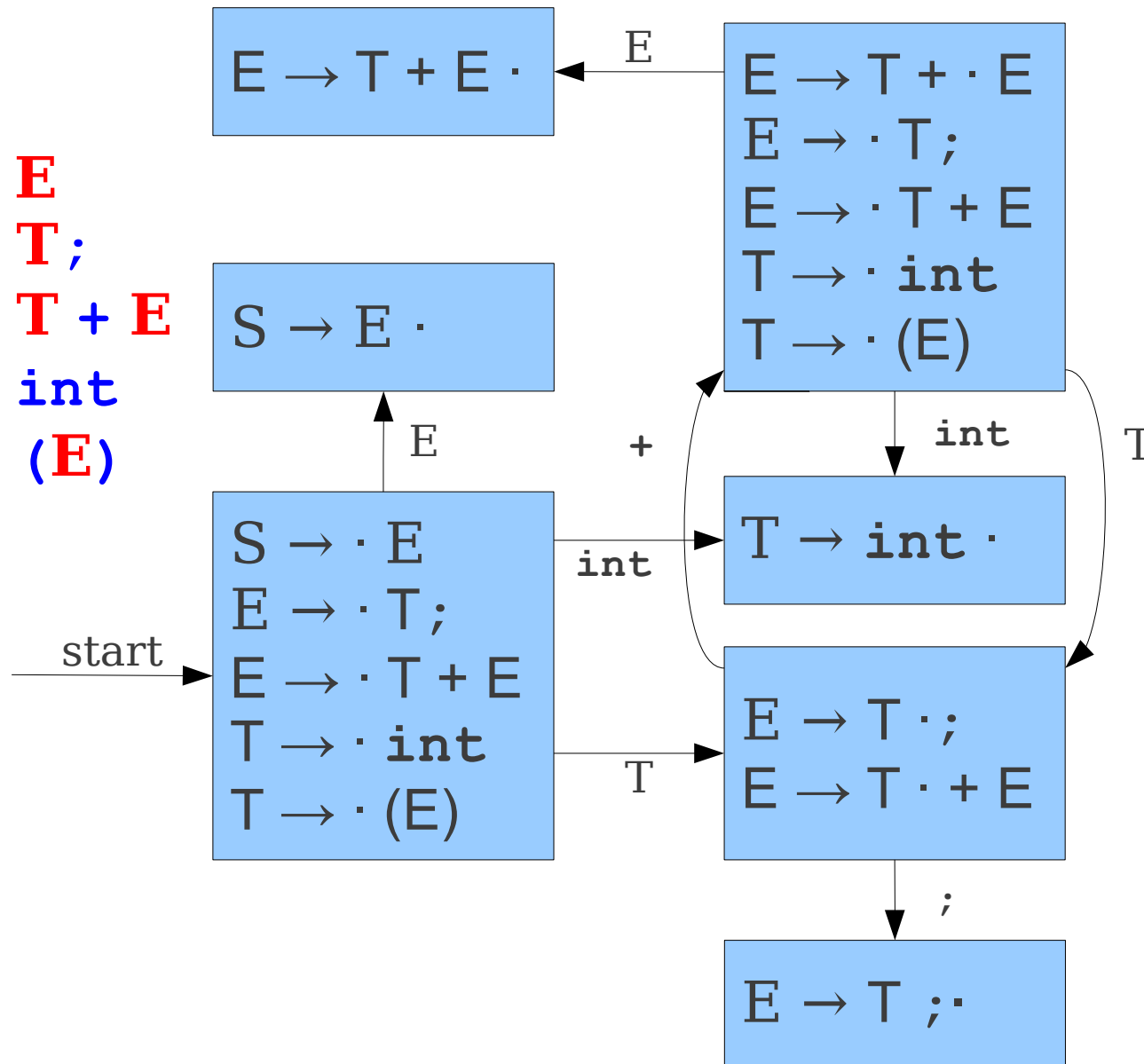
A Deterministic Automaton

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E → **T**;
E → **T** + **E**
T → **int**
T → (**E**)



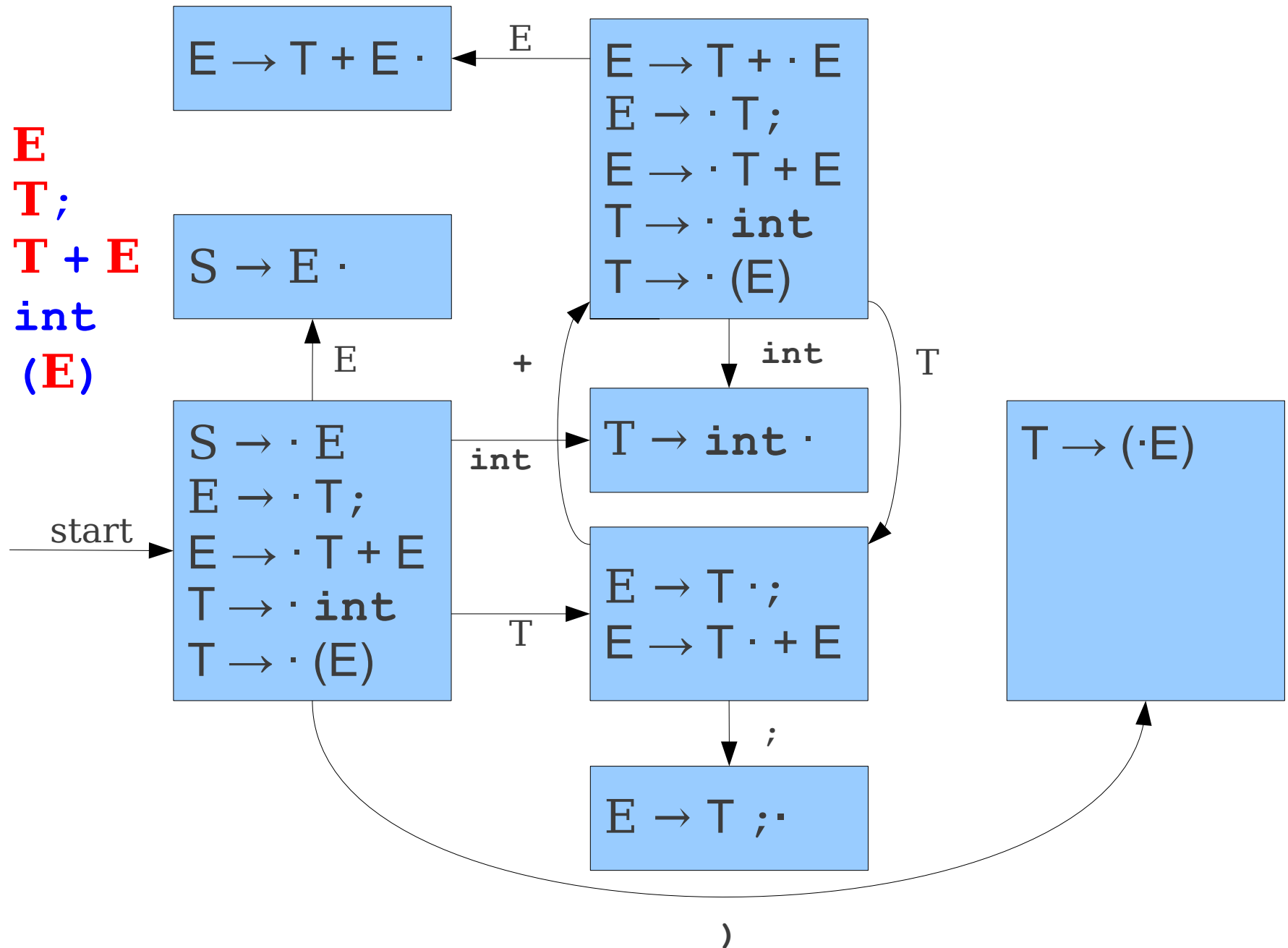
A Deterministic Automaton

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T → (**E**)



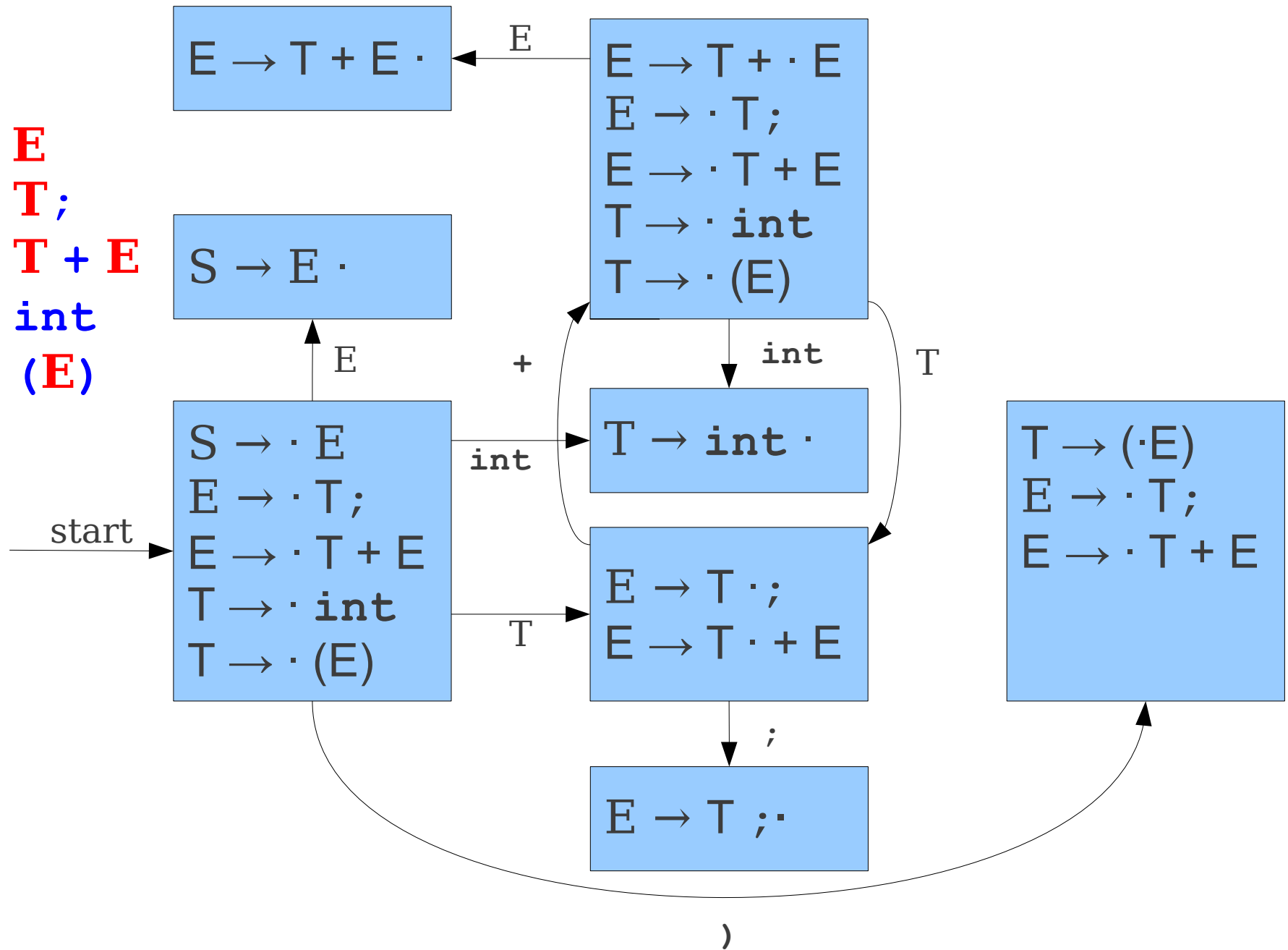
A Deterministic Automaton

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E → **T** + **E**
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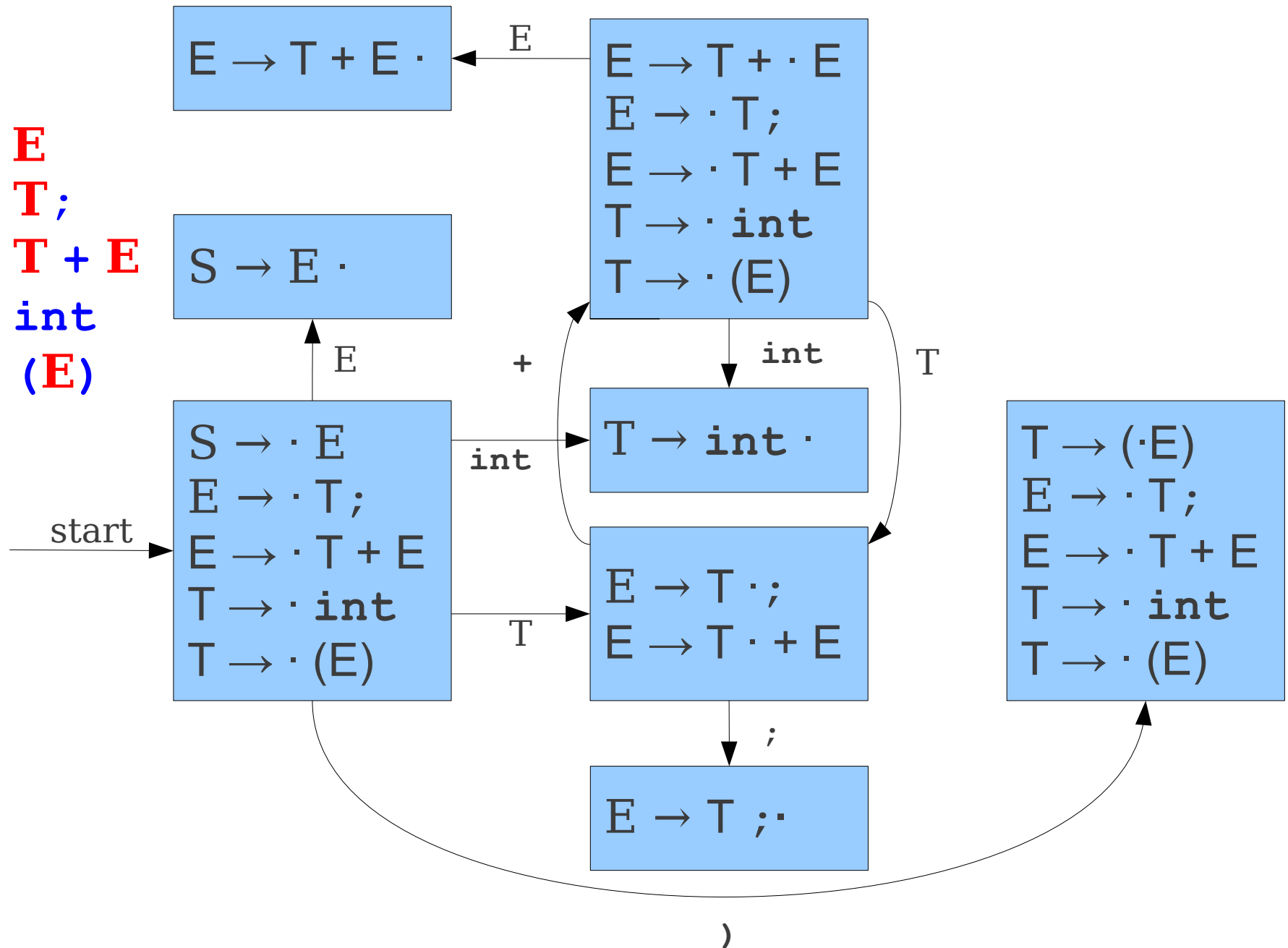
A Deterministic Automaton

S → **E**
E → **T**;
E → **T** + **E**
T → **int**
T → (**E**)



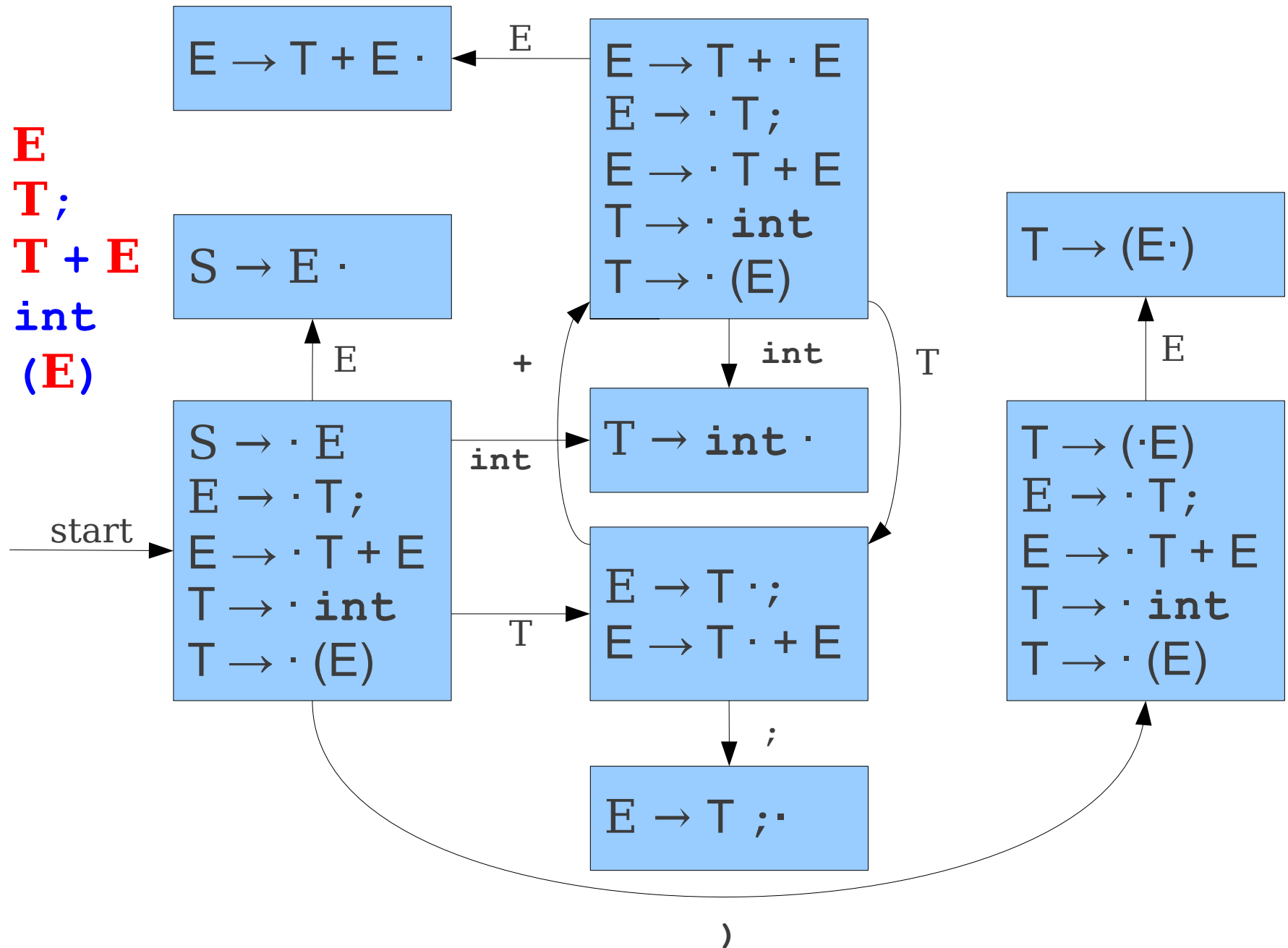
A Deterministic Automaton

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 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



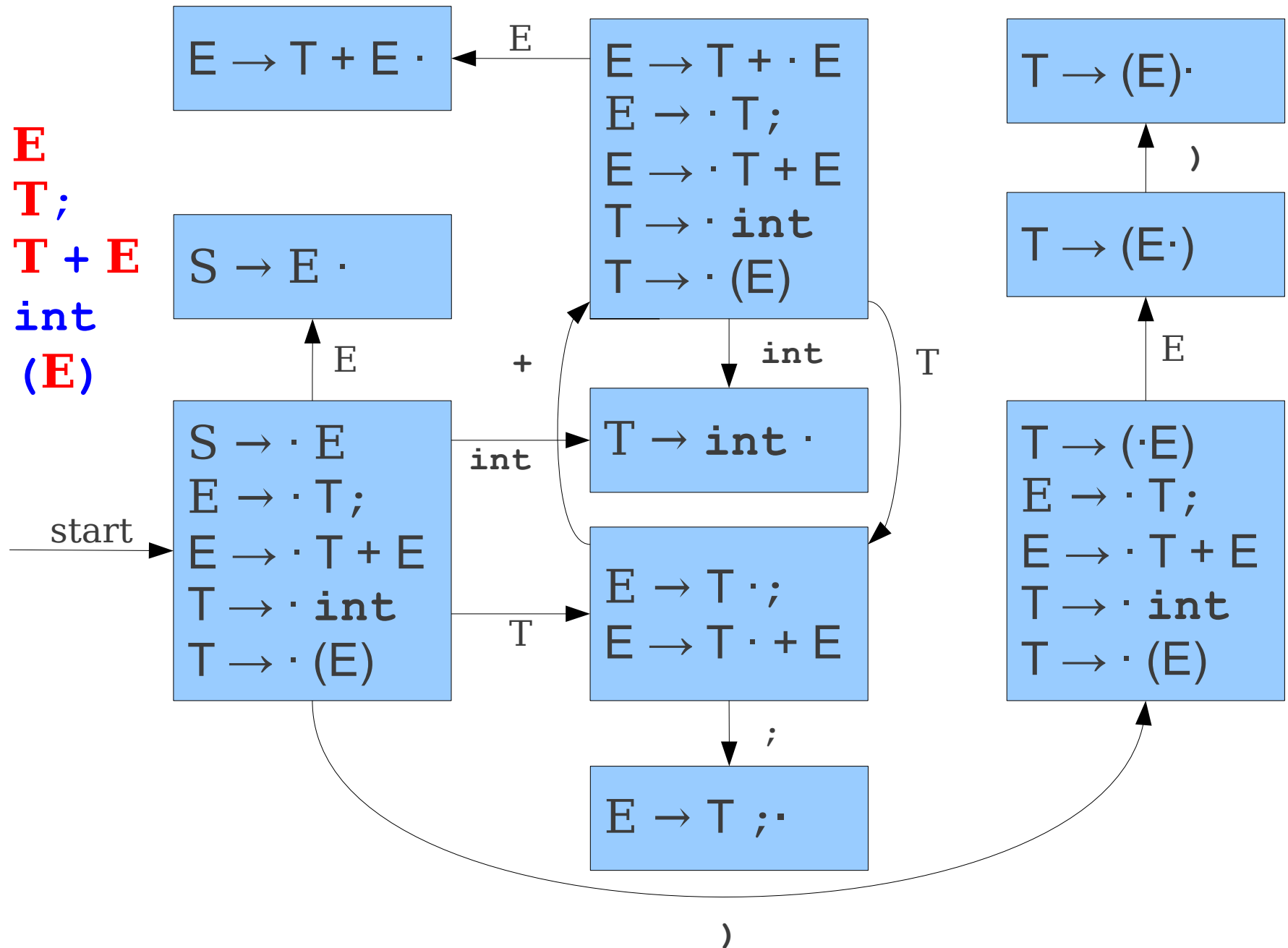
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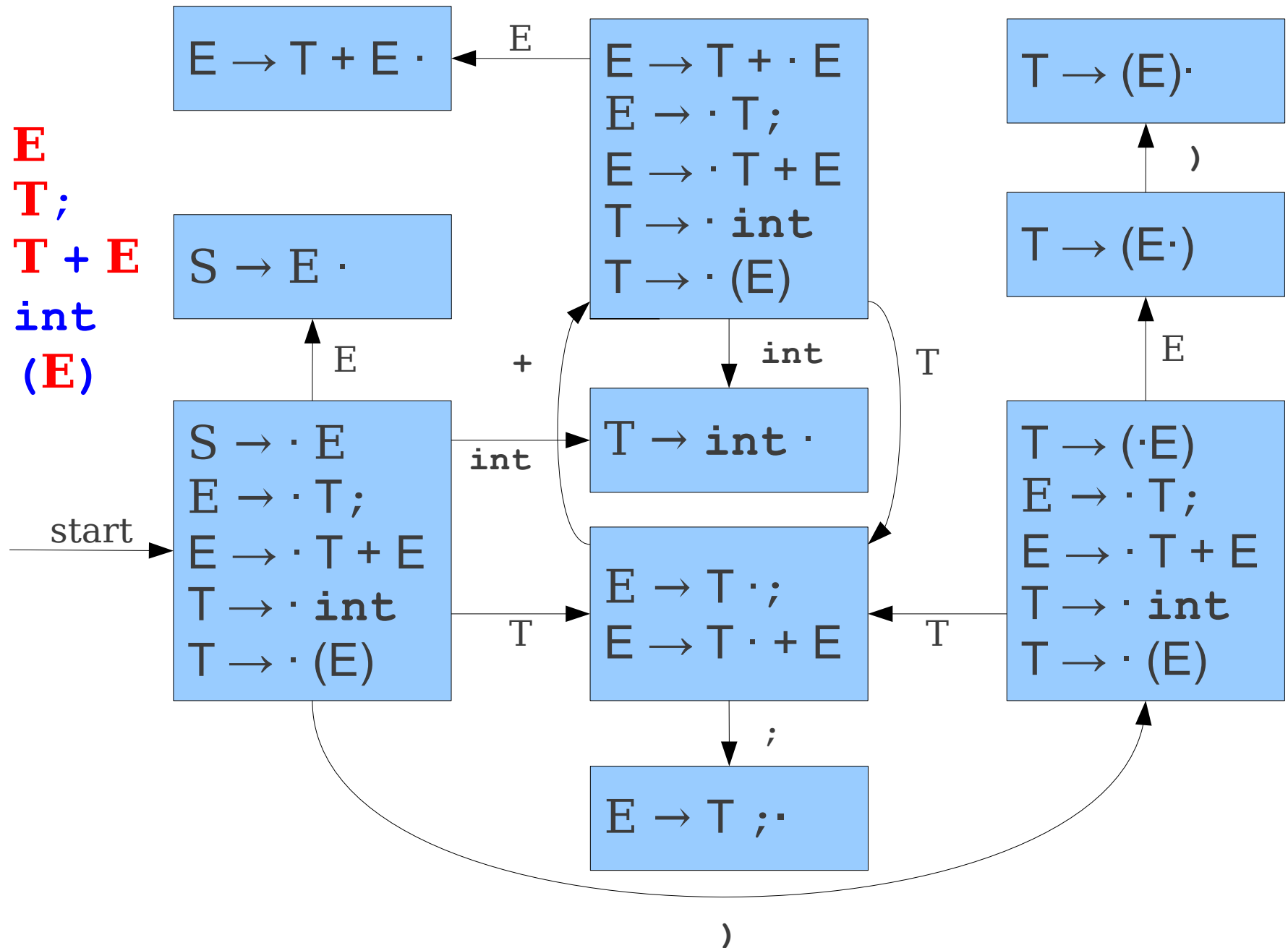
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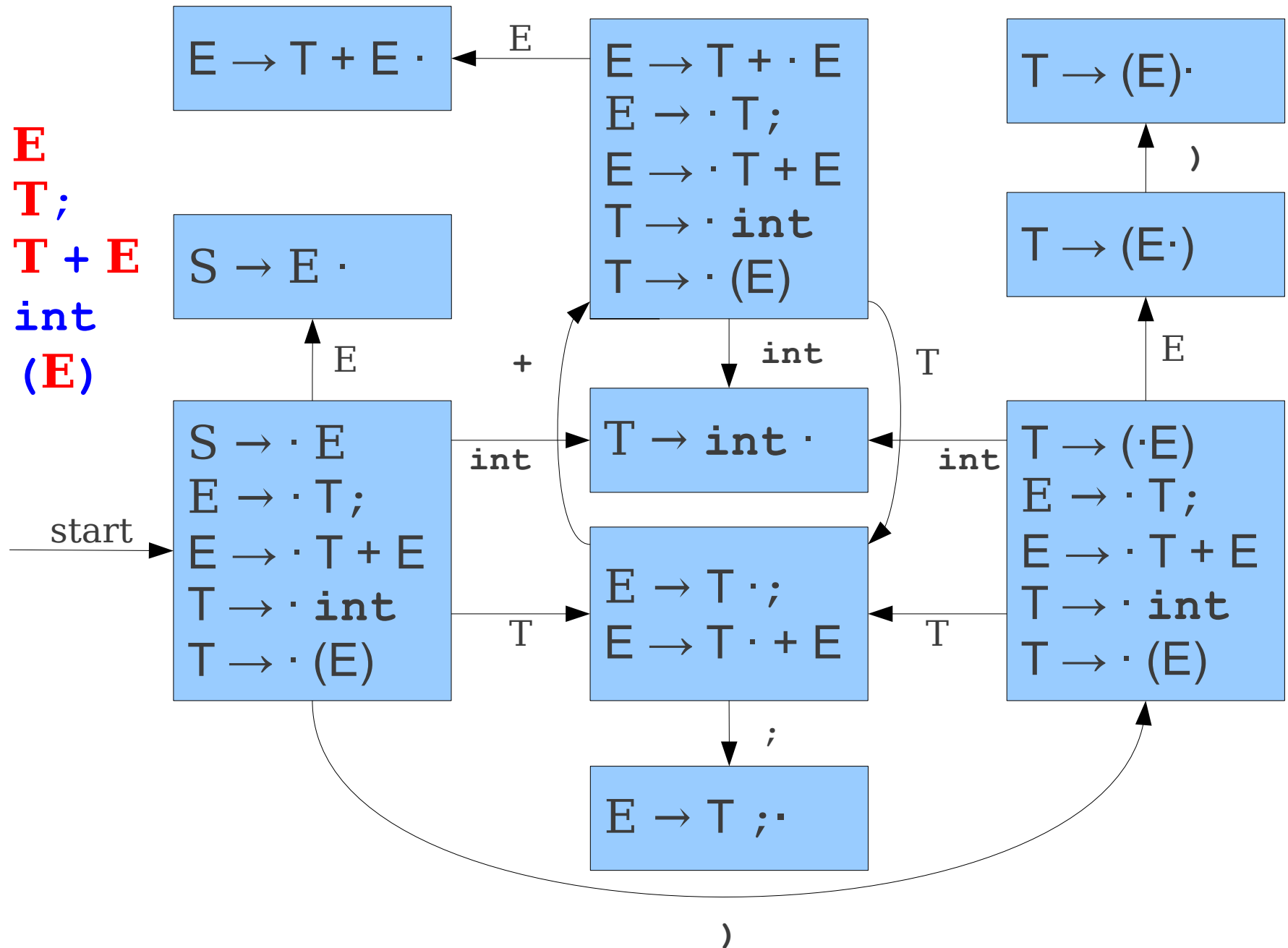
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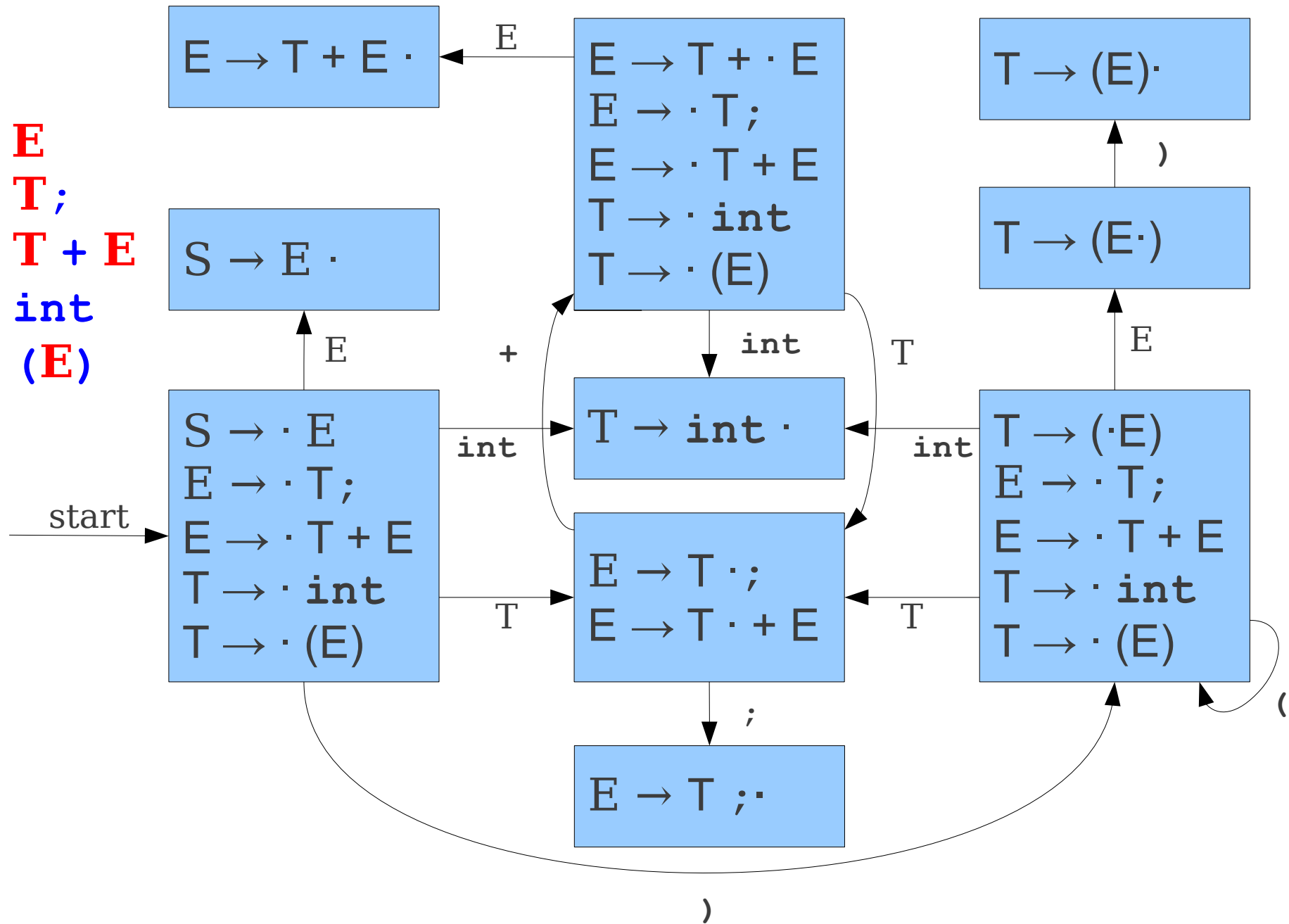
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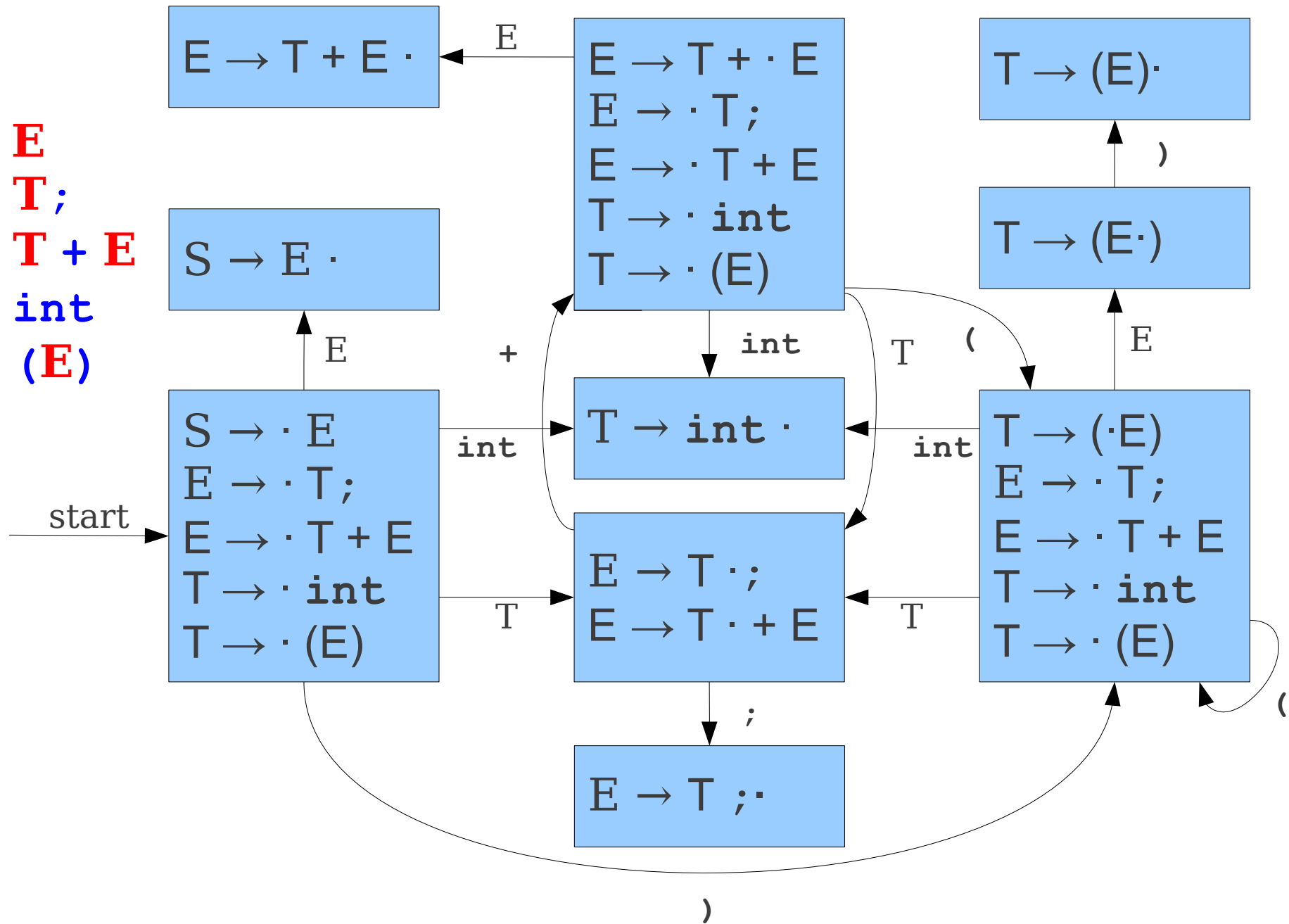
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A Deterministic Automaton

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



Constructing the Automaton II

- Begin in a state containing $S \rightarrow \cdot A$, where S is the augmented start symbol.
- Compute the **closure** of the state:
 - If $A \rightarrow \alpha \cdot B\omega$ is in the state, add $B \rightarrow \cdot \gamma$ to the state for each production $B \rightarrow \gamma$.
 - Yet another fixed-point iteration!
- Repeat until no new states are added:
 - If a state contains a production $A \rightarrow \alpha \cdot x\omega$ for symbol x , add a transition on x from that state to the state containing the closure of $A \rightarrow \alpha x \cdot \omega$
- This is equivalent to a subset construction on the NFA.

Handle-Finding Automata

- Handling-finding automata can be very large.
- NFA has states proportional to the size of the grammar, so DFA can have size exponential in the size of the grammar.
 - There are grammars that can exhibit this worst-case.
- Automata are almost always generated by tools like **bison**.

Finding Handles

- Where do we look for handles?
 - **At the top of the stack.**
- How do we search for handles?
 - **Build a handle-finding automaton.**
- How do we recognize handles?
 - Once we've found a possible handle, how do we confirm that it's correct?

Question Three:

How do we recognize handles?

Handle Recognition

- Our automaton will tell us all places where a handle might be.
- However, if the automaton says that there might be a handle at a given point, we need a way to confirm this.
- We'll thus use **predictive bottom-up parsing**:
 - Have a deterministic procedure for guessing where handles are.
- There are many predictive algorithms, each of which recognize different grammars.

Our First Algorithm: **LR(0)**

- Bottom-up predictive parsing with:
 - **L**: Left-to-right scan of the input.
 - **R**: Rightmost derivation.
 - (**0**): Zero tokens of lookahead.
- Use the handle-finding automaton, without any lookahead, to predict where handles are.

LR(0) Parsing

S → **E**

E → **T**;

E → **T** + **E**

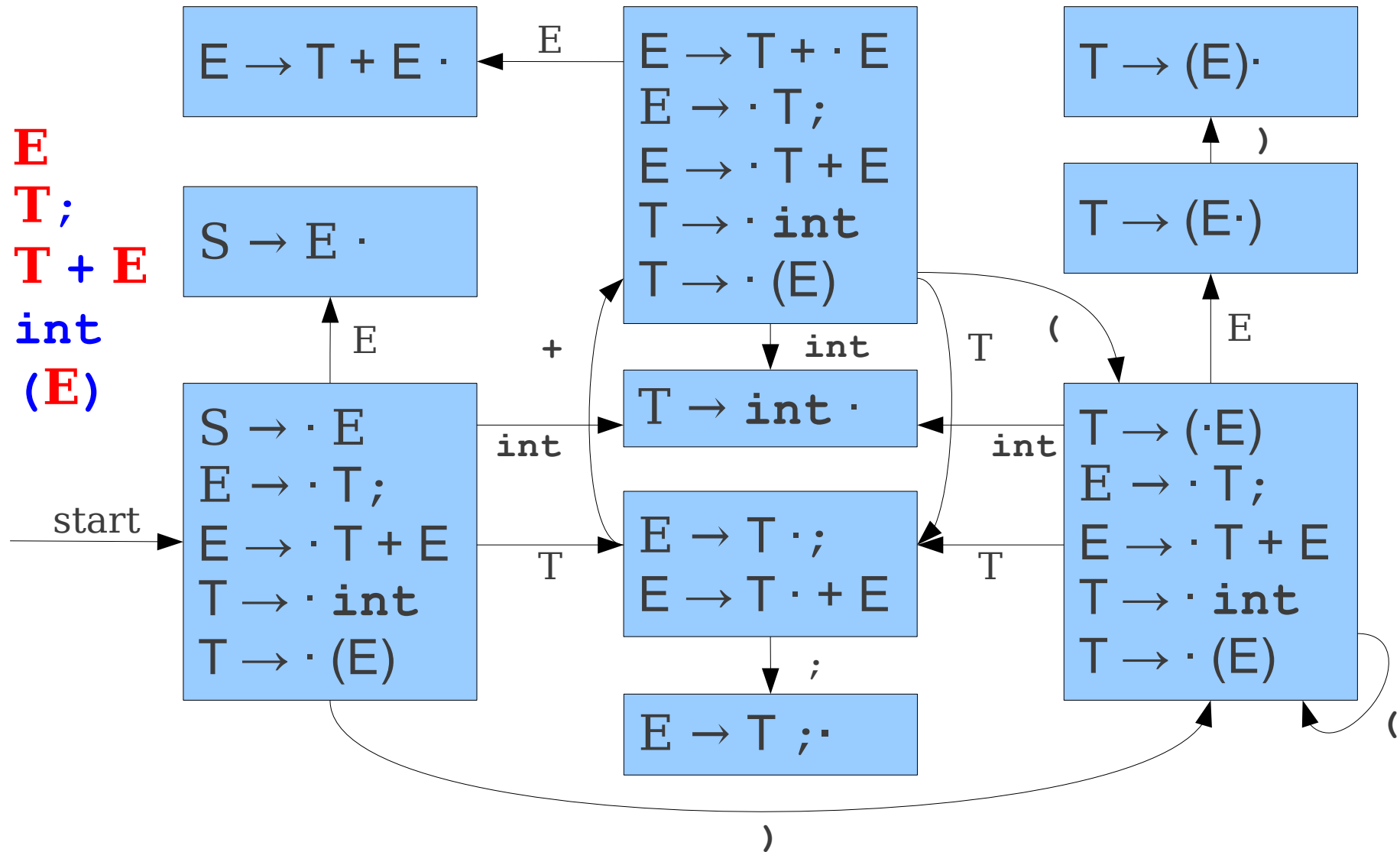
T → **int**

T → (**E**)

int	+	(	int	+	int	;	)	;
-----	---	---	-----	---	-----	---	---	---

LR(0) Parsing

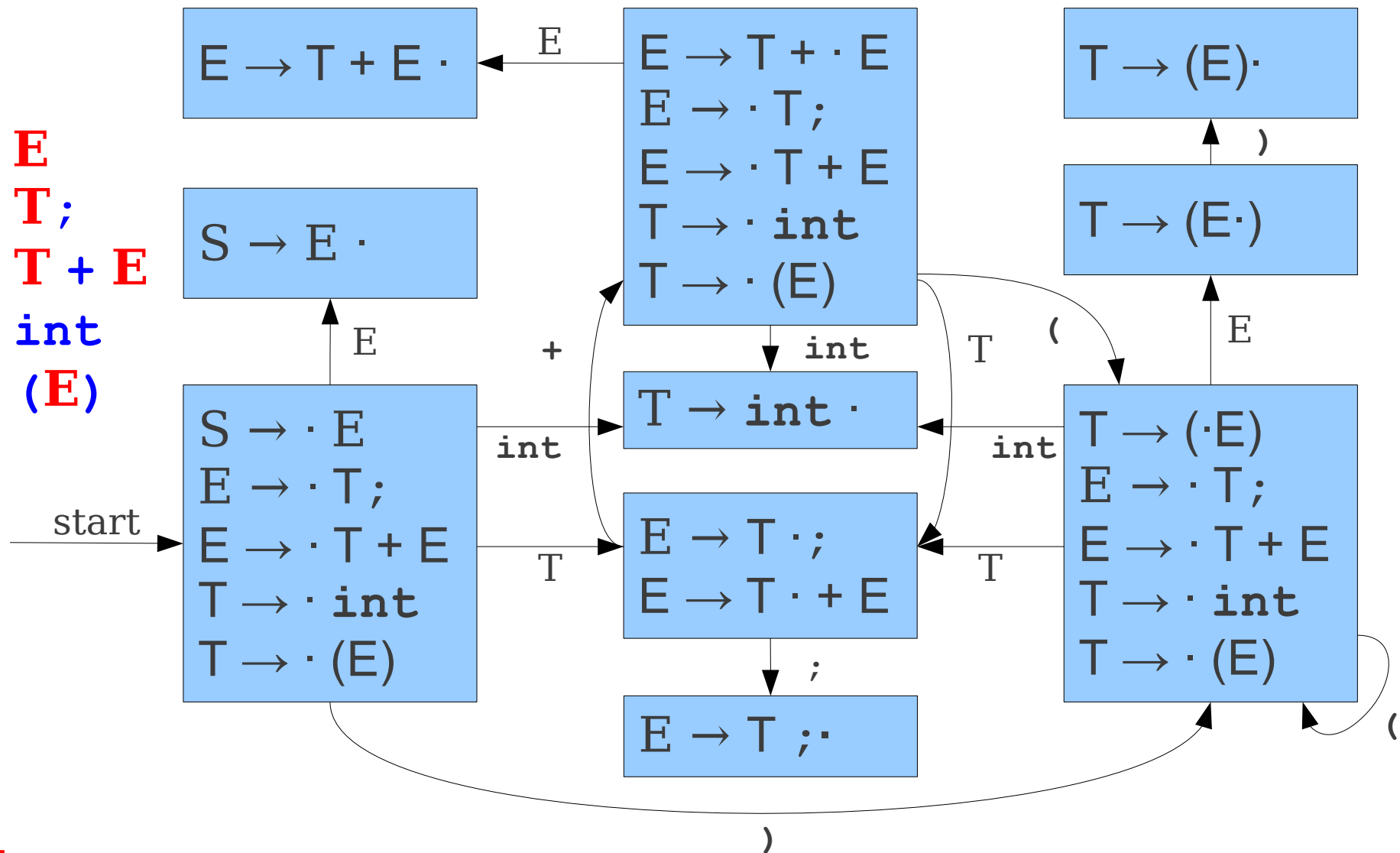
$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



int	+	(	int	+	int	;	)	;
-----	---	---	-----	---	-----	---	---	---

LR(0) Parsing

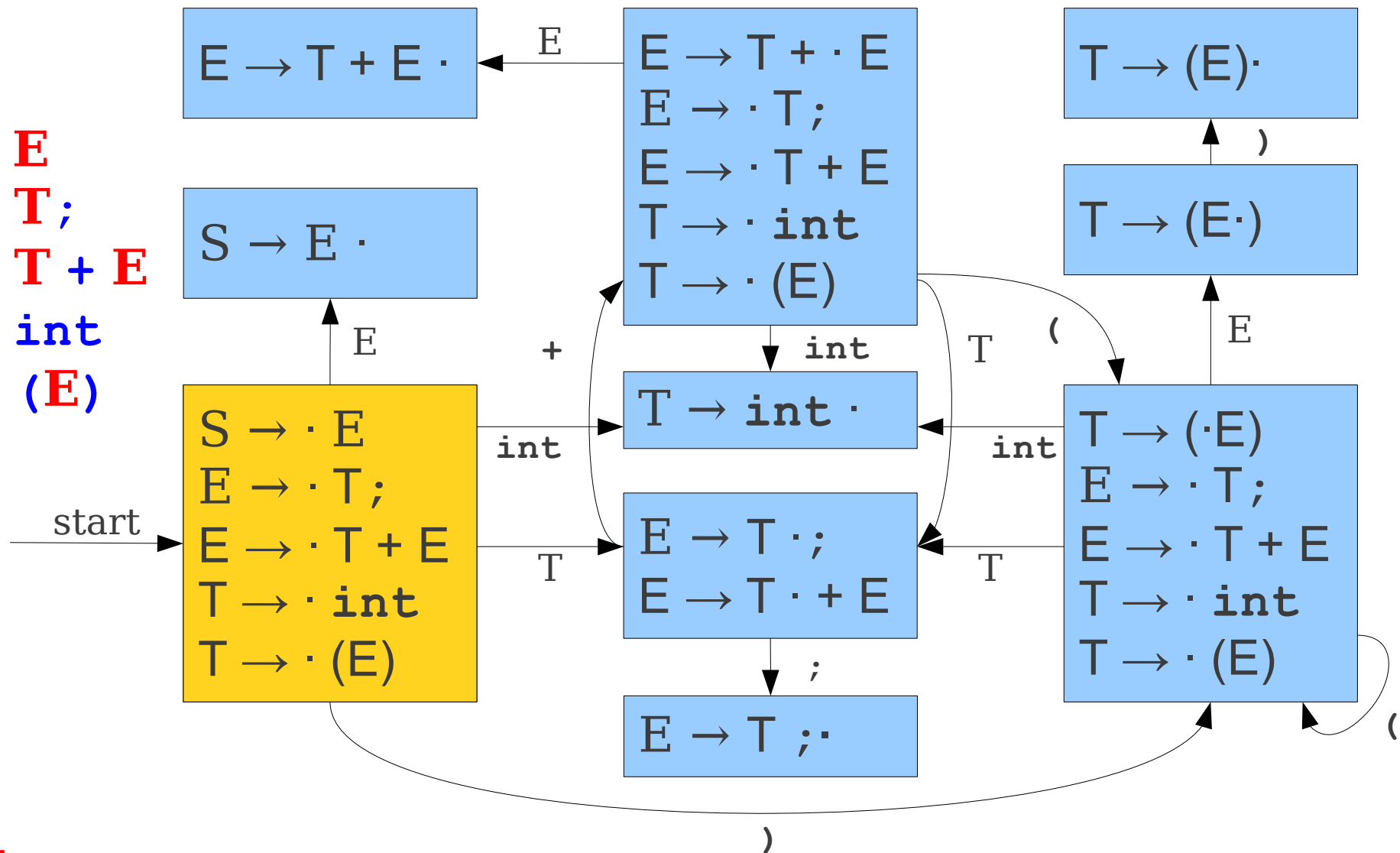
$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



int	+	(	int	+	int	;	)	;
-----	---	---	-----	---	-----	---	---	---

LR(0) Parsing

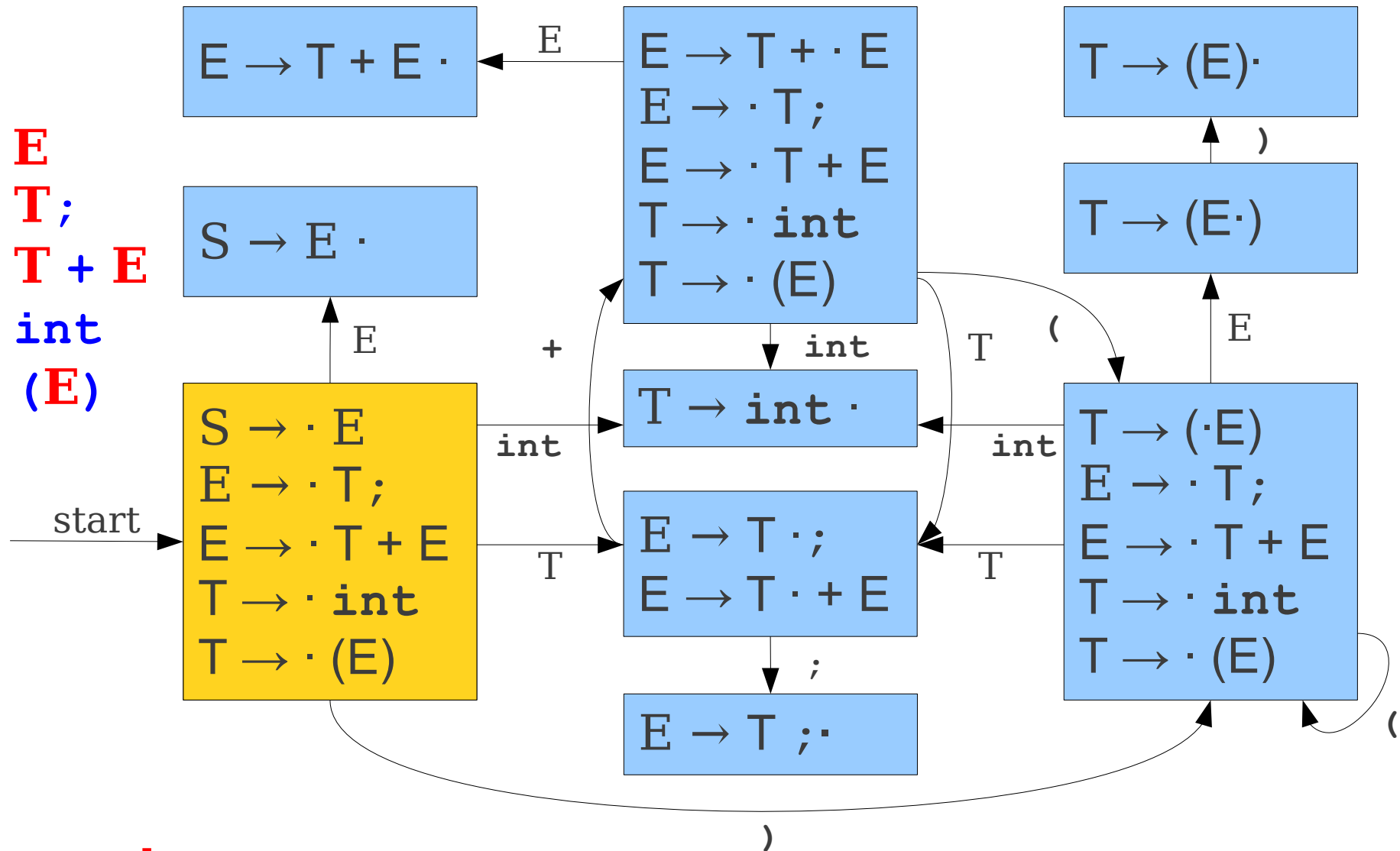
$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



int	+	(	int	+	int	;	)	;
-----	---	---	-----	---	-----	---	---	---

LR(0) Parsing

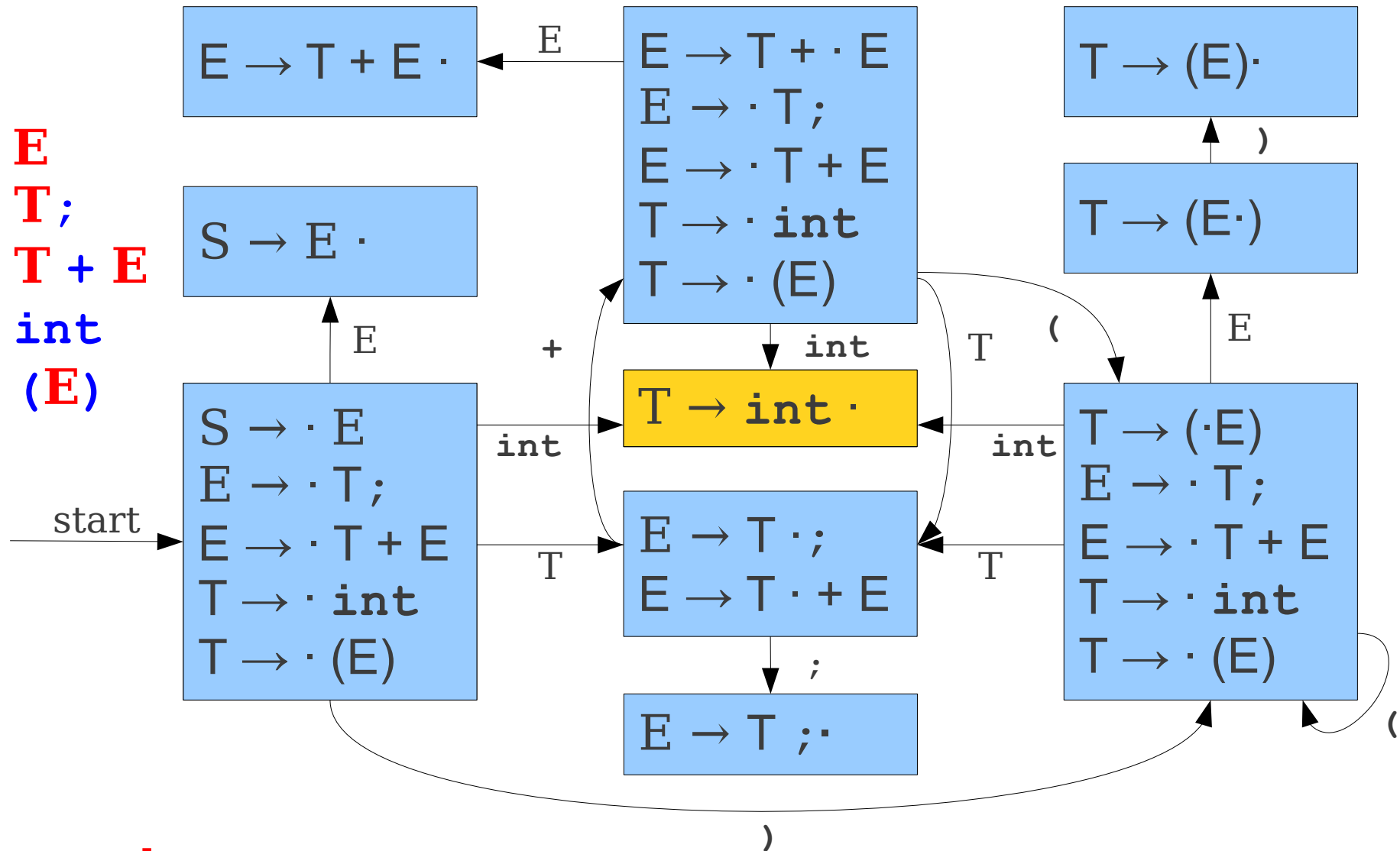
$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



int	+	(	int	+	int	;	)	;
-----	---	---	-----	---	-----	---	---	---

LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
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 $T \rightarrow \text{int}$
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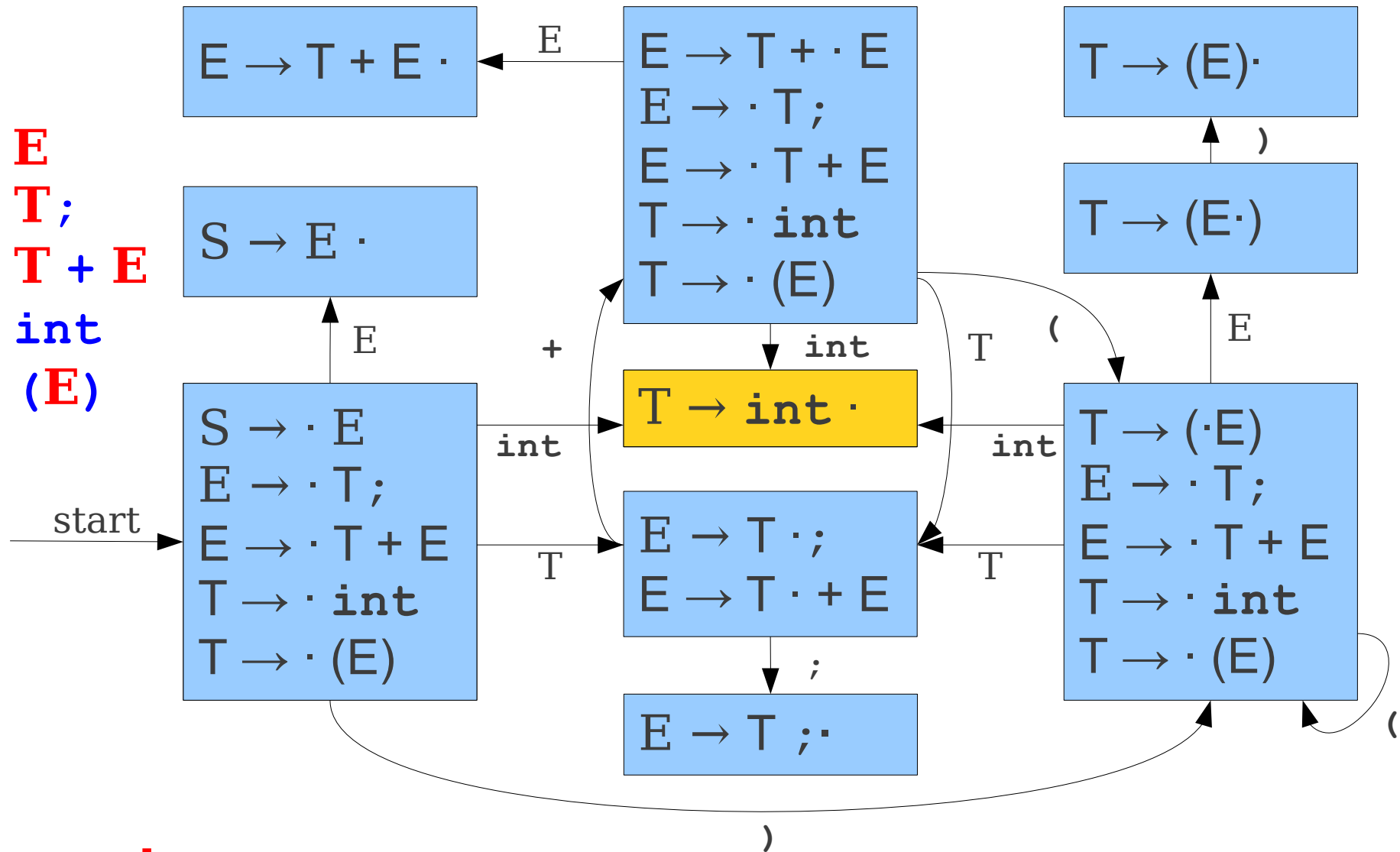


int

+ (int + int ;) ;

LR(0) Parsing

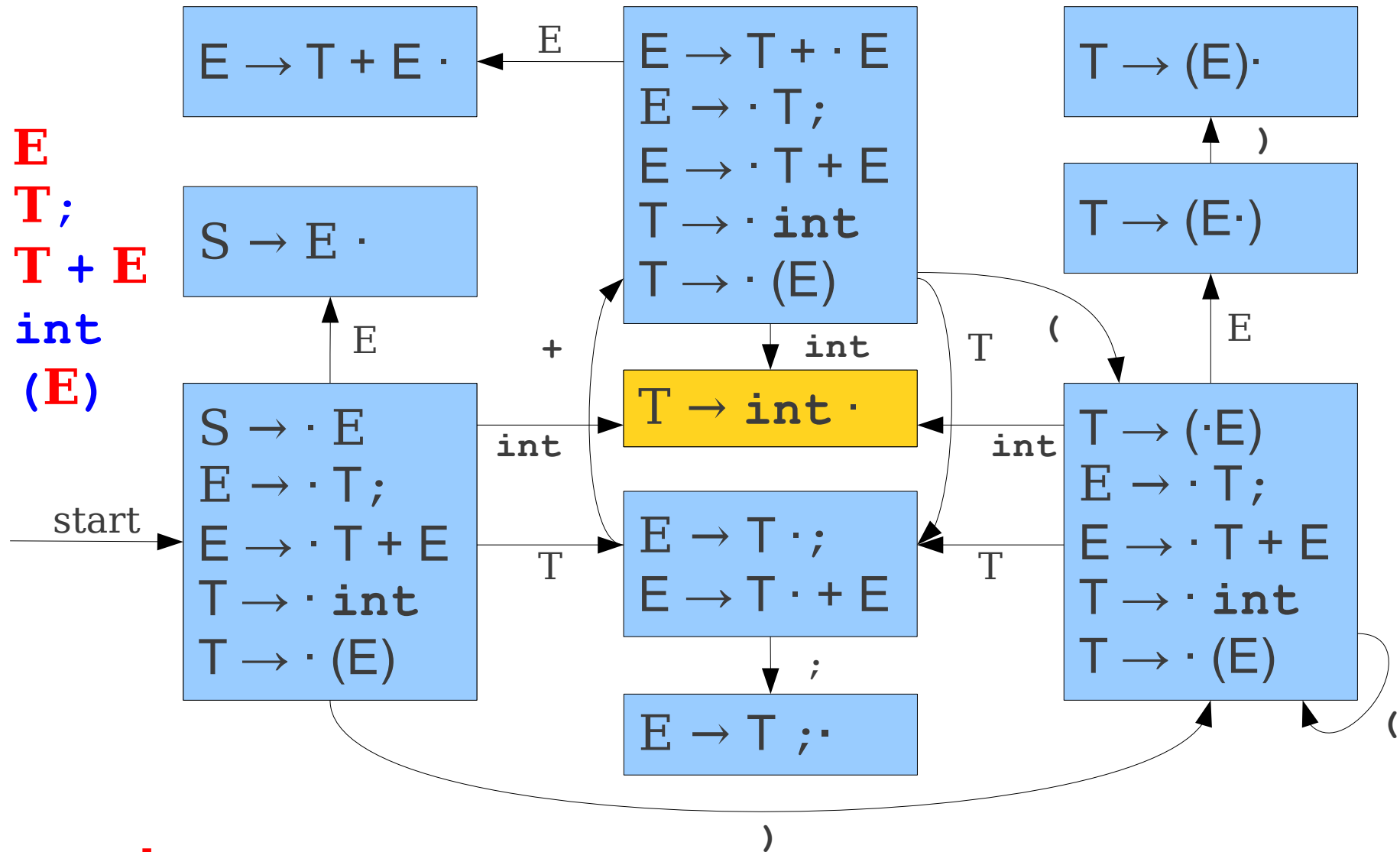
$S \rightarrow E$
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 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



+	(	int	+	int	;	)	;
---	---	-----	---	-----	---	---	---

LR(0) Parsing

$S \rightarrow E$
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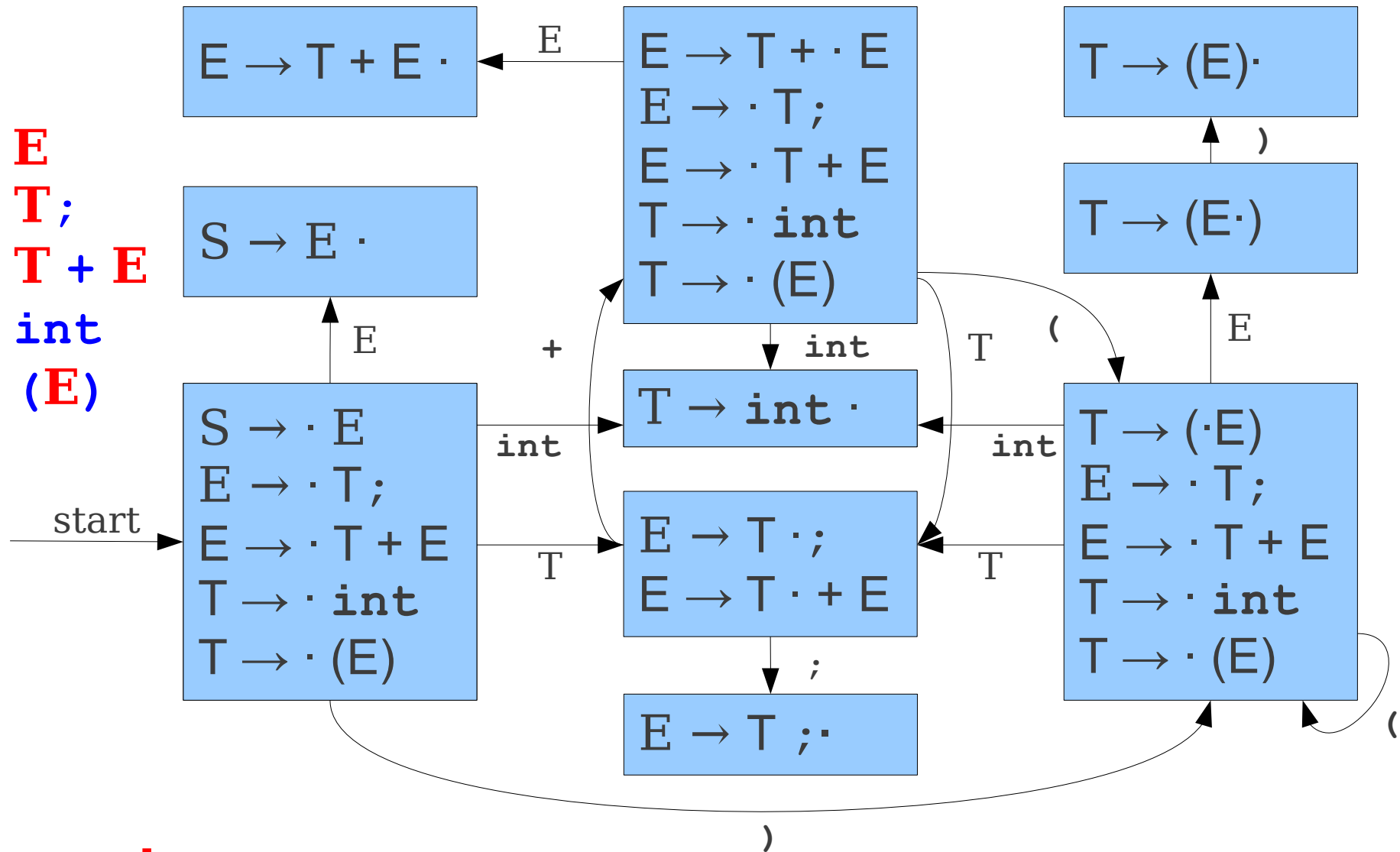


T

+	(	int	+	int	;	)	;
---	---	-----	---	-----	---	---	---

LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
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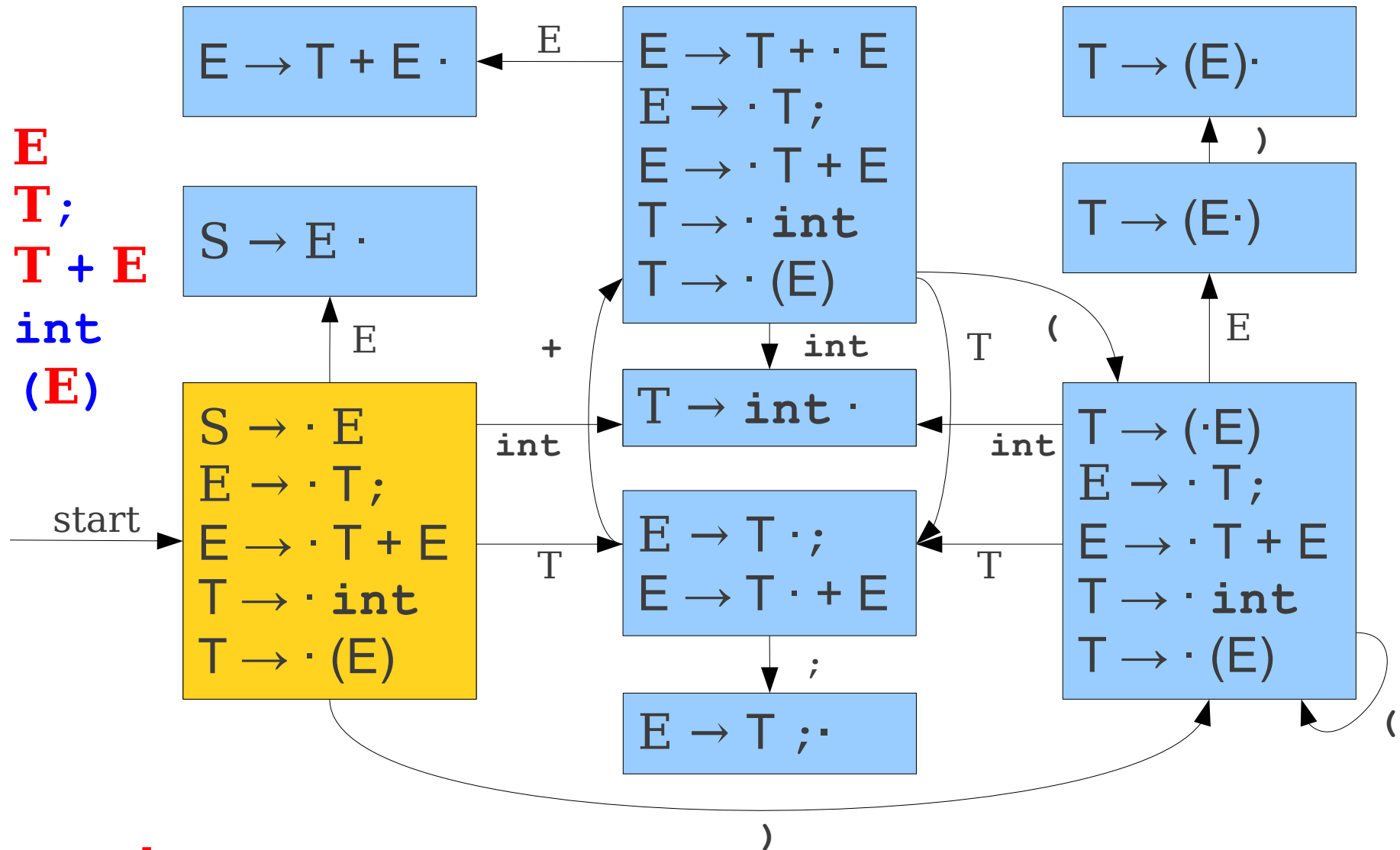


T

+	(	int	+	int	;	)	;
---	---	-----	---	-----	---	---	---

LR(0) Parsing

$S \rightarrow E$
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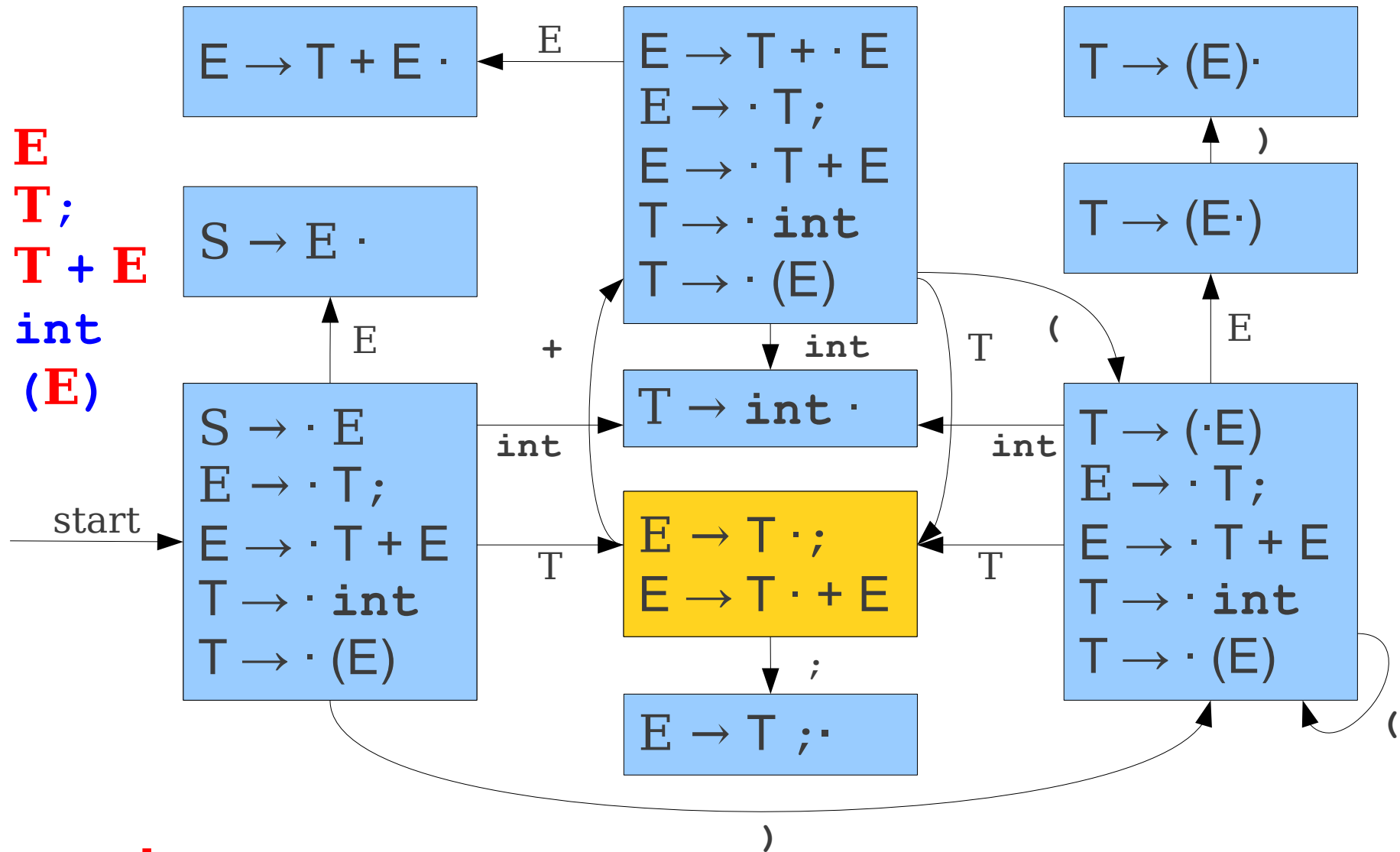


T

+	(	int	+	int	;	)	;
---	---	-----	---	-----	---	---	---

LR(0) Parsing

$S \rightarrow E$
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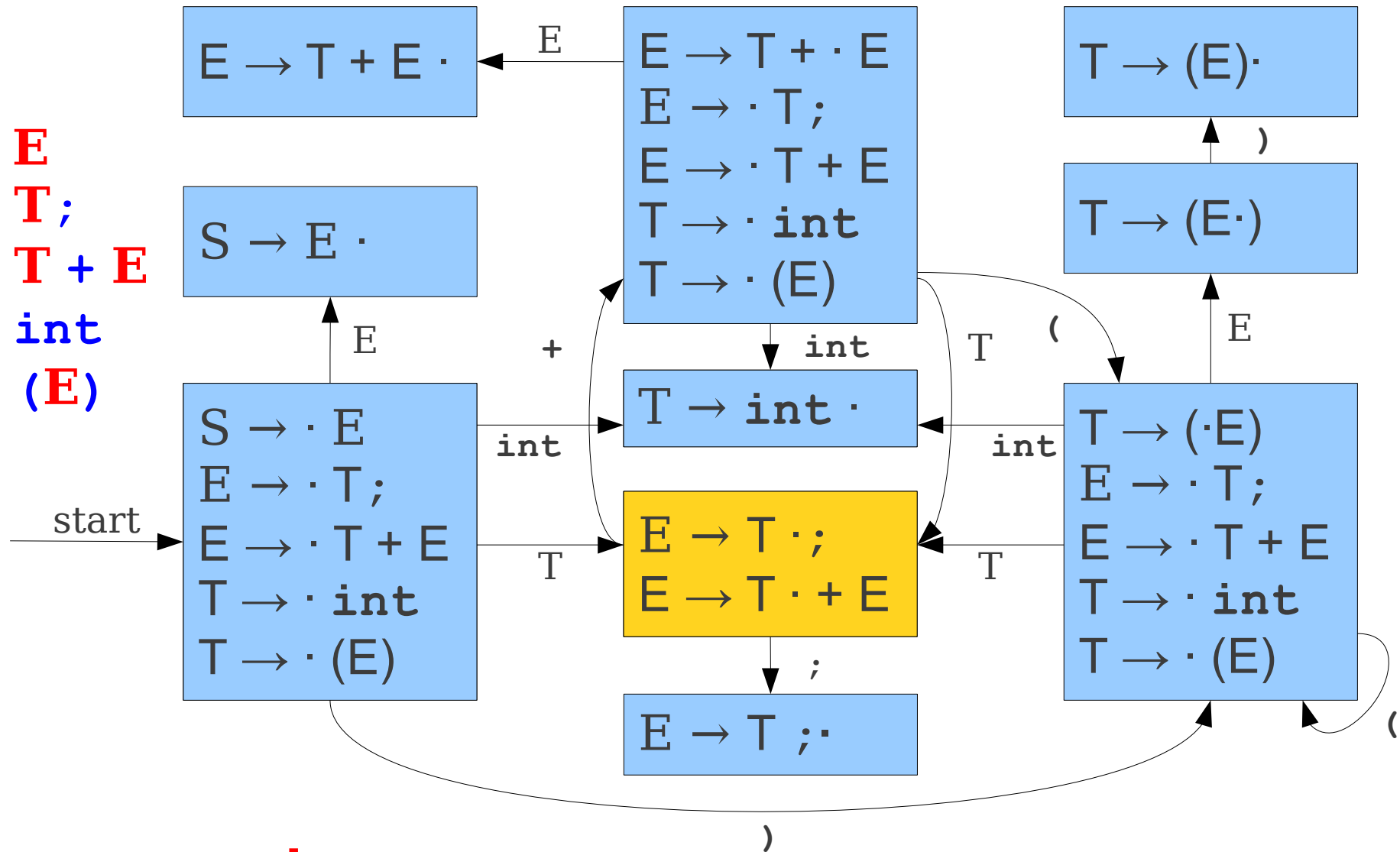


T

+	(	int	+	int	;	)	;
---	---	-----	---	-----	---	---	---

LR(0) Parsing

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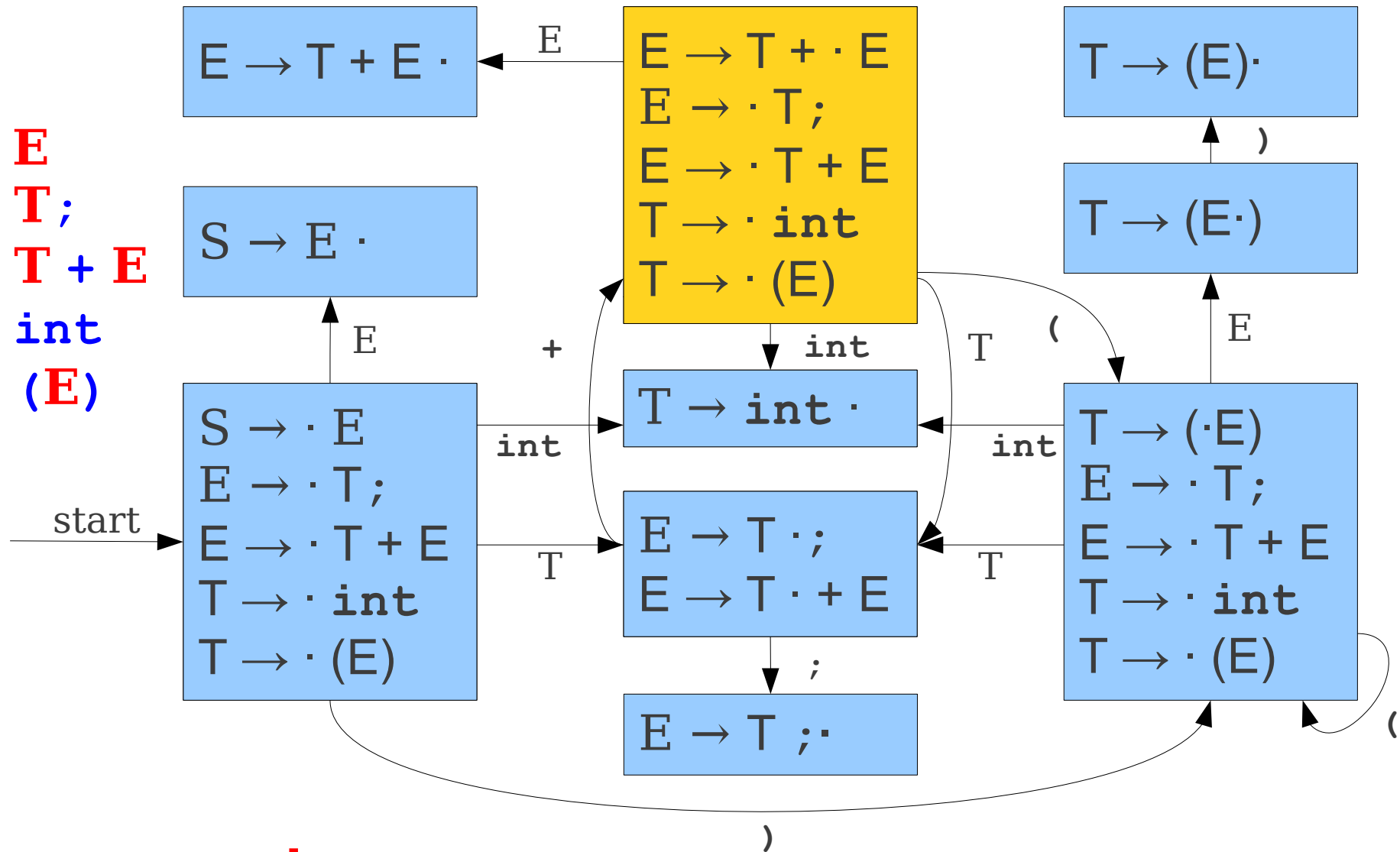


T	+
---	---

(	int	+	int	;	)	;
---	-----	---	-----	---	---	---

LR(0) Parsing

$S \rightarrow E$
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 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
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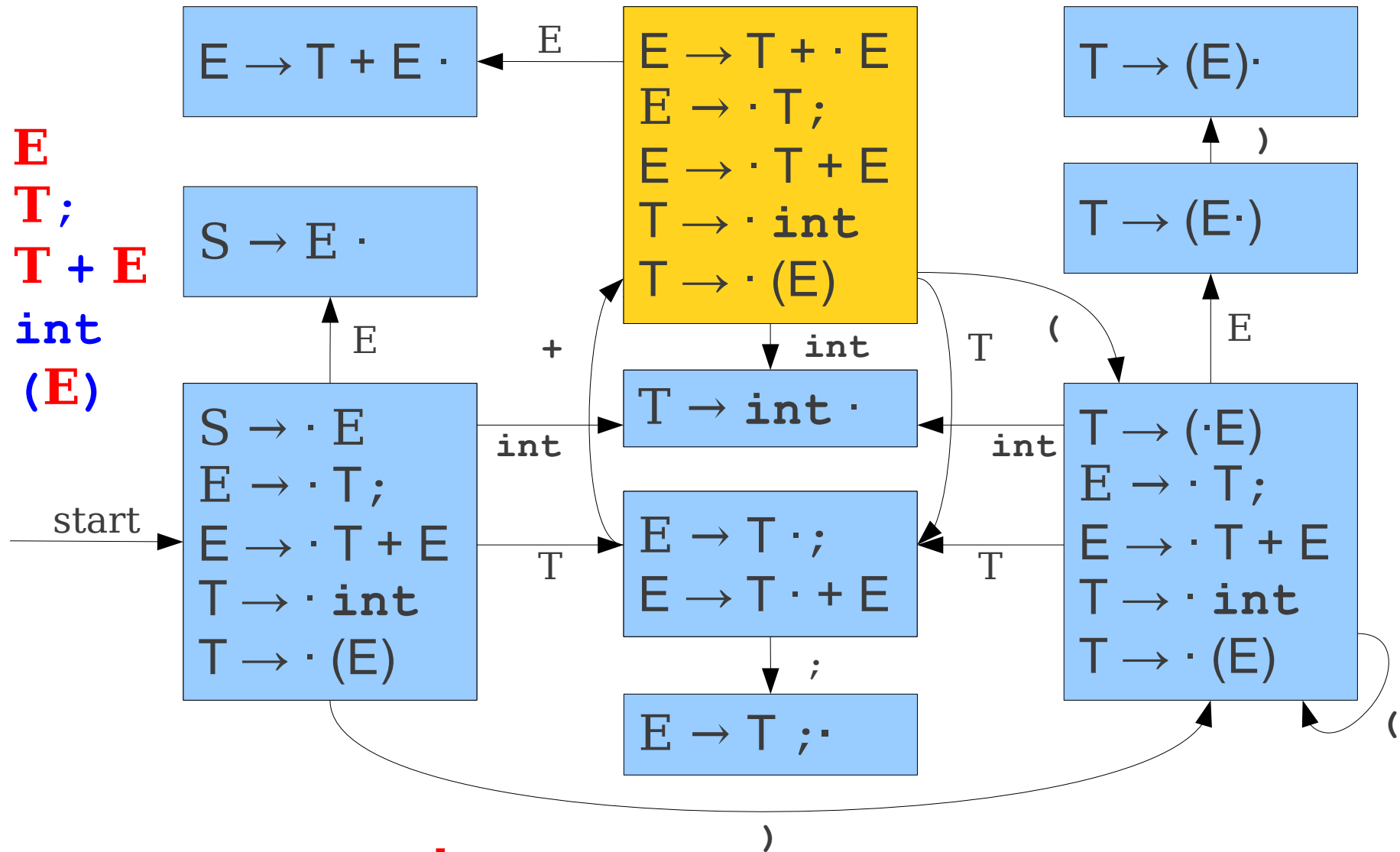


T	+
---	---

(	int	+	int	;	)	;
---	-----	---	-----	---	---	---

LR(0) Parsing

$S \rightarrow E$
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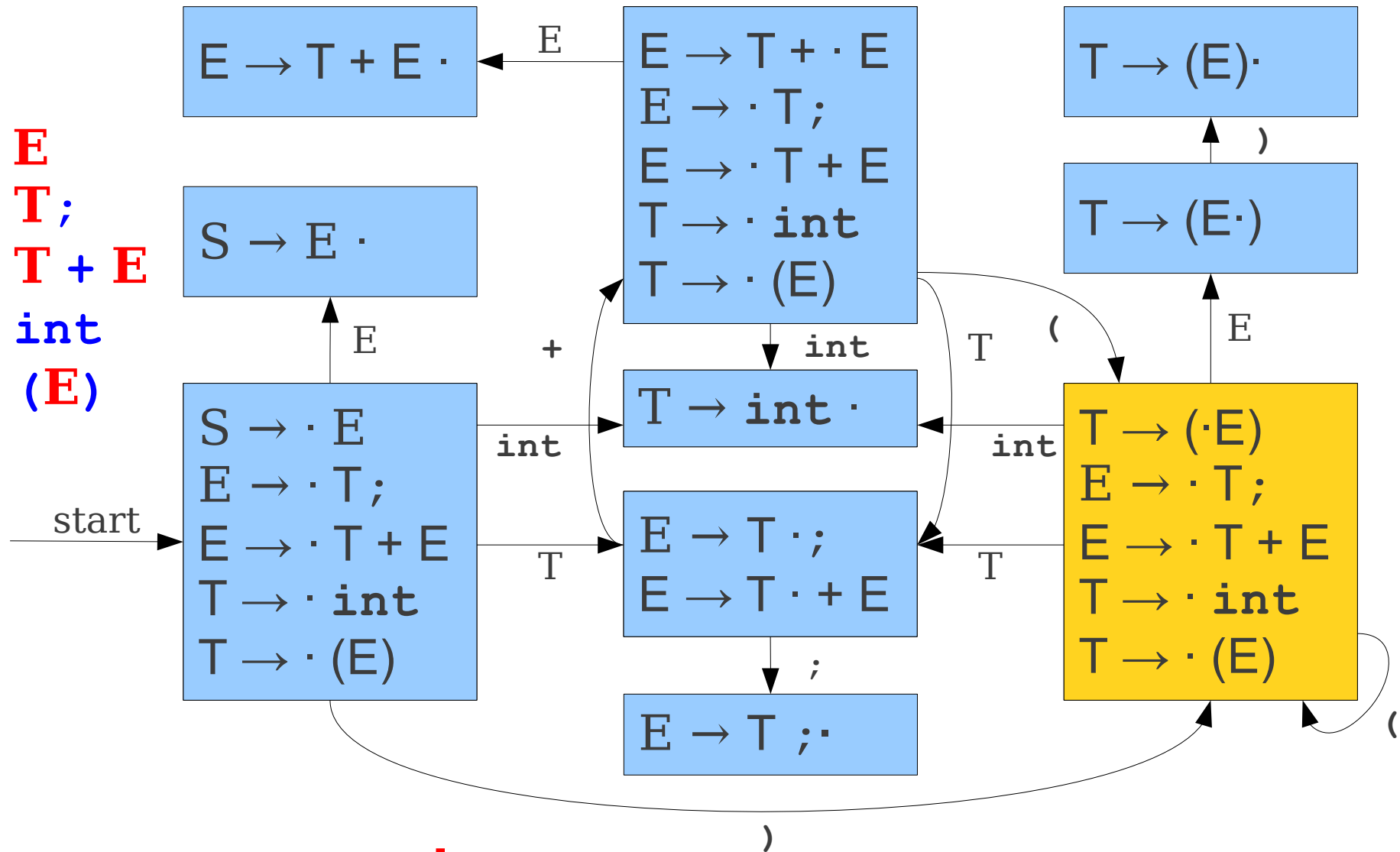


T	+	(
---	---	---

int	+	int	;	)	;
-----	---	-----	---	---	---

LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
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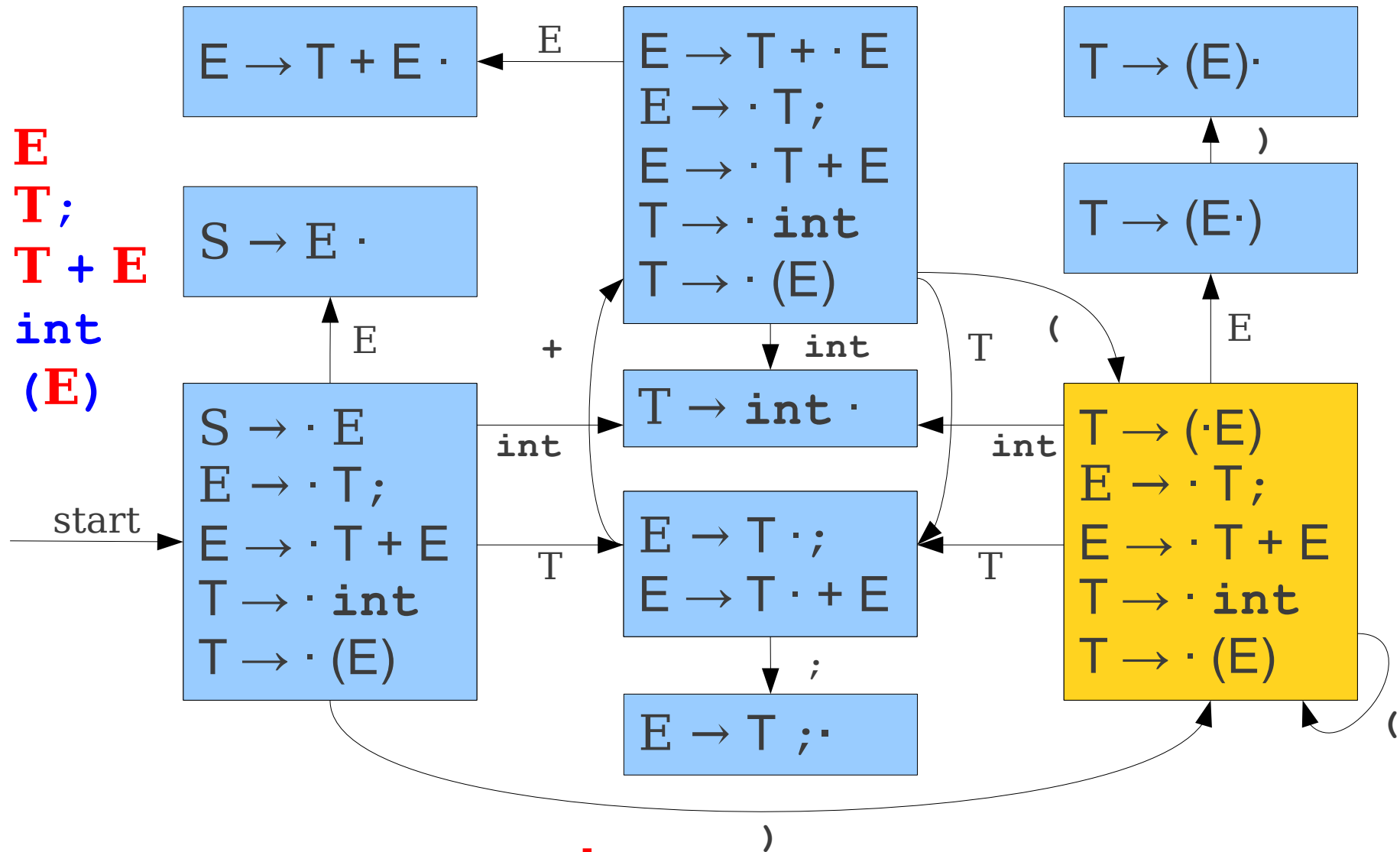


T	+	(
---	---	---

int	+	int	;	)	;
-----	---	-----	---	---	---

LR(0) Parsing

$S \rightarrow E$
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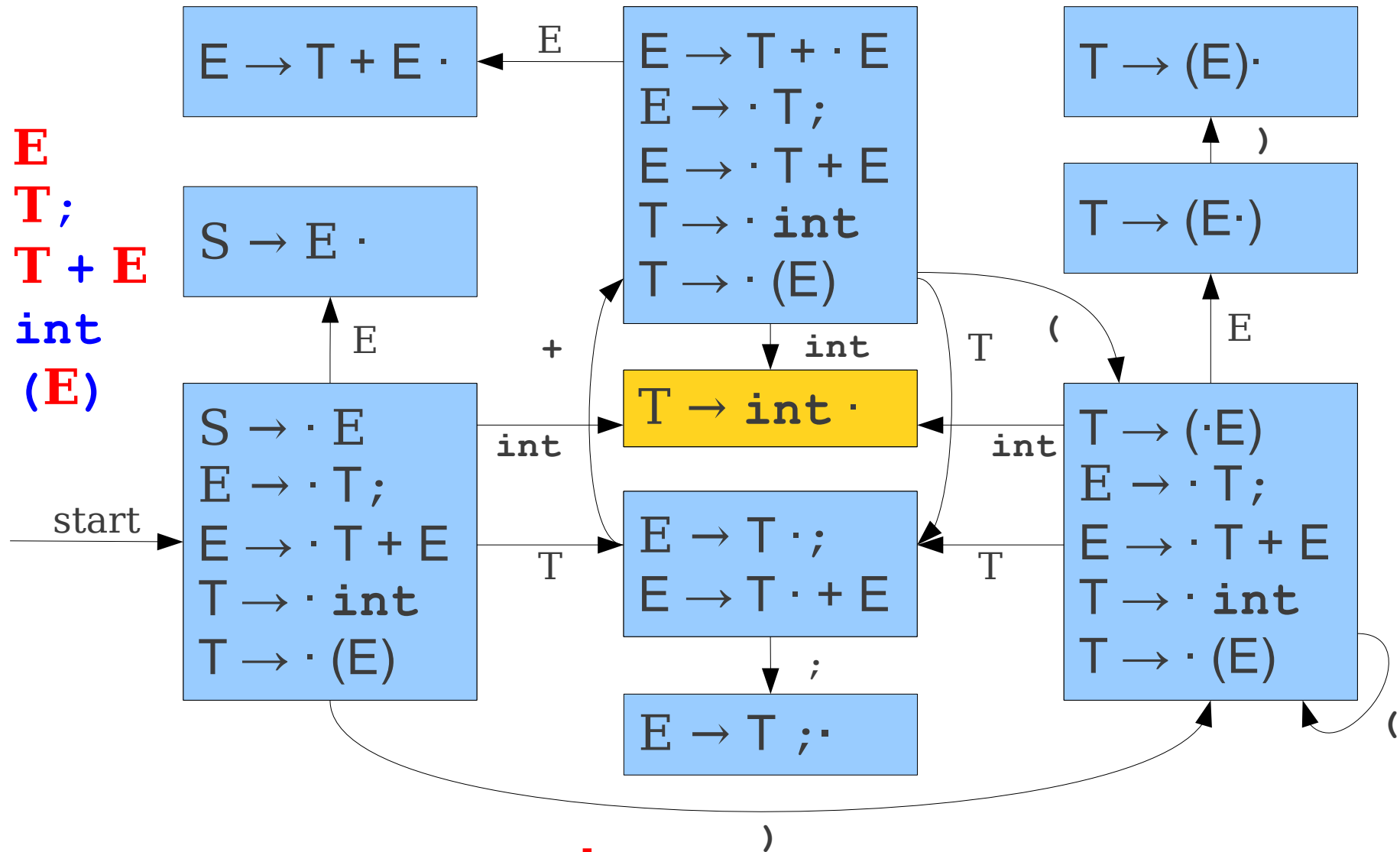


T	+	(	int
---	---	---	-----

+	int	;	)	;
---	-----	---	---	---

LR(0) Parsing

$S \rightarrow E$
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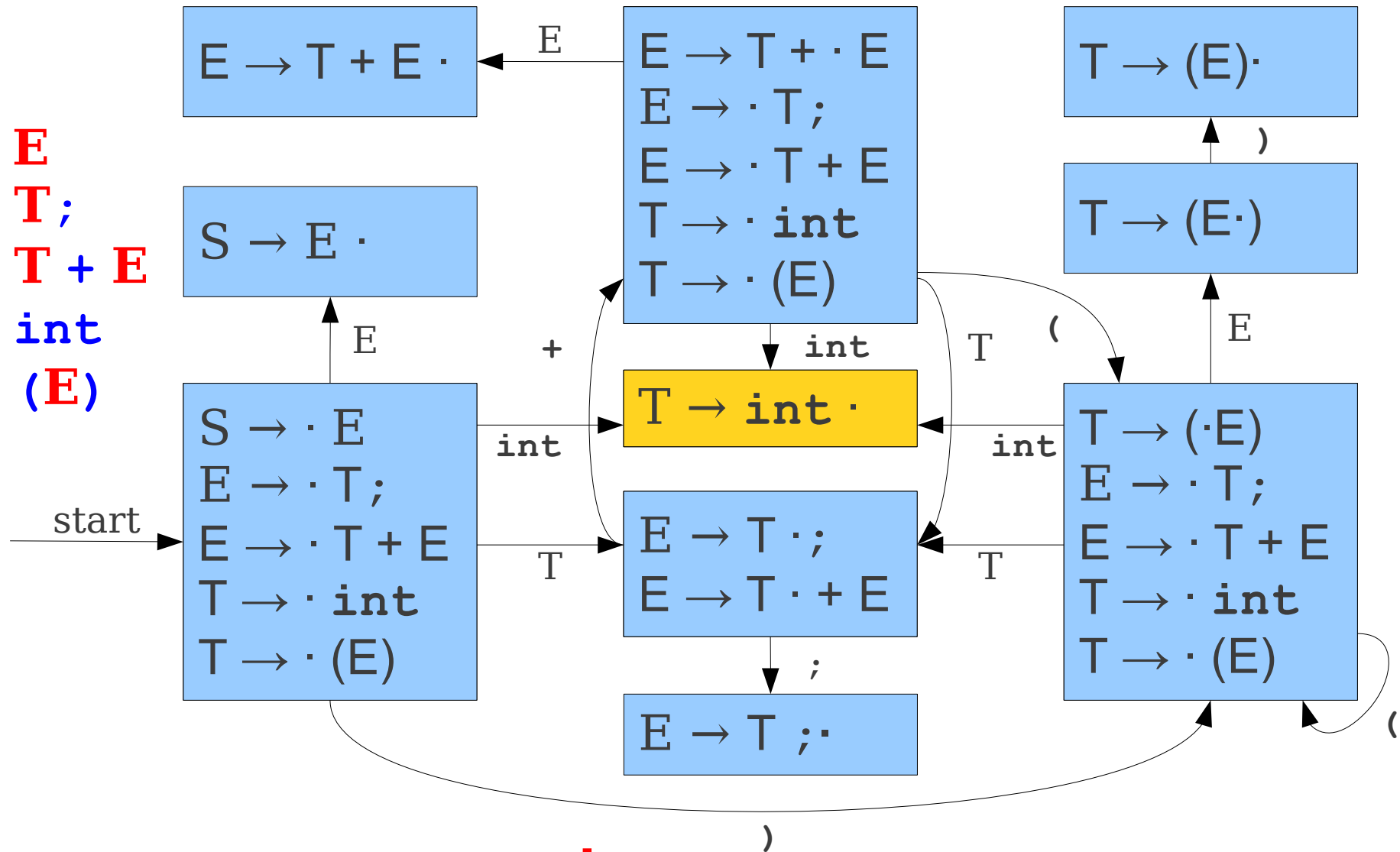


T	+	(	int
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+	int	;	)	;
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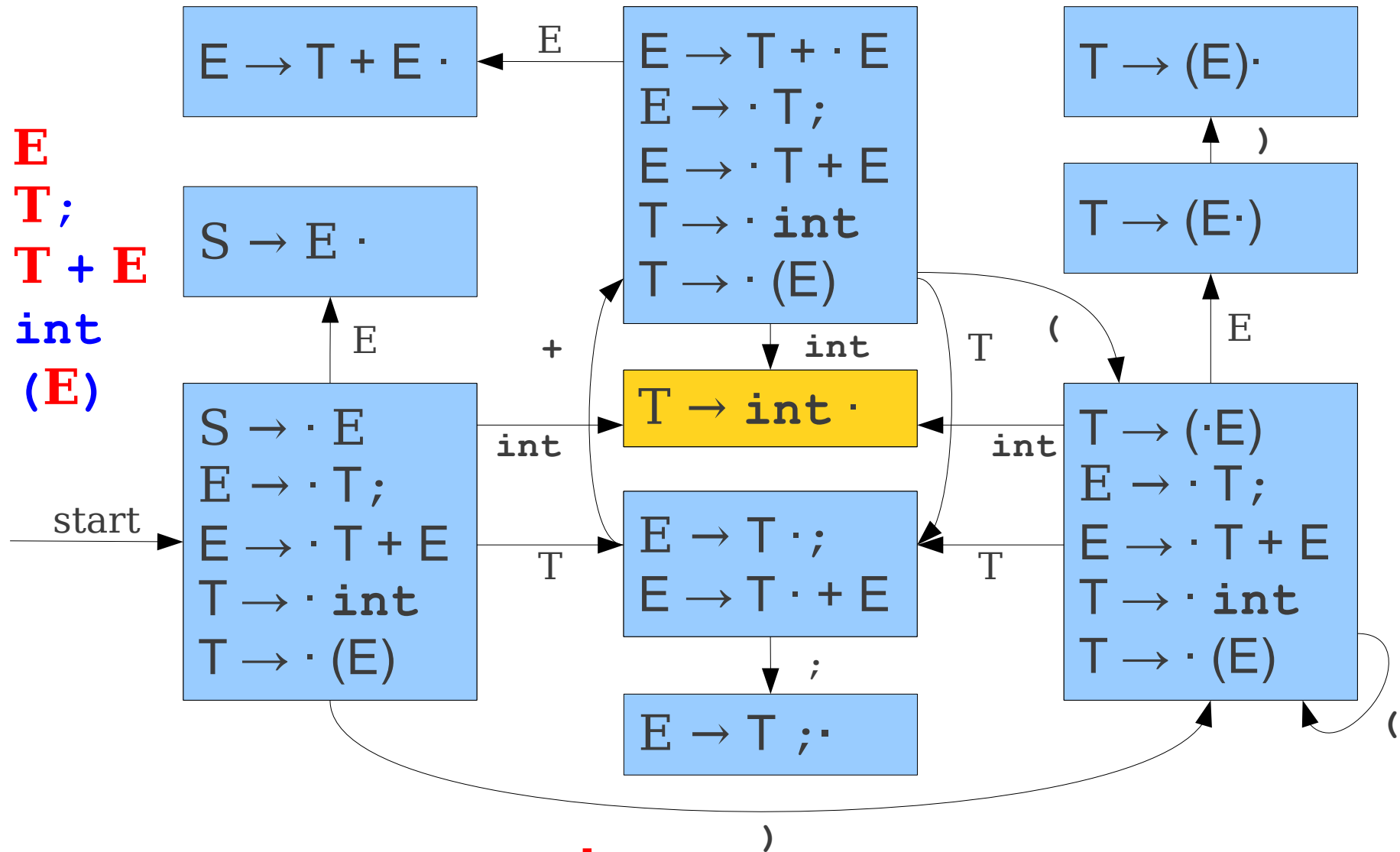


T	+	(
---	---	---

+	int	;	)	;
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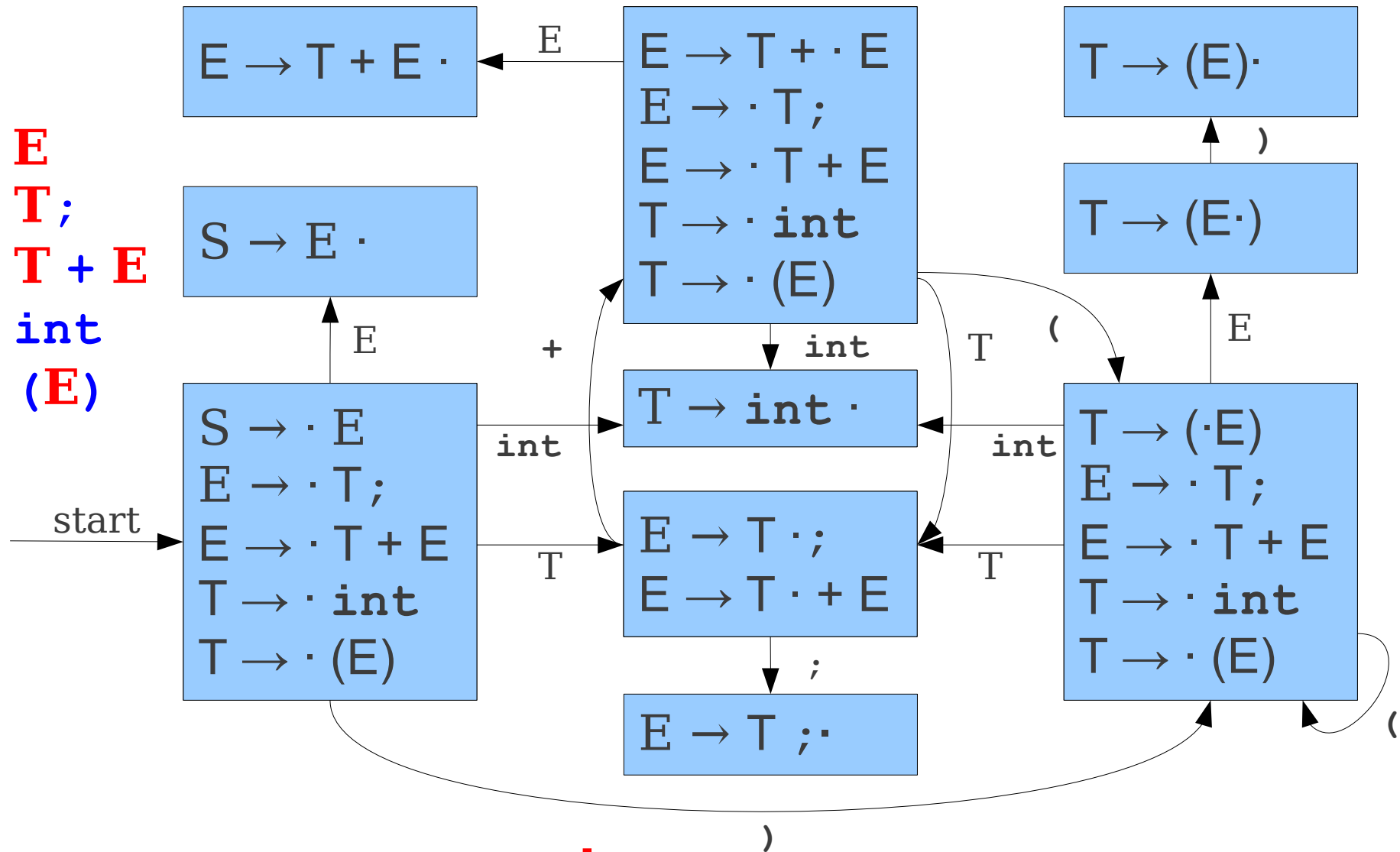


T	+	(	T
---	---	---	---

+	int	;	)	;
---	-----	---	---	---

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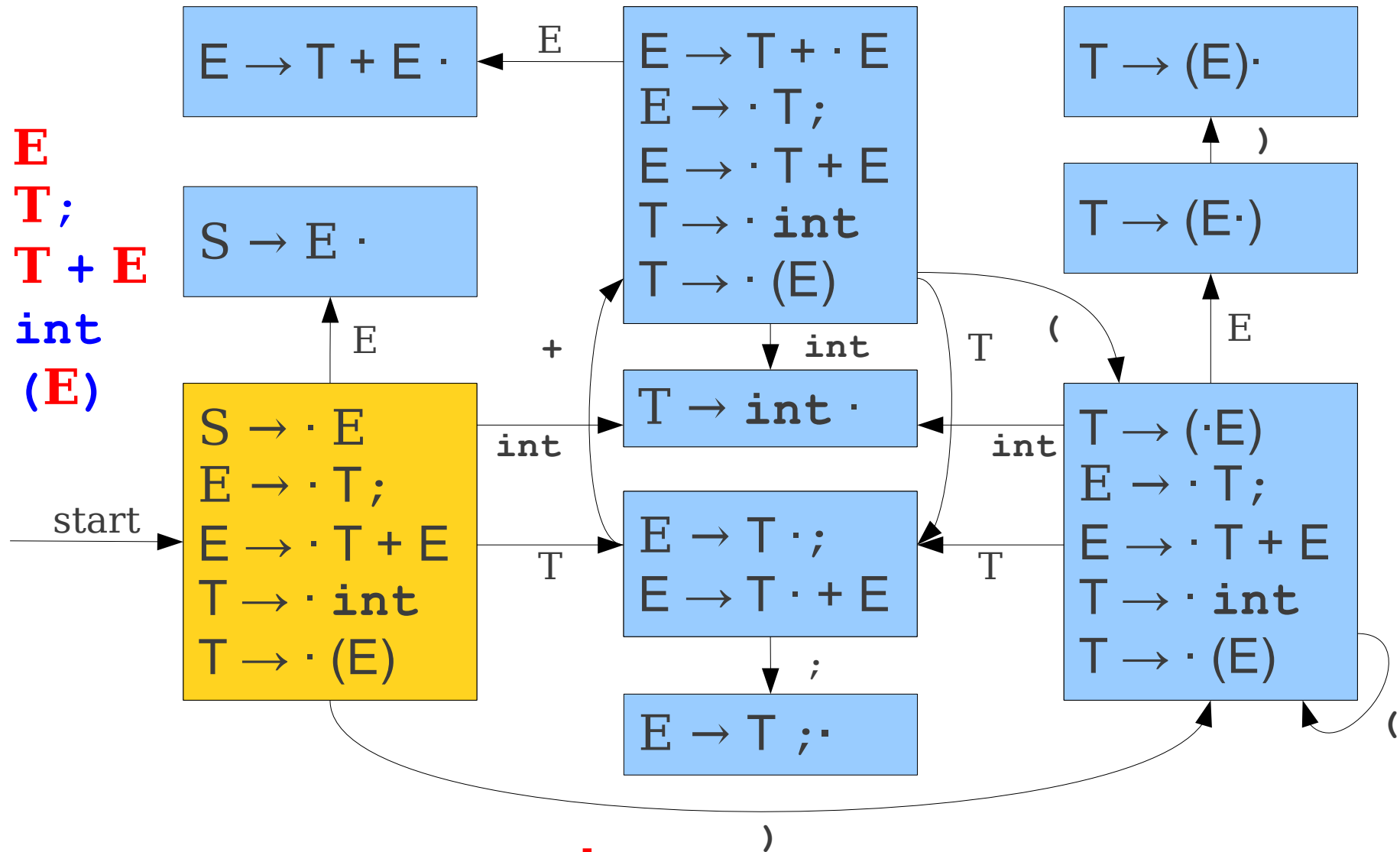


T	+	(	T
---	---	---	---

+	int	;	)	;
---	-----	---	---	---

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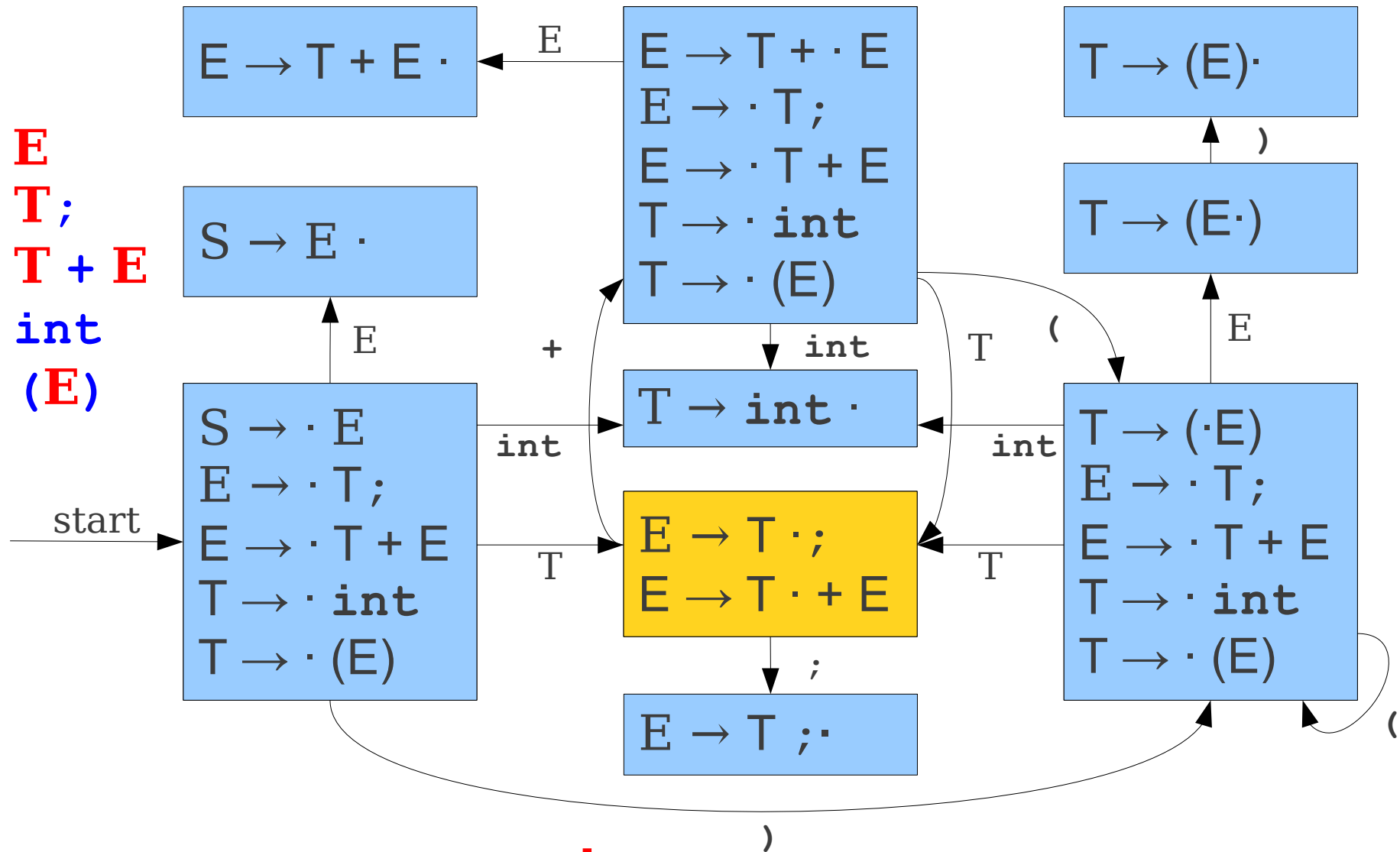


T	+	(	T
---	---	---	---

+	int	;	)	;
---	-----	---	---	---

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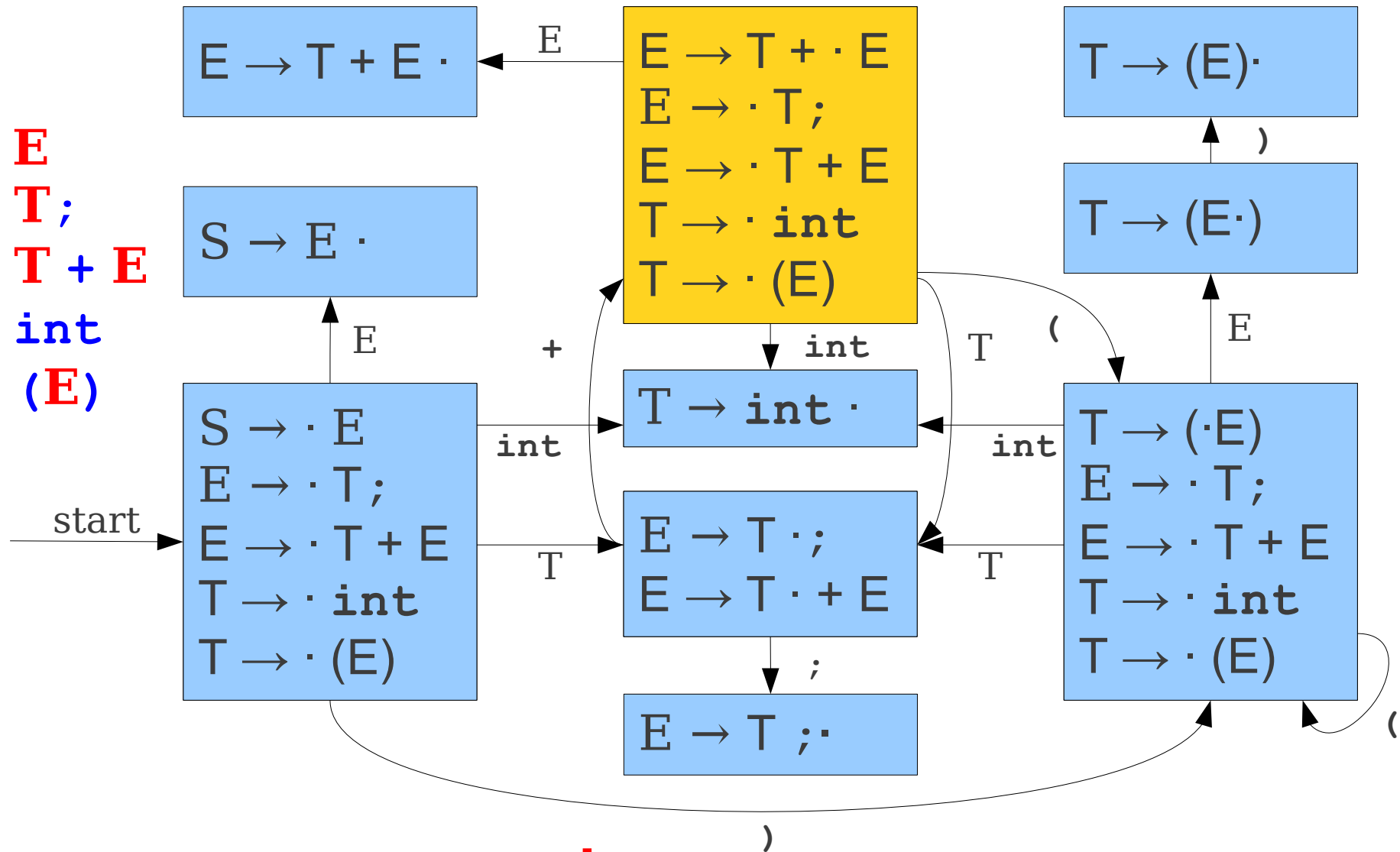


T	+	(	T
---	---	---	---

+	int	;	)	;
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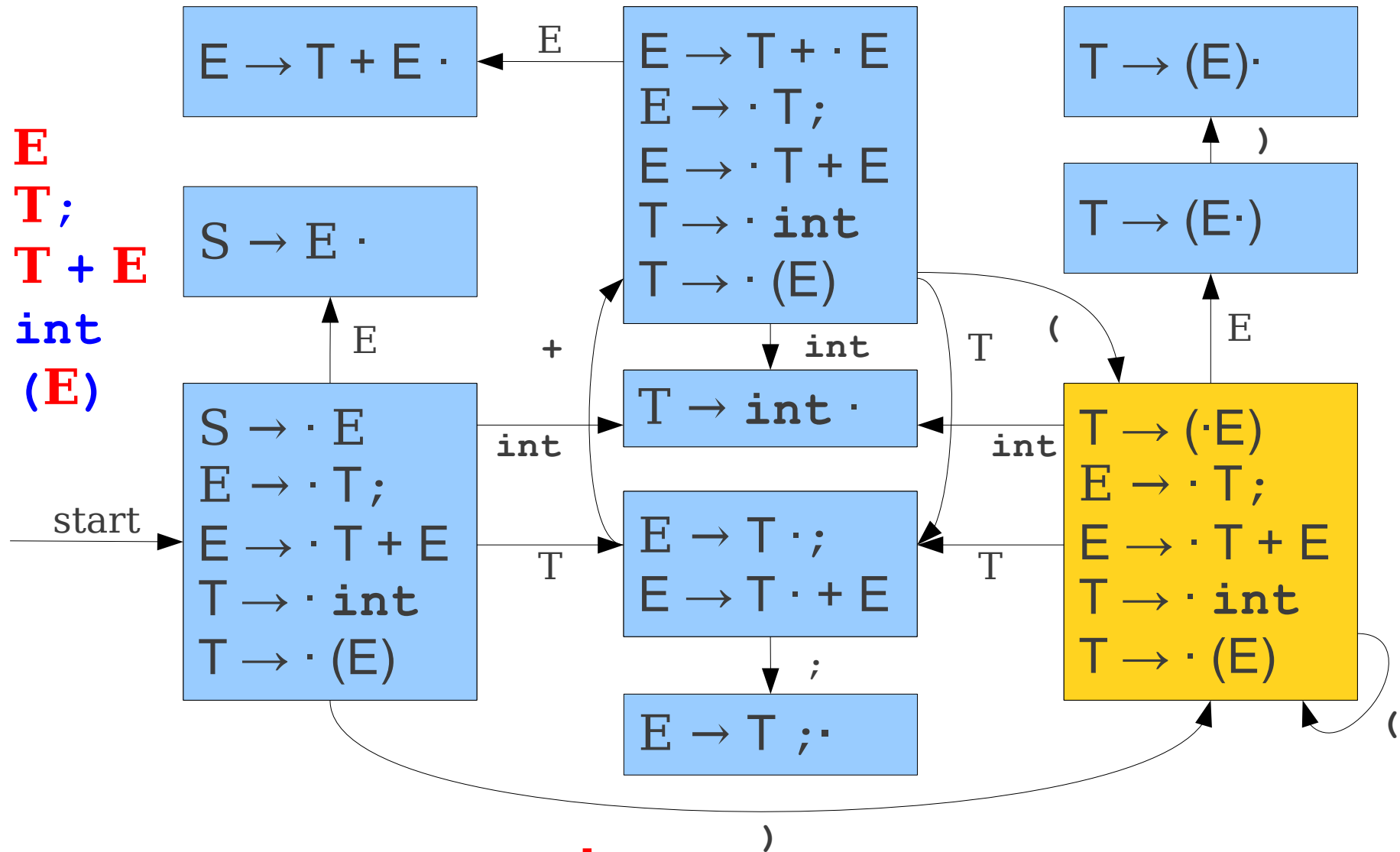


T	+	(	T
---	---	---	---

+	int	;	)	;
---	-----	---	---	---

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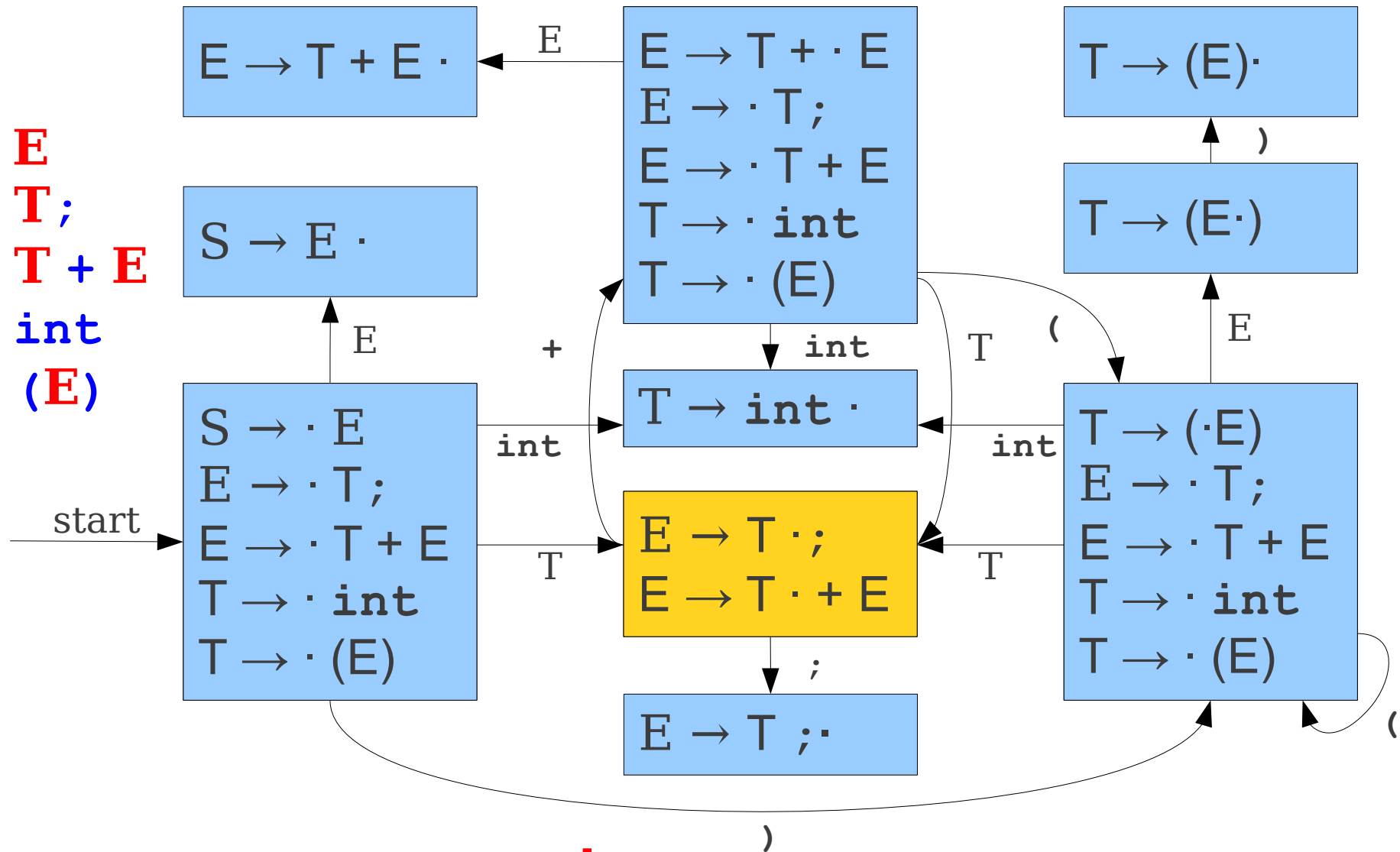


T	+	(	T
---	---	---	---

+	int	;	)	;
---	-----	---	---	---

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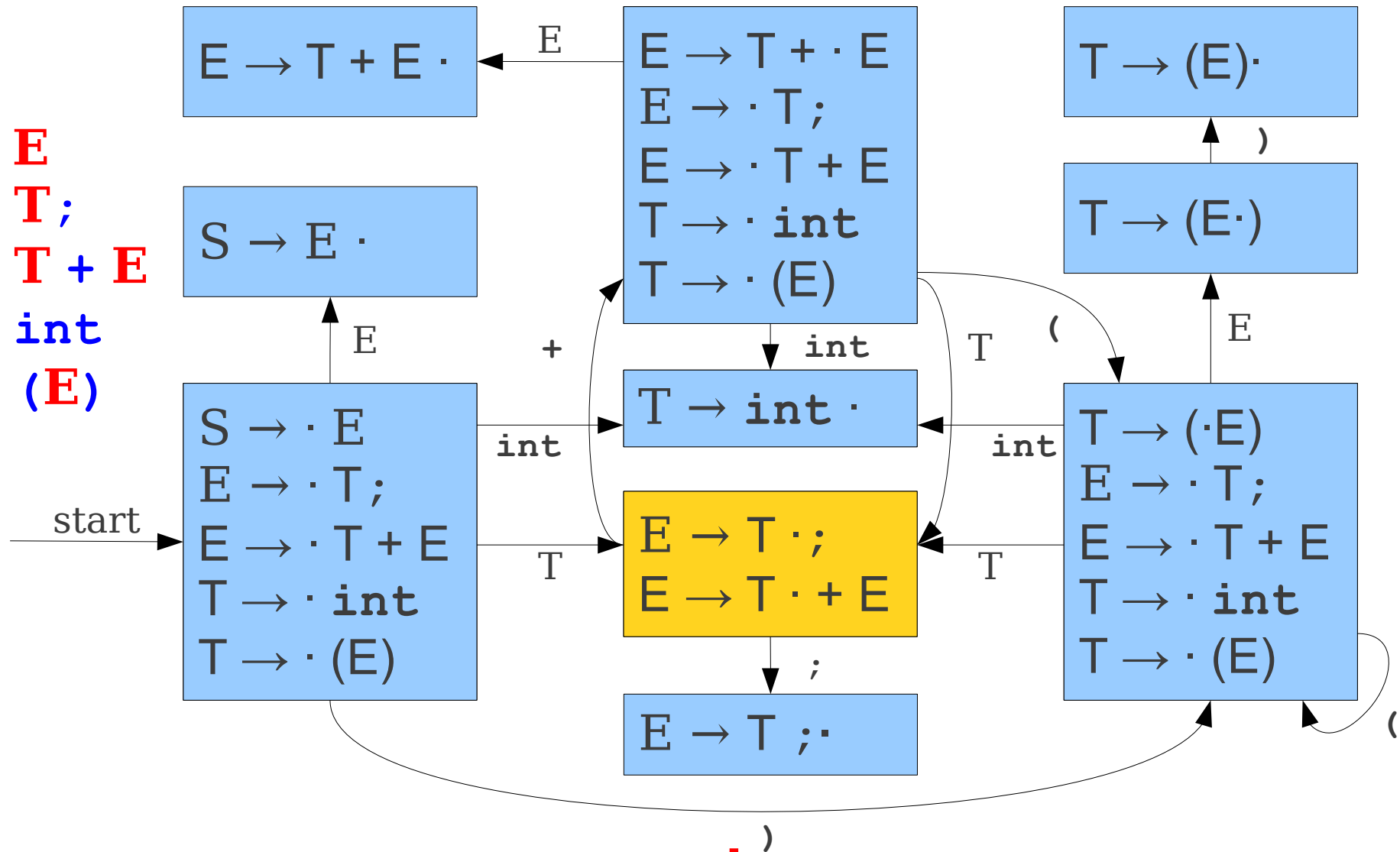


T	+	(	T
---	---	---	---

+	int	;	)	;
---	-----	---	---	---

LR(0) Parsing

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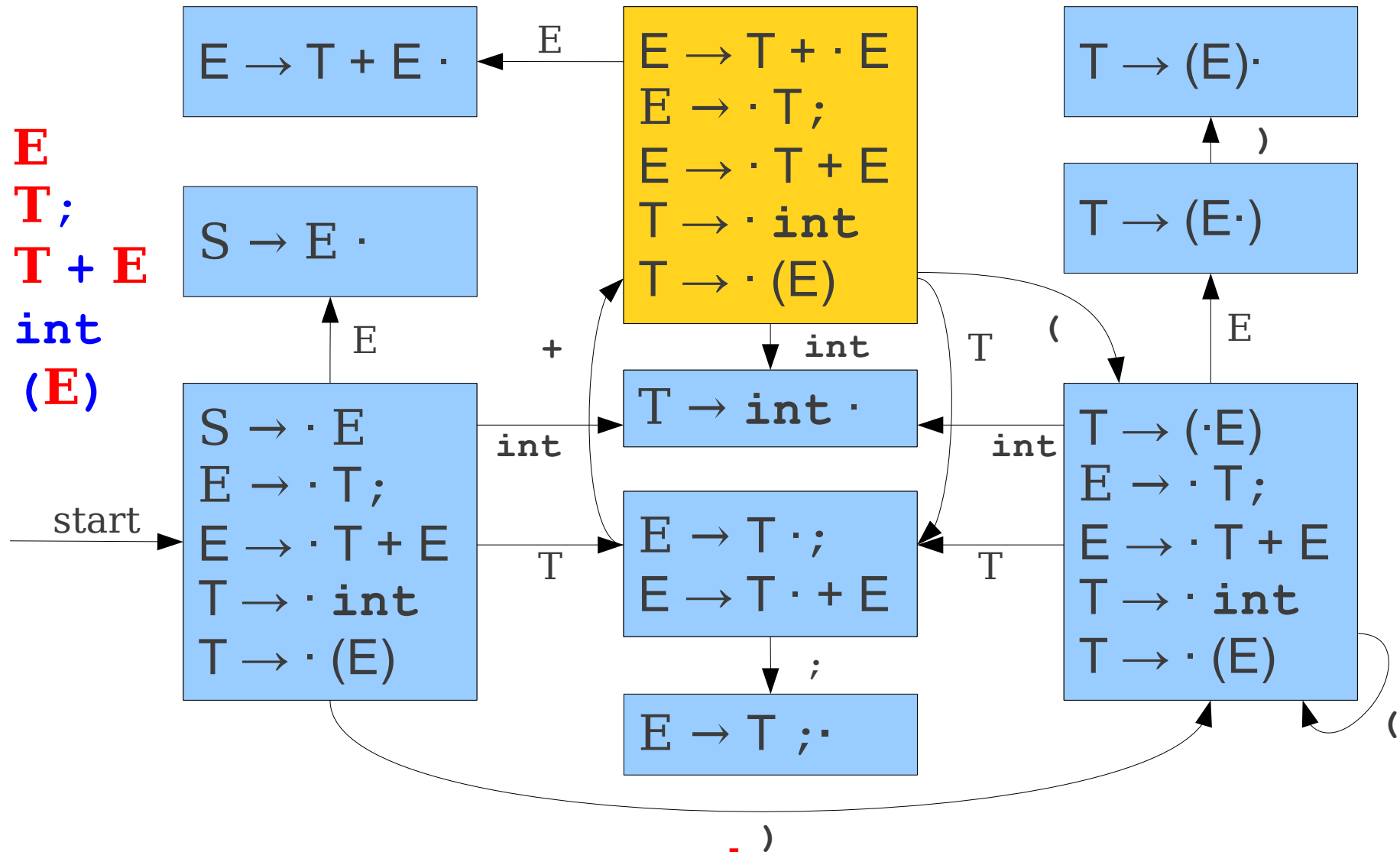


T	+	(	T	+
---	---	---	---	---

int	;	)	;
-----	---	---	---

LR(0) Parsing

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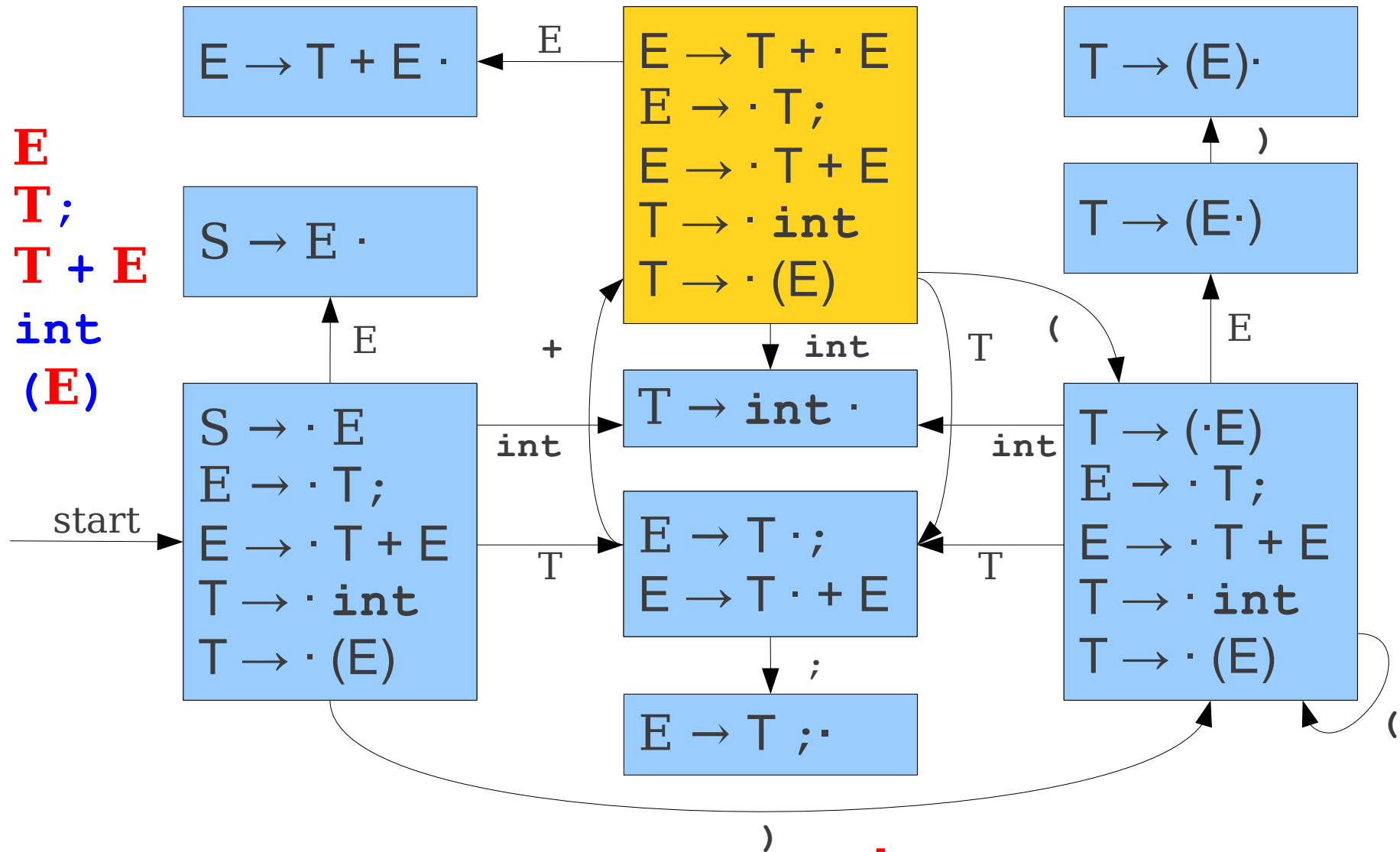


T	+	(	T	+
---	---	---	---	---

int	;	)	;
-----	---	---	---

LR(0) Parsing

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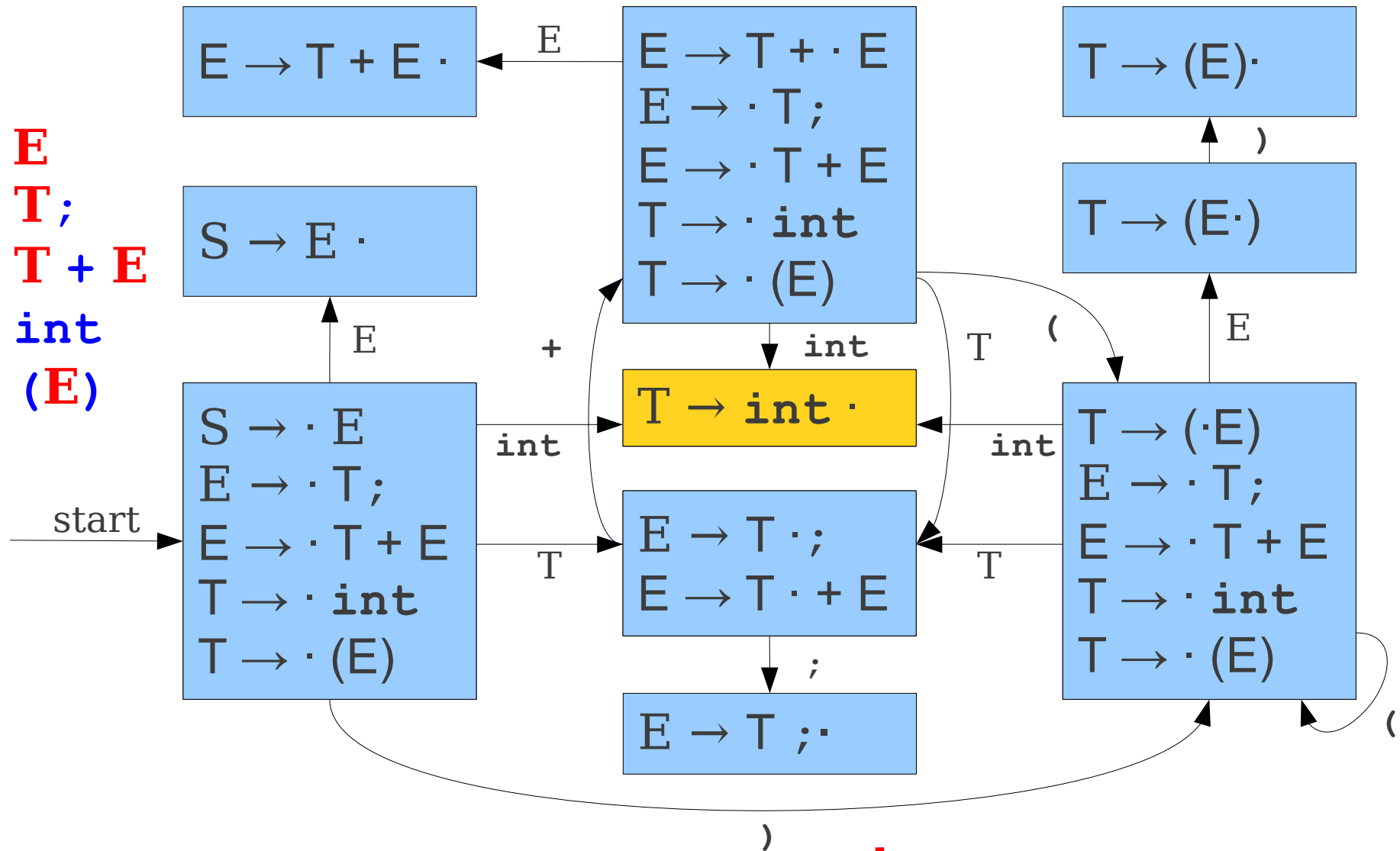


T	+	(	T	+	int
---	---	---	---	---	-----

;	)	;
---	---	---

LR(0) Parsing

$S \rightarrow E$
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 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
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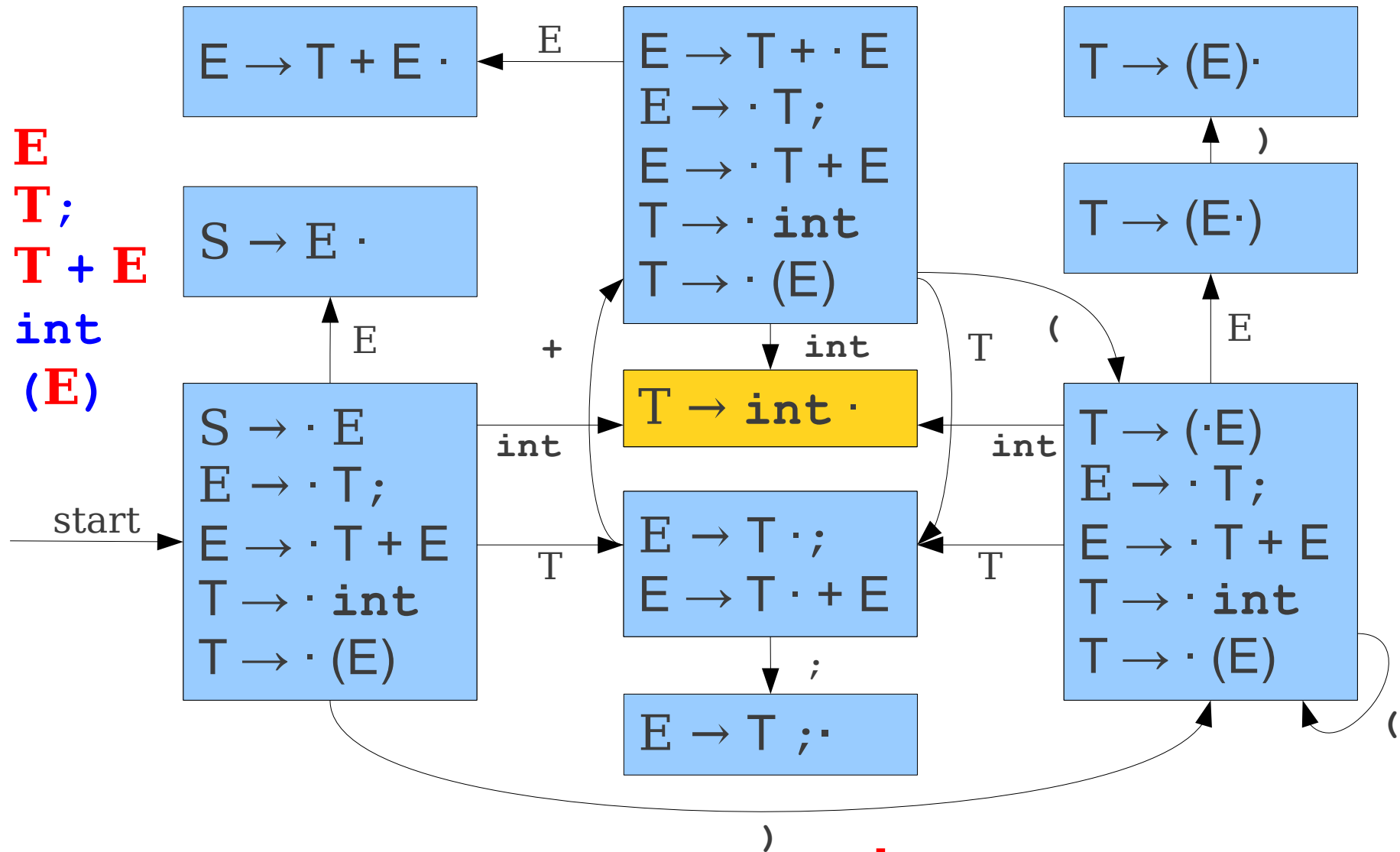


T	+	(	T	+	int
---	---	---	---	---	-----

;	)	;
---	---	---

LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

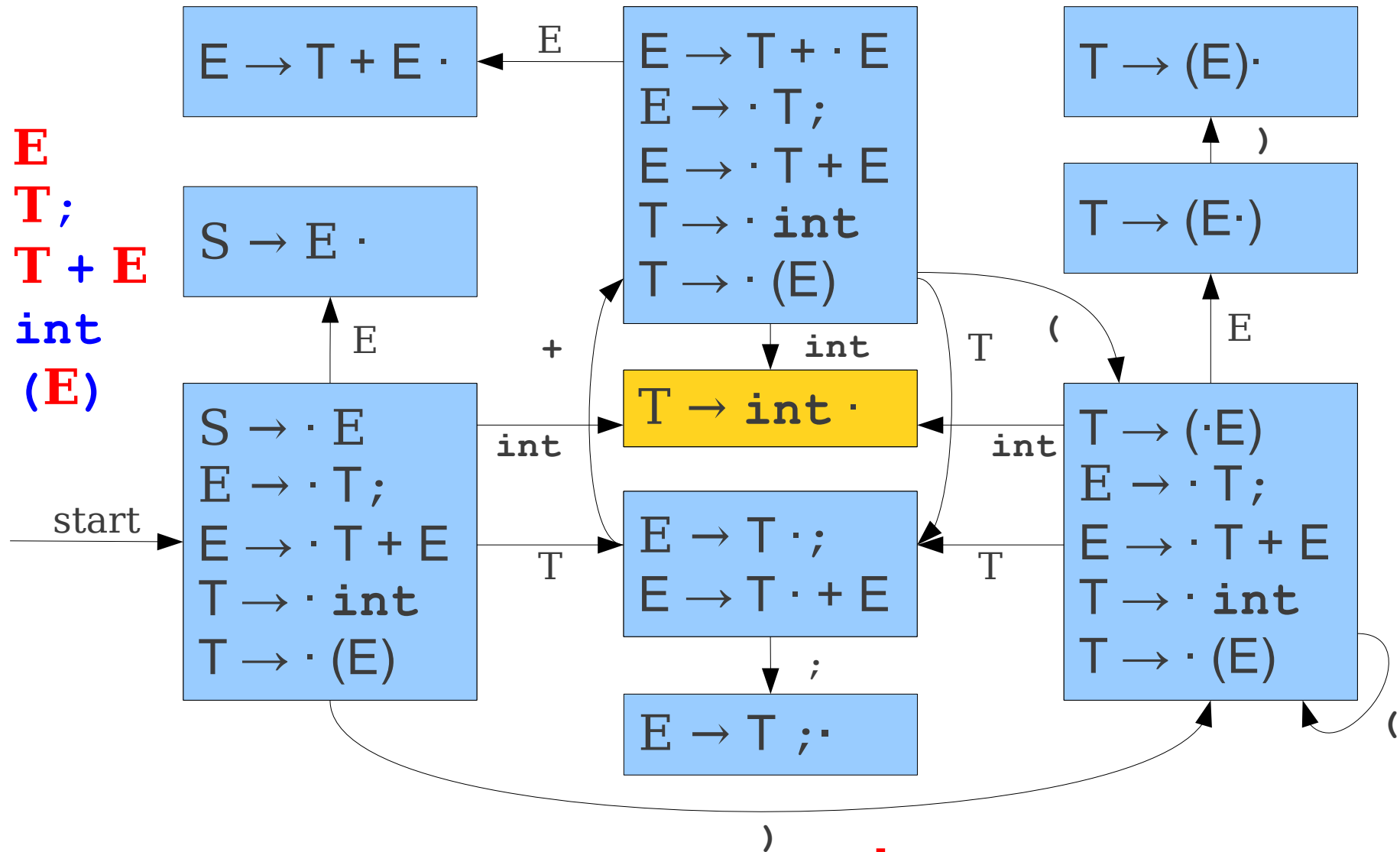


T	+	(	T	+
---	---	---	---	---

;	)	;
---	---	---

LR(0) Parsing

$S \rightarrow E$
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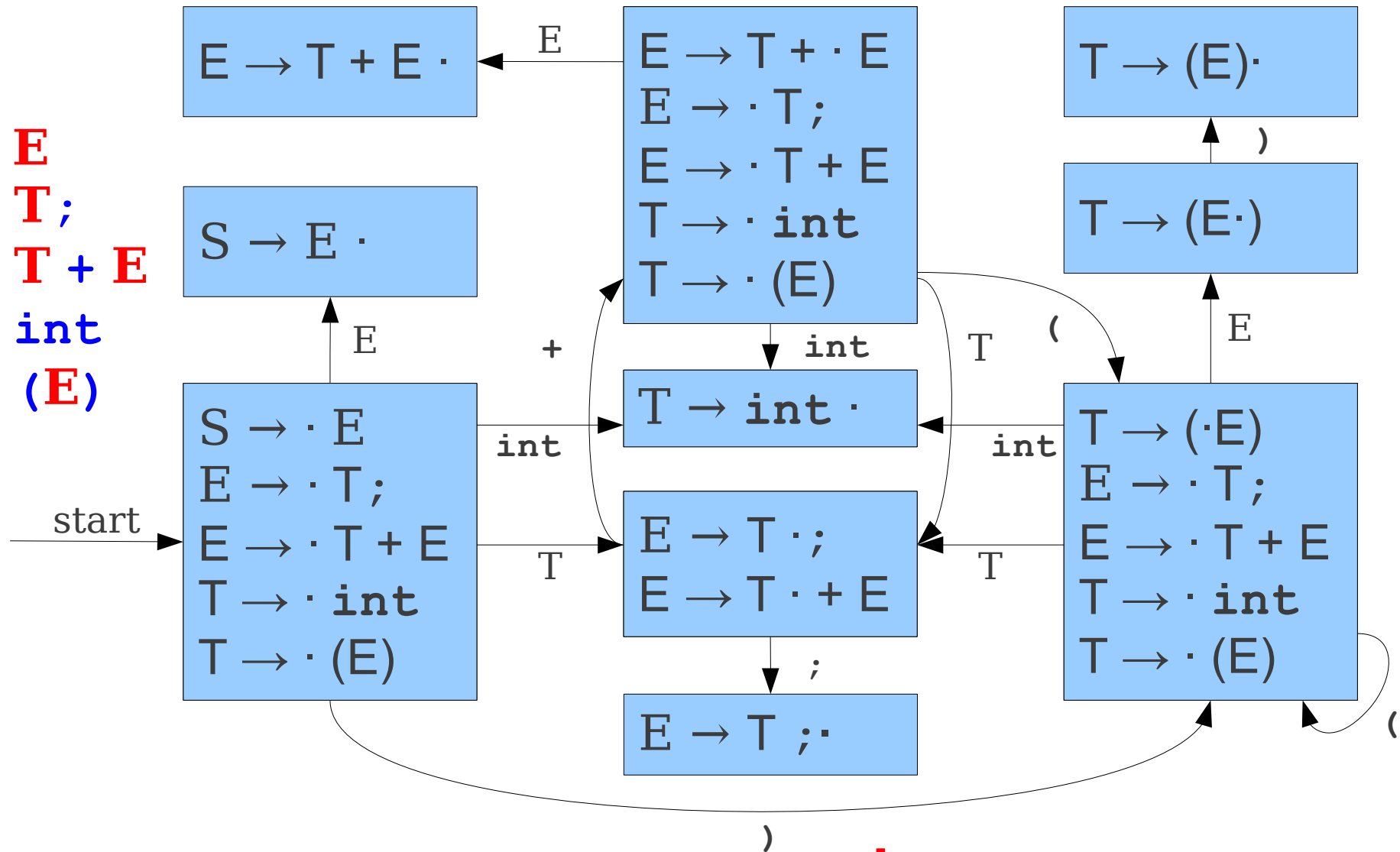


T	+	(	T	+	T
---	---	---	---	---	---

;	)	;
---	---	---

LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
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T	+	(	T	+	T
---	---	---	---	---	---

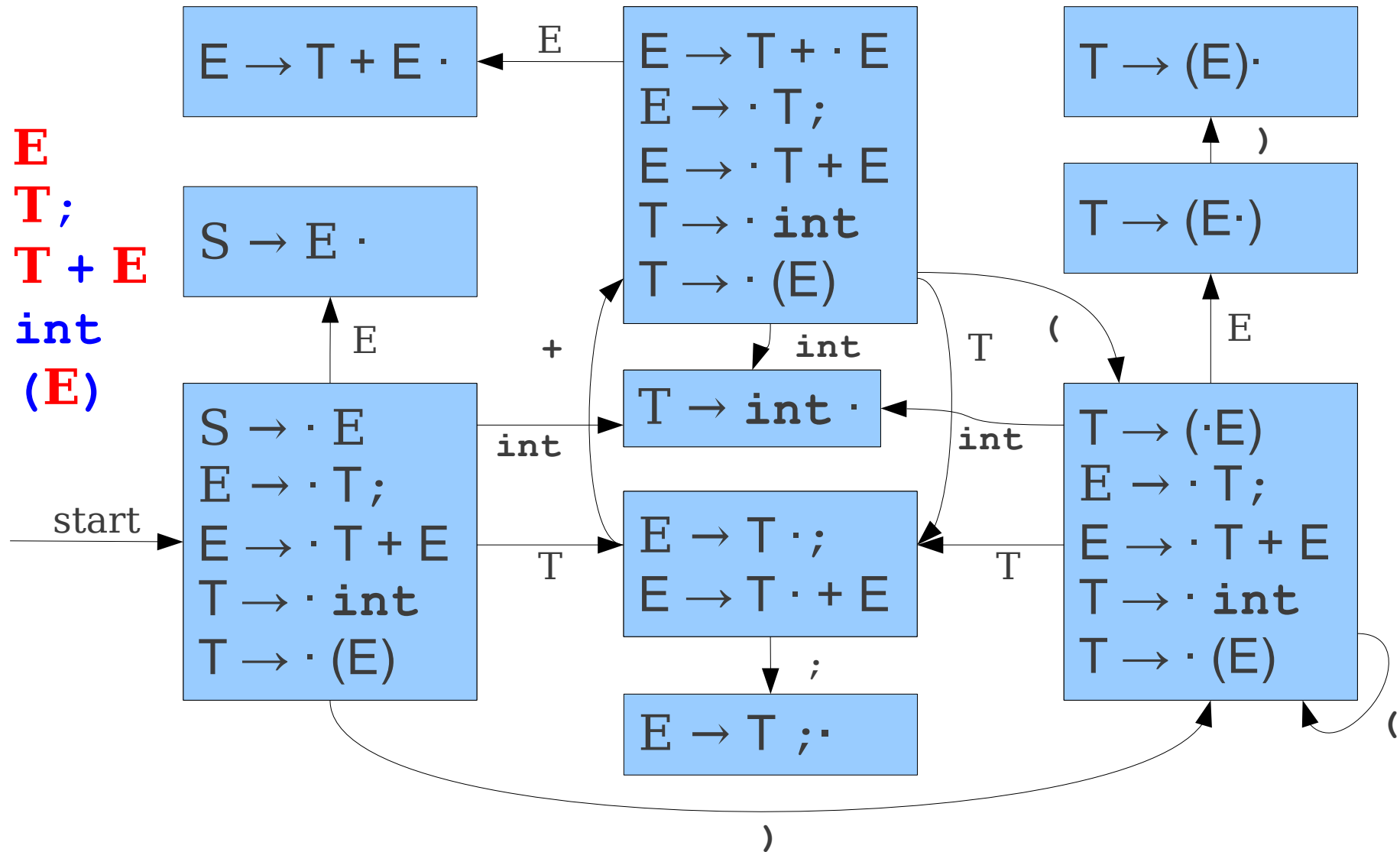
;	)	;
---	---	---

An Optimization

- Rather than restart the automaton on each reduction, remember what state we were in for each symbol.
- When applying a reduction, restart the automaton from the last known good state.

LR(0) Parsing

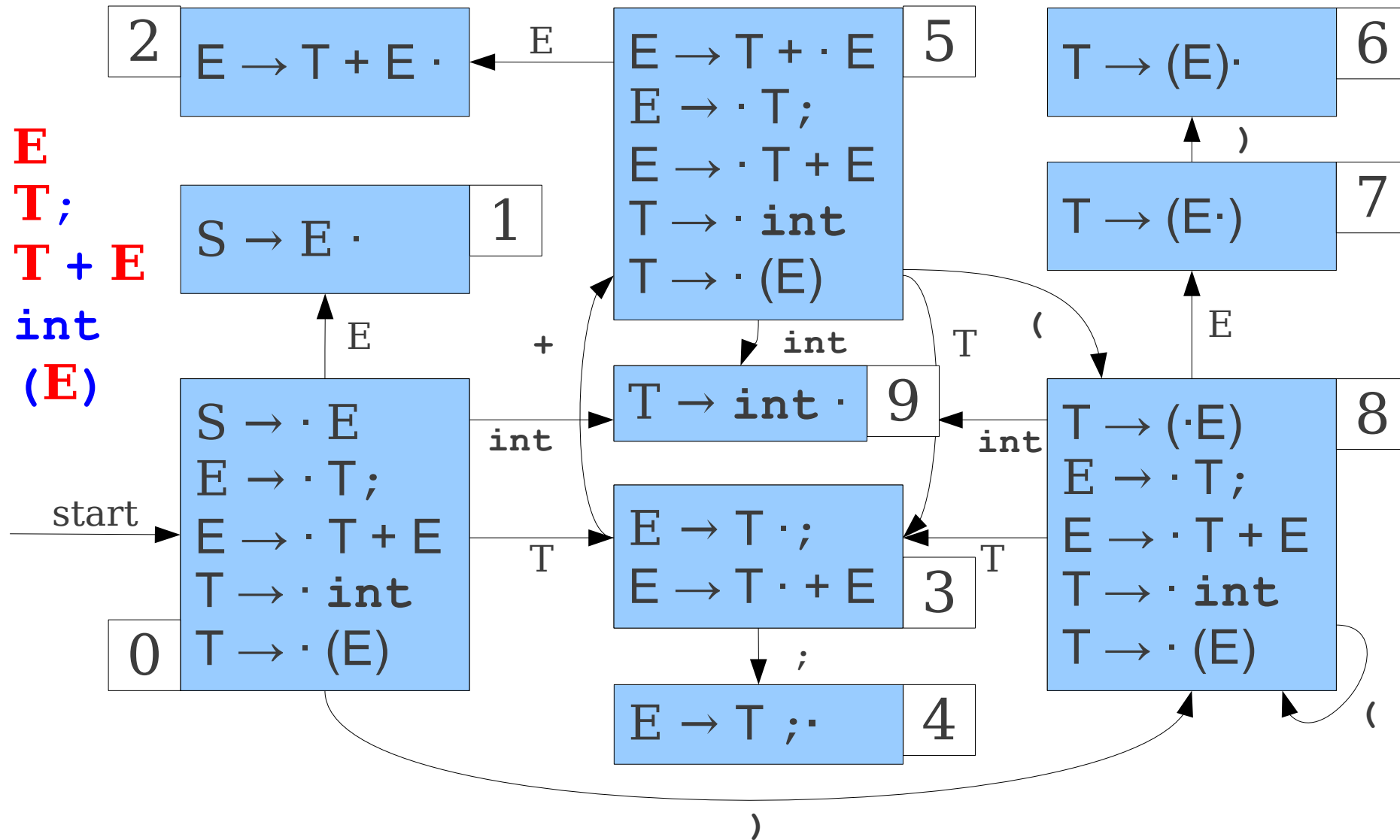
$S \rightarrow E$
 $E \rightarrow T;$
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 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



int	+	(	int	+	int	;	)	;
-----	---	---	-----	---	-----	---	---	---

LR(0) Parsing

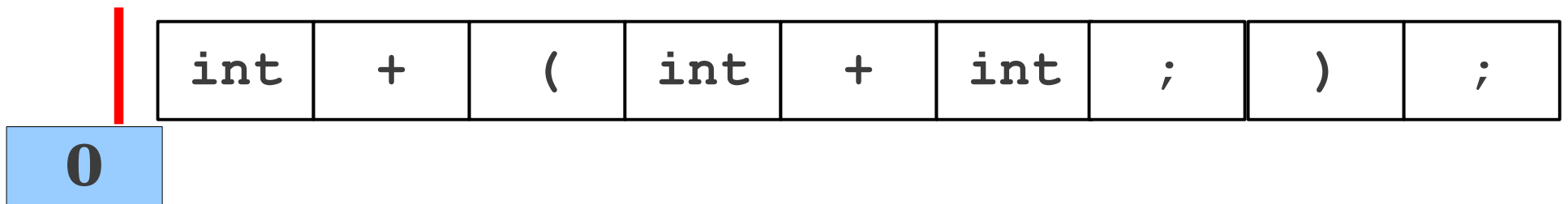
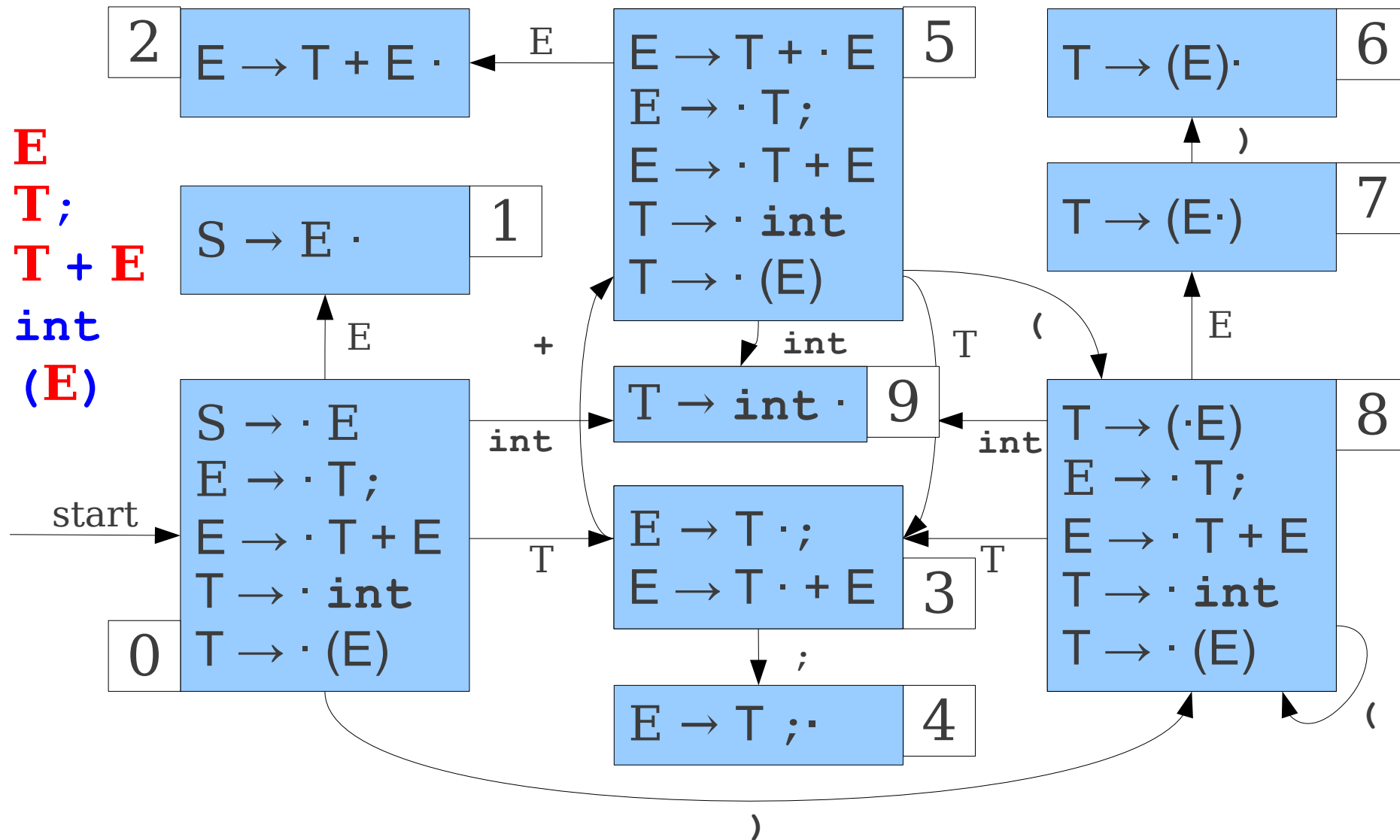
$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



int	+	(	int	+	int	;	)	;
-----	---	---	-----	---	-----	---	---	---

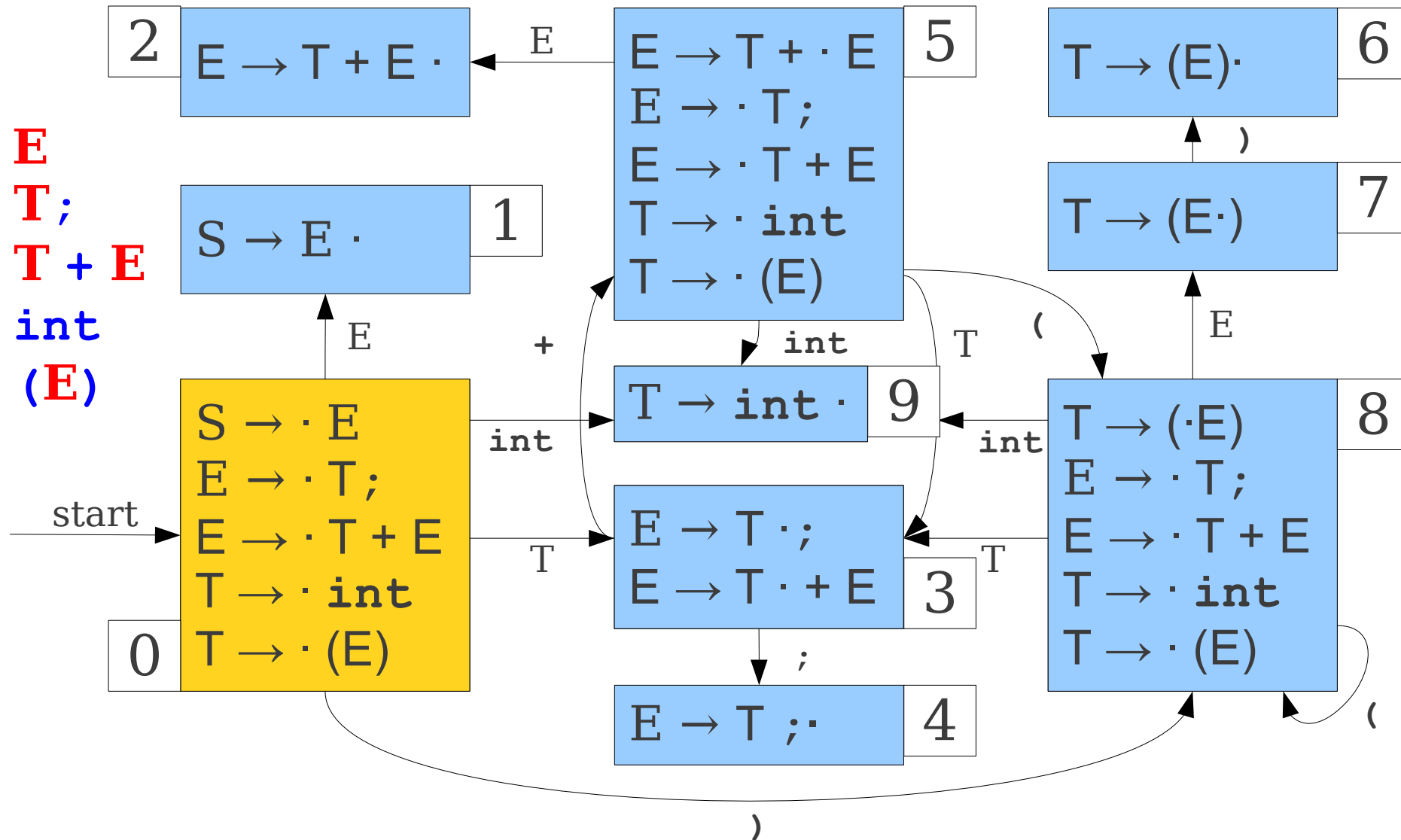
LR(0) Parsing

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LR(0) Parsing

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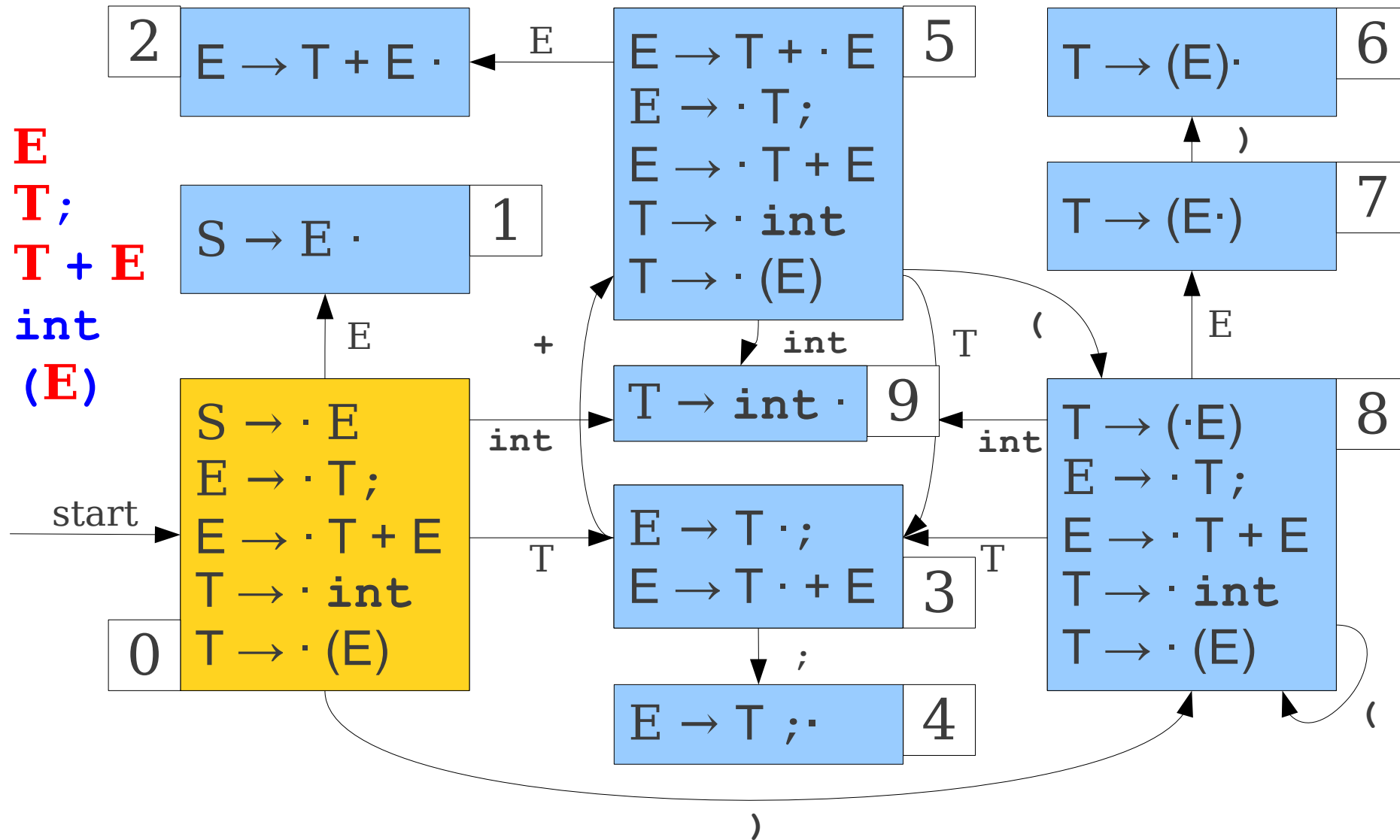


int	+	(	int	+	int	;	)	;
-----	---	---	-----	---	-----	---	---	---

0

LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

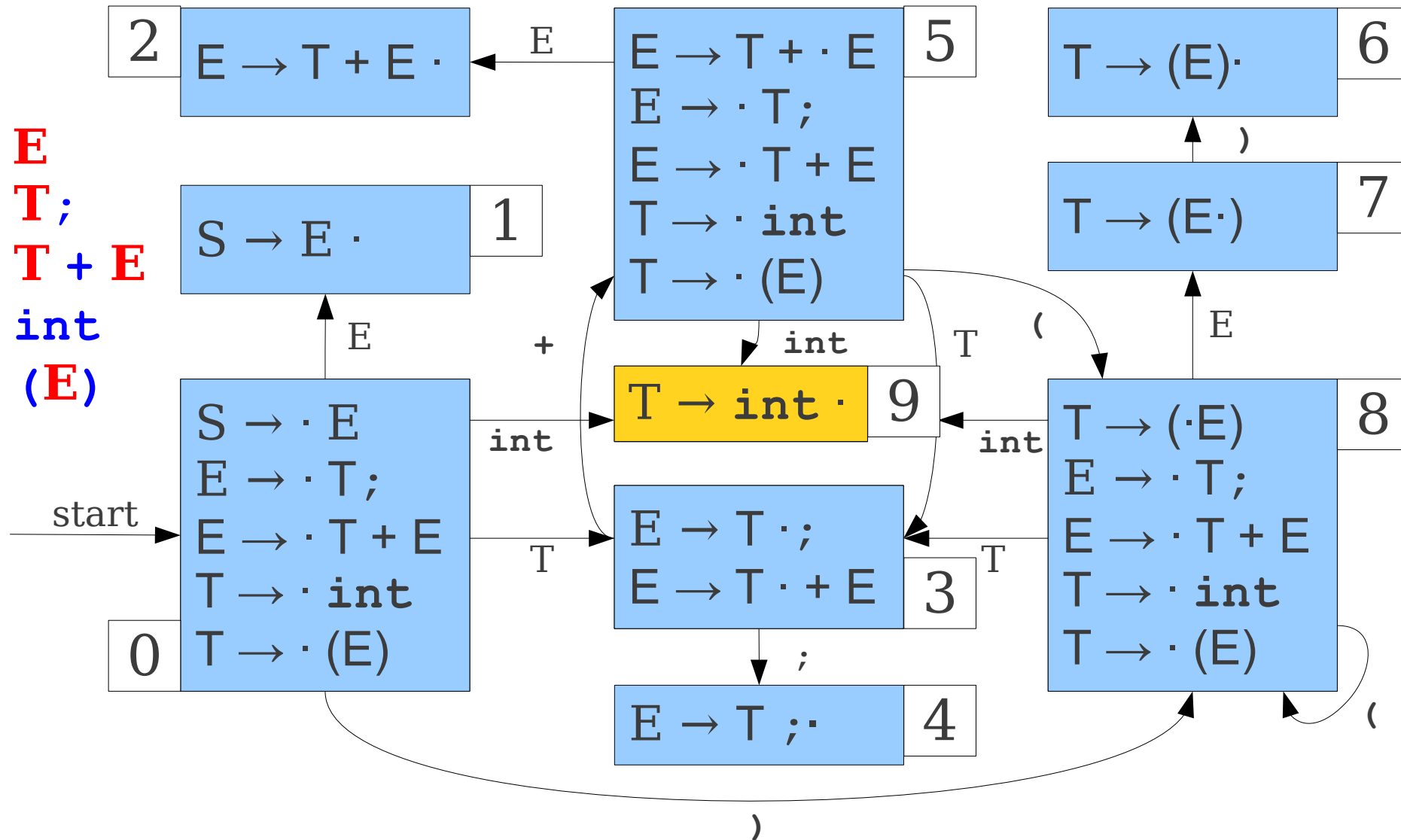


int		+	(	int	+	int	;	)	;
-----	--	---	---	-----	---	-----	---	---	---

0

LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
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 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

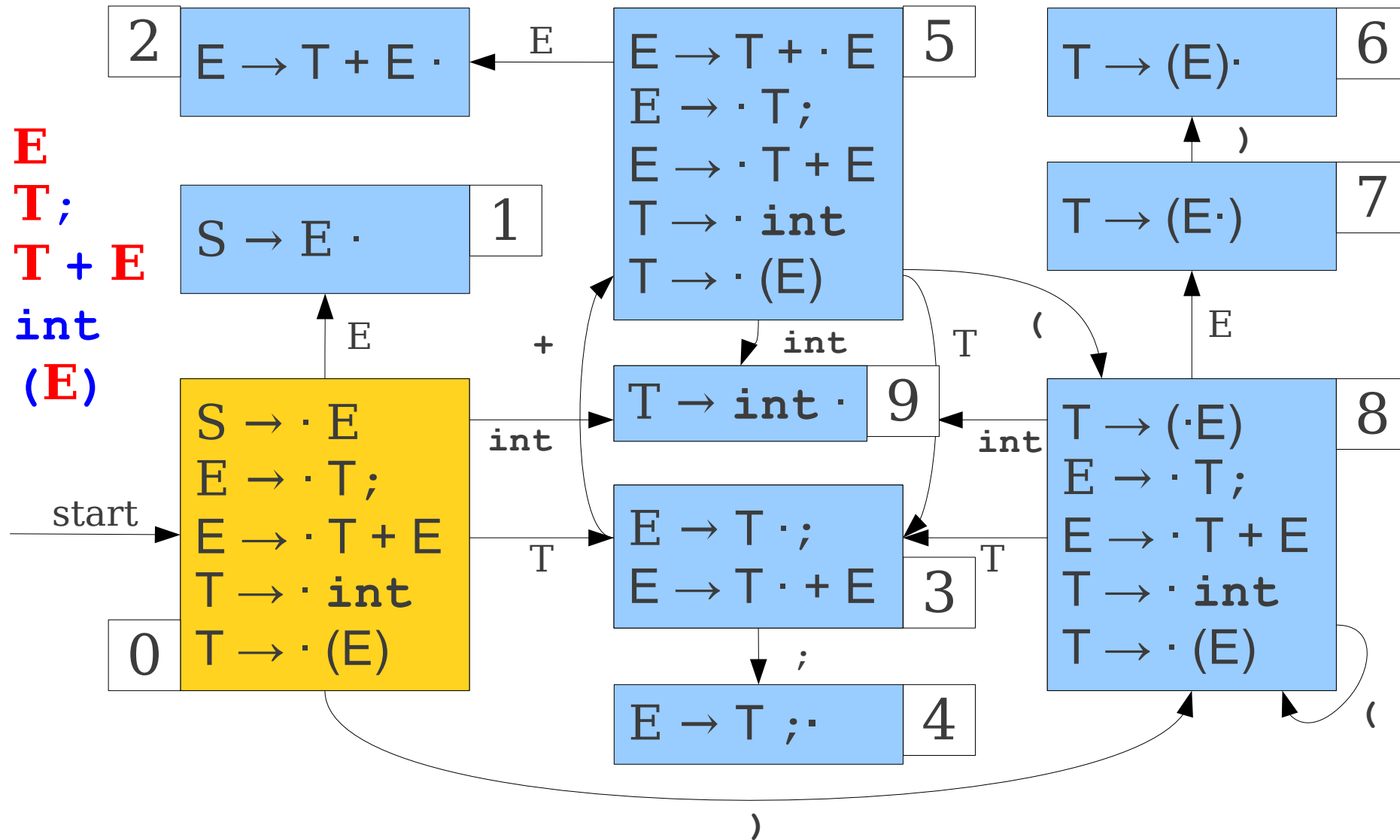


int		+	(	int	+	int	;	)	;
-----	--	---	---	-----	---	-----	---	---	---

0

LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$

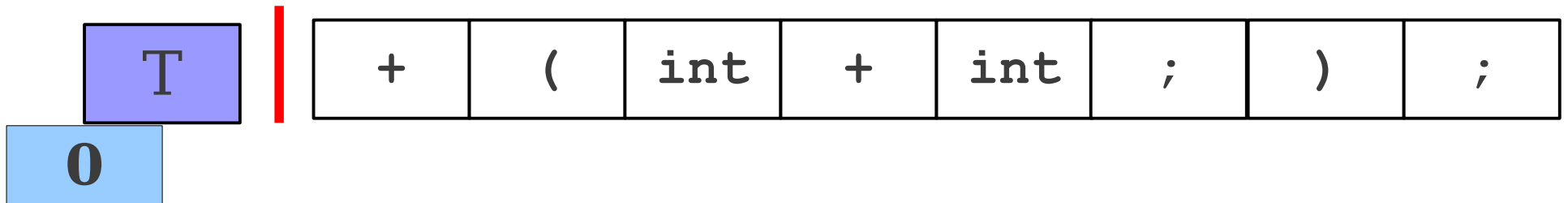
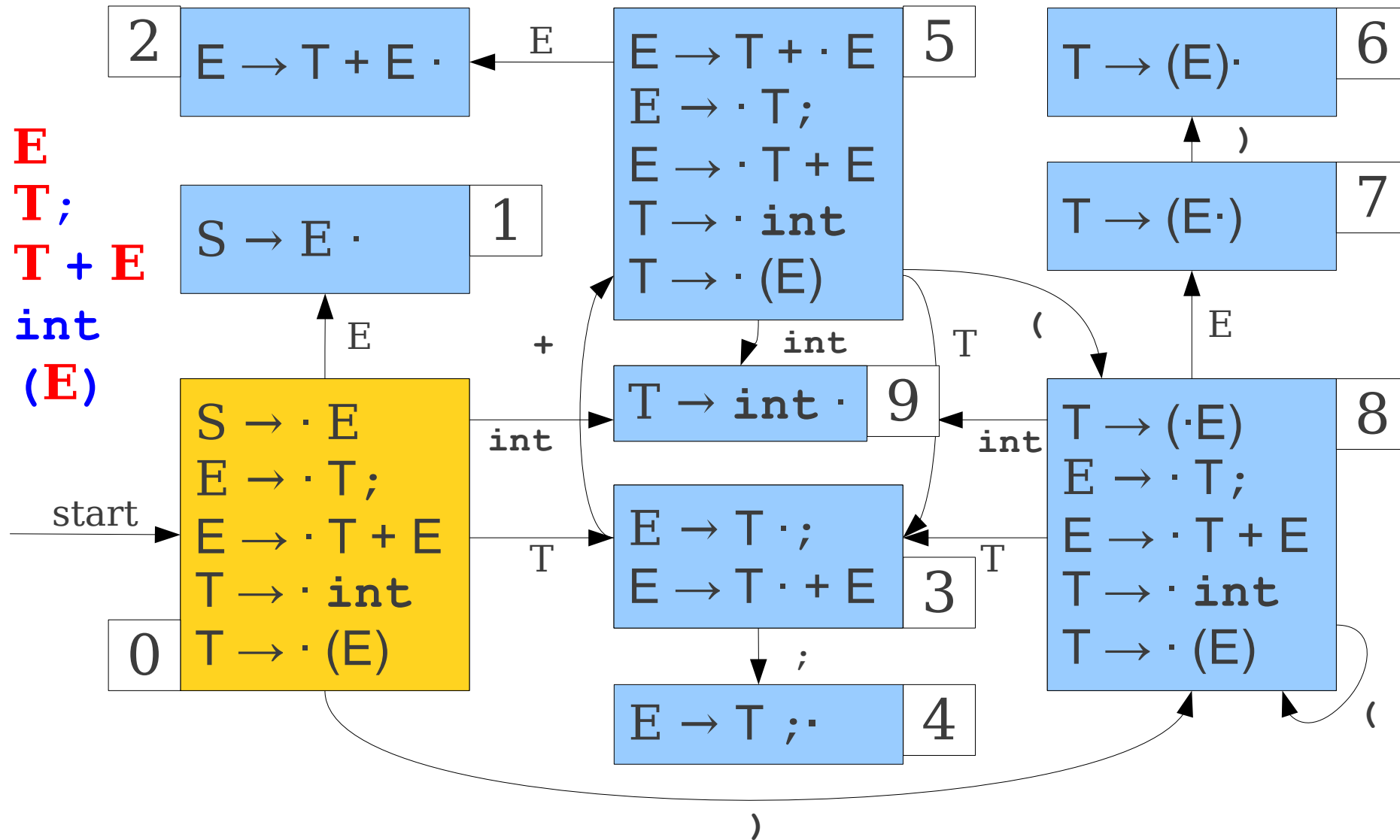


+	(	int	+	int	;	)	;
---	---	-----	---	-----	---	---	---

0

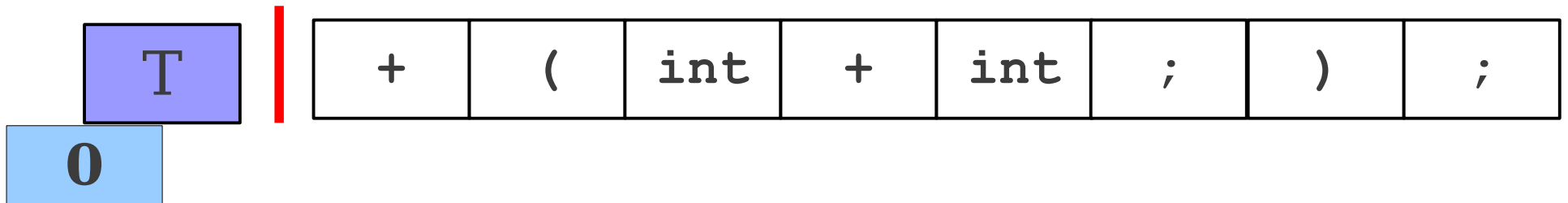
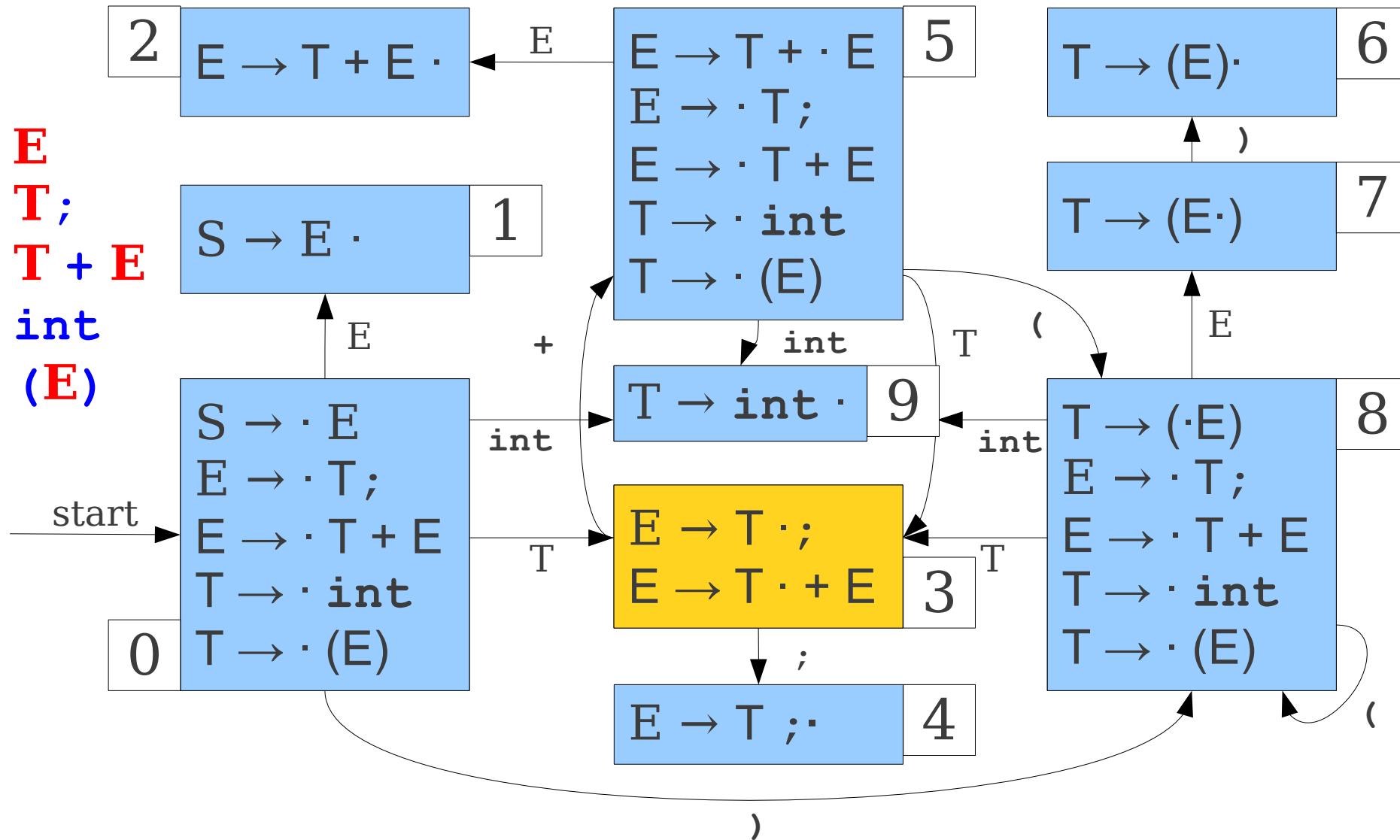
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



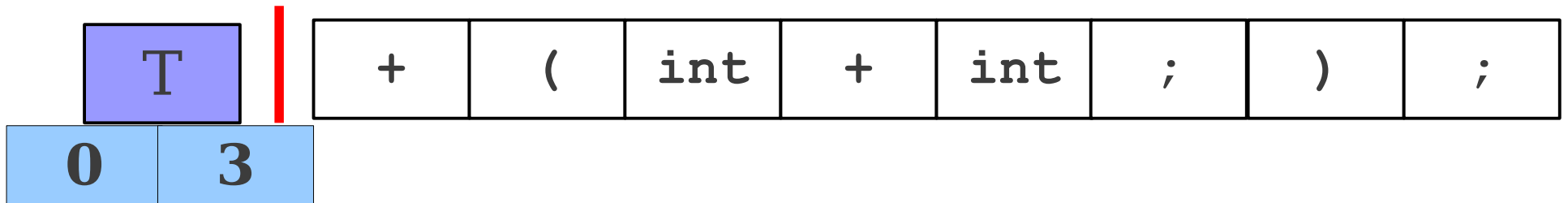
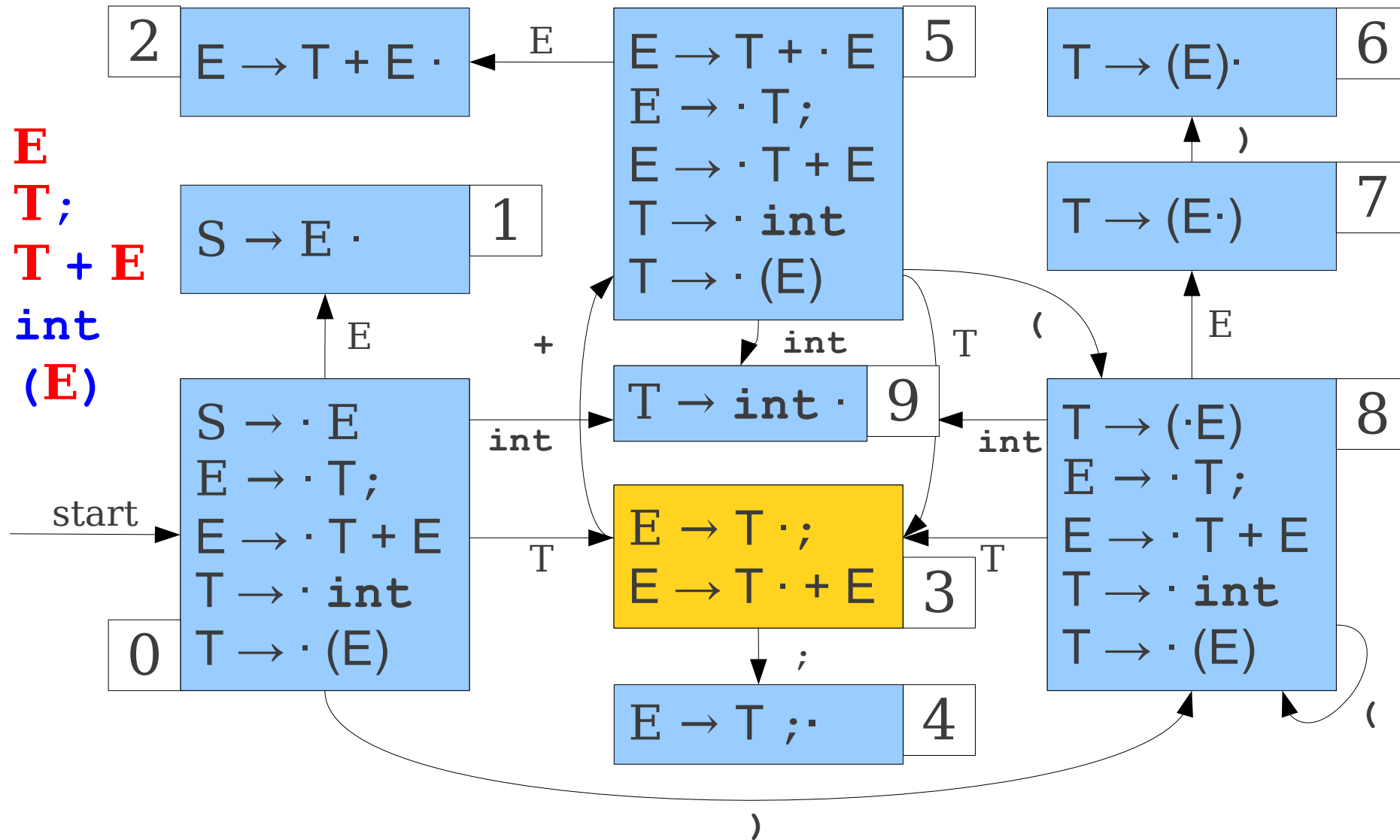
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



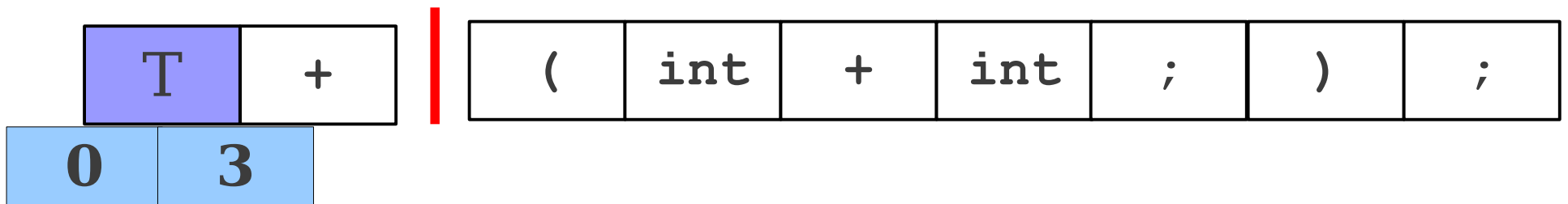
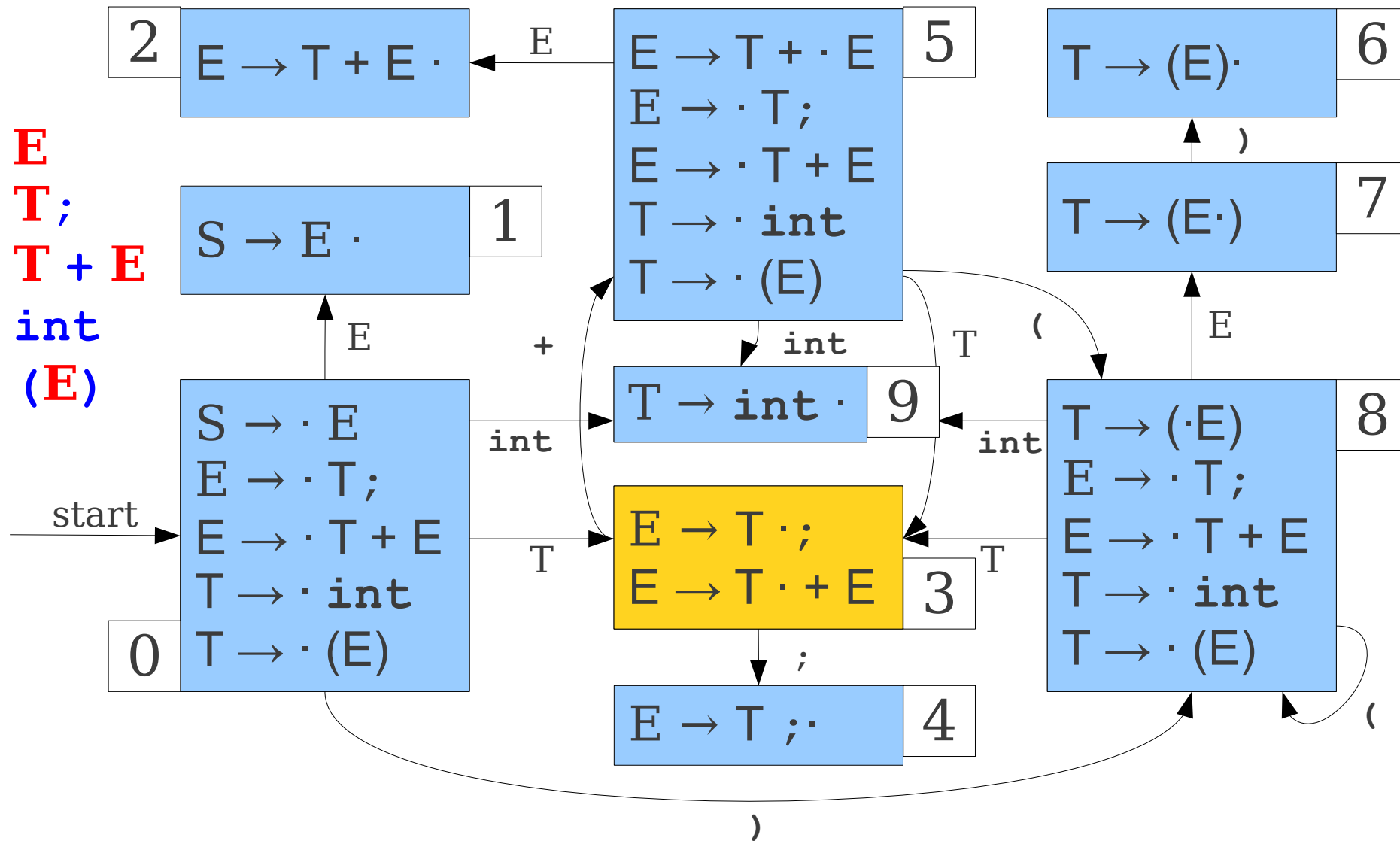
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



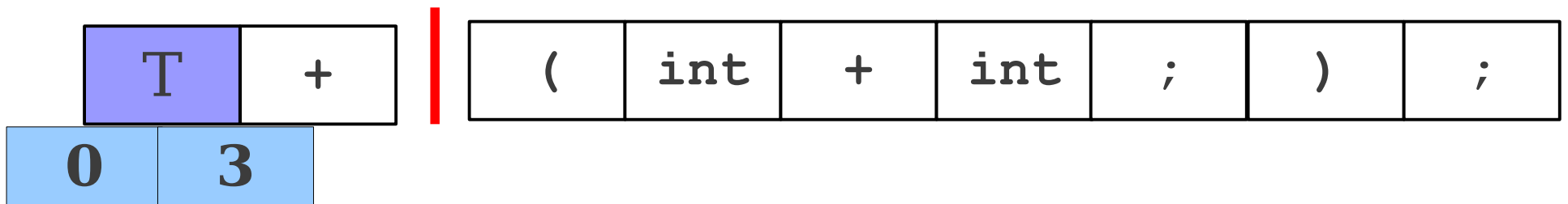
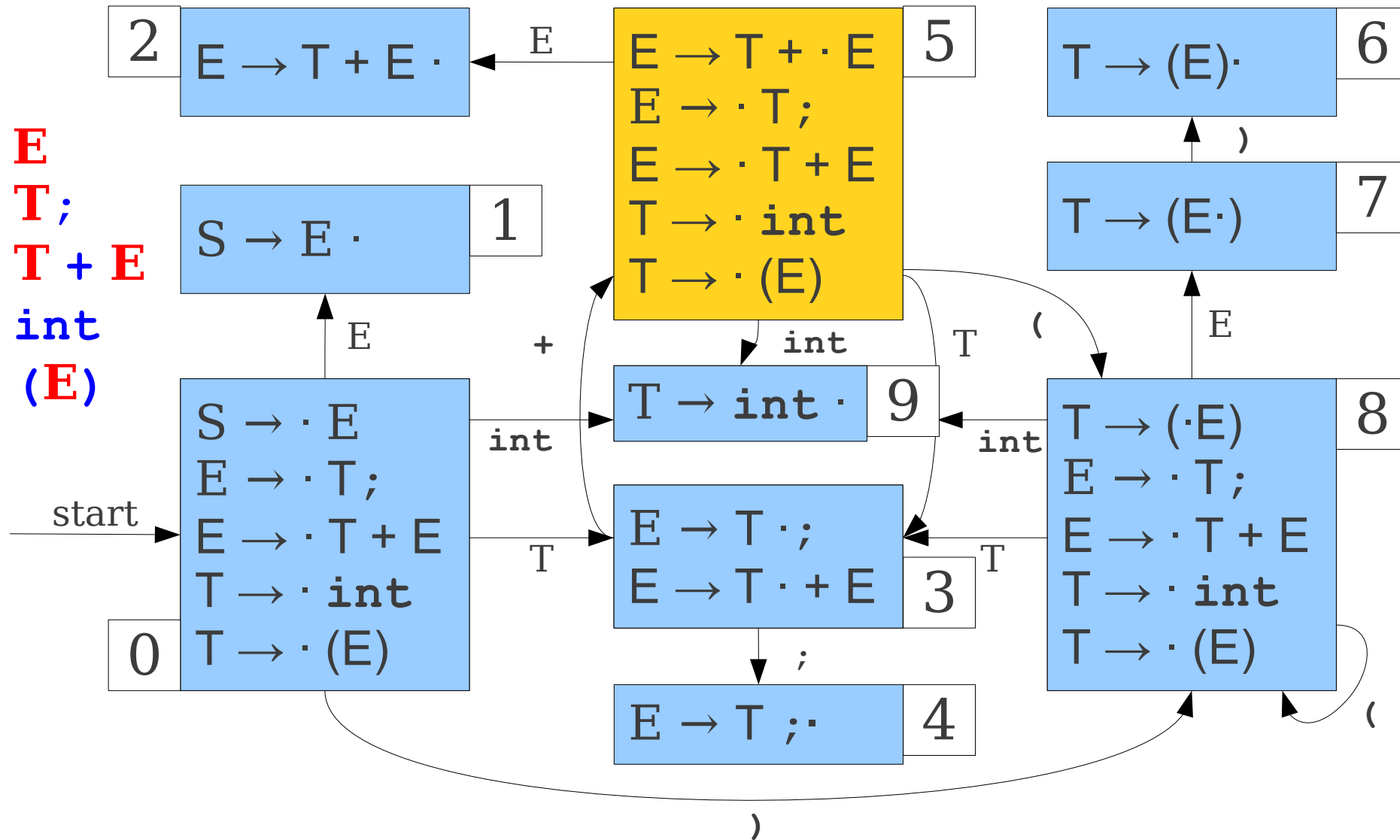
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



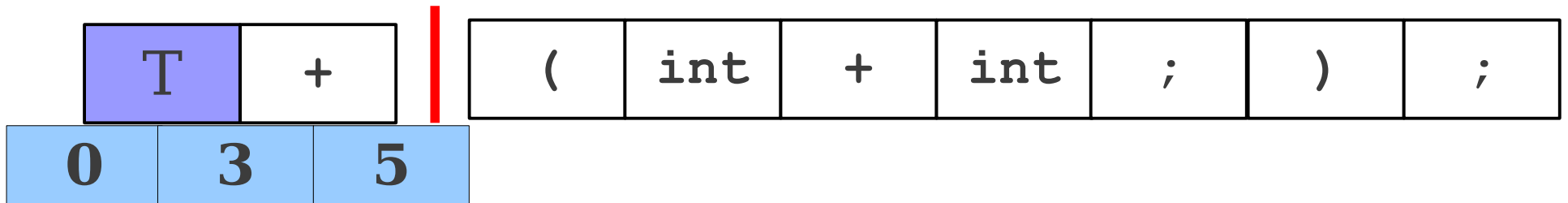
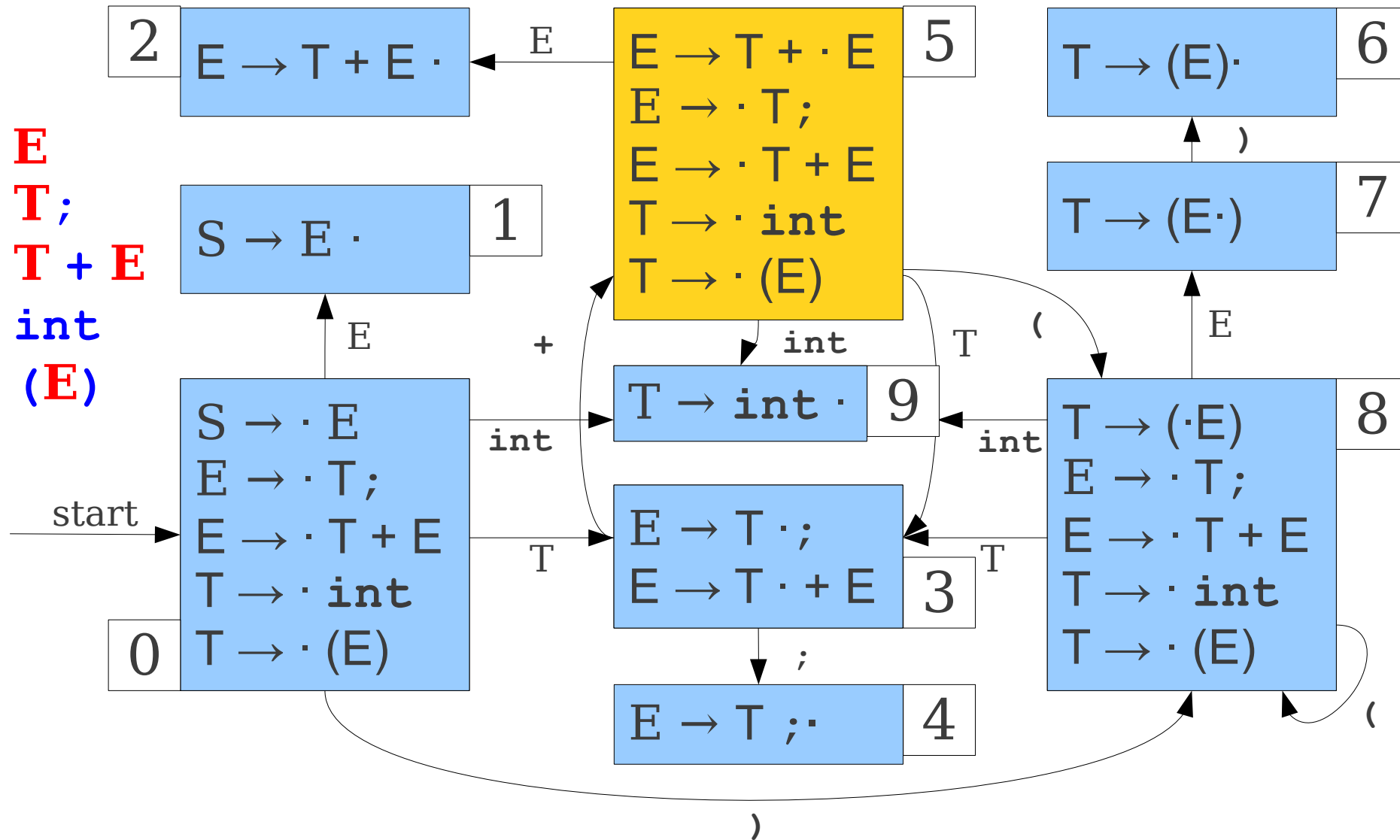
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



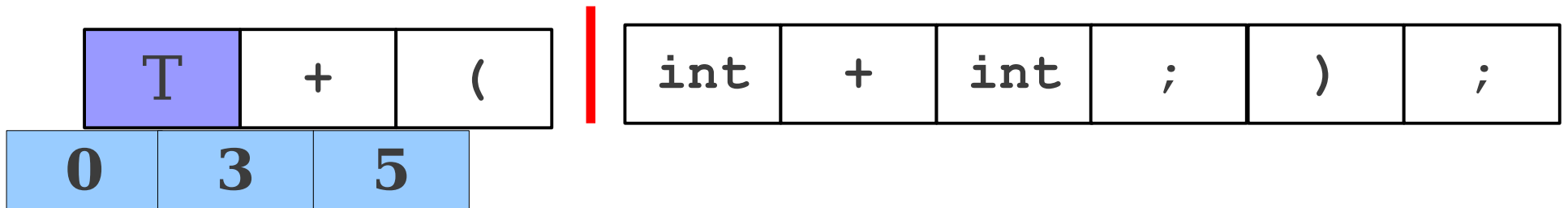
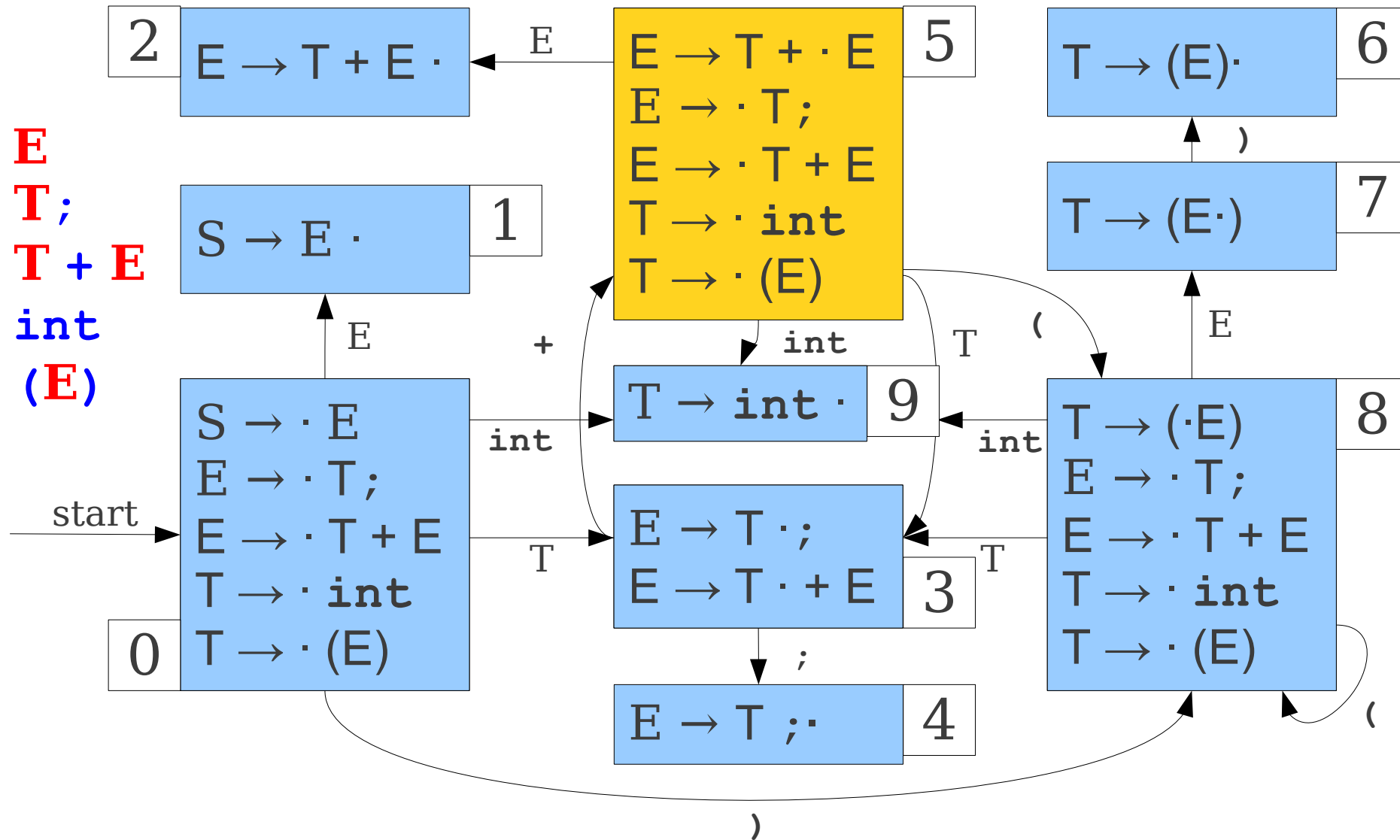
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



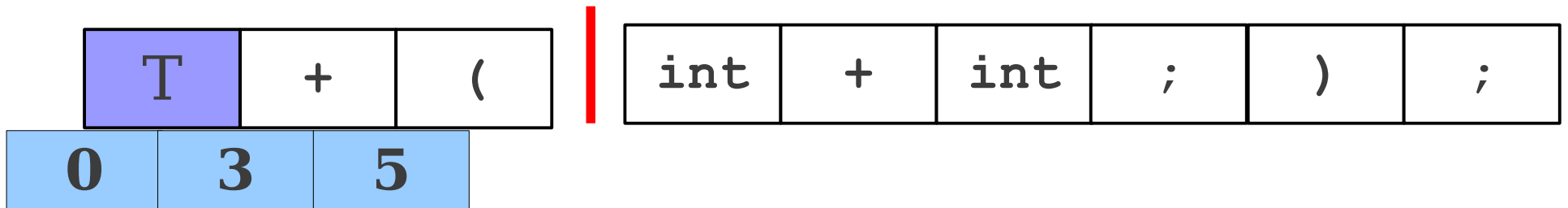
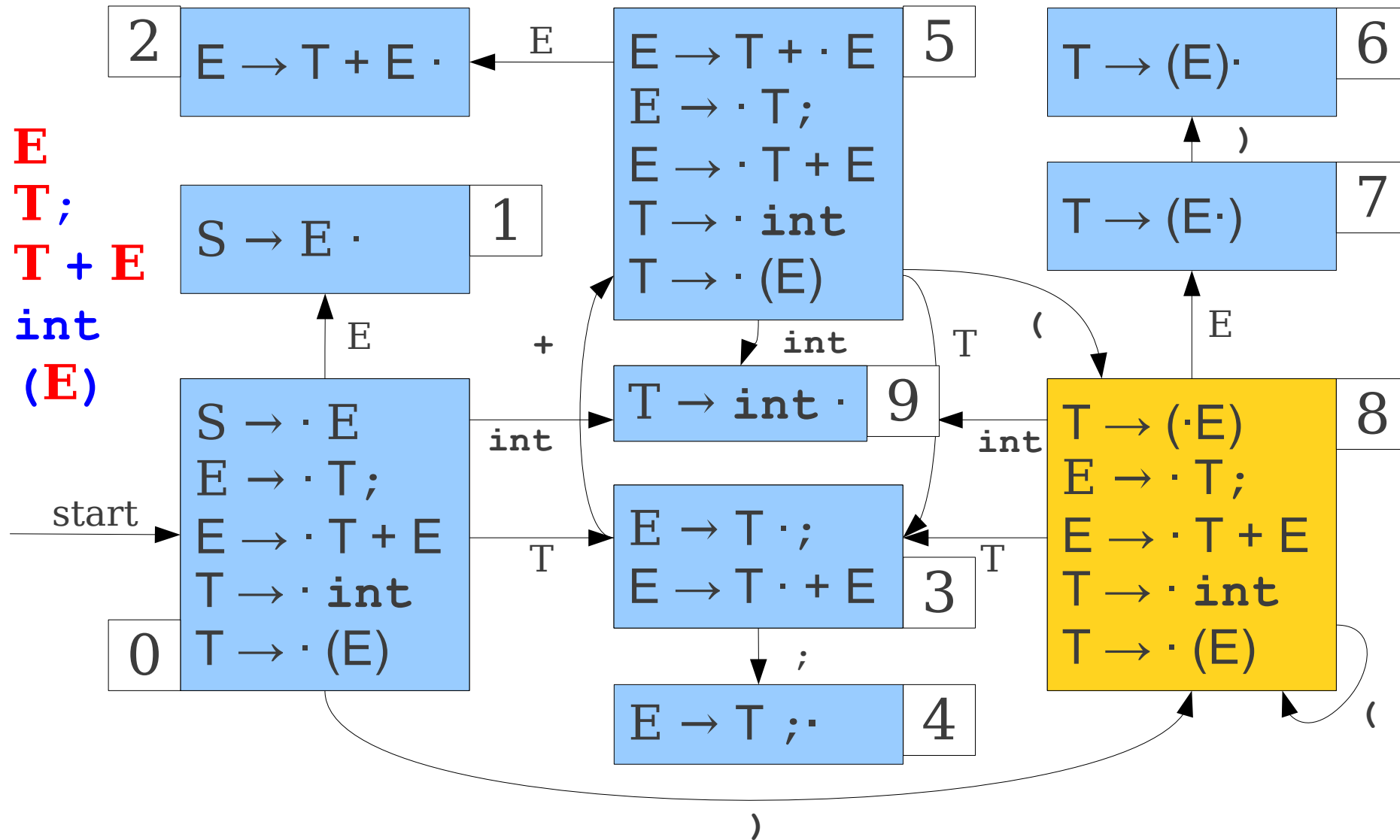
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



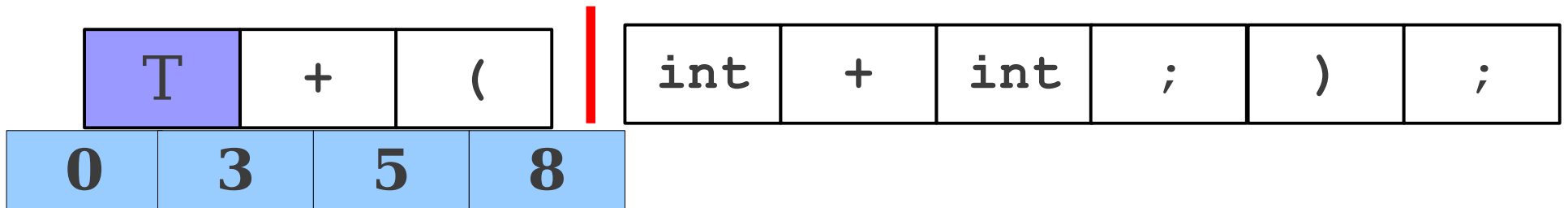
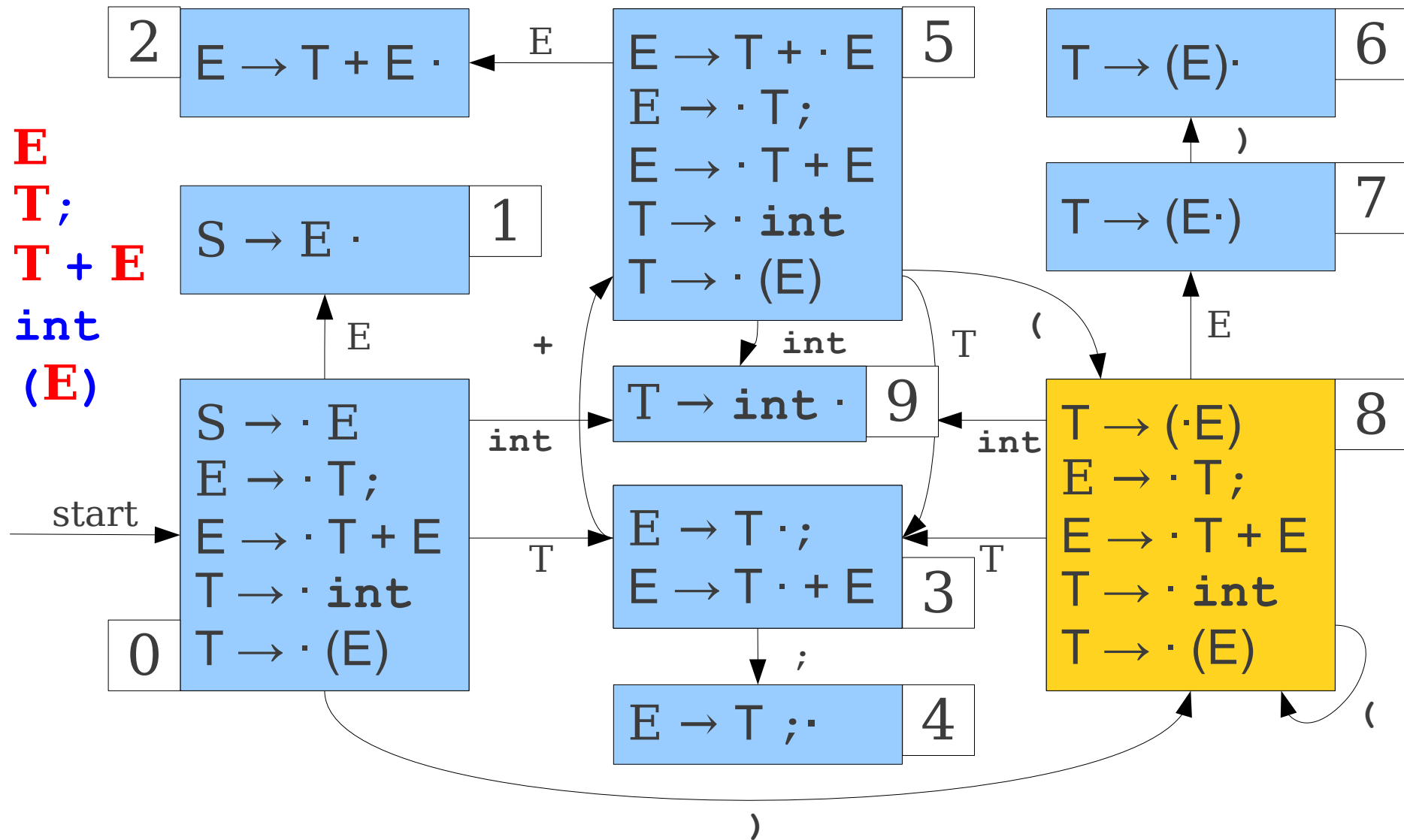
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



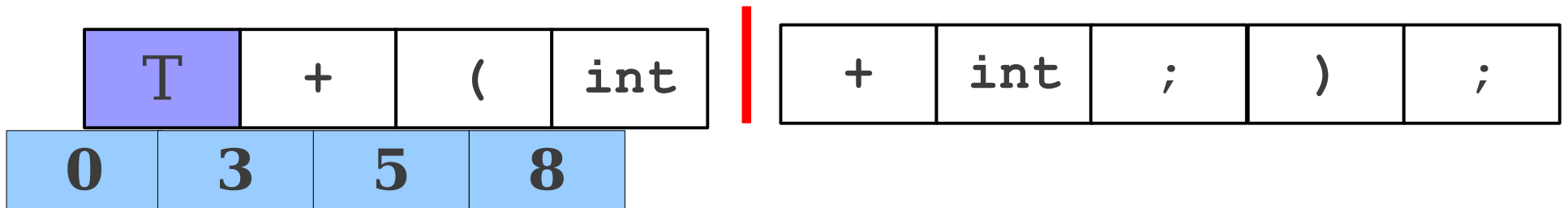
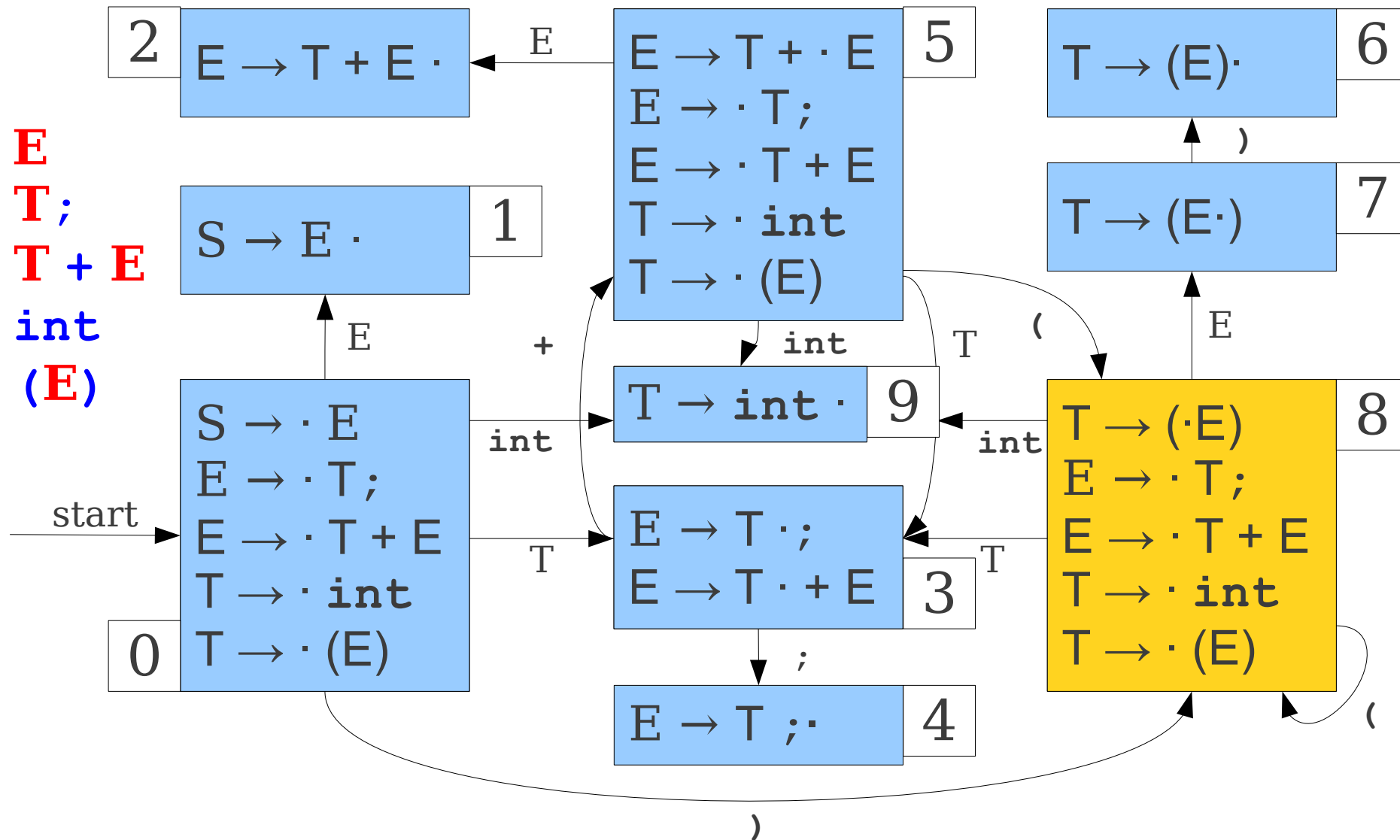
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



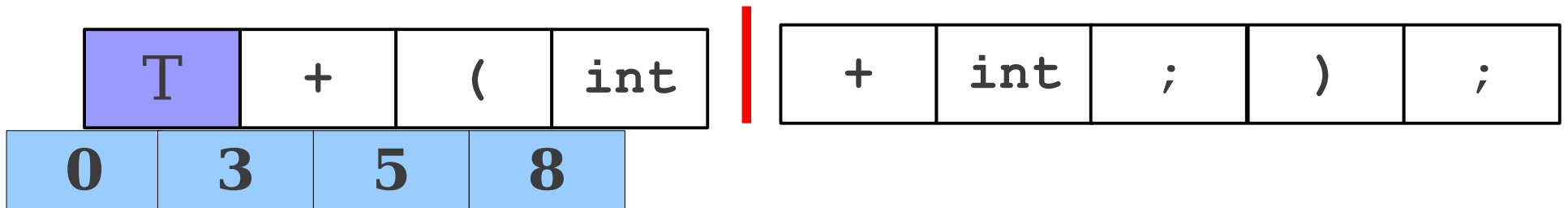
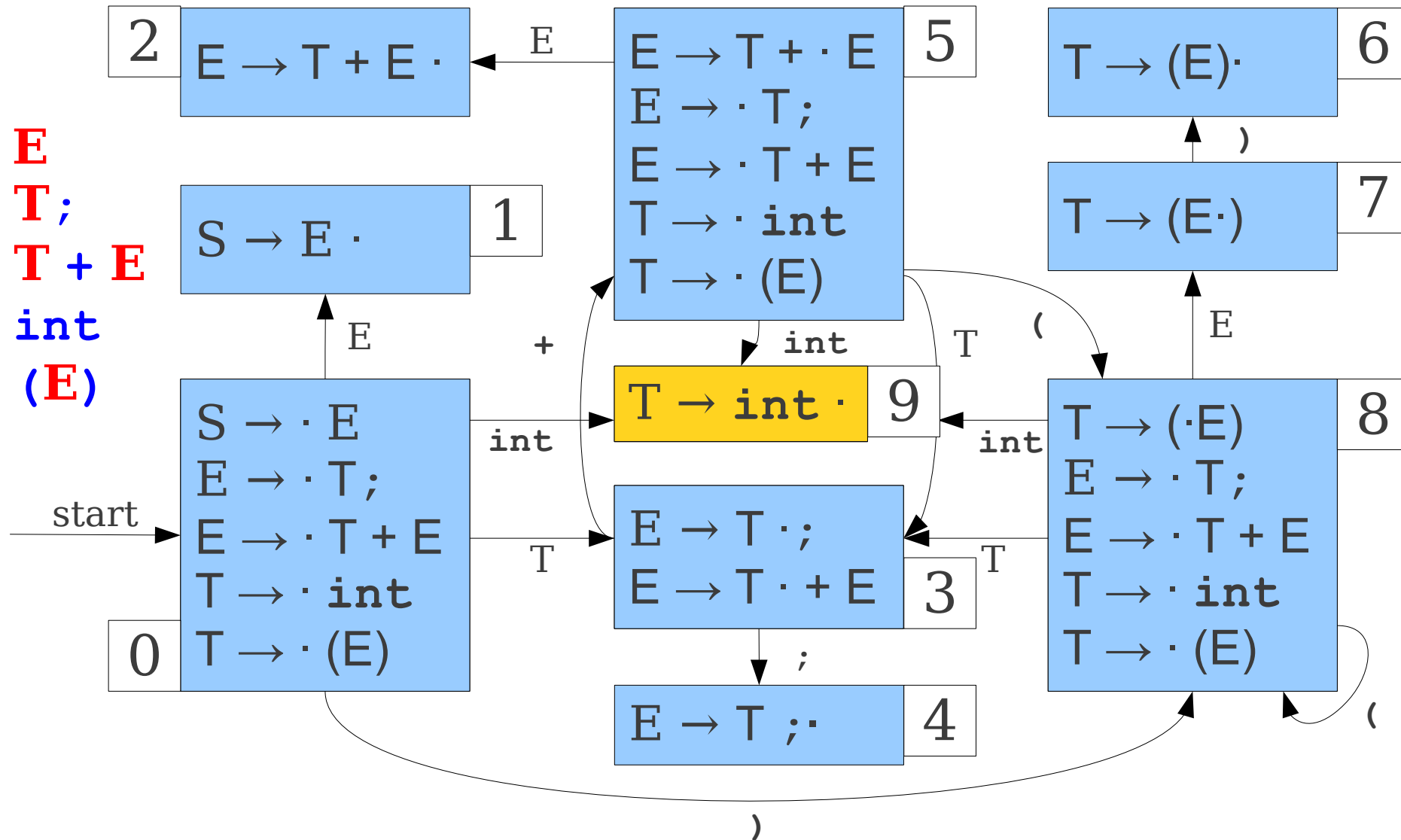
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



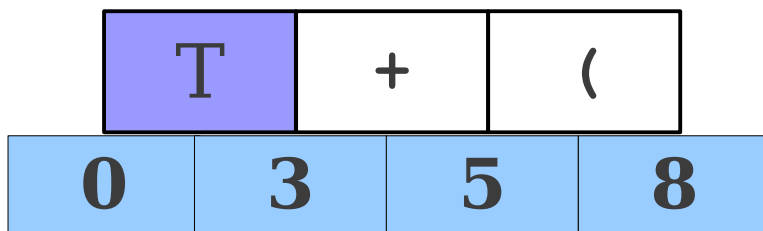
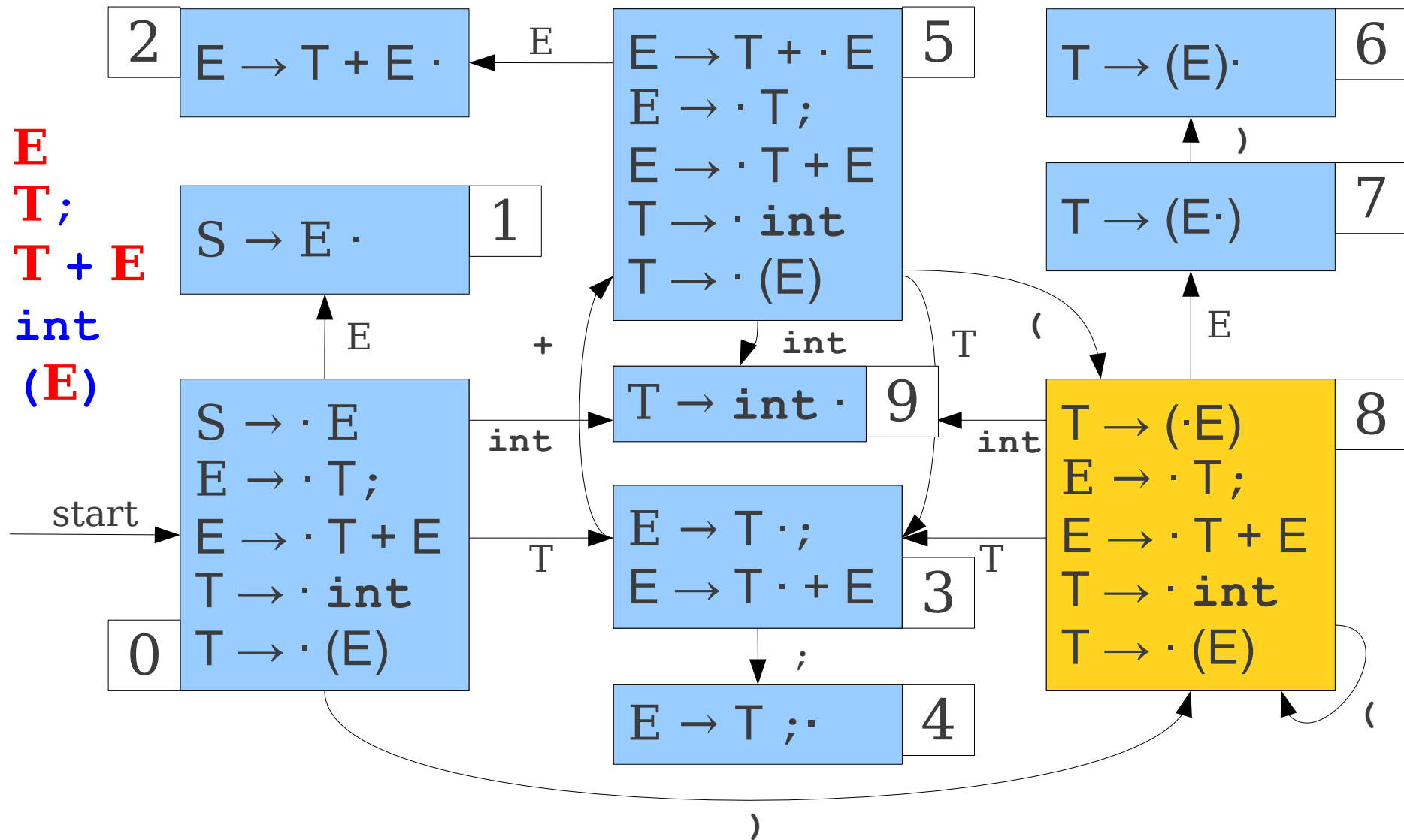
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



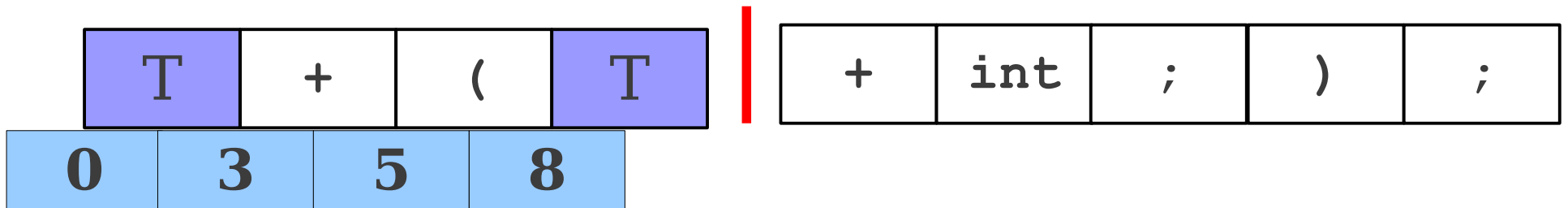
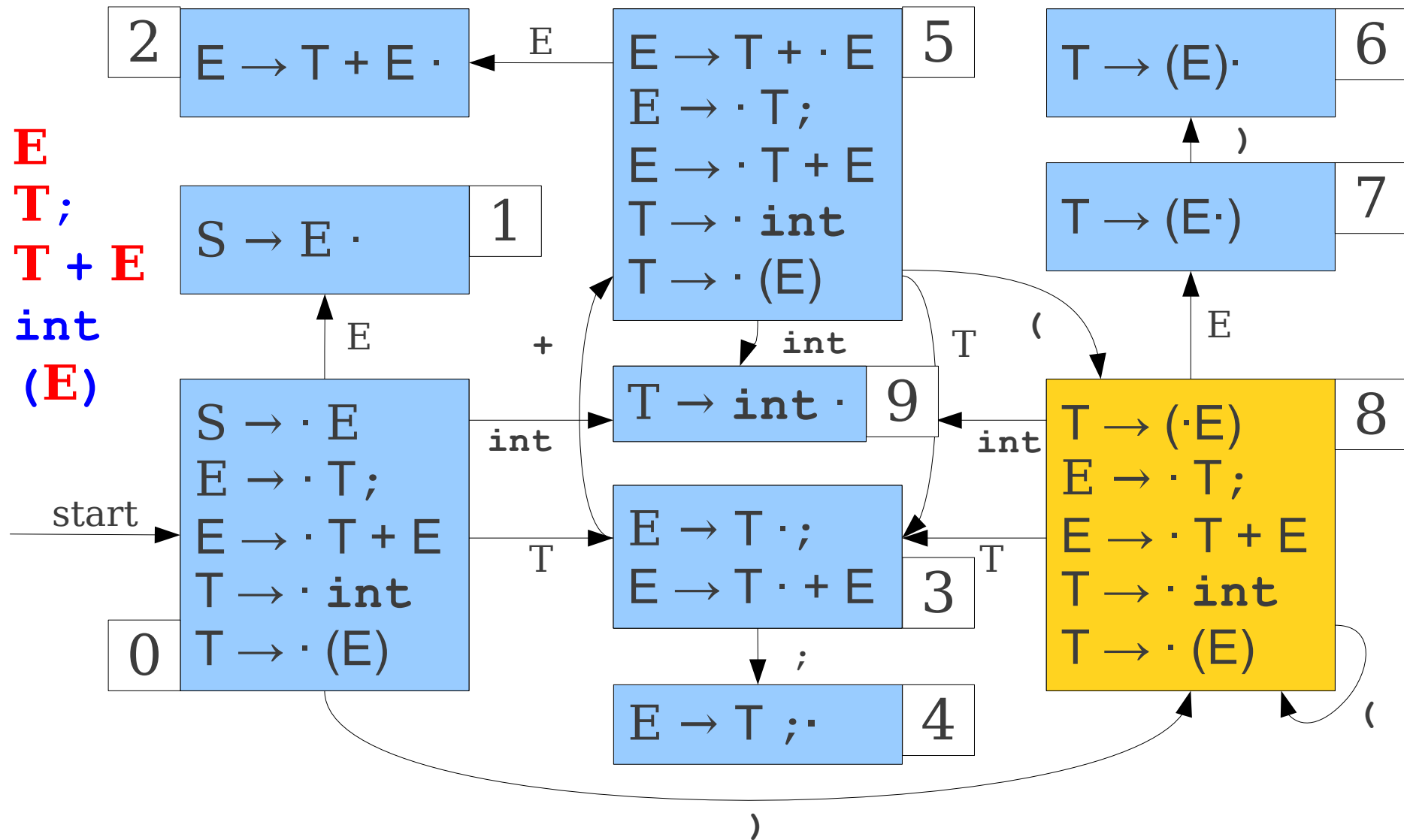
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



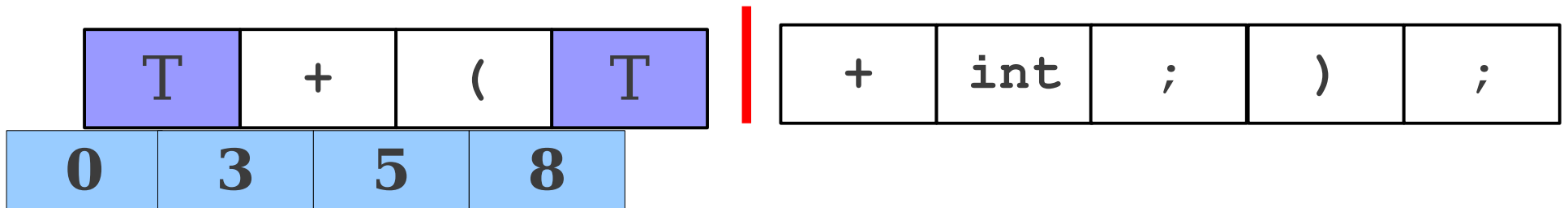
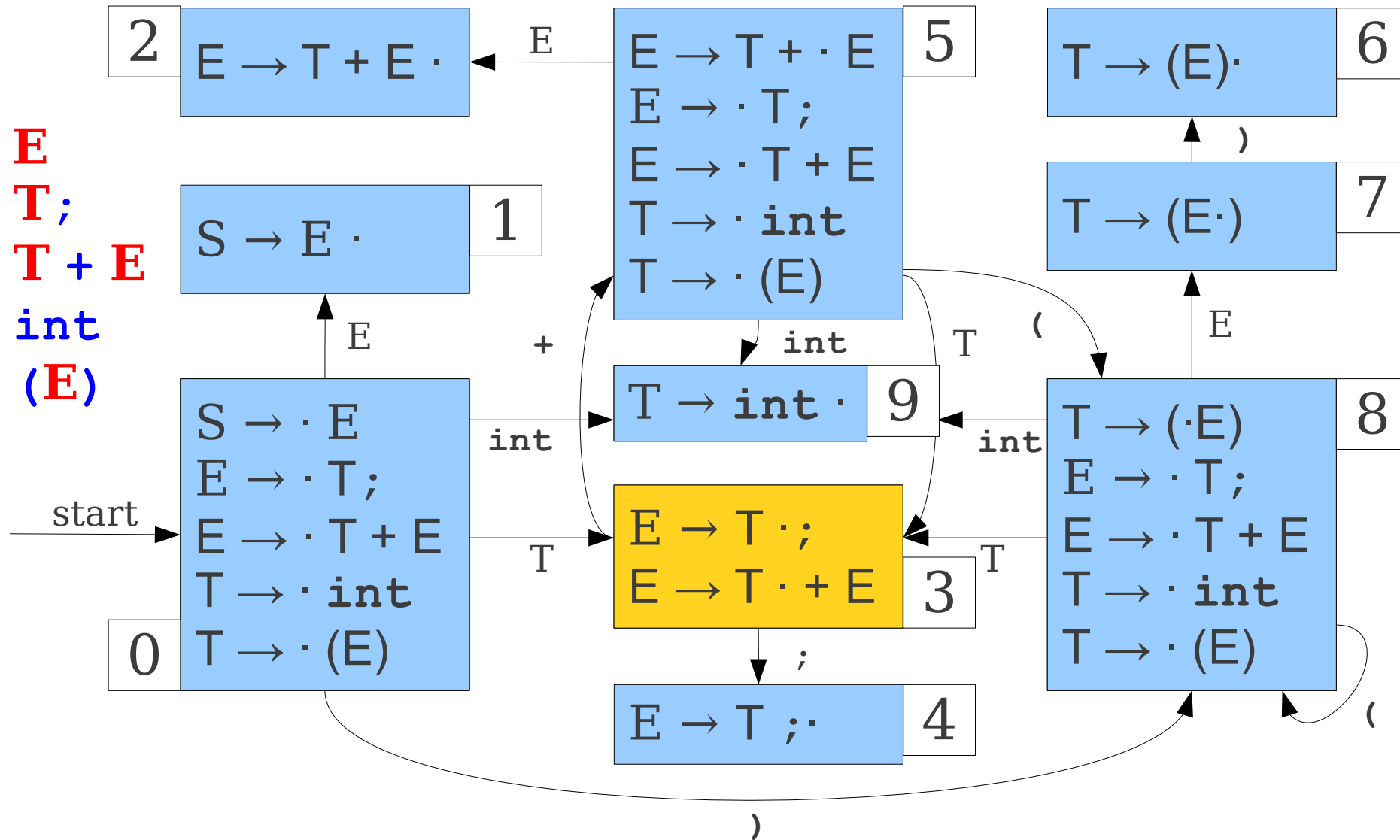
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



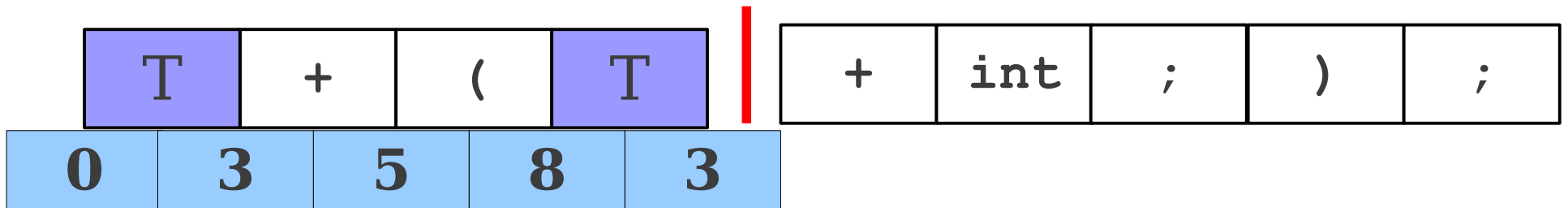
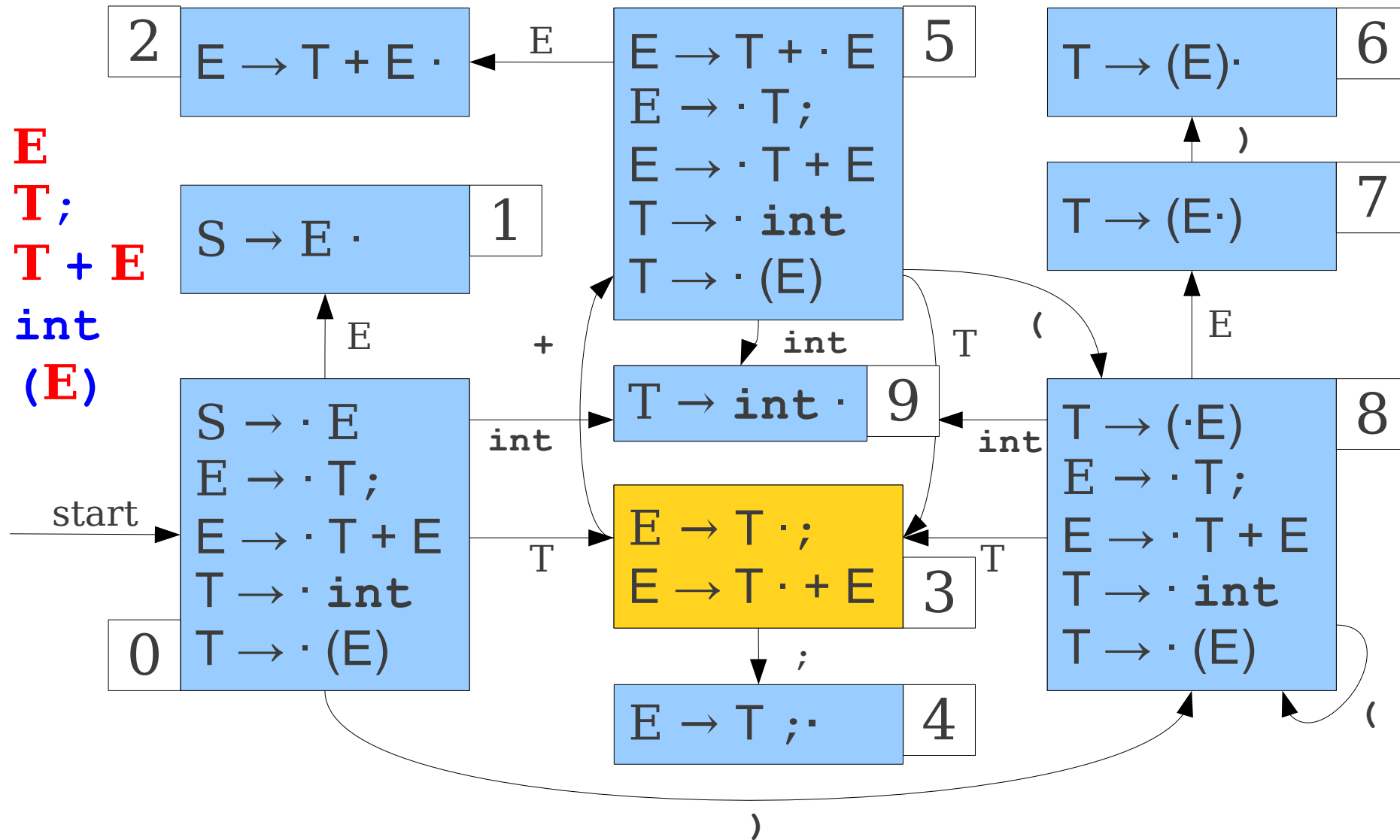
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



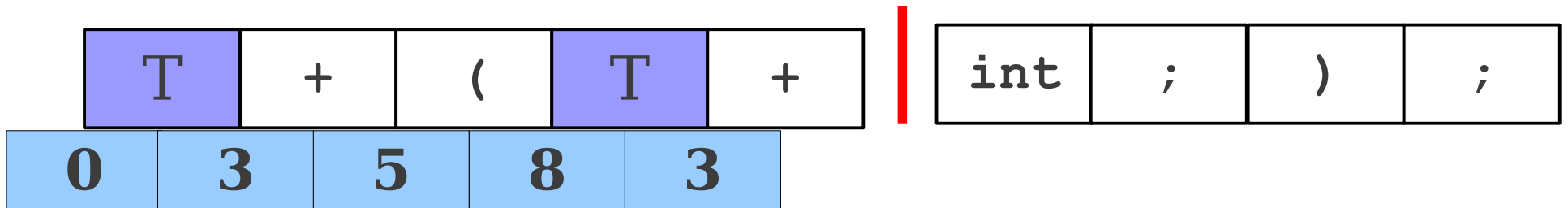
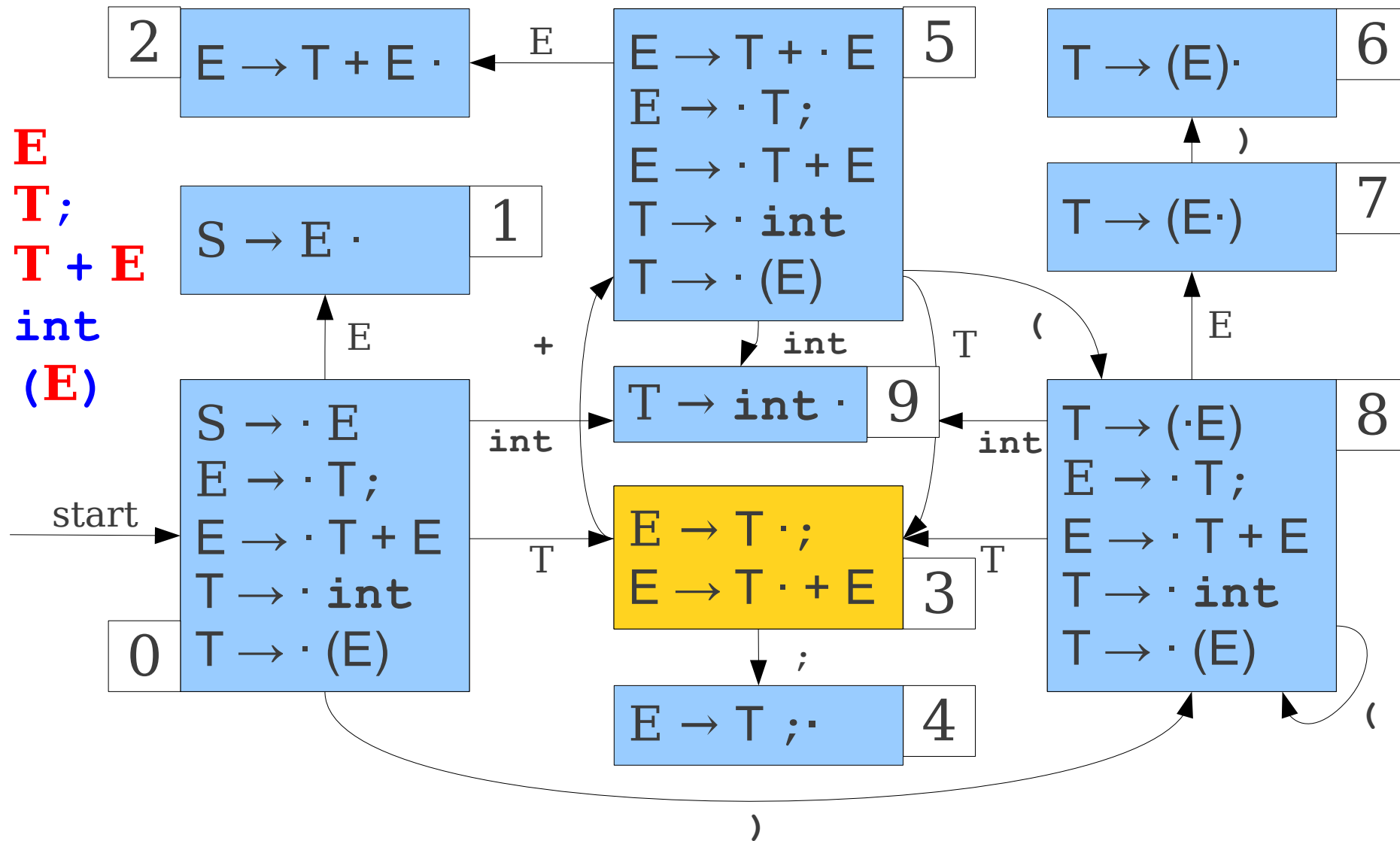
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



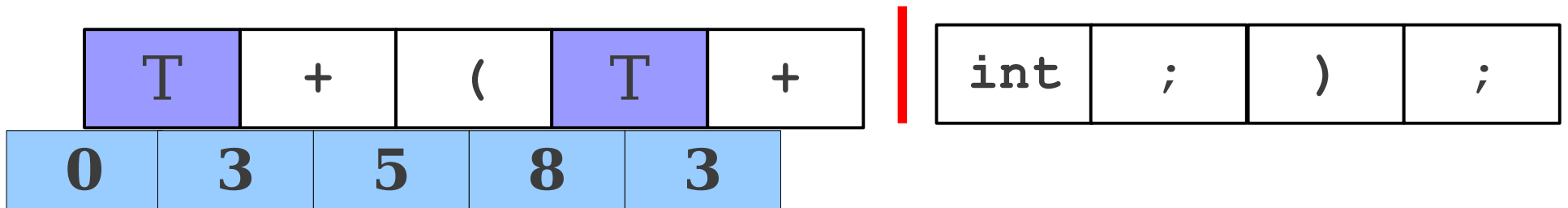
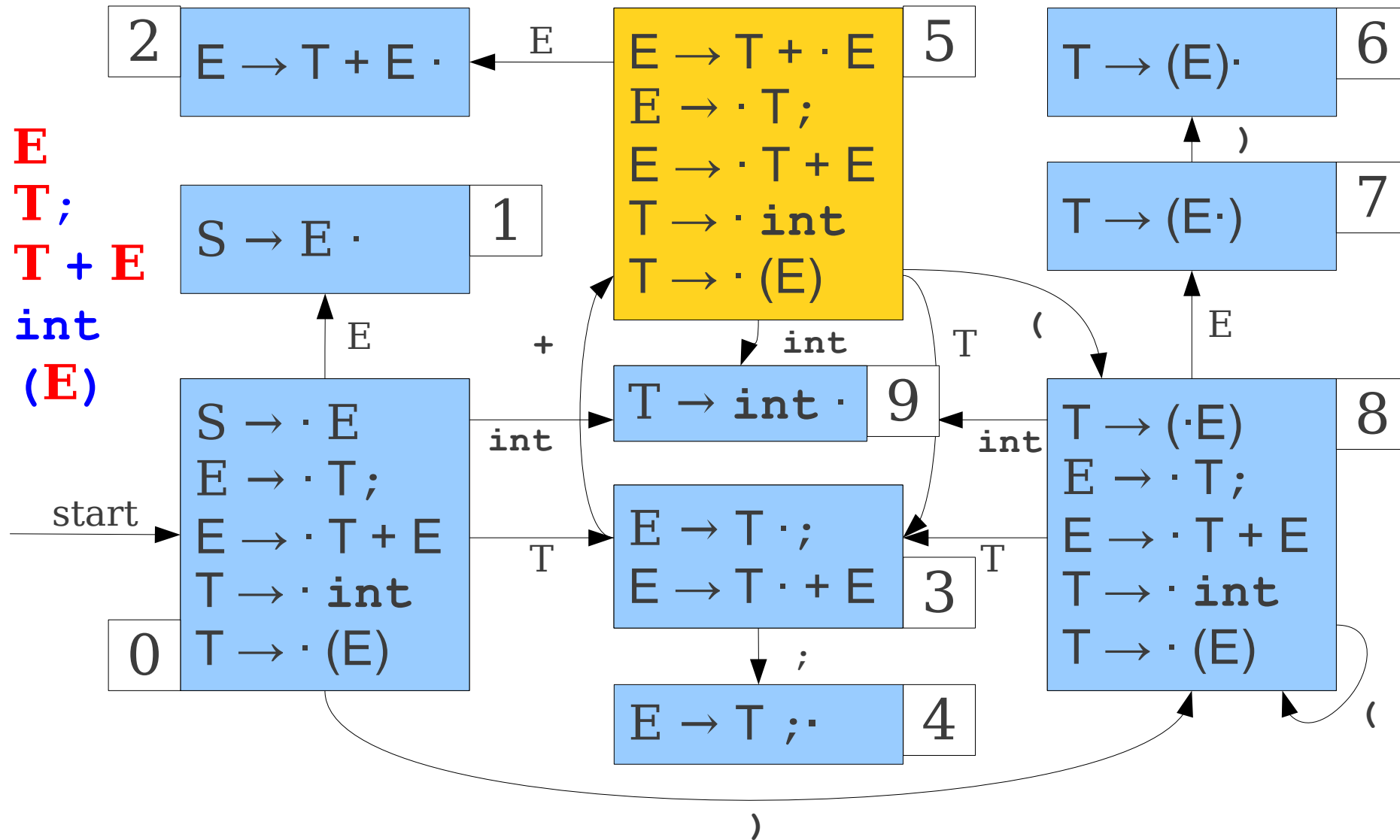
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



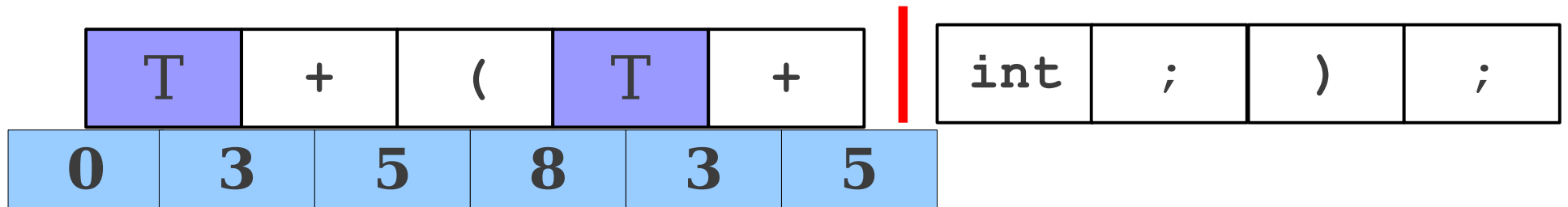
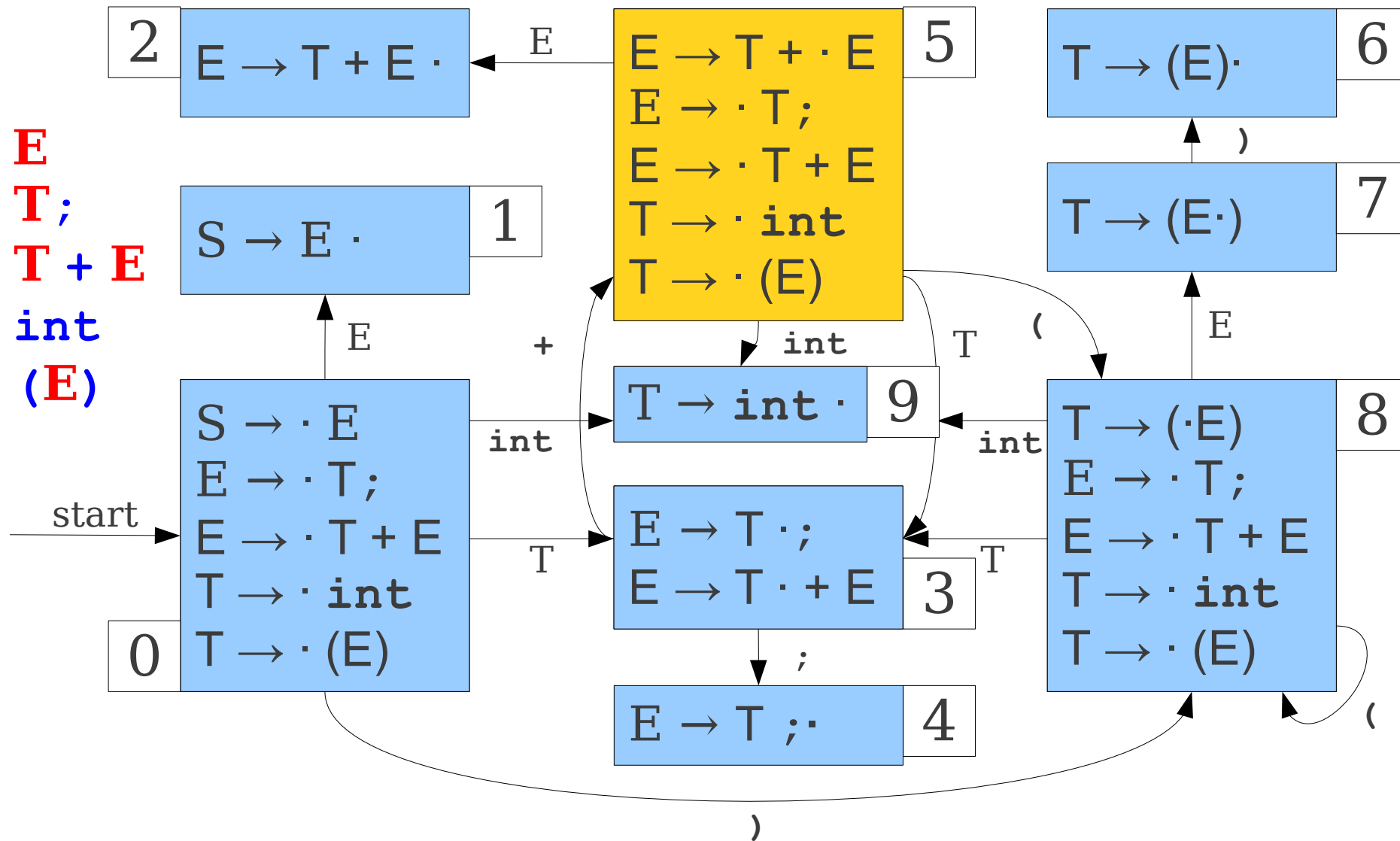
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



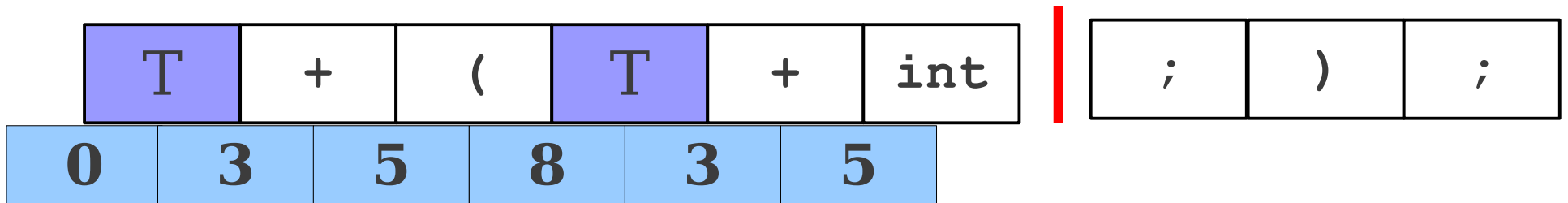
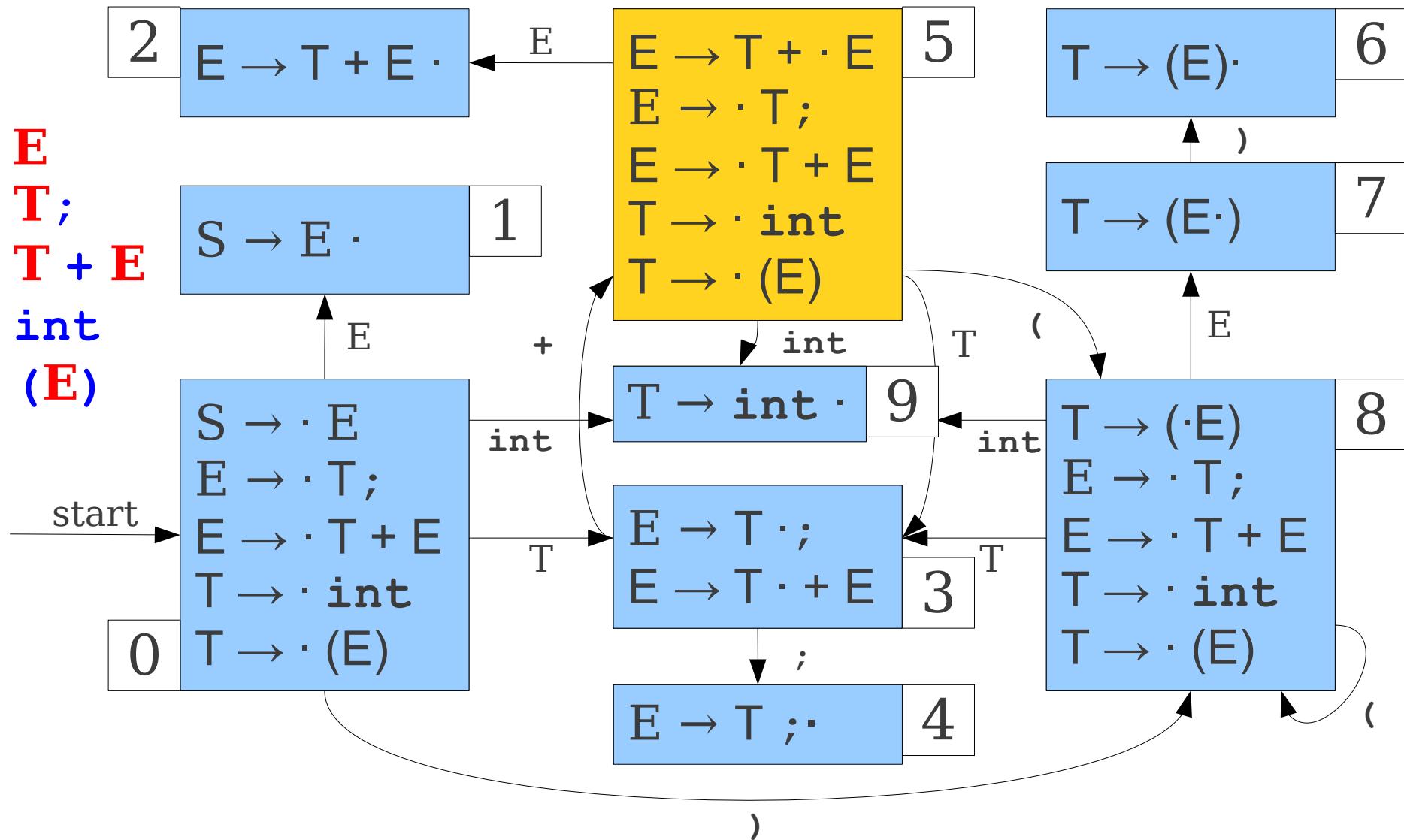
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



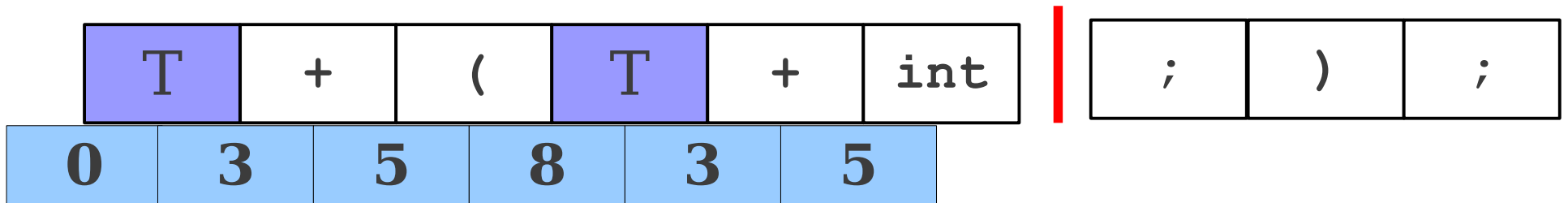
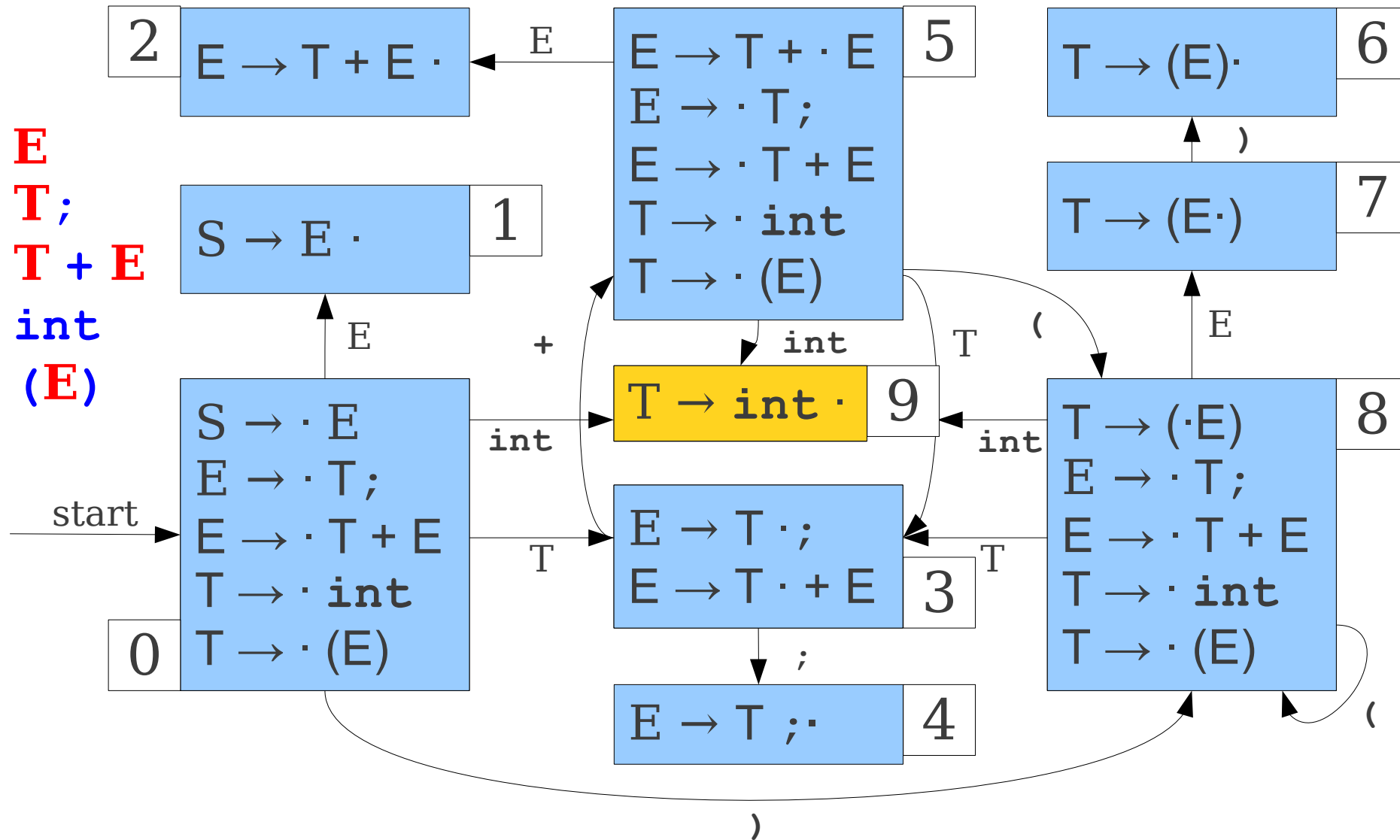
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



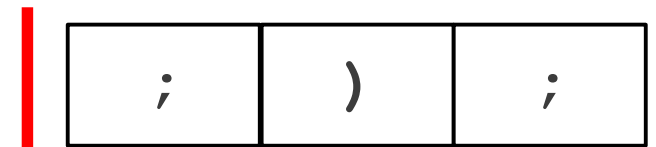
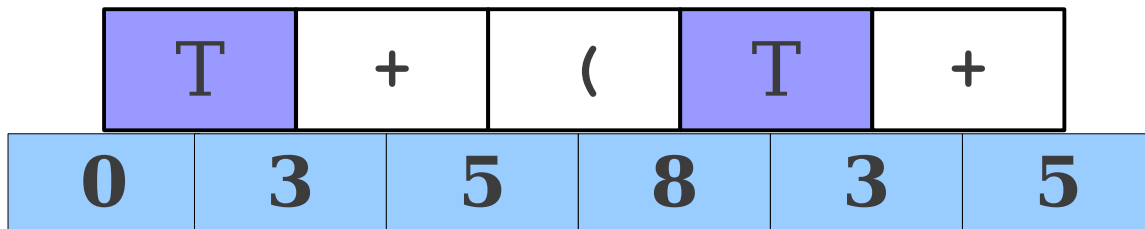
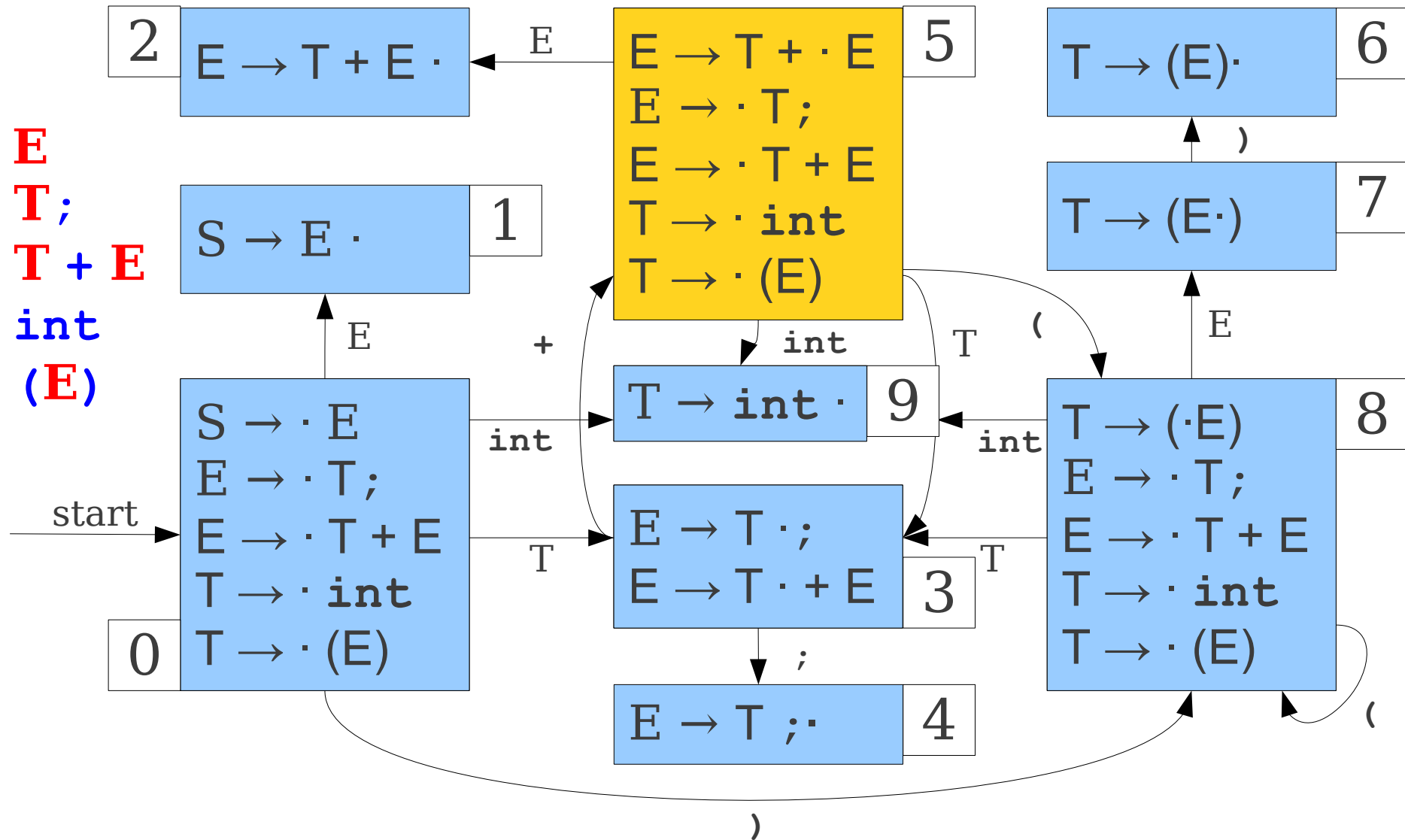
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



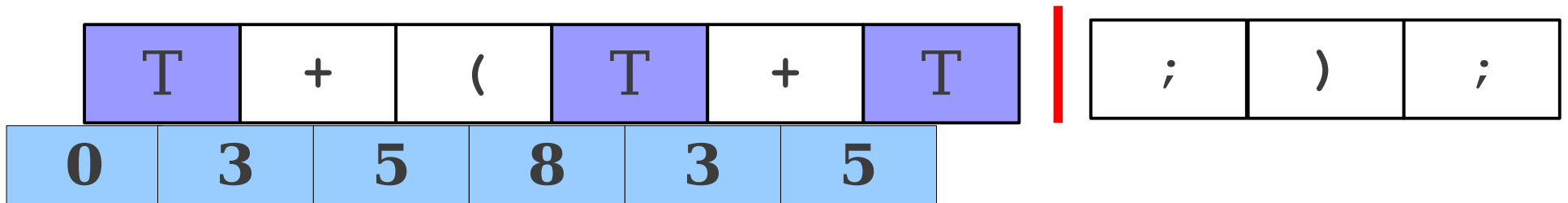
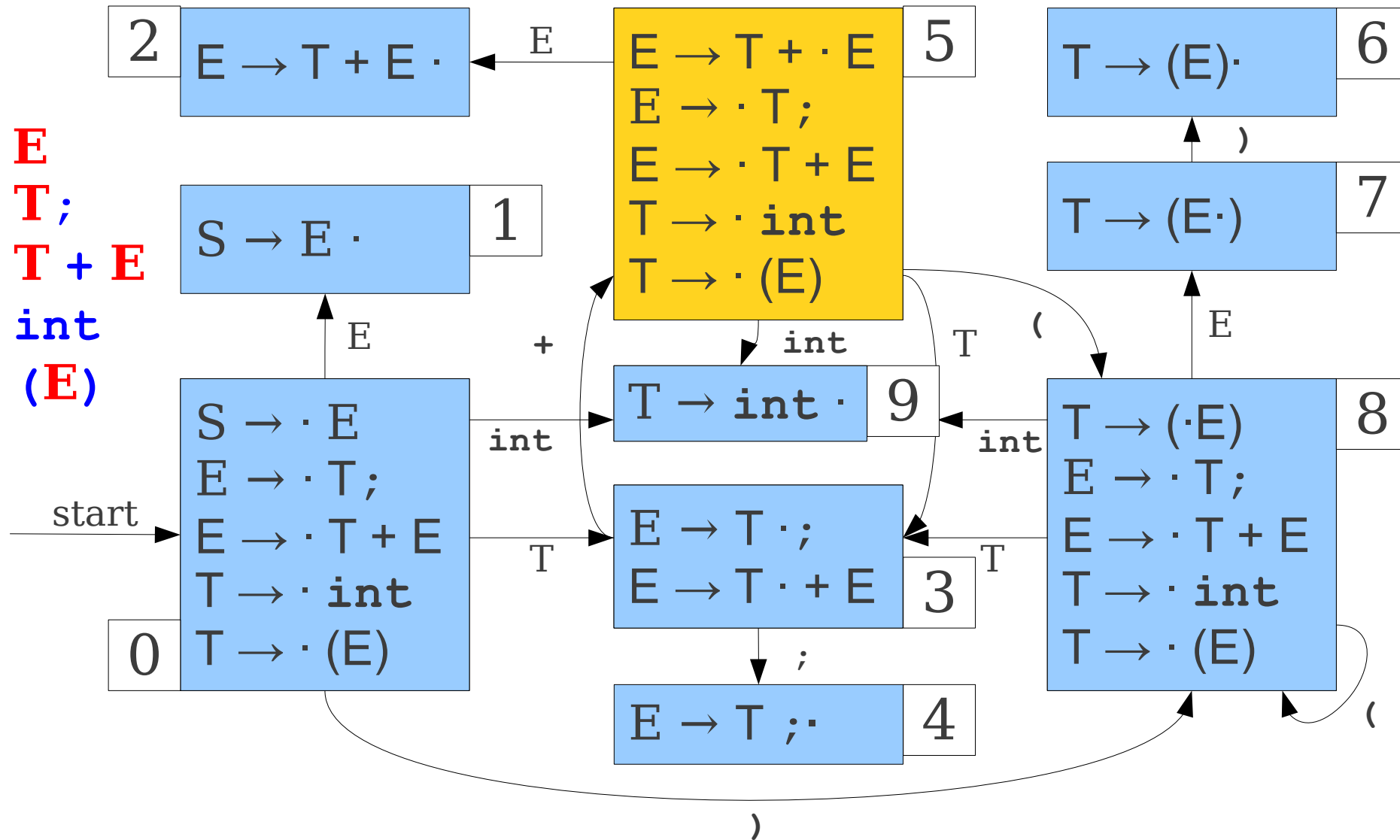
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



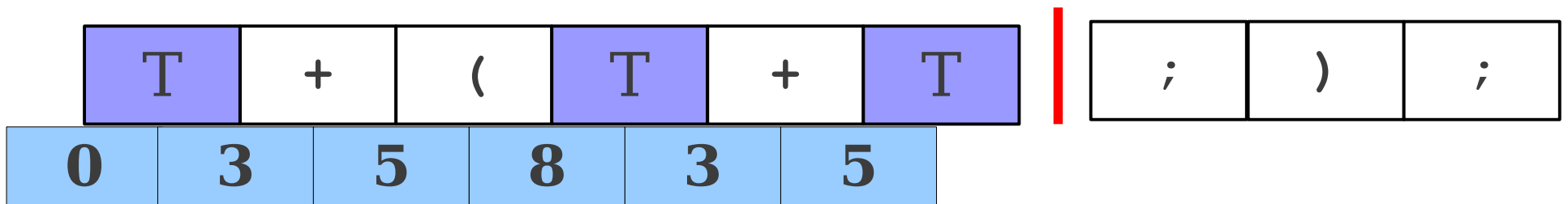
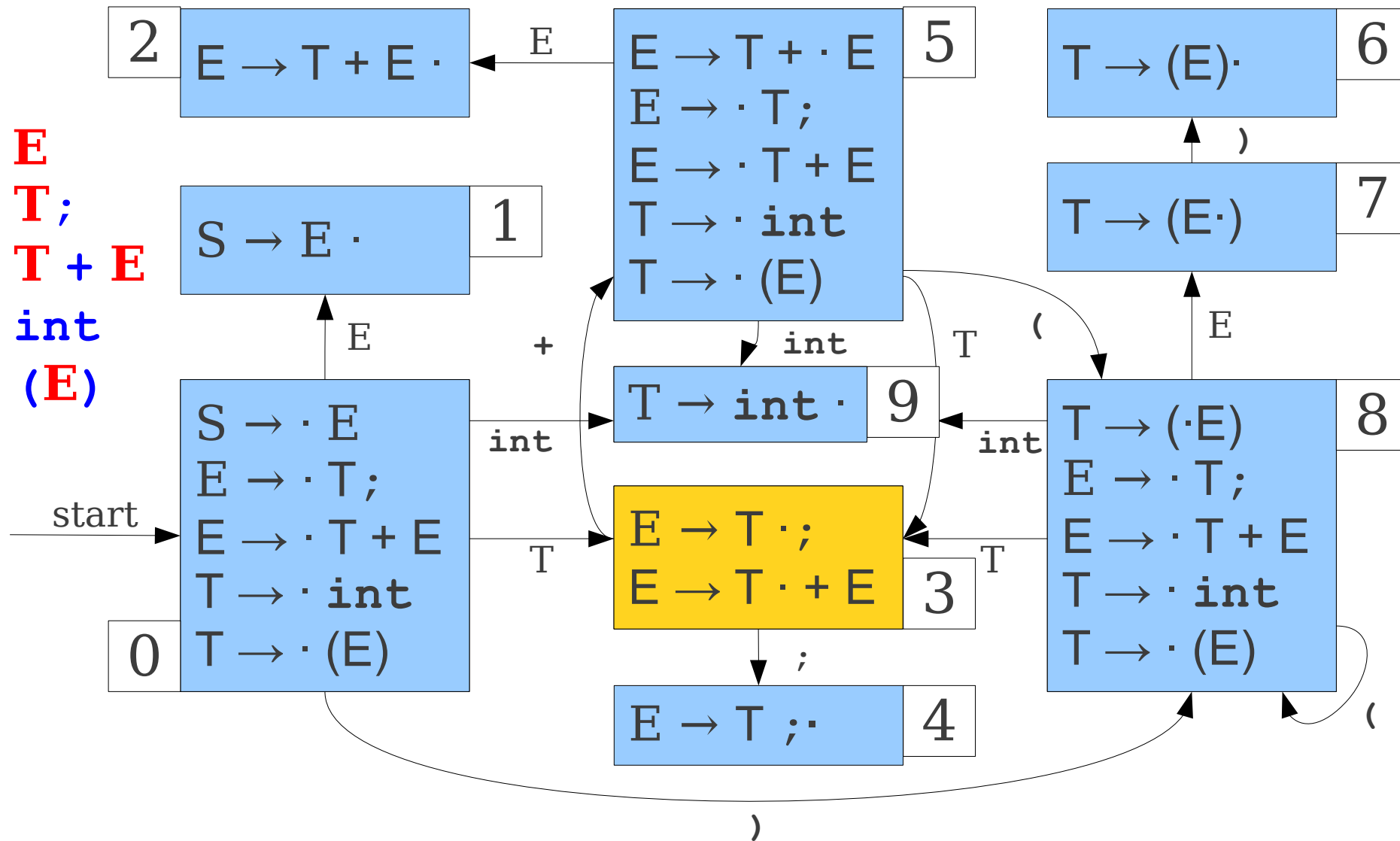
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



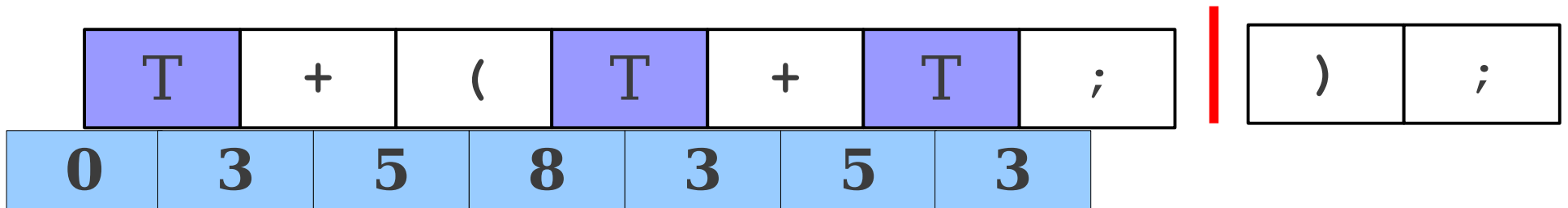
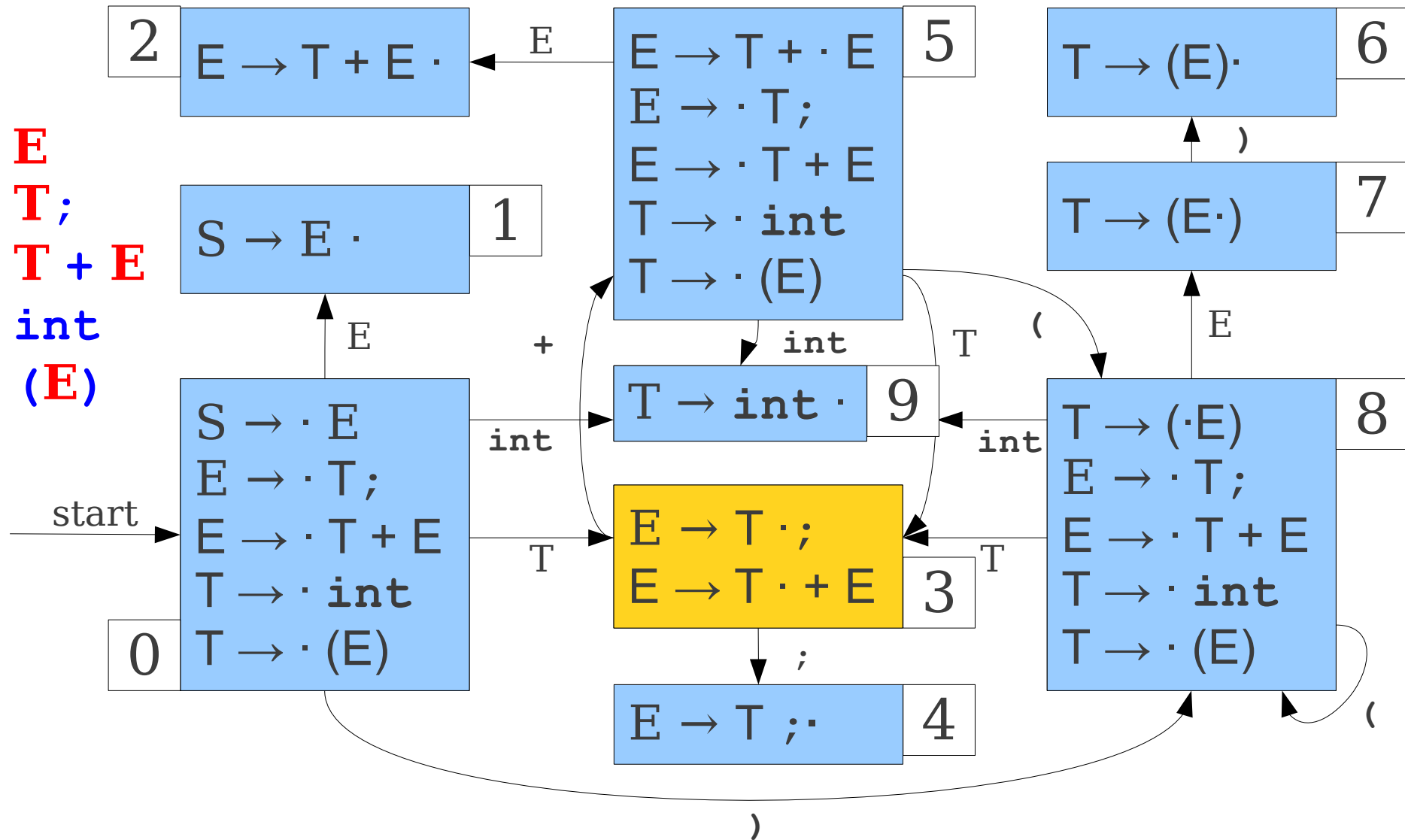
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



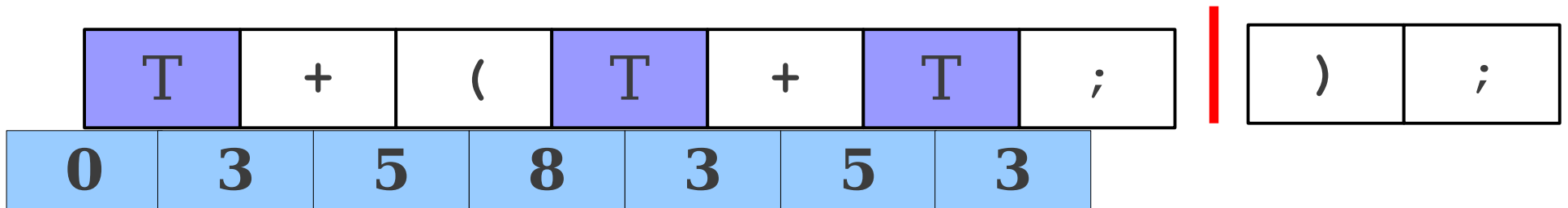
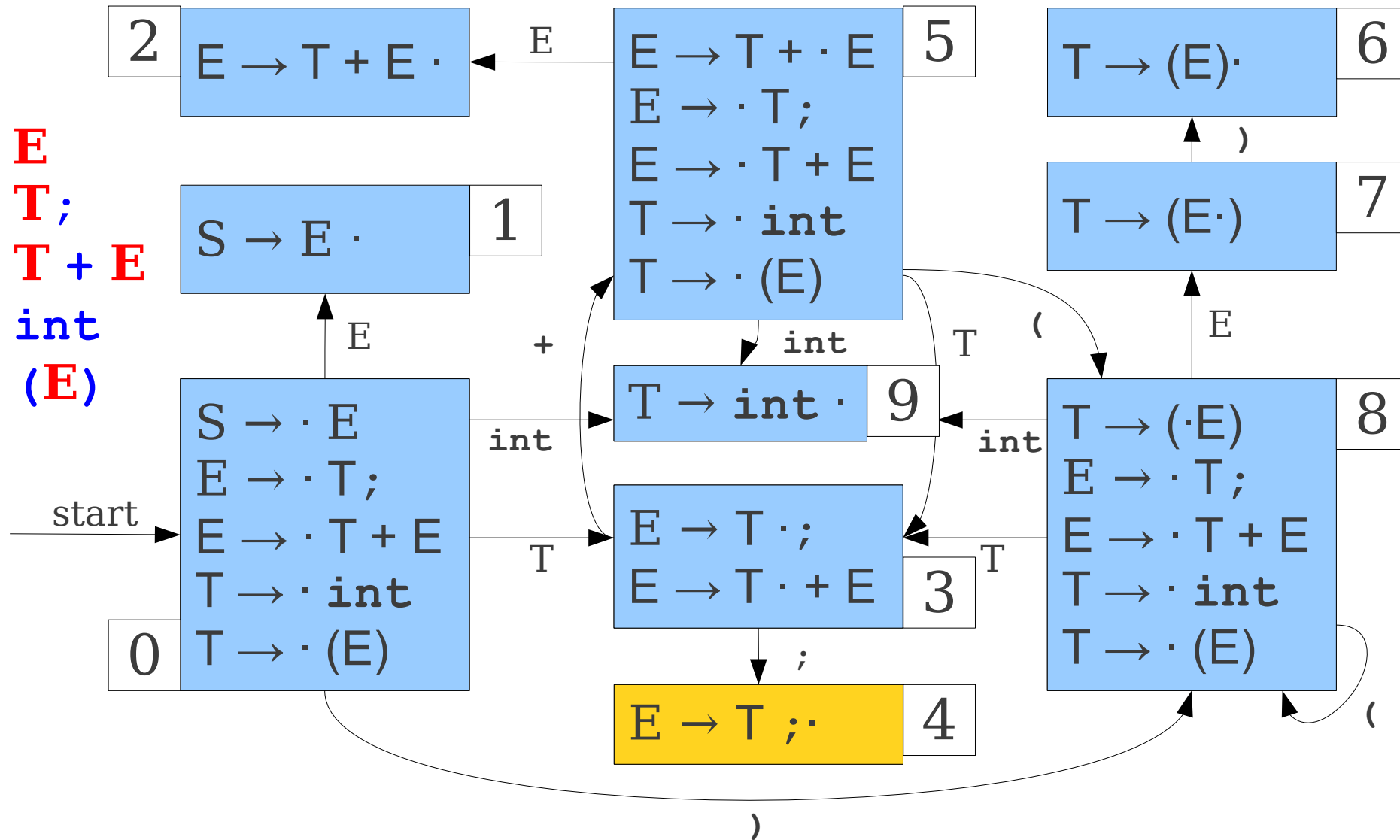
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



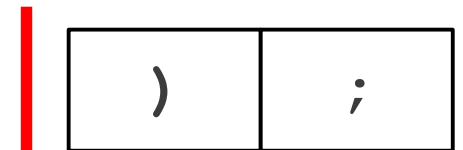
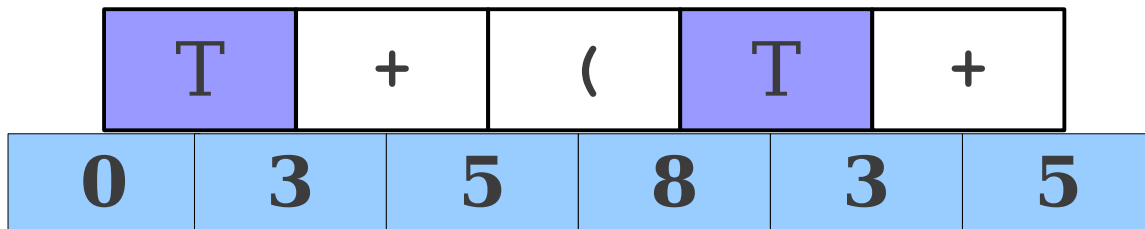
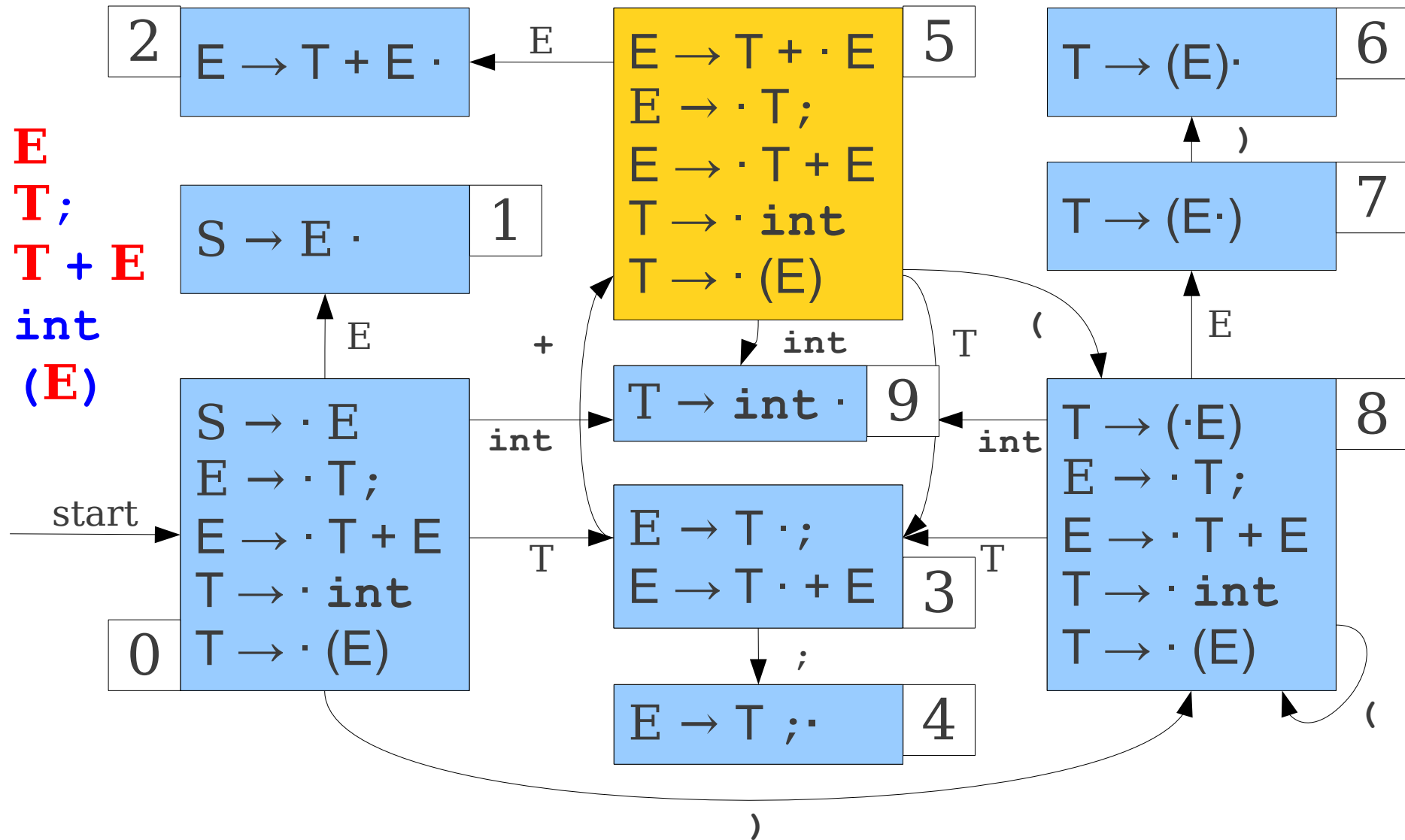
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



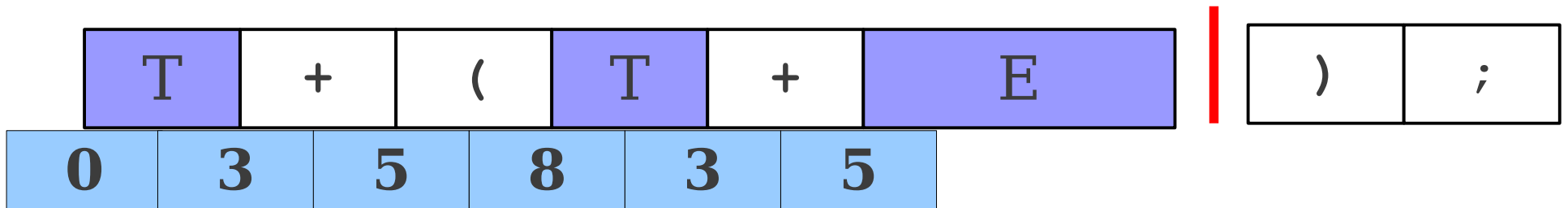
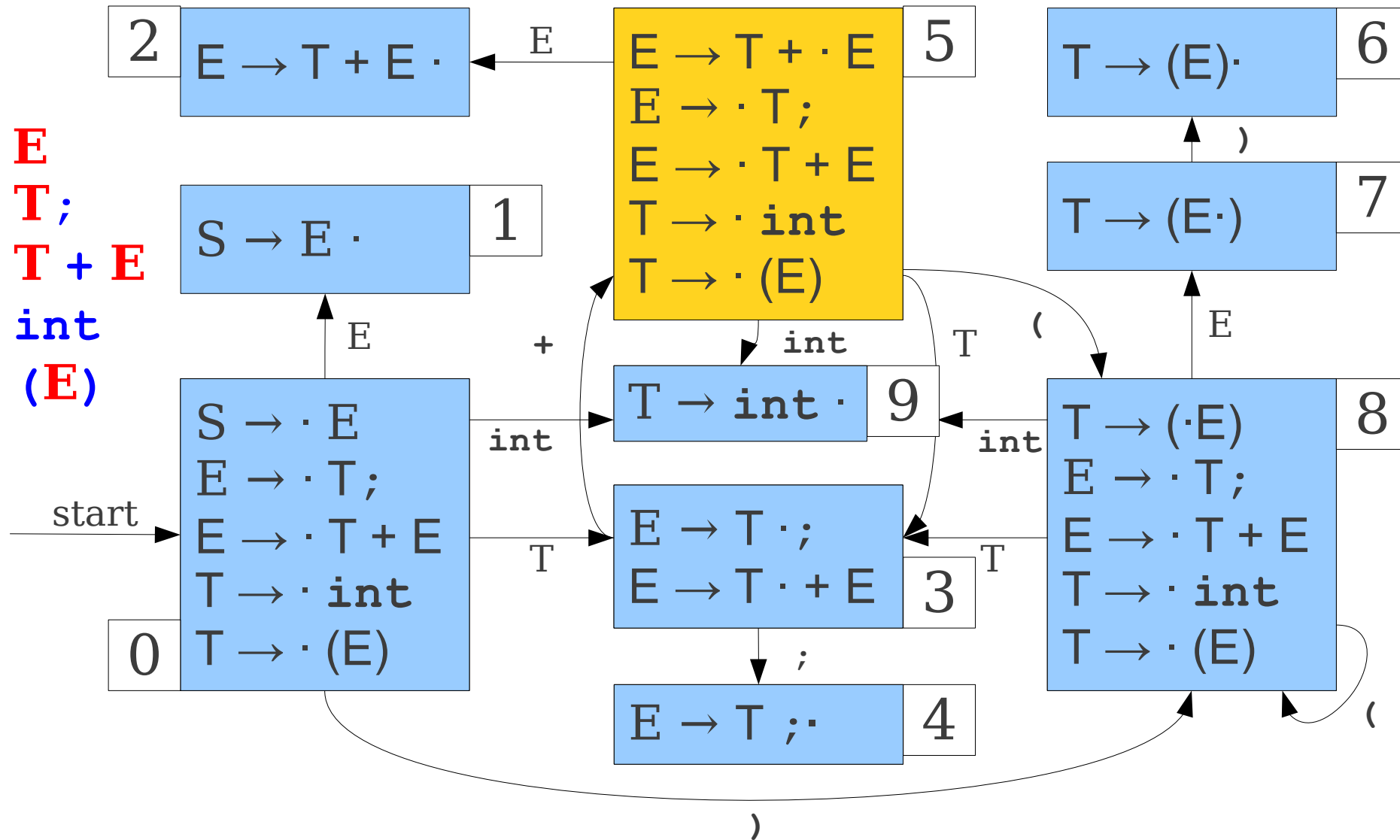
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



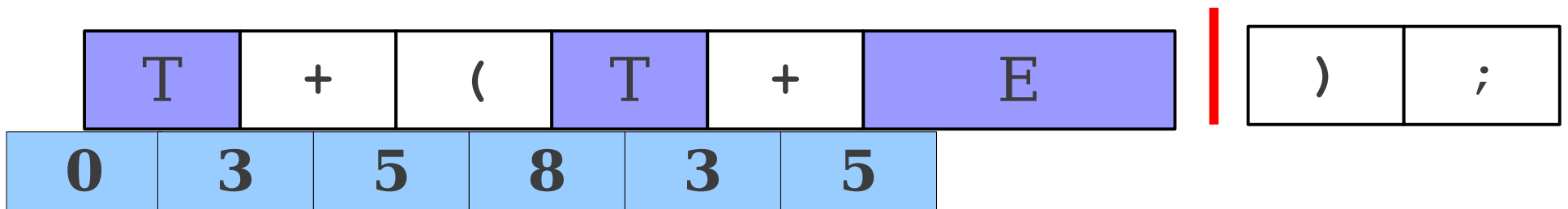
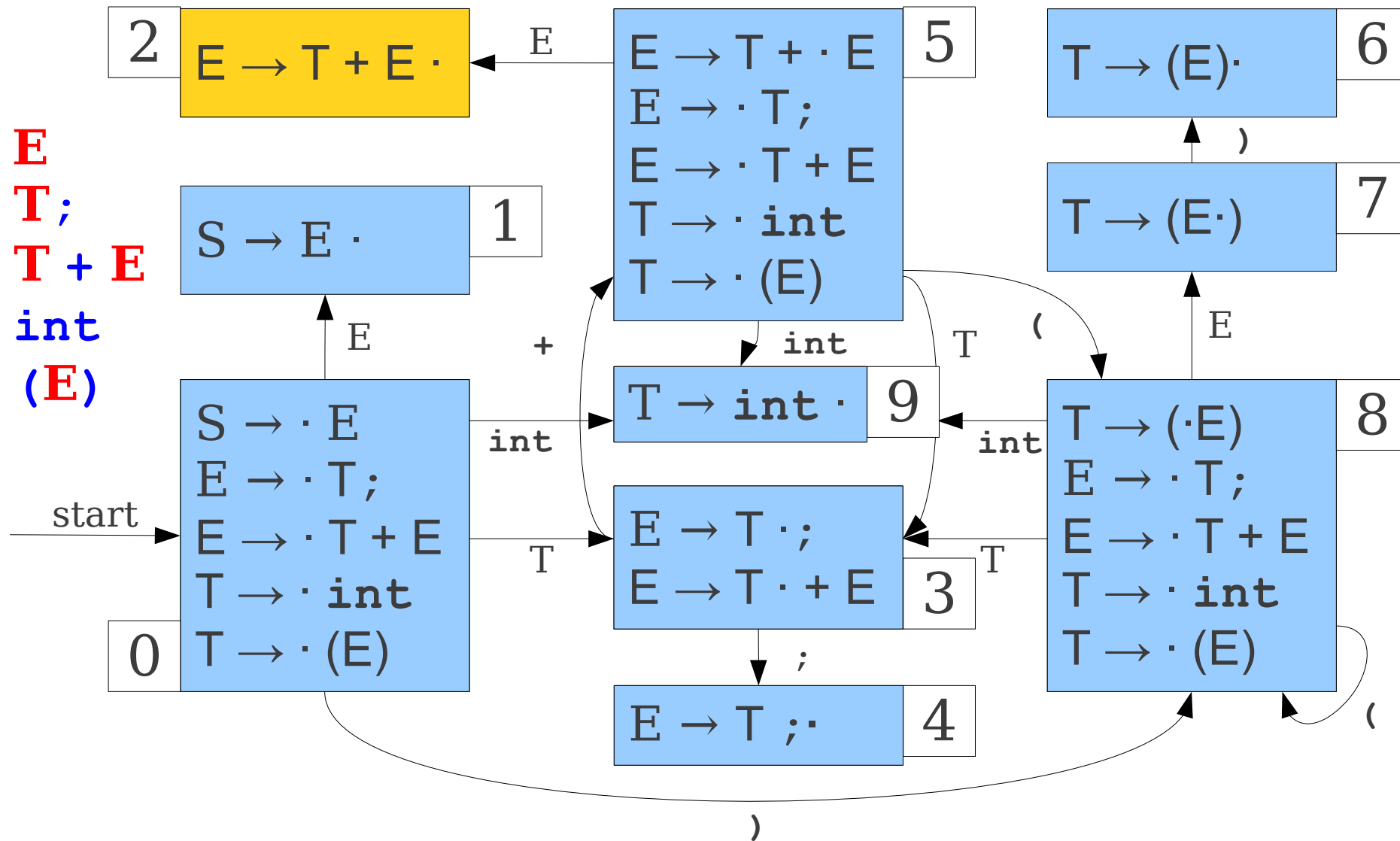
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
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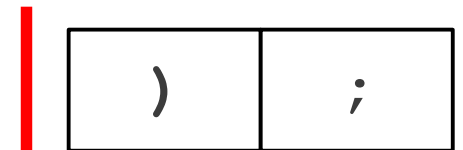
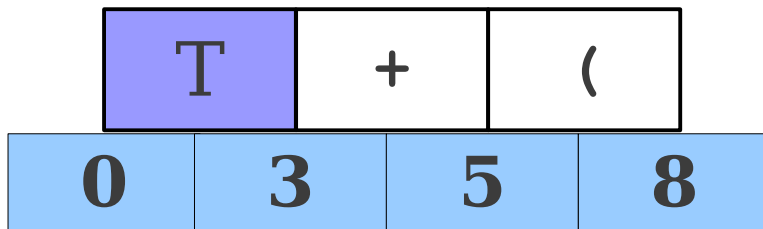
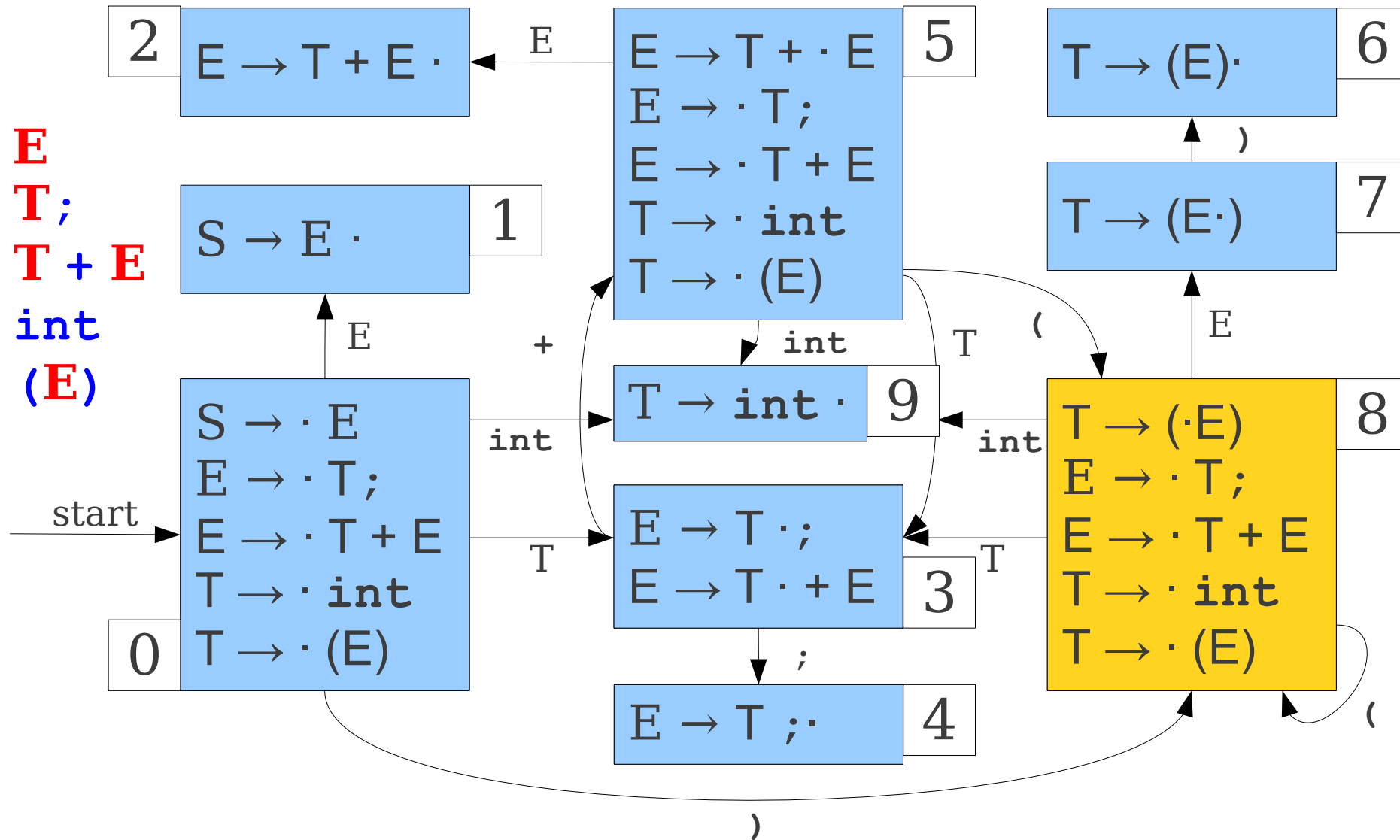
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



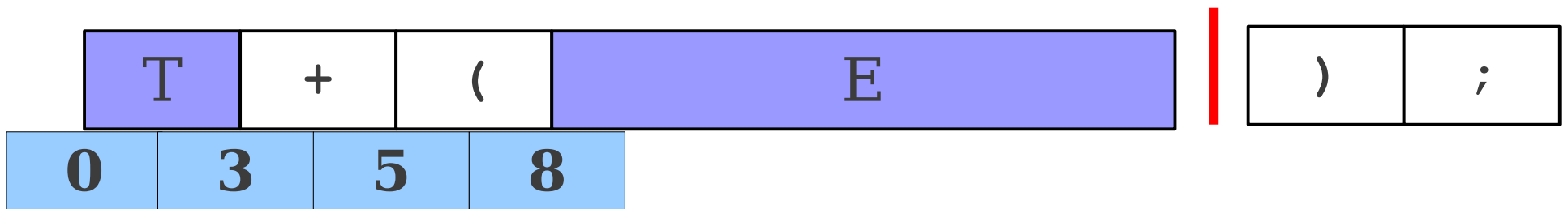
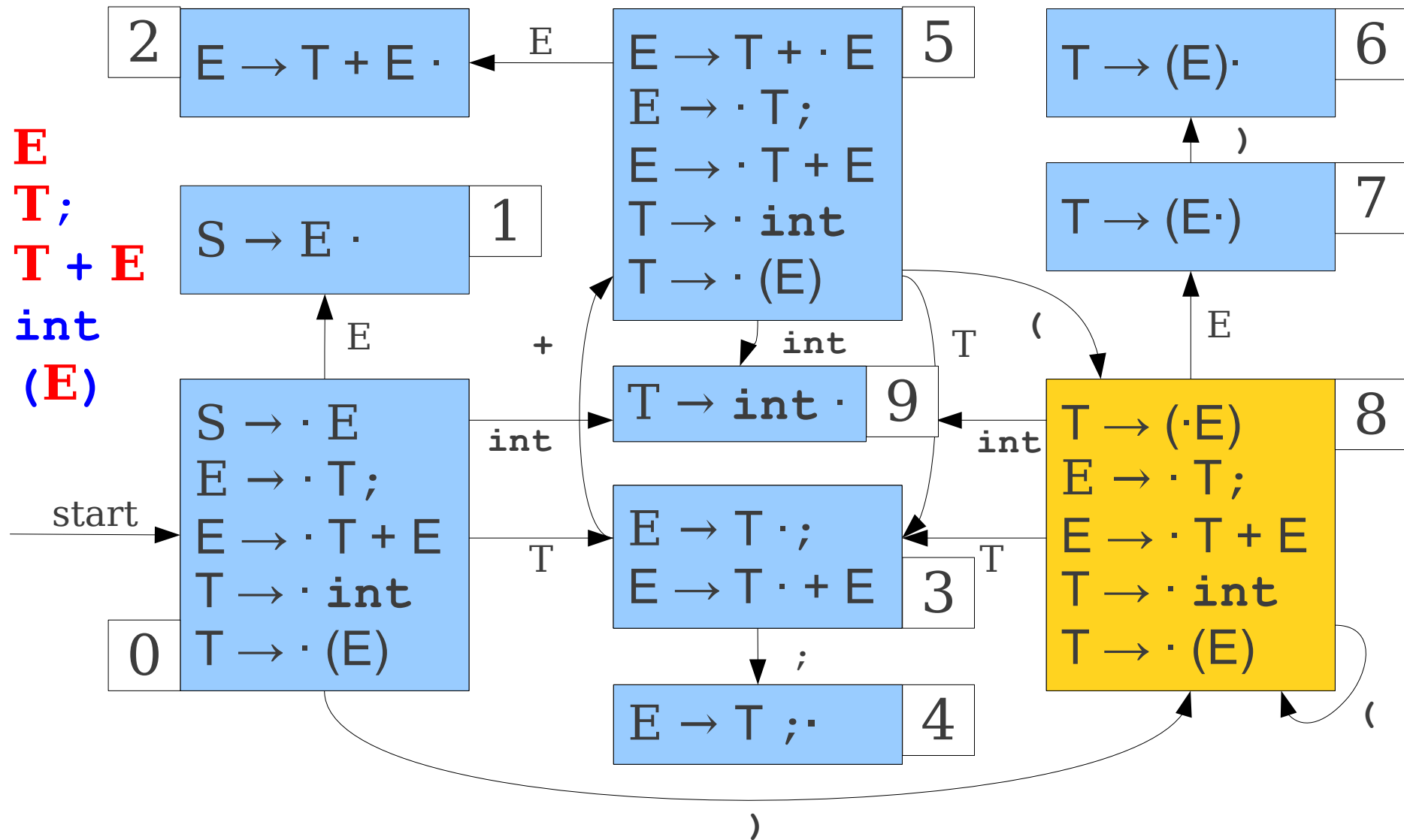
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



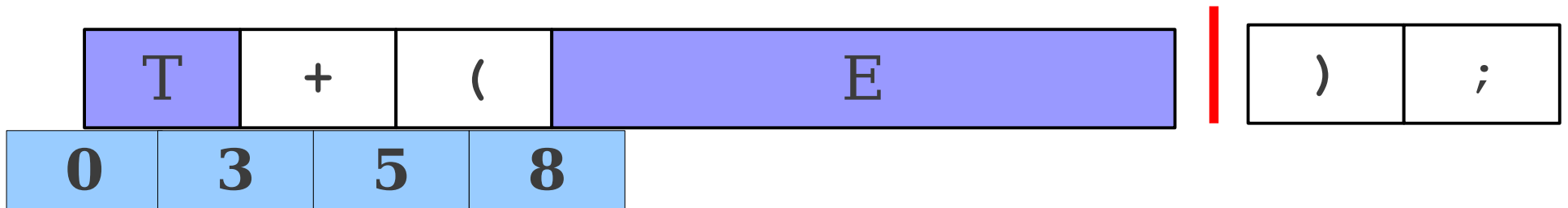
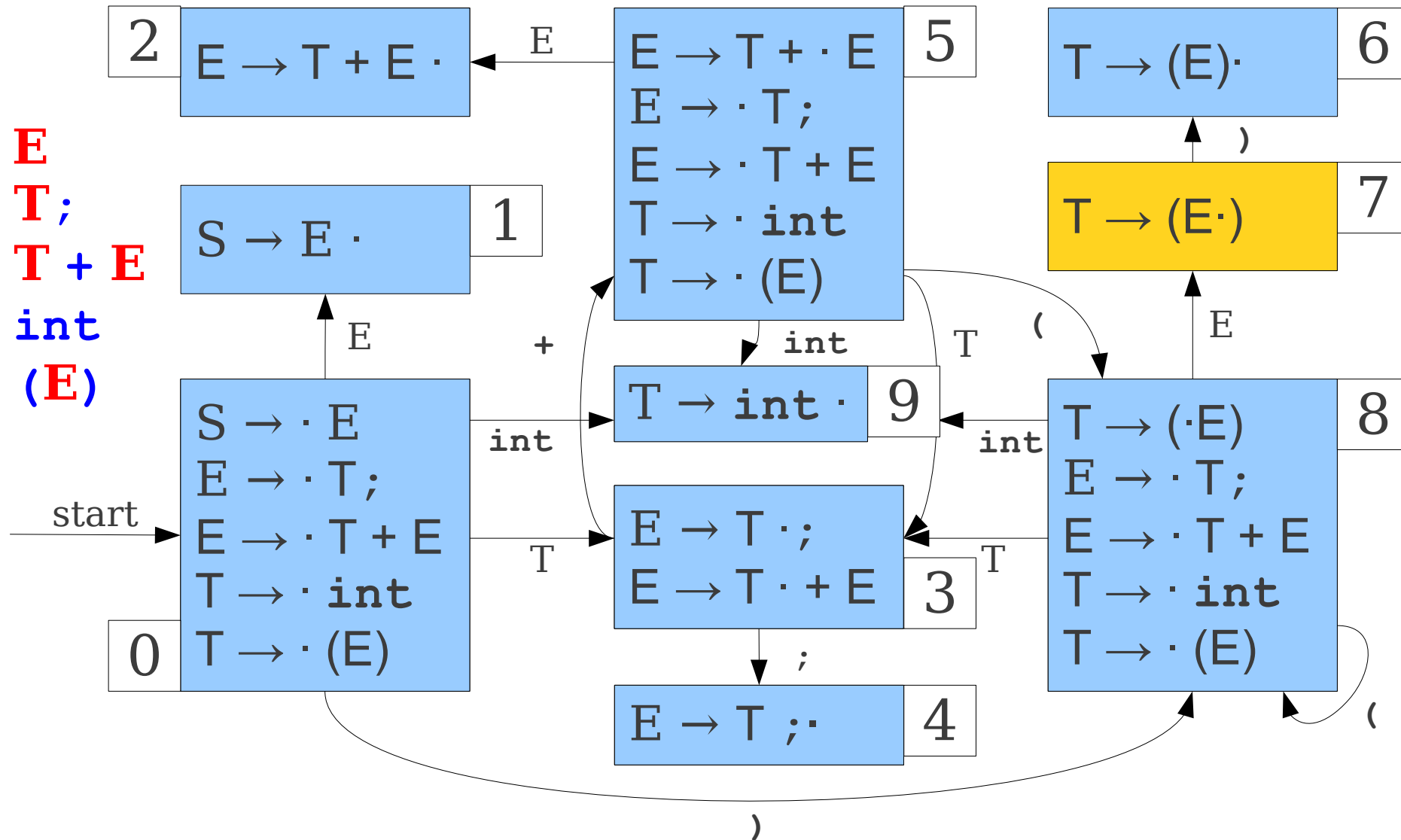
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



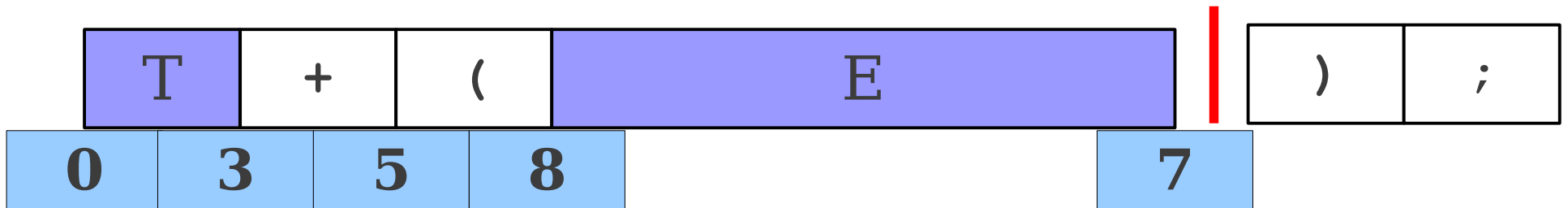
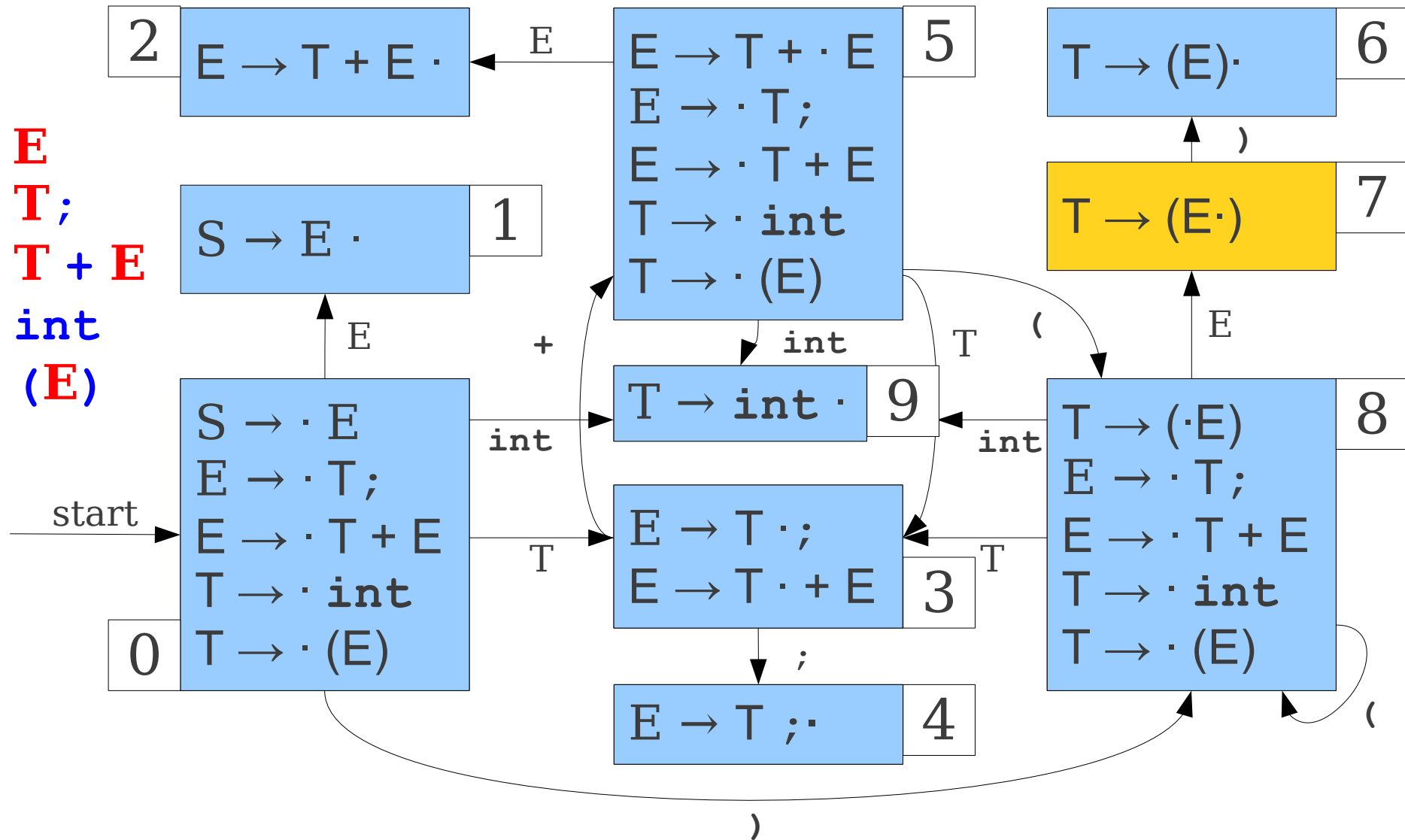
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



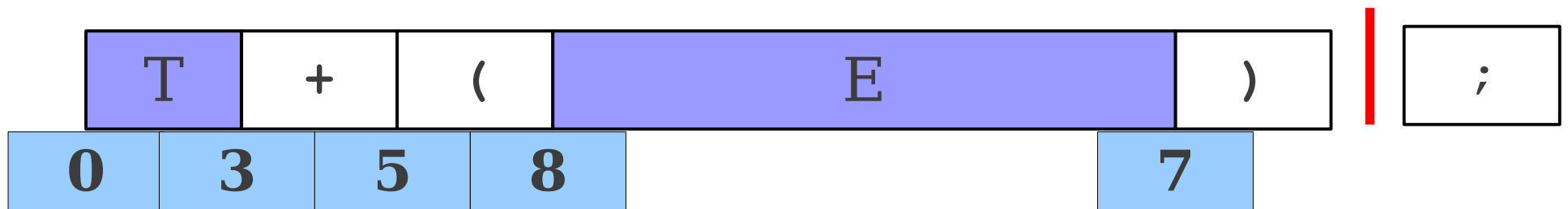
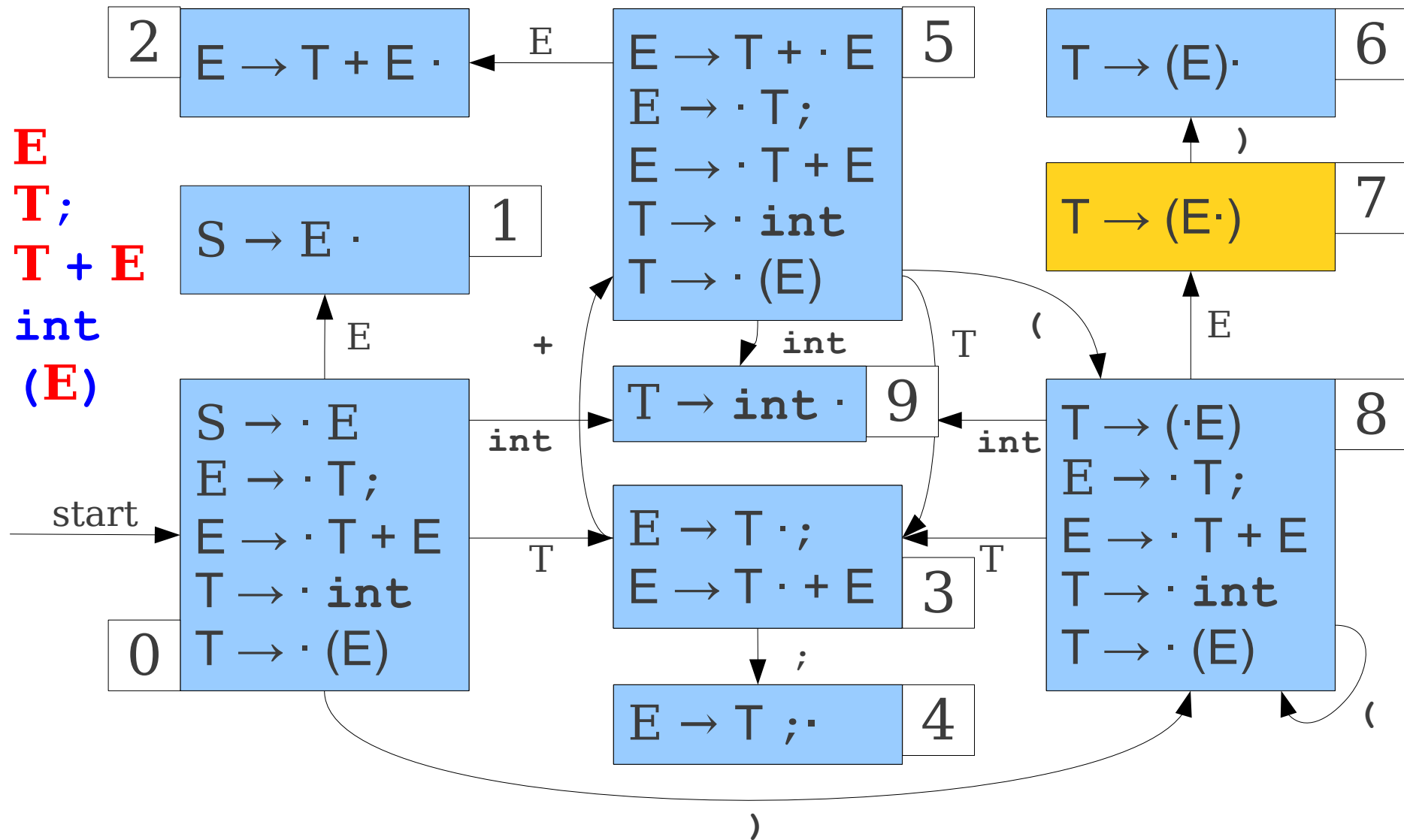
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



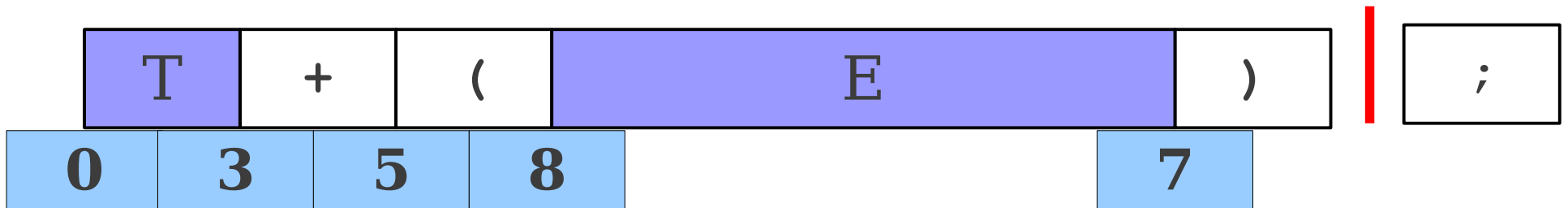
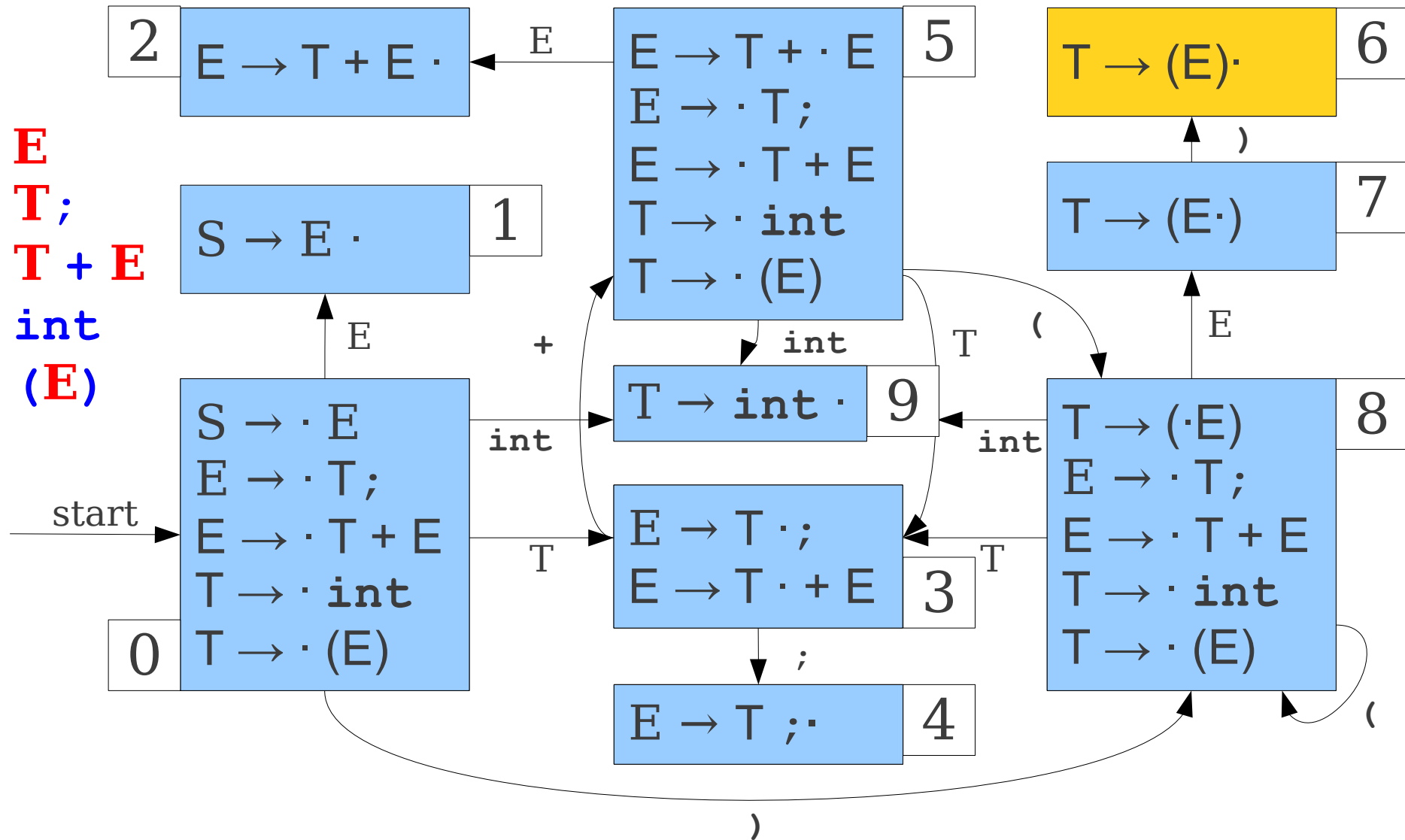
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



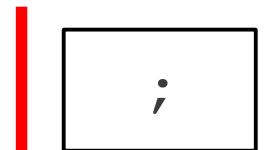
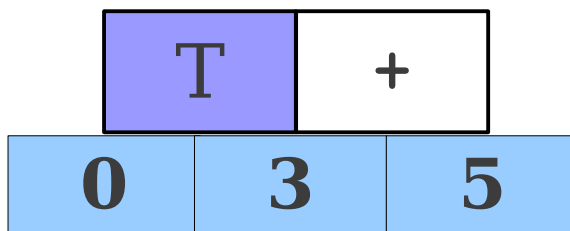
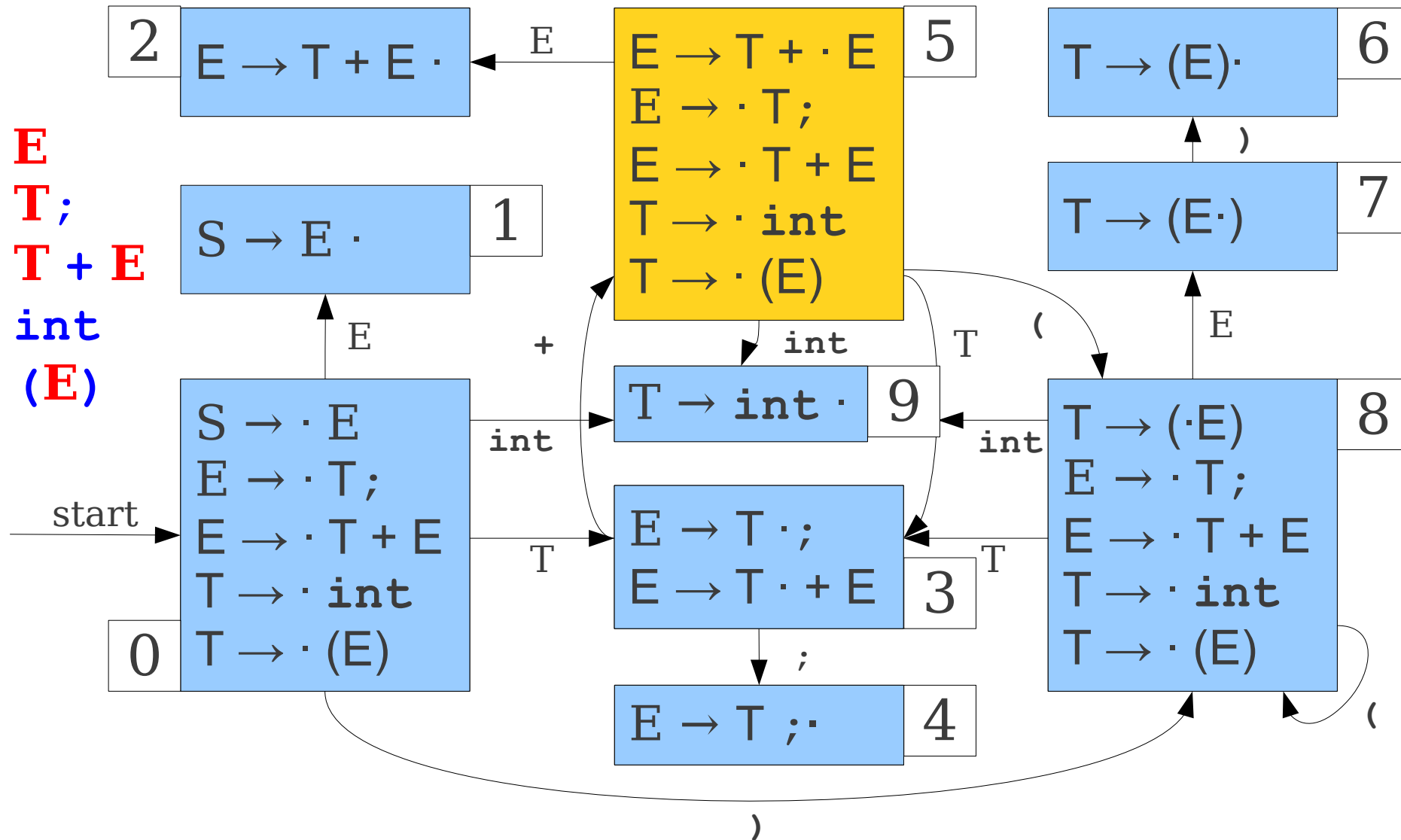
LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



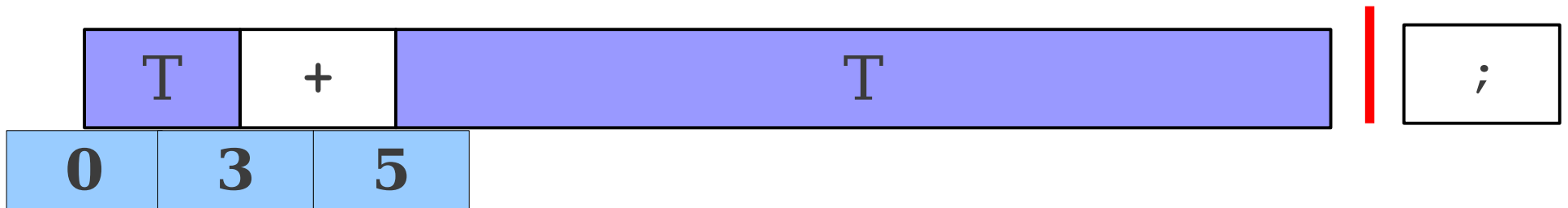
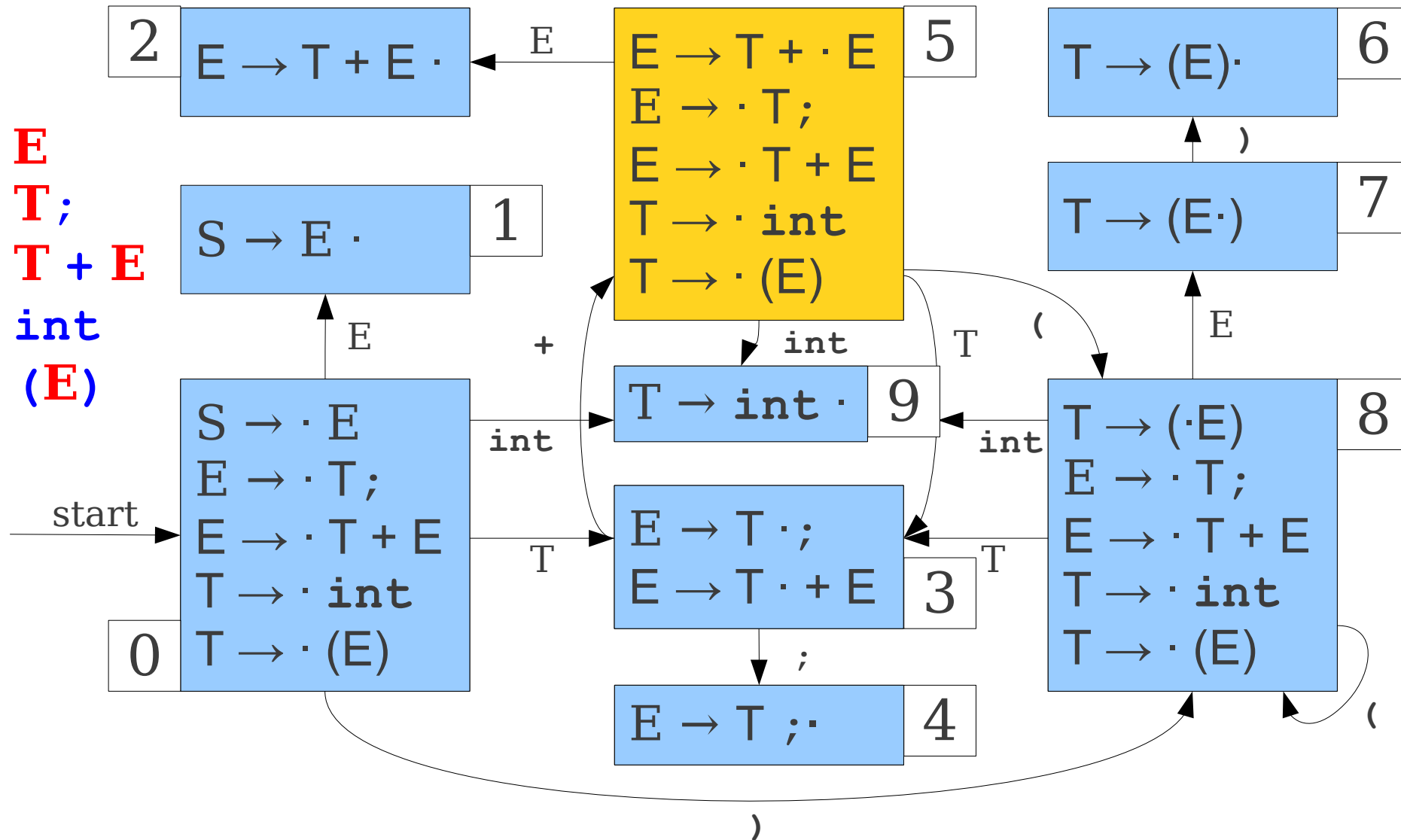
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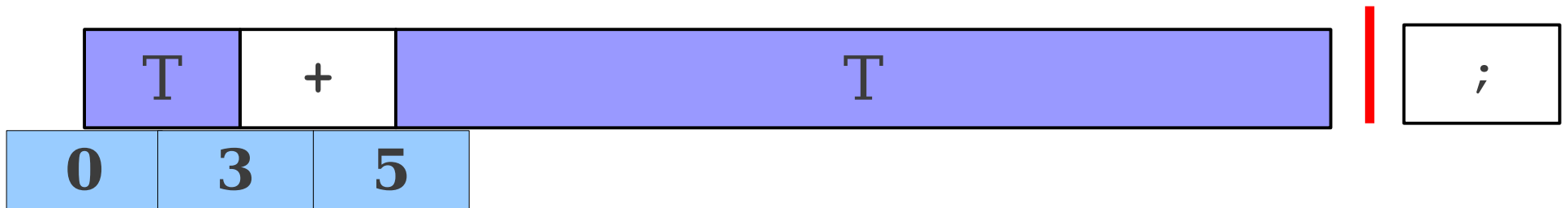
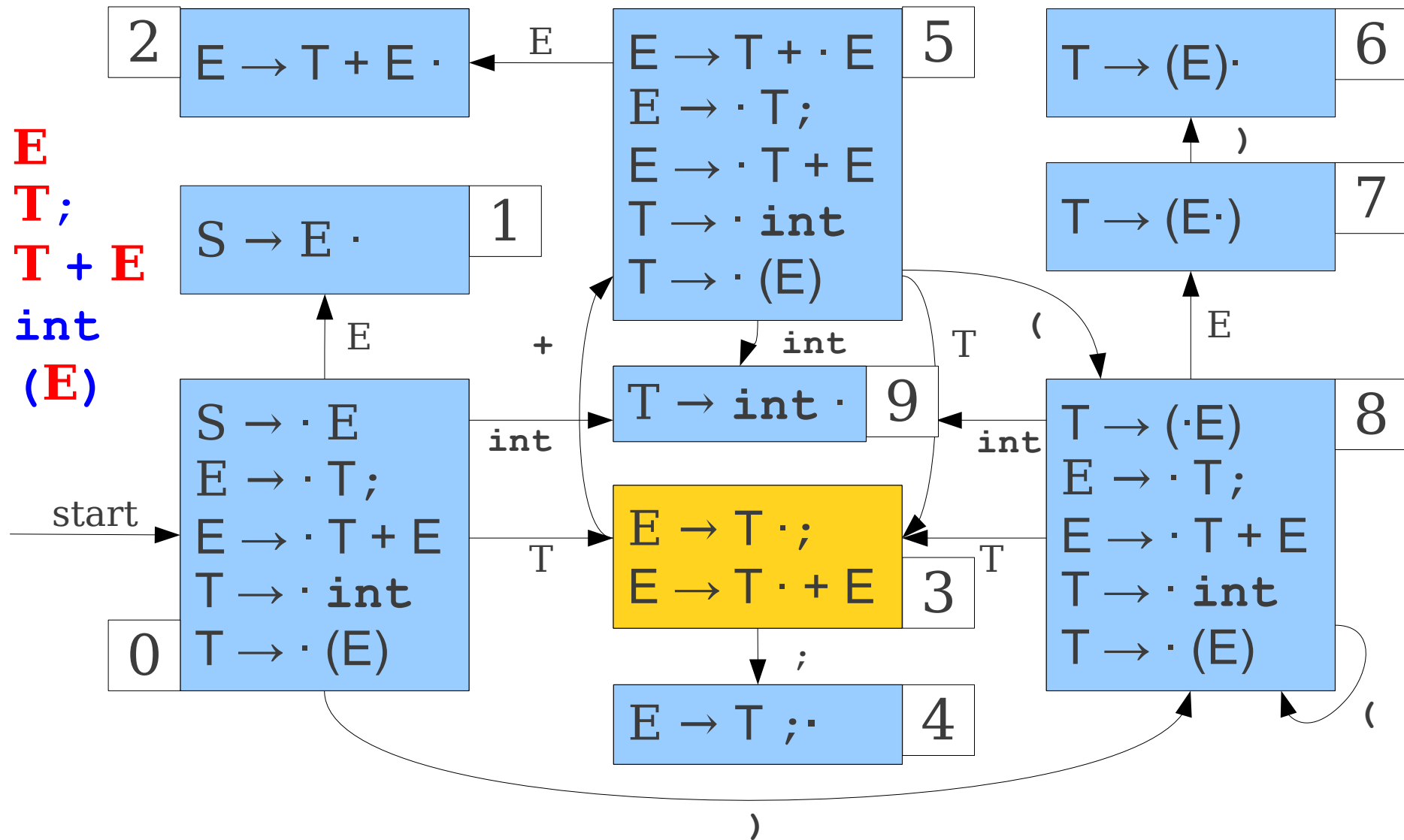
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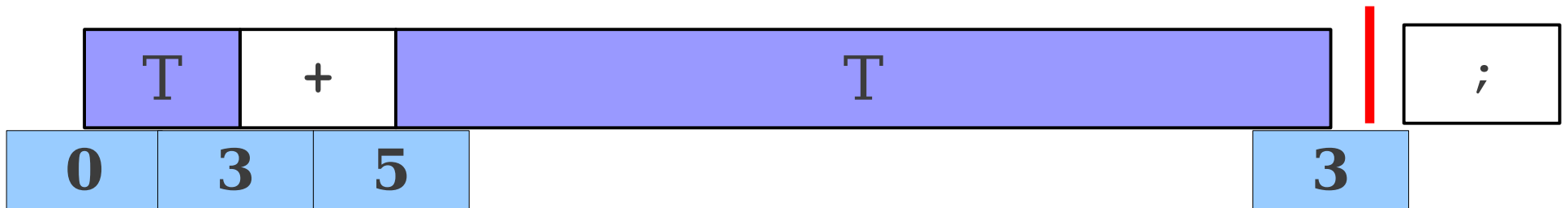
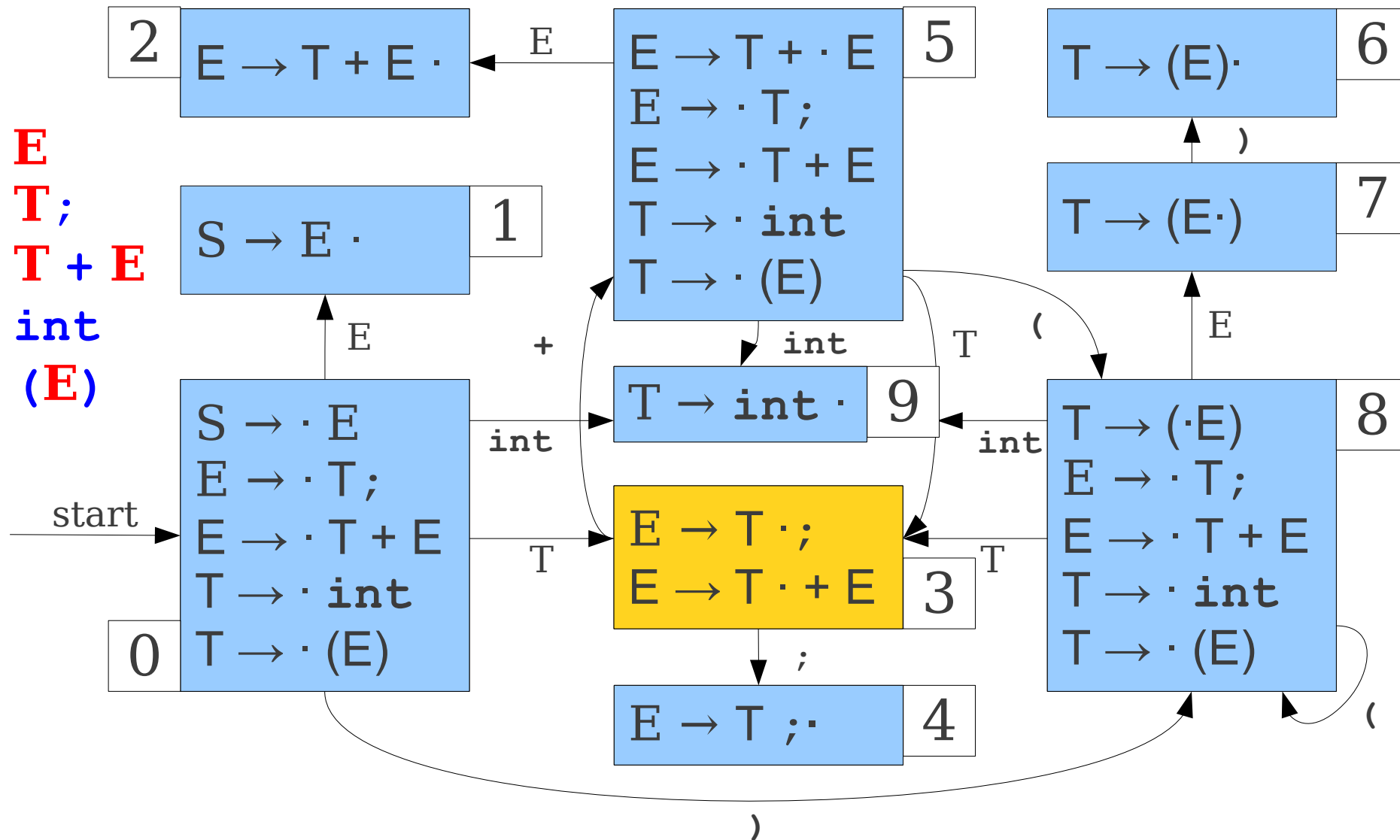
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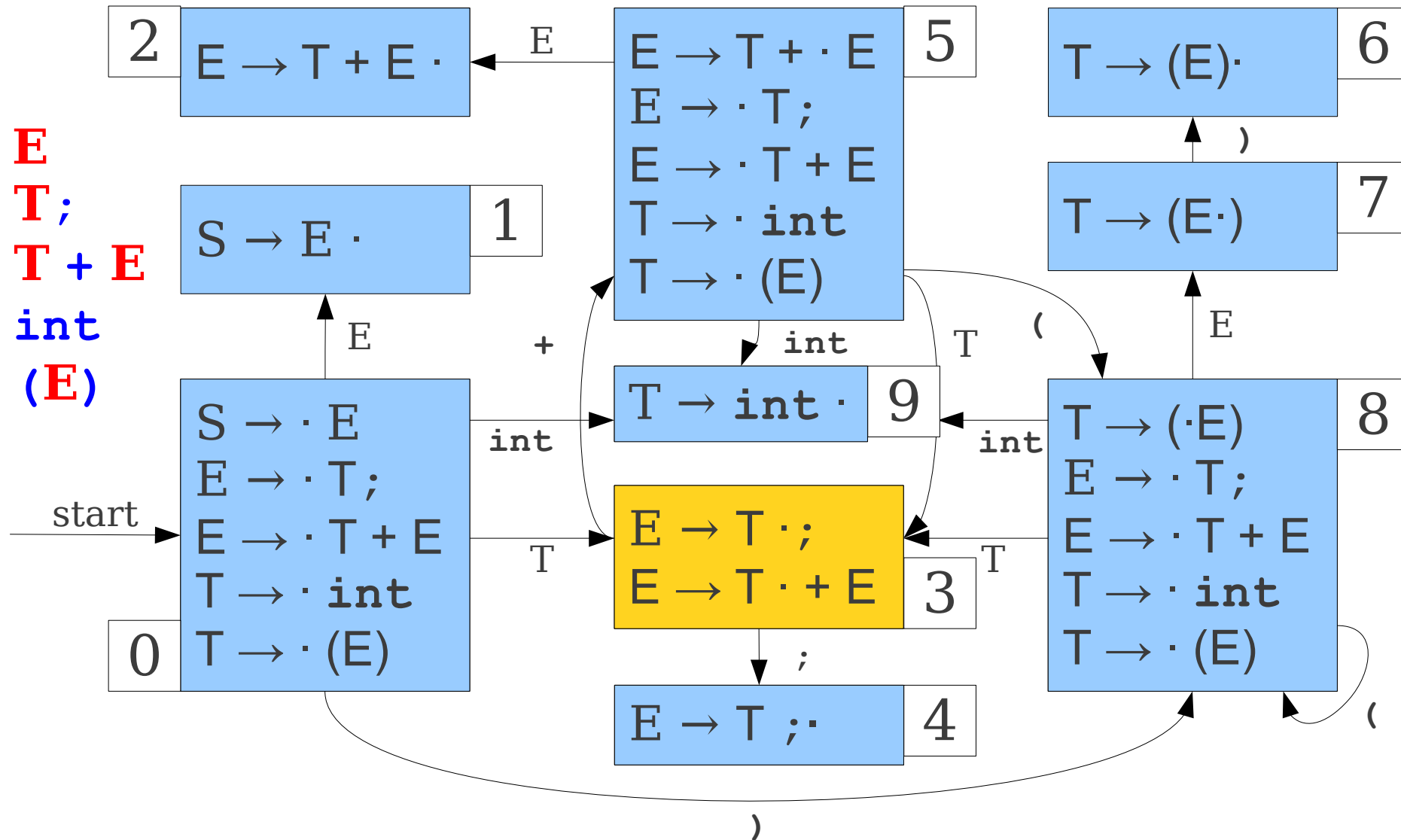
LR(0) Parsing

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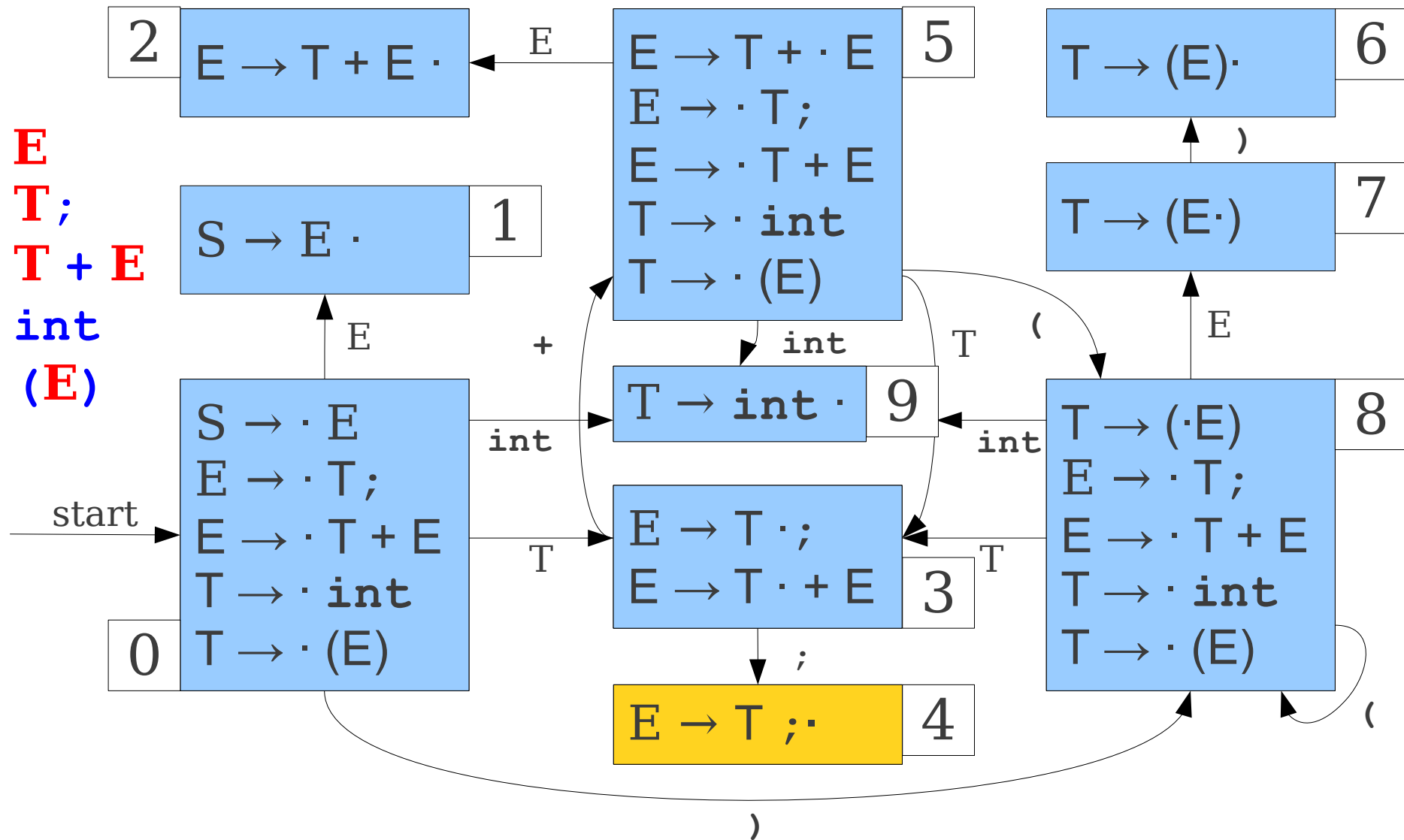
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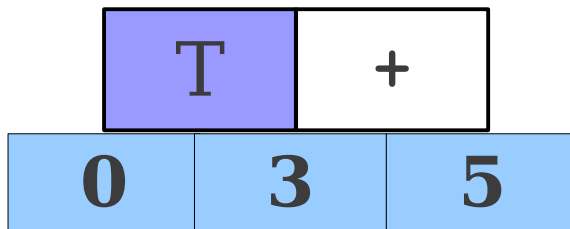
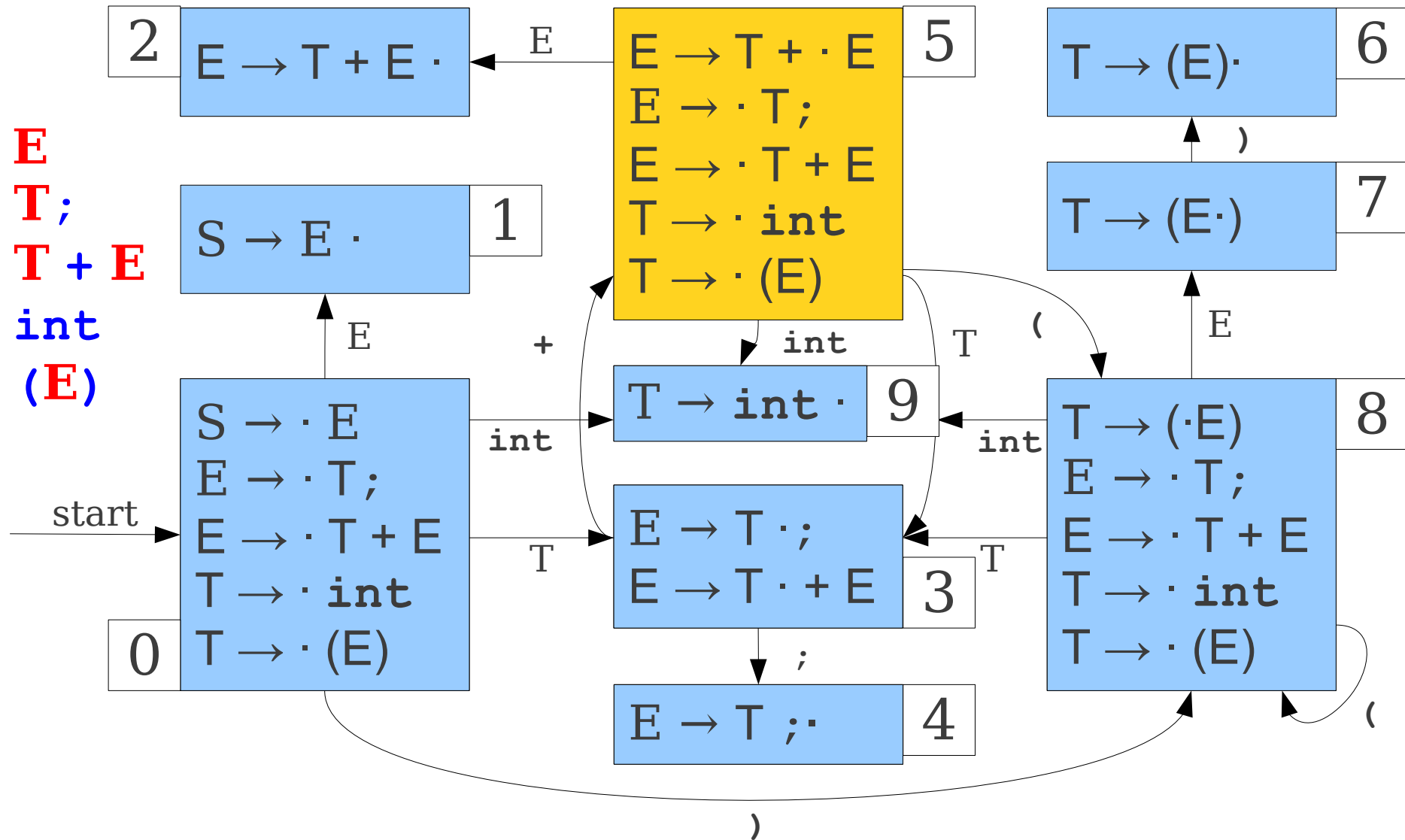
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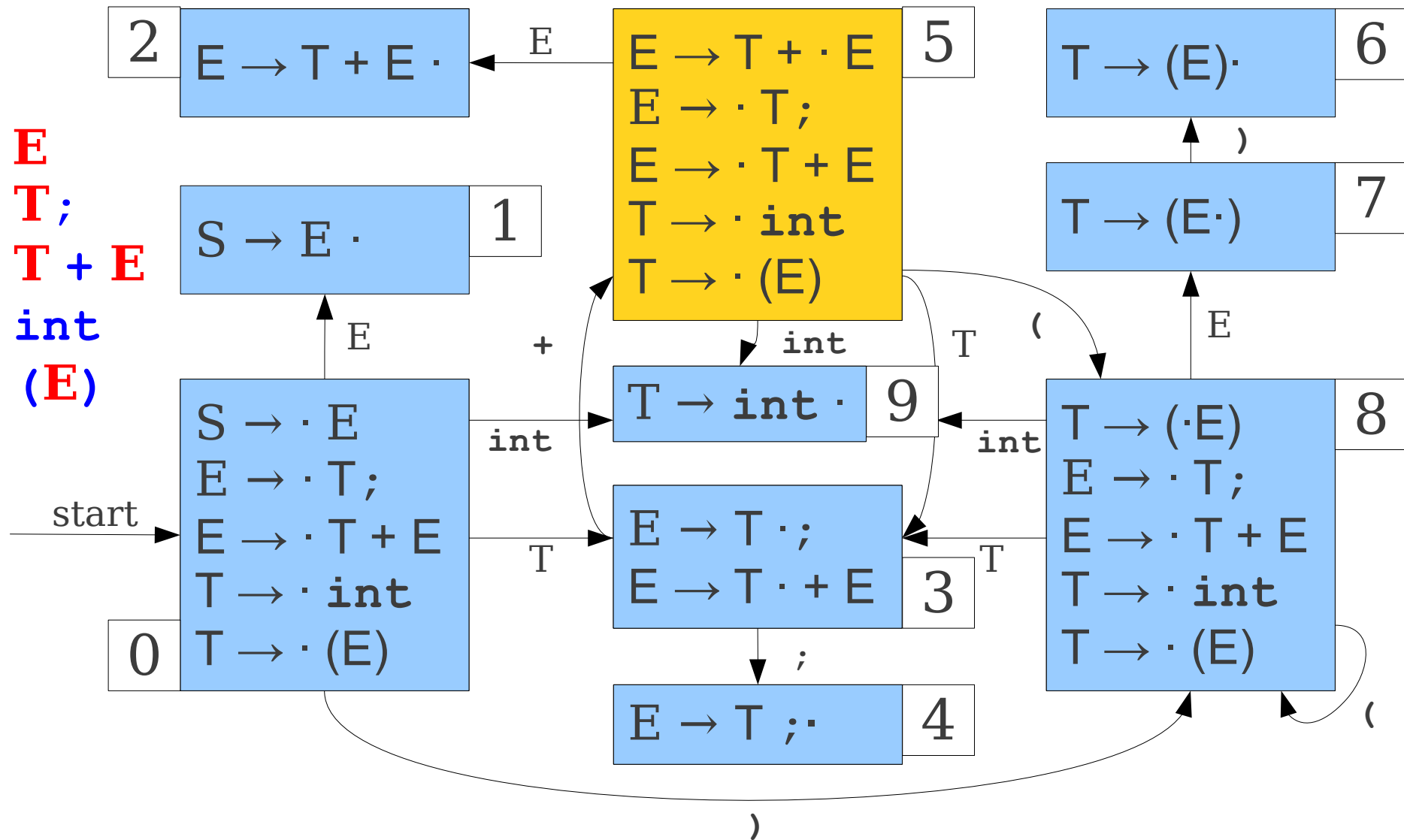
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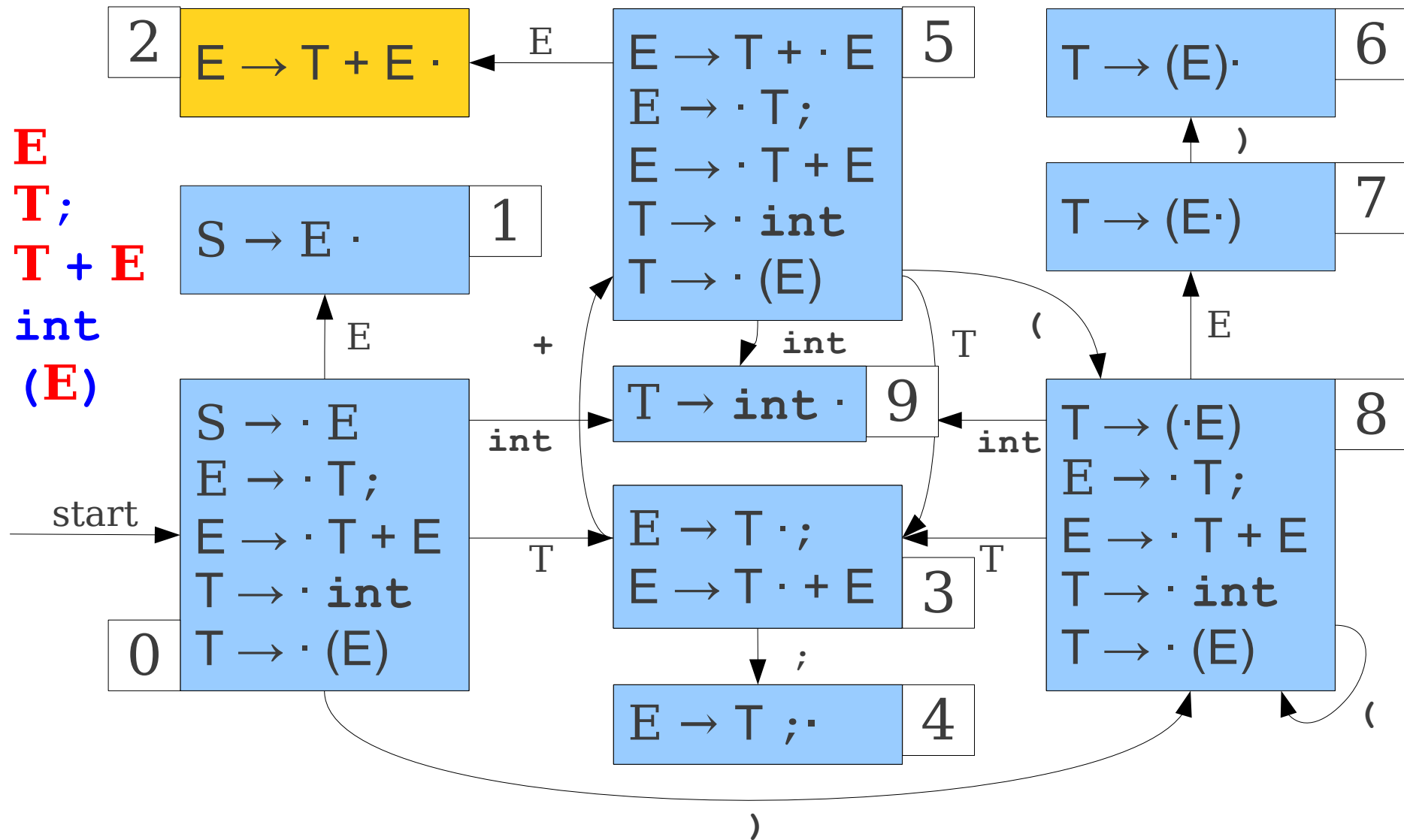
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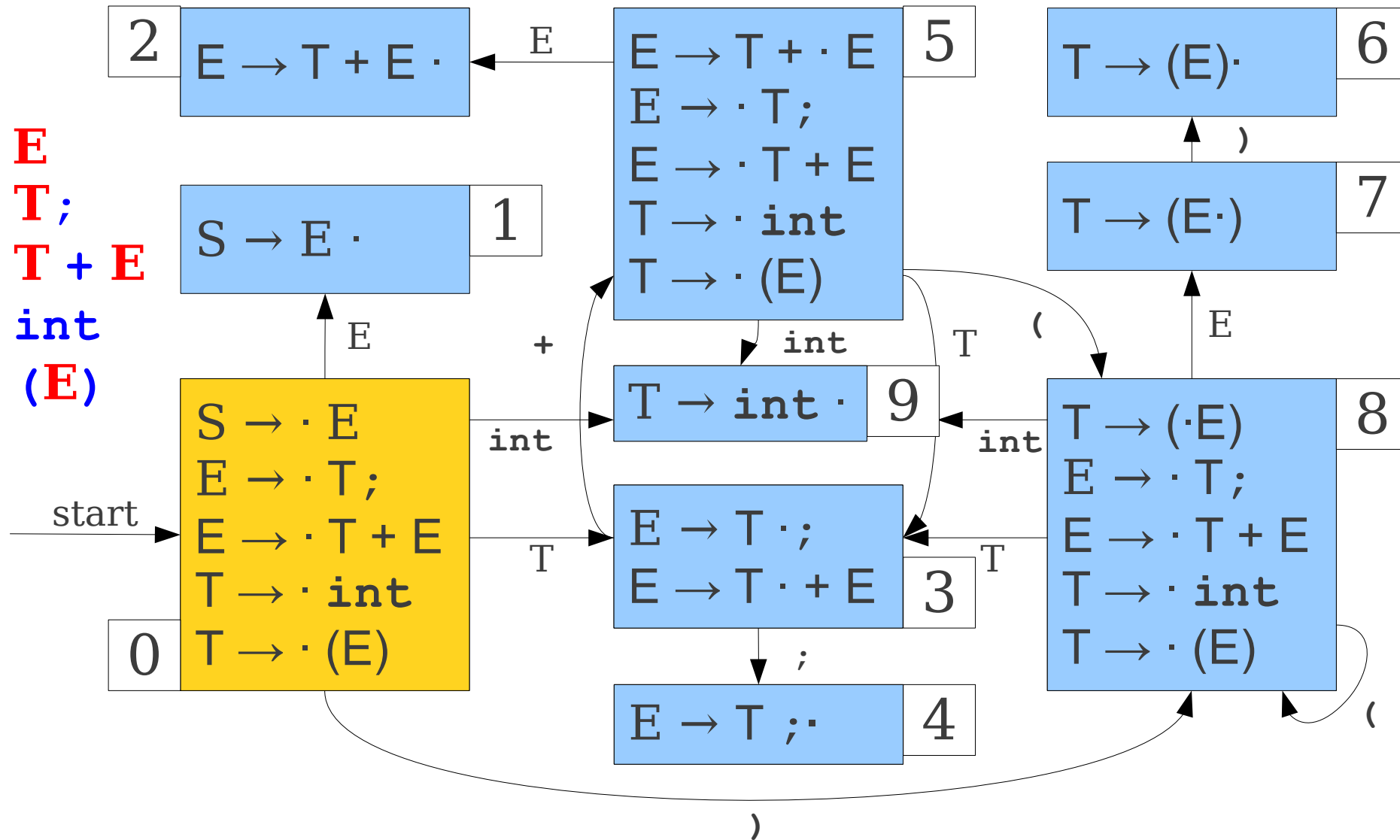
LR(0) Parsing

$S \rightarrow E$
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 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



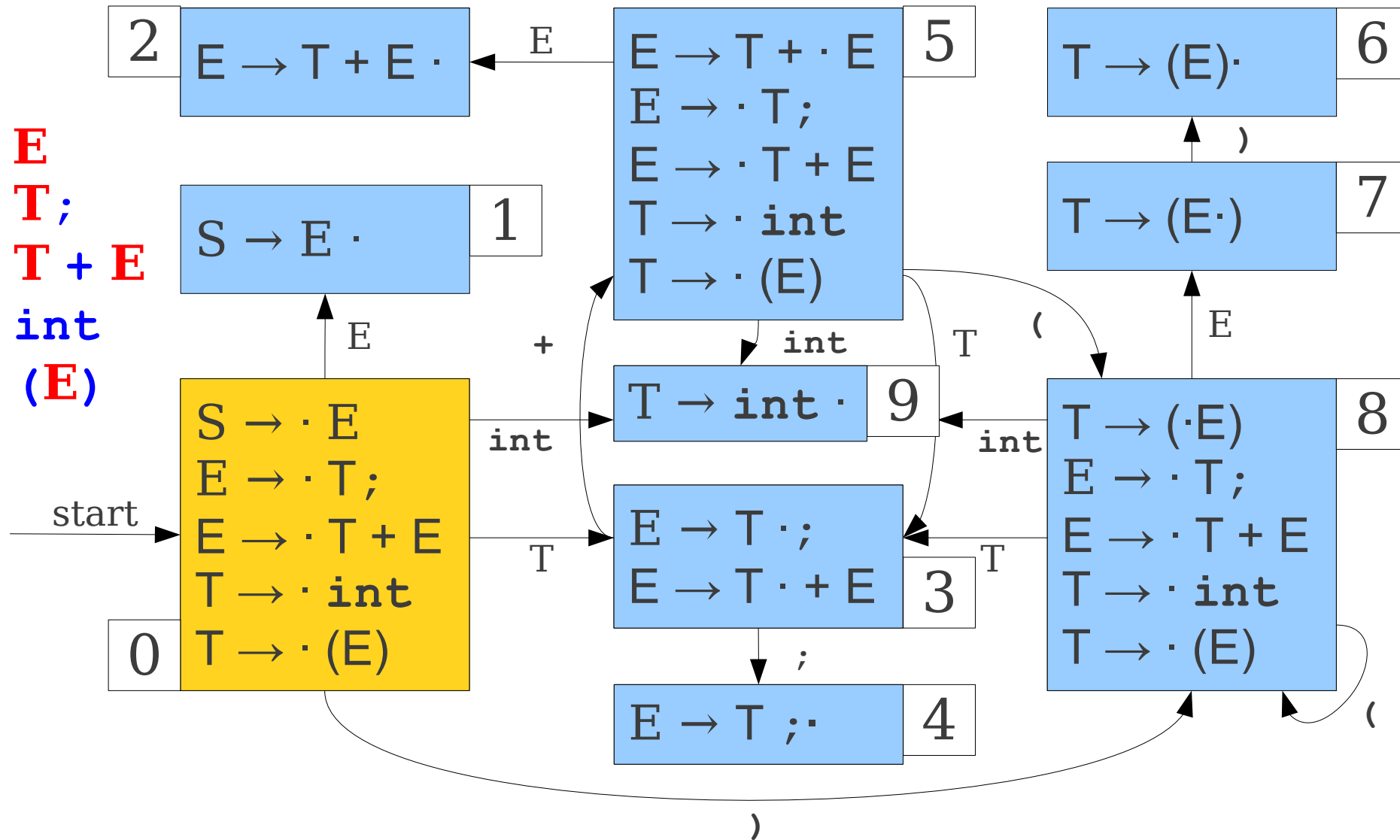
LR(0) Parsing

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 $E \rightarrow T + E$
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LR(0) Parsing

$S \rightarrow E$
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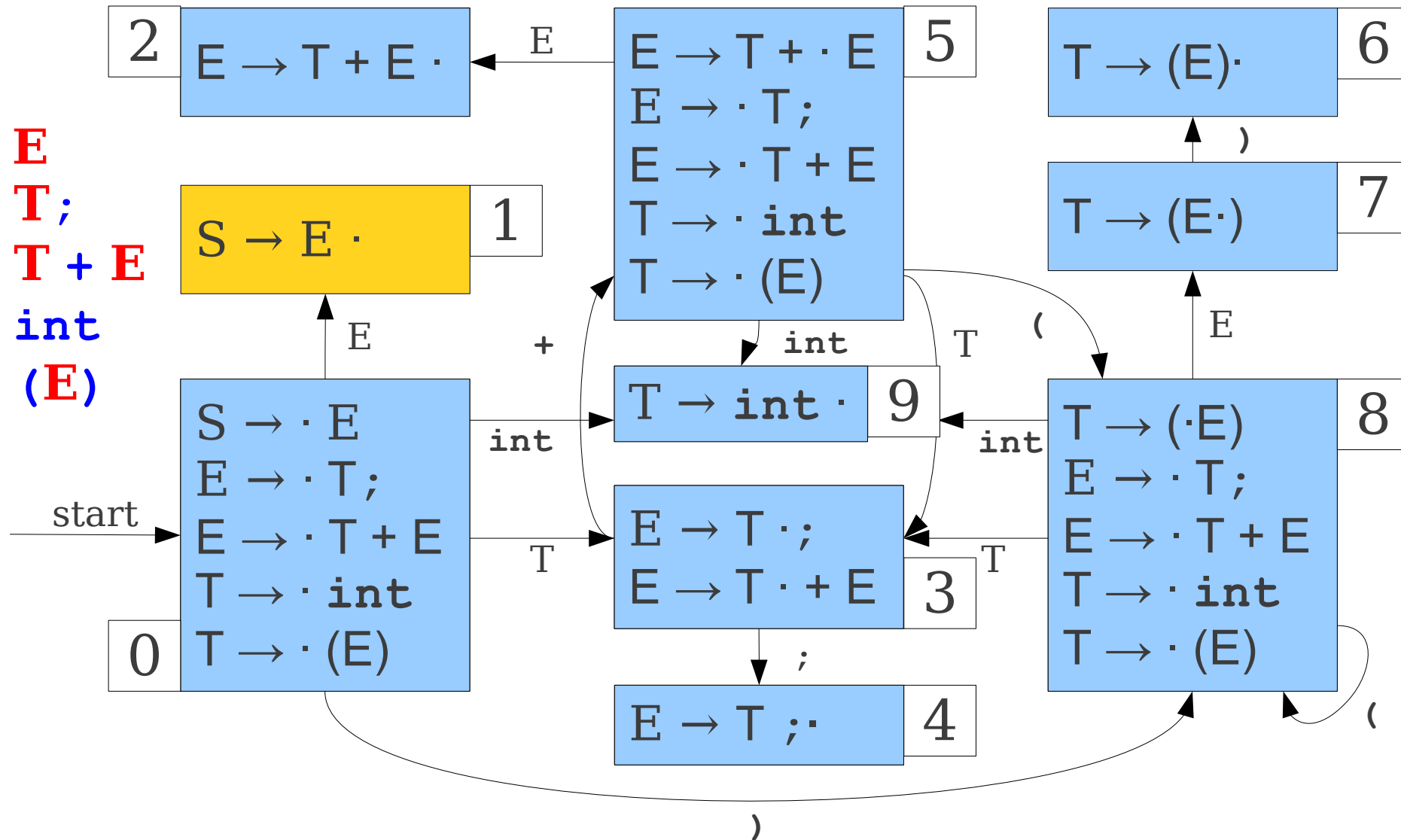


E

0

LR(0) Parsing

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



E

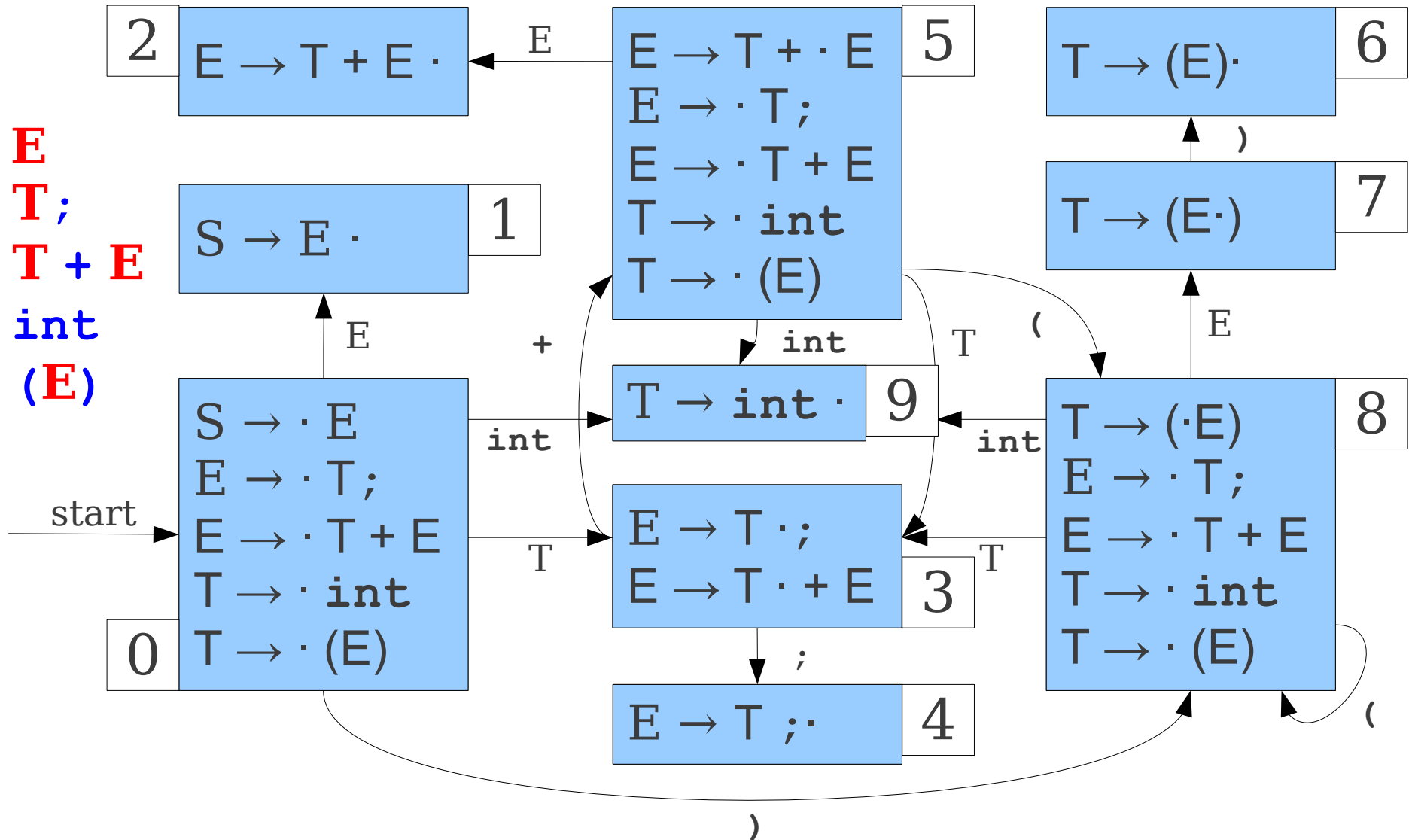
0

Representing the Automaton

- LR(0) parsers are usually represented via two tables: an **action** table and a **goto** table.
- The **action** table maps each state to an action:
 - **shift**, which shifts the next terminal, and
 - **reduce** $A \rightarrow \omega$, which performs reduction $A \rightarrow \omega$.
 - Any state of the form $A \rightarrow \omega \cdot$ does that reduction; everything else shifts.
- The **goto** table maps state/symbol pairs to a next state.
 - This is just the transition table for the automaton.

Building LR(0) Tables

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



LR(0) Tables

	int	+	;	(	)	E	T	Action
0	9			8		1	3	Shift
1								Accept
2								Reduce E → T + E
3		5	4					Shift
4								Reduce E → T ;
5	9			8		2	3	Shift
6								Reduce T → (E)
7					6			Shift
8	9			8		7	3	Shift
9								Reduce T → int

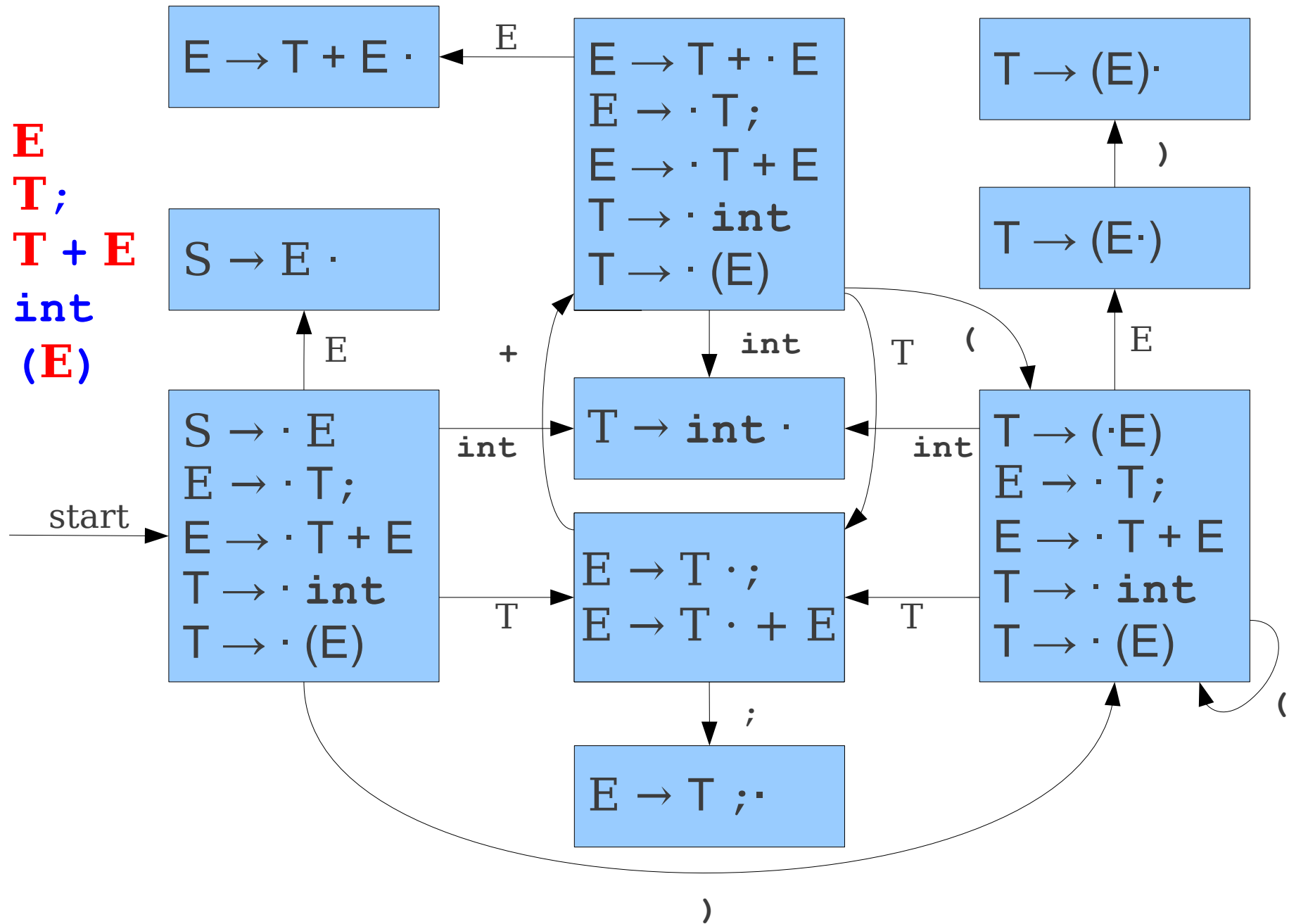
The LR(0) Algorithm

- Maintain a stack of (symbol, state) pairs, which is initially (ϵ , 1) for some dummy symbol ϵ .
- While the stack is not empty:
 - Let **state** be the top state.
 - If **action[state]** is **shift**:
 - Let t be the next symbol in the input.
 - Push (t , **goto[state, t]**) atop the stack.
 - If **action[state]** is **reduce** $A \rightarrow \omega$:
 - Remove $|\omega|$ symbols from the top of the stack.
 - Let **top-state** be the state on top of the stack.
 - Push (A , **goto[top-state, A]**) atop the stack.
 - Otherwise, report an error.

The Limits of LR(0)

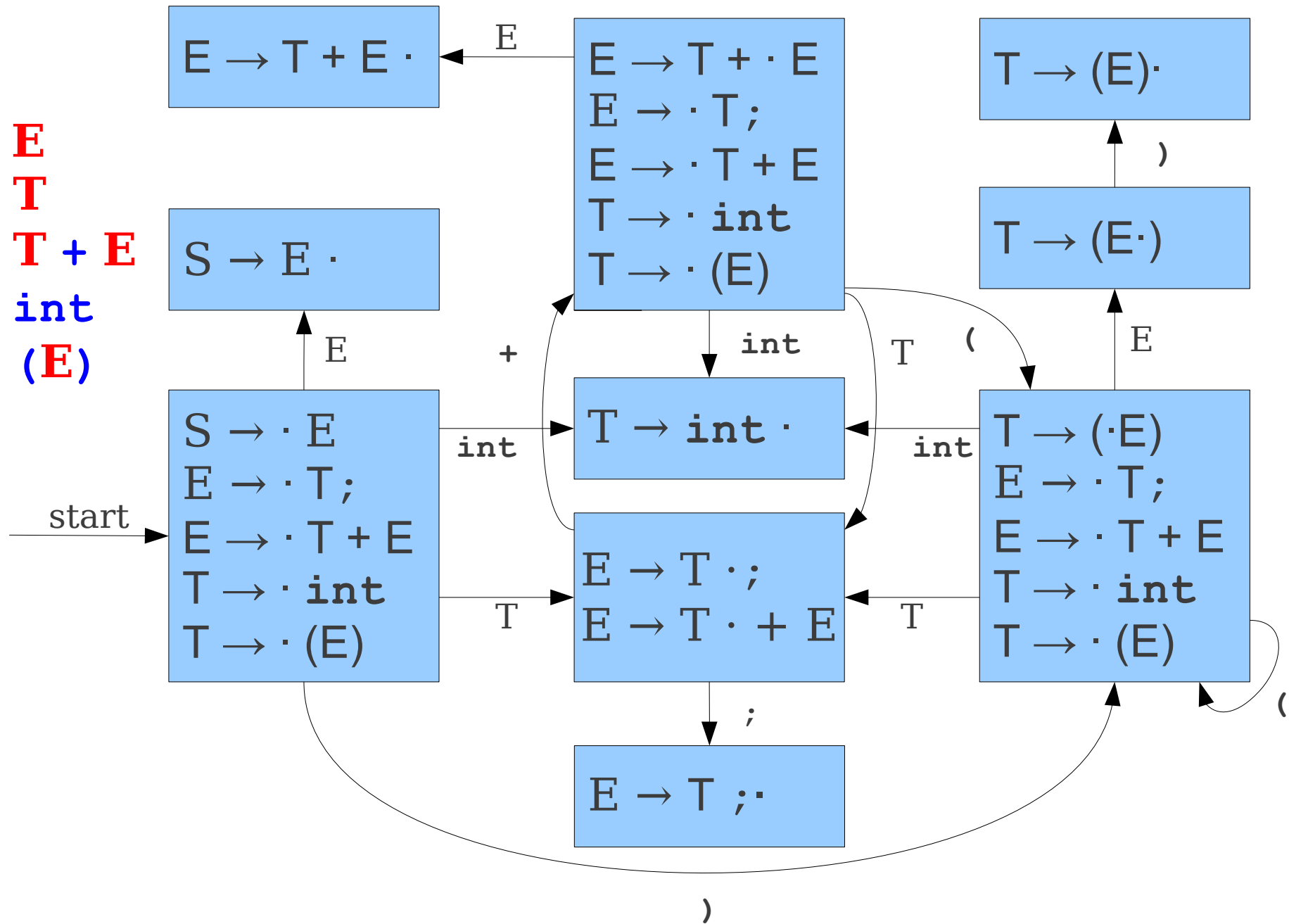
A Non-LR(0) Grammar

$S \rightarrow E$
 $E \rightarrow T;$
 $E \rightarrow T + E$
 $T \rightarrow \text{int}$
 $T \rightarrow (E)$



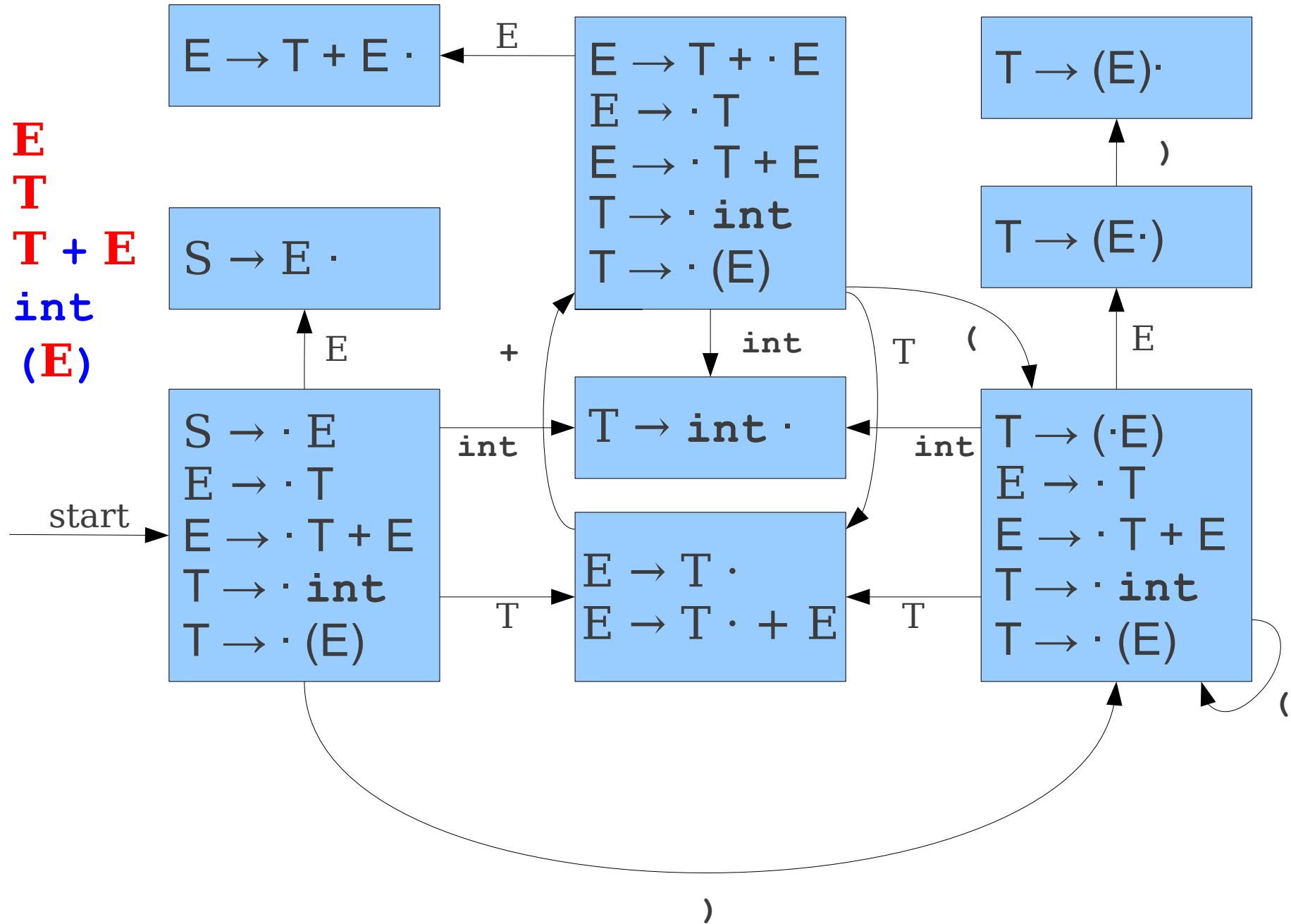
A Non-LR(0) Grammar

S → **E**
E → **T**
E → **T** + **E**
T → **int**
T → (**E**)



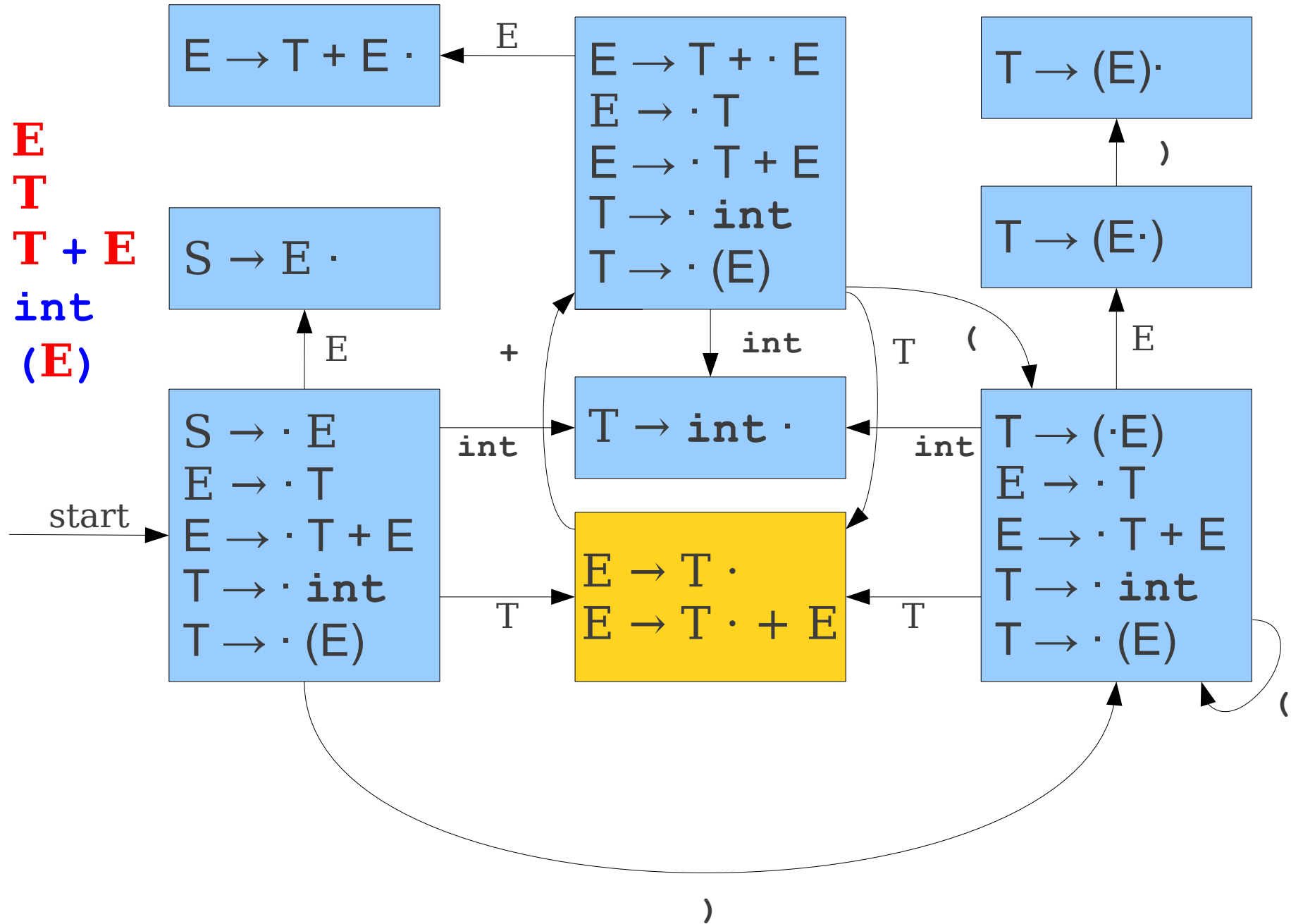
A Non-LR(0) Grammar

S → **E**
E → **T**
E → **T** + **E**
T → **int**
T → (**E**)



A Non-LR(0) Grammar

S → **E**
E → **T**
E → **T** + **E**
T → **int**
T → (**E**)

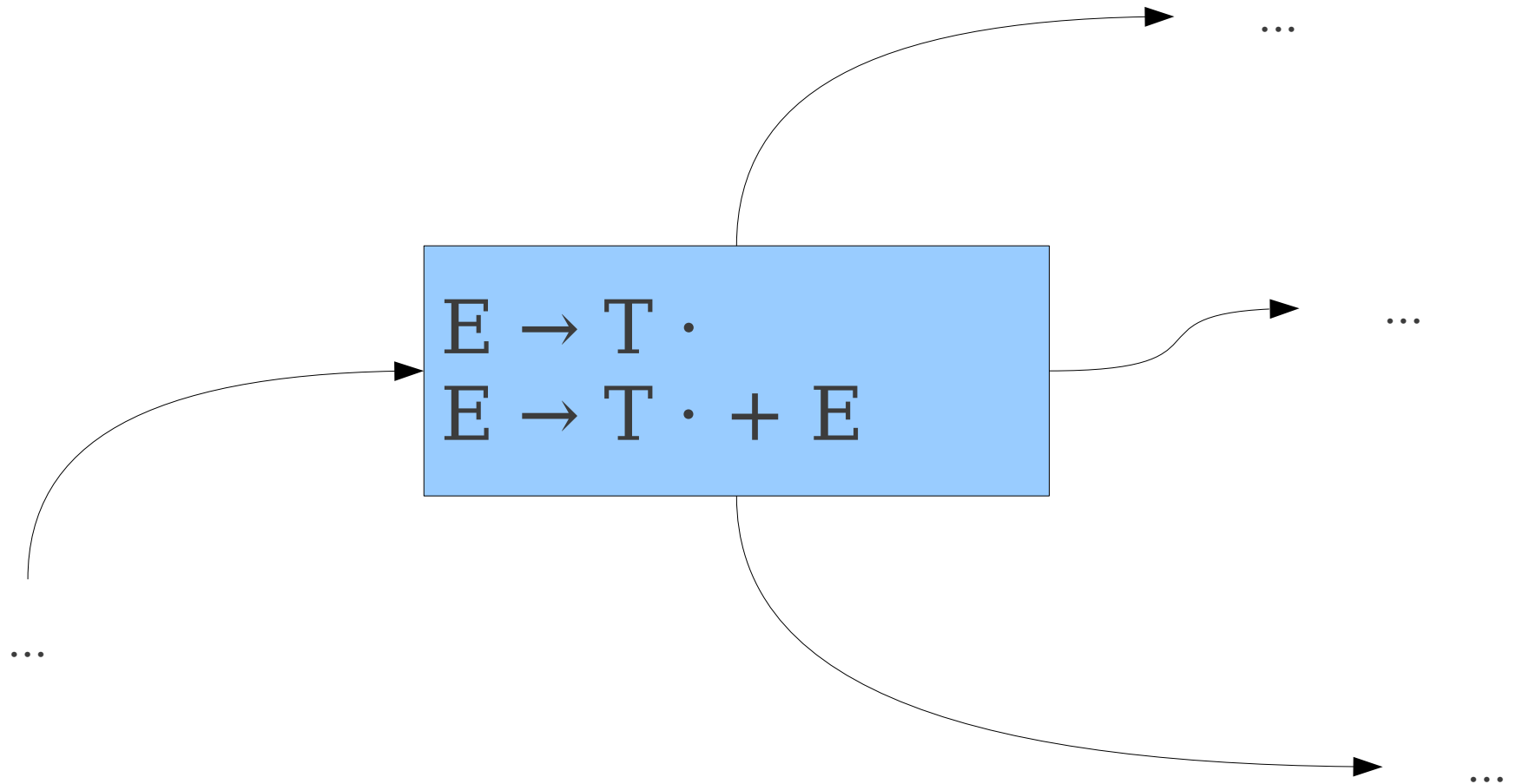


LR Conflicts

- A **shift/reduce conflict** is an error where a shift/reduce parser cannot tell whether to shift a token or perform a reduction.
 - Often happens when two productions overlap.
- A **reduce/reduce conflict** is an error where a shift/reduce parser cannot tell which of many reductions to perform.
 - Often the result of ambiguous grammars.
- A grammar whose handle-finding automaton contains a shift/reduce conflict or a reduce/reduce conflict is not LR(0).
- Can you have a shift/shift conflict?

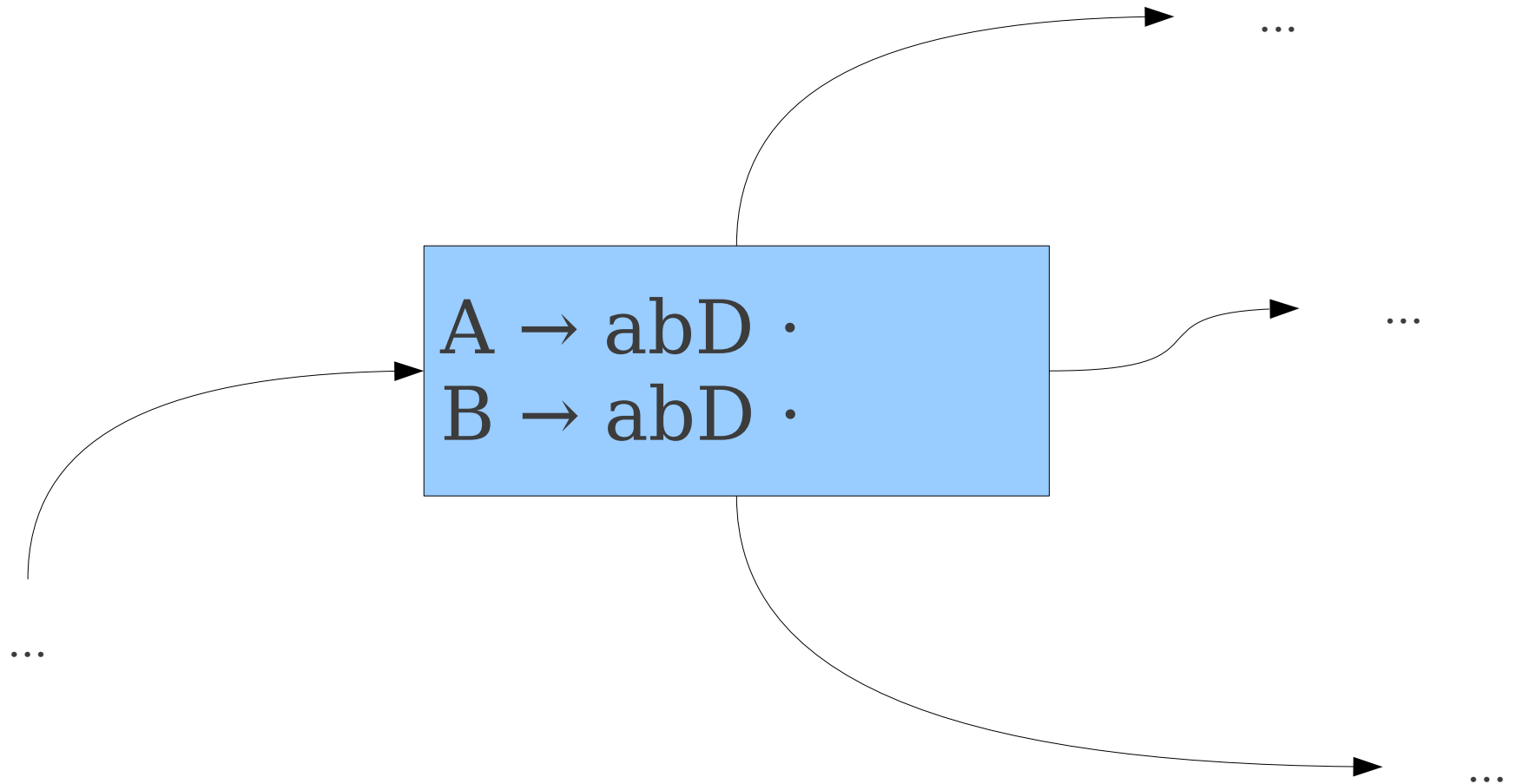
What error is this?

What error is this?



What about this?

What about this?



What do these conflicts mean?

- Recall: our automaton was constructed by looking for viable prefixes.
- Each accepting state represents a point where the handle might occur.
- A **shift/reduce** conflict is a state where the handle might occur, but we might actually need to keep searching.
- A **reduce/reduce** conflict is a state where we know we have found the handle, but can't tell which reduction to apply.

Next Time

- Add Lookahead to parse a more powerful subset of CFGs than LR(0).

